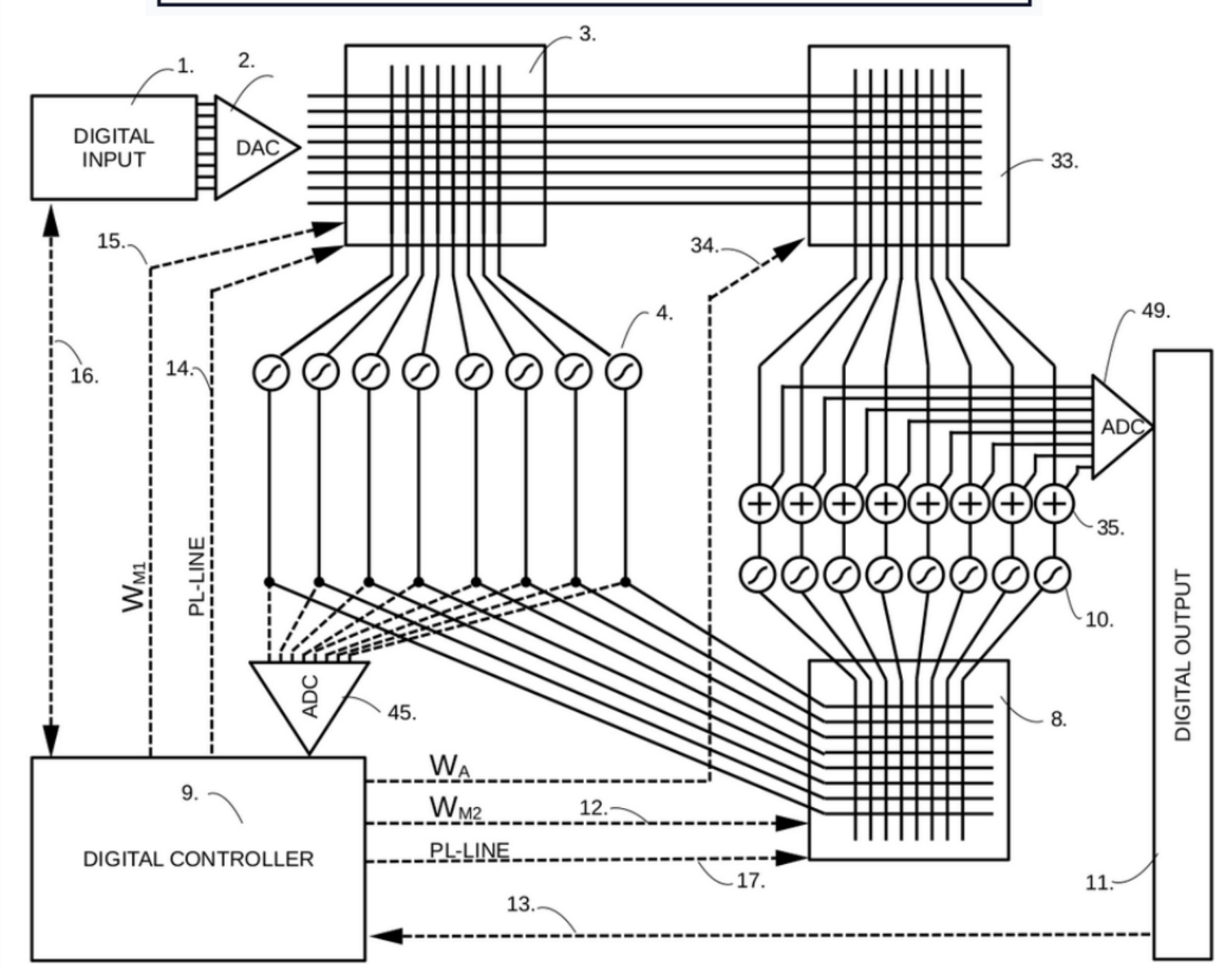
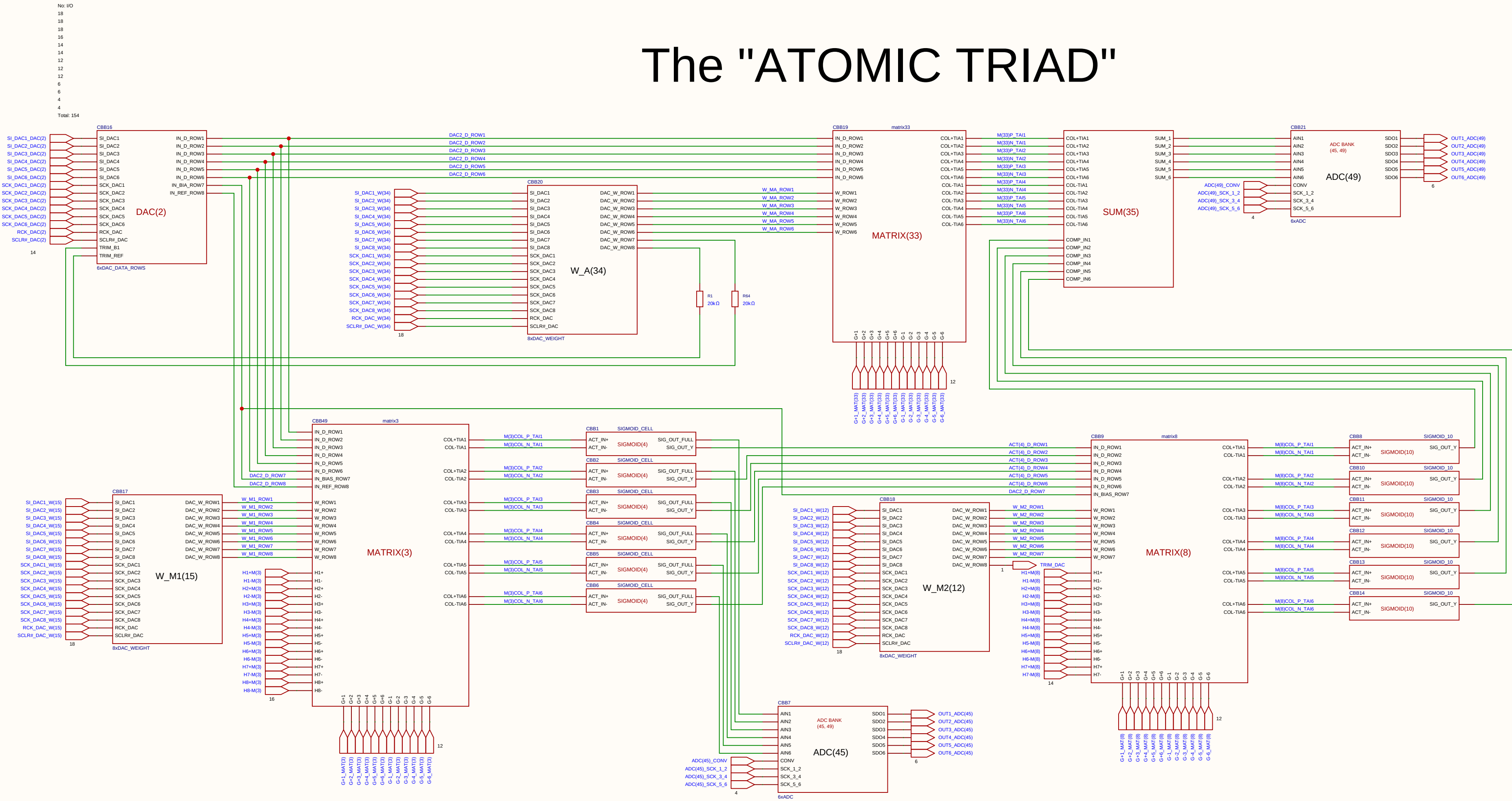


TITLE: INVERTED MAML FOR ANALOG IMC MESH
PATENT APP NO: SE 2630397-4
STATUS: PATENT PENDING
INVENTOR: OLLE WELIN
PRIORITY DATE: 2026-06-03



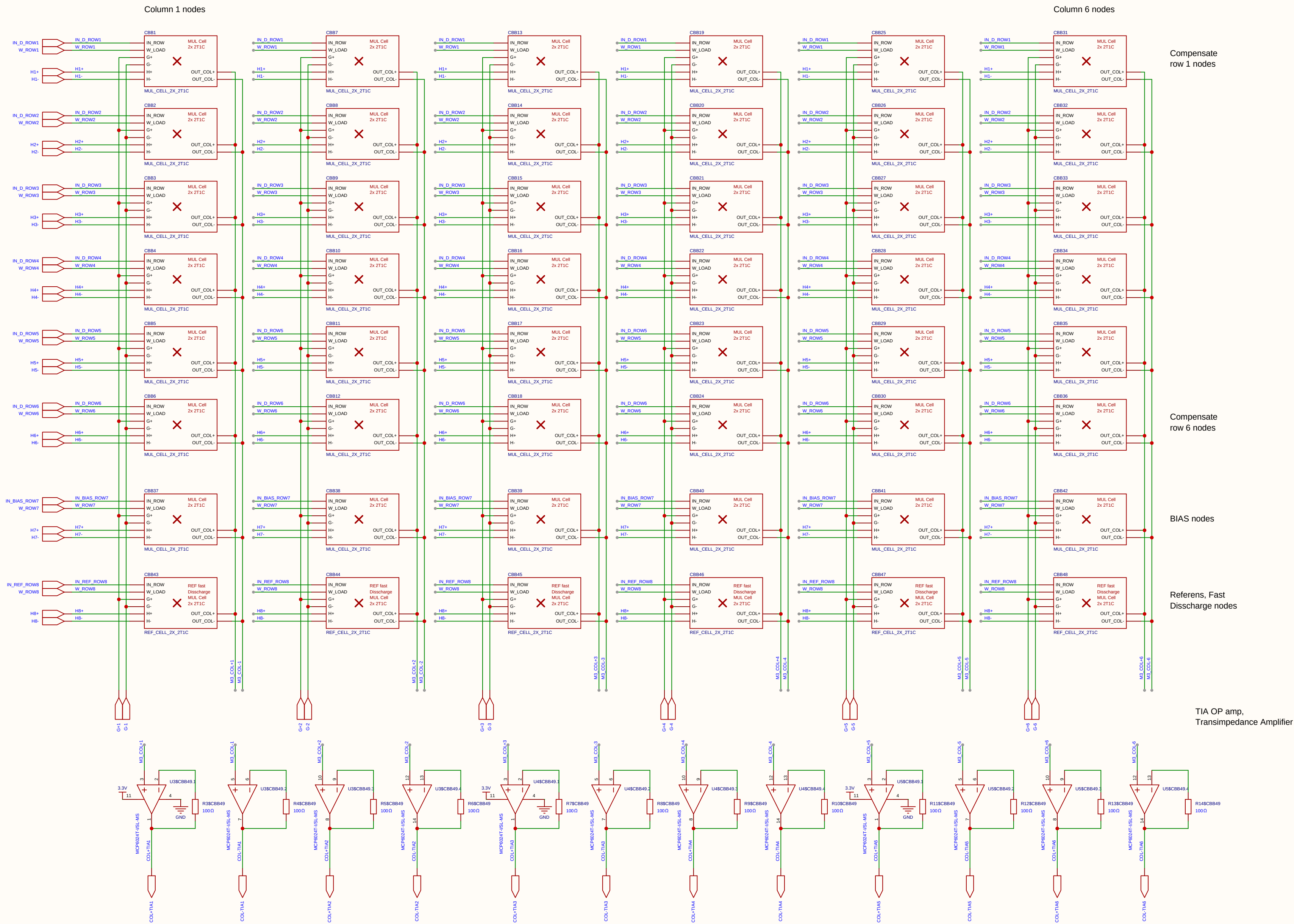
Schematic	Inverted_MAML			Create at	2026-06-19
				Update at	2026-06-19
Board	Analog_Matrix			Page	cover
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 3	
		V1.0	A4	EasyEDA.com	

The "ATOMIC TRIAD"



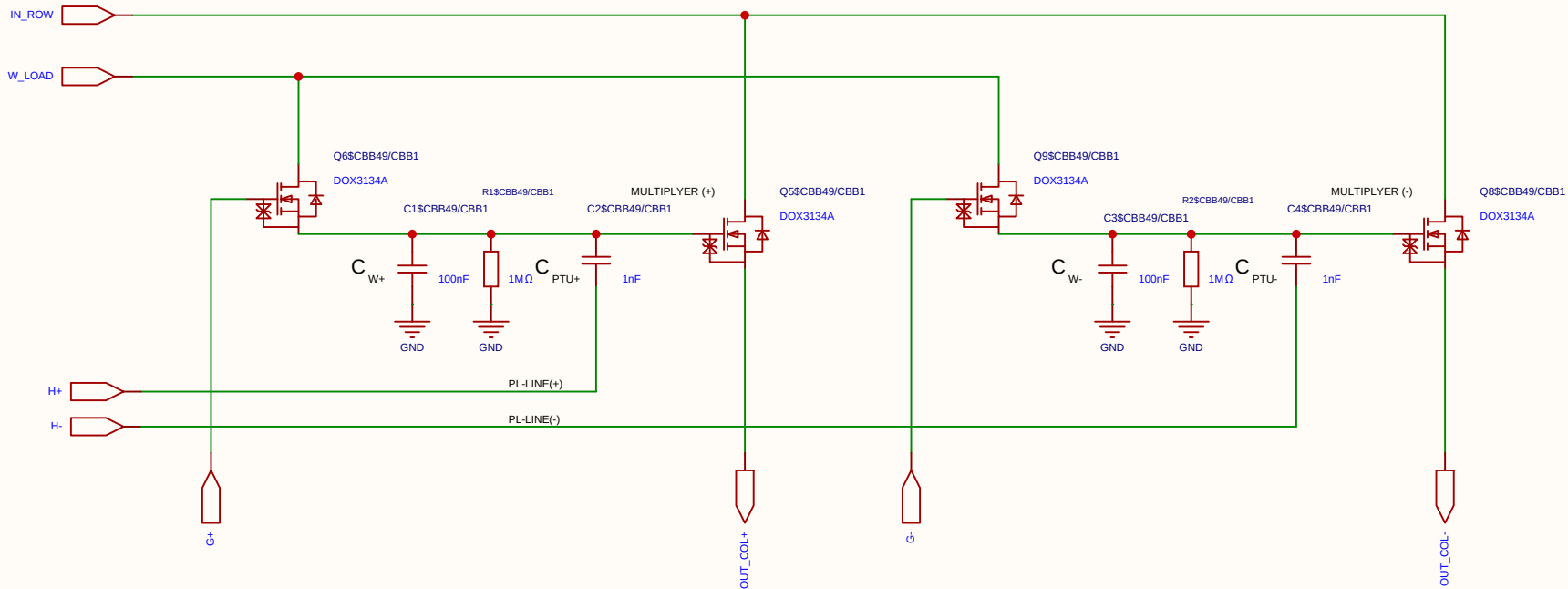
Schematic	Inverted_MAML			Create at	2026-06-25
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Board	Analog_Matrix			Page	Atomic_Triad
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 2 Total 3	
		V1.0	A4	EasyEDA.com	

$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$



Schematic	matrix3			Create at	2026-06-25
				Update at	2026-06-27
Board				Page	matx3
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

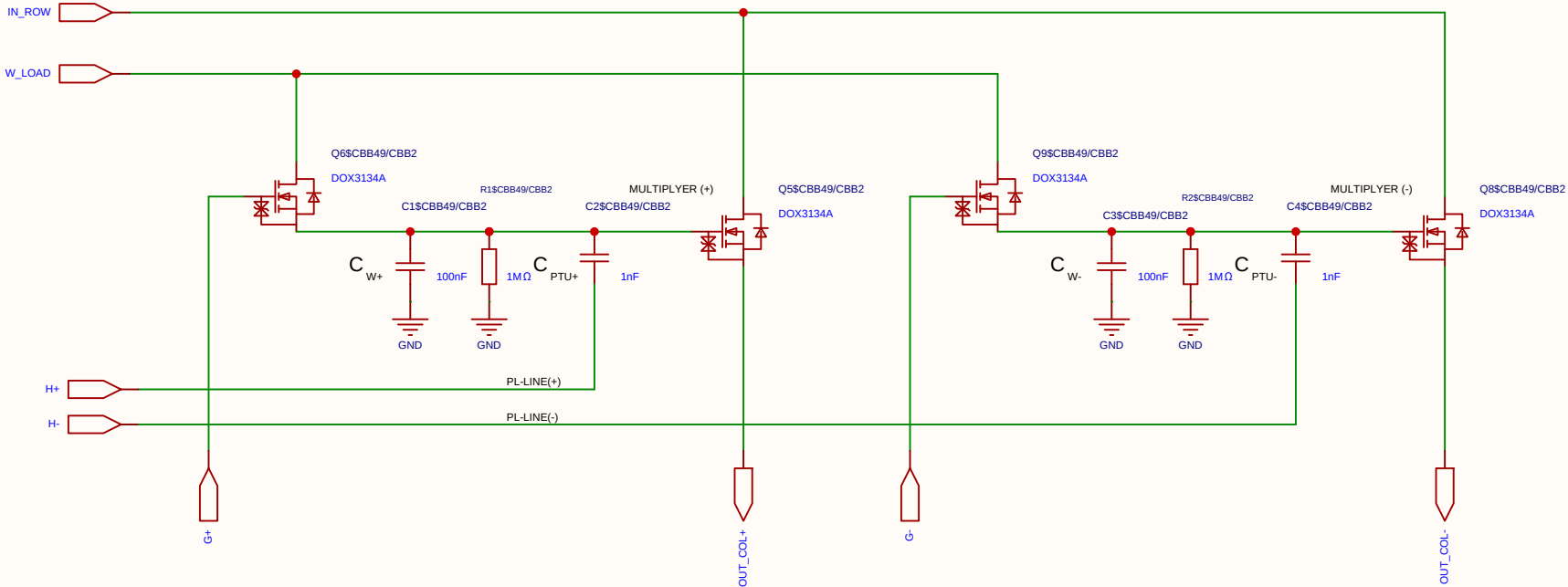
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

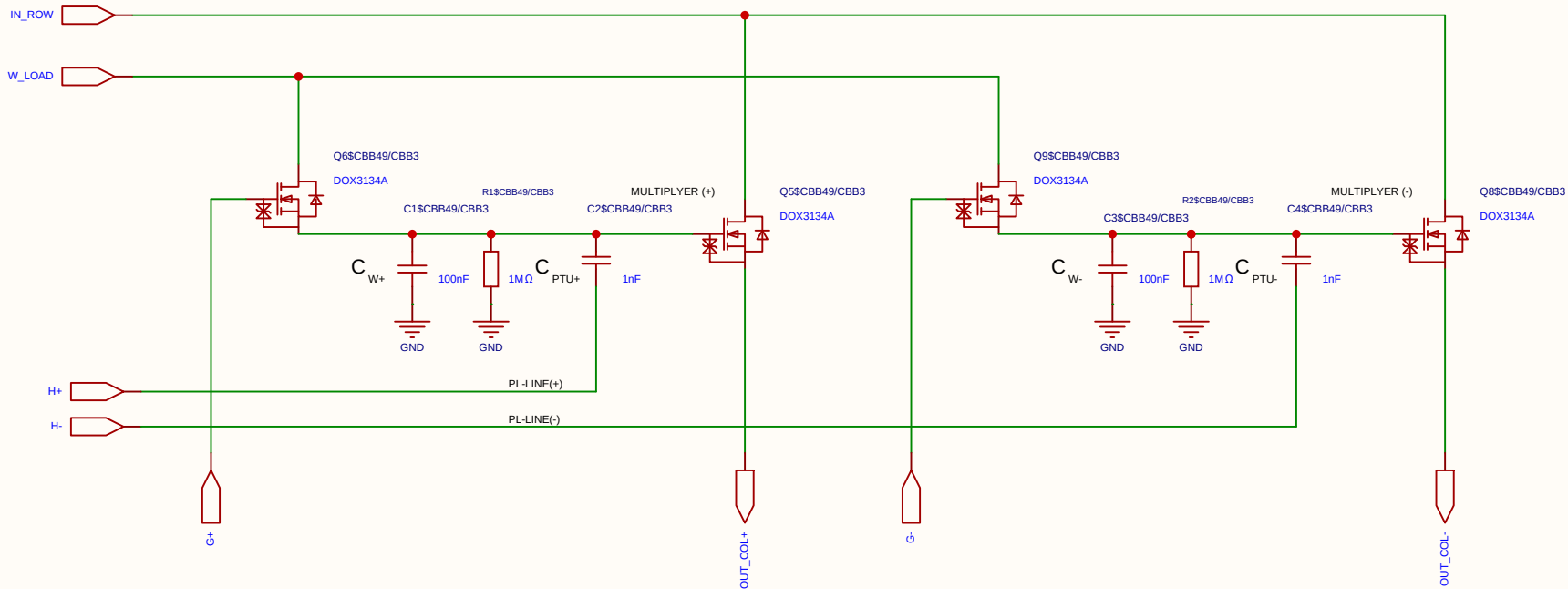
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

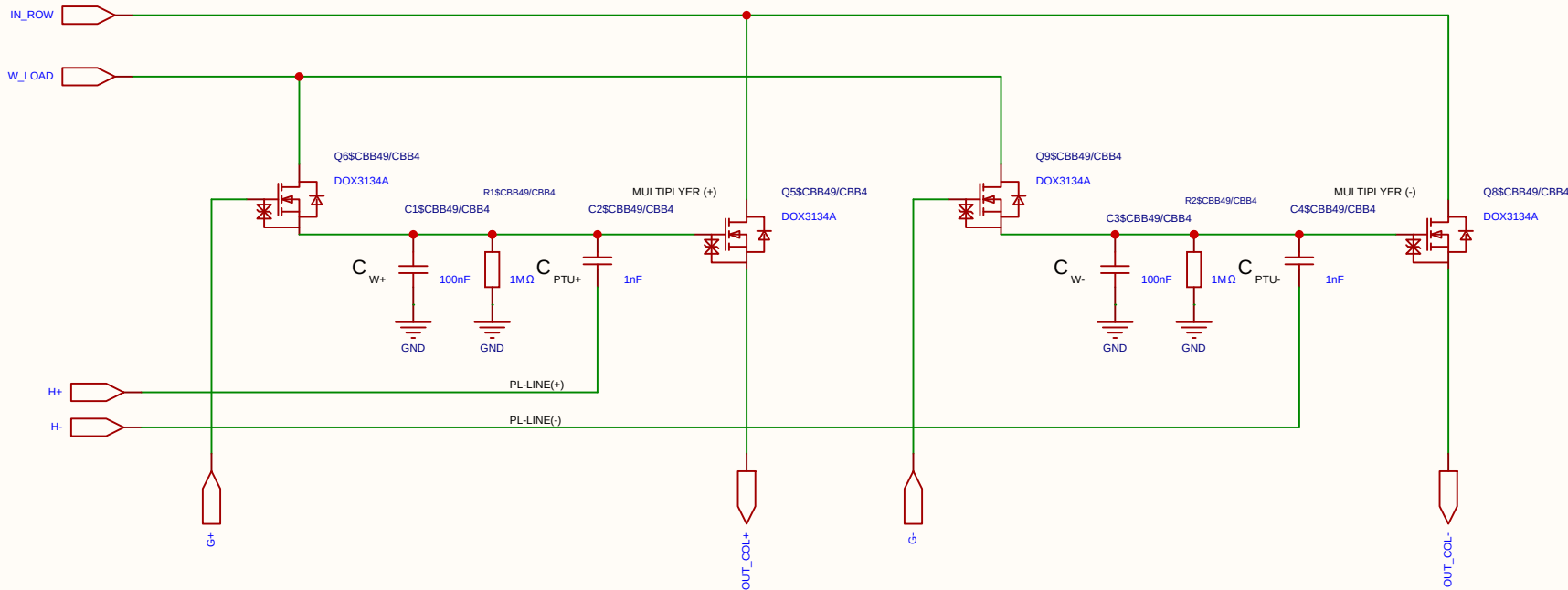
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

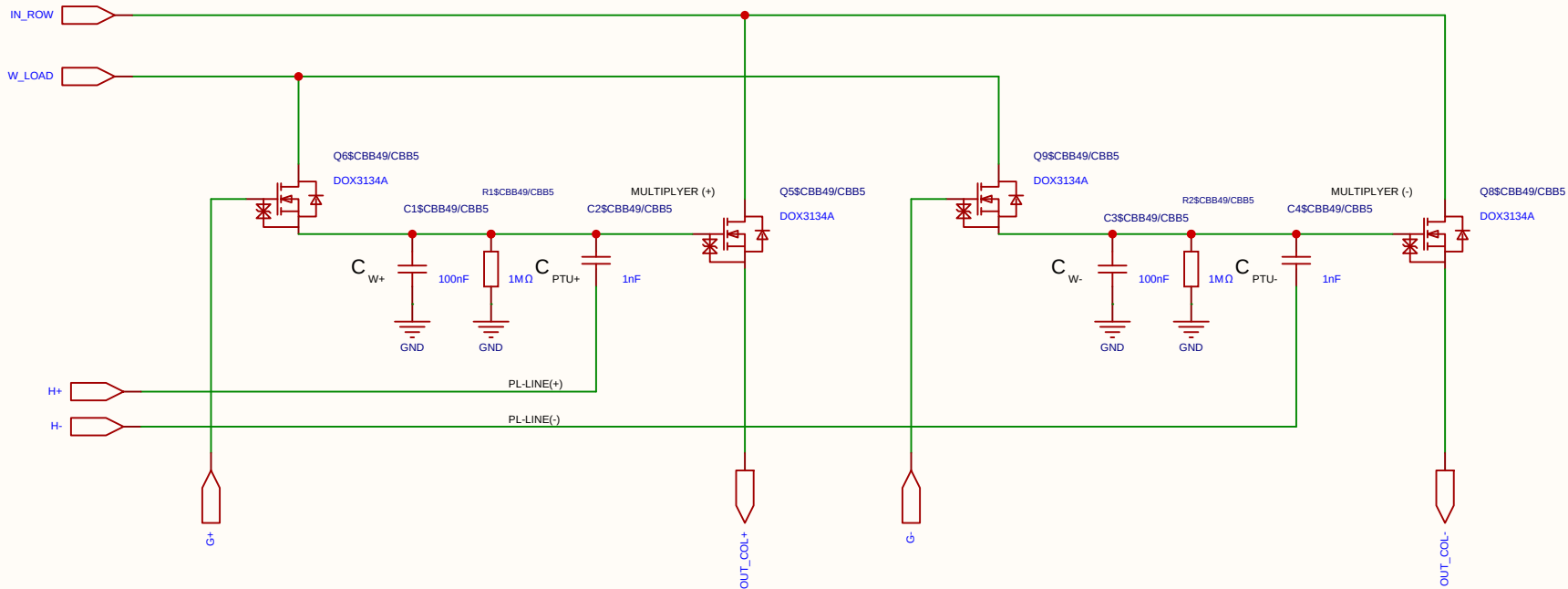
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

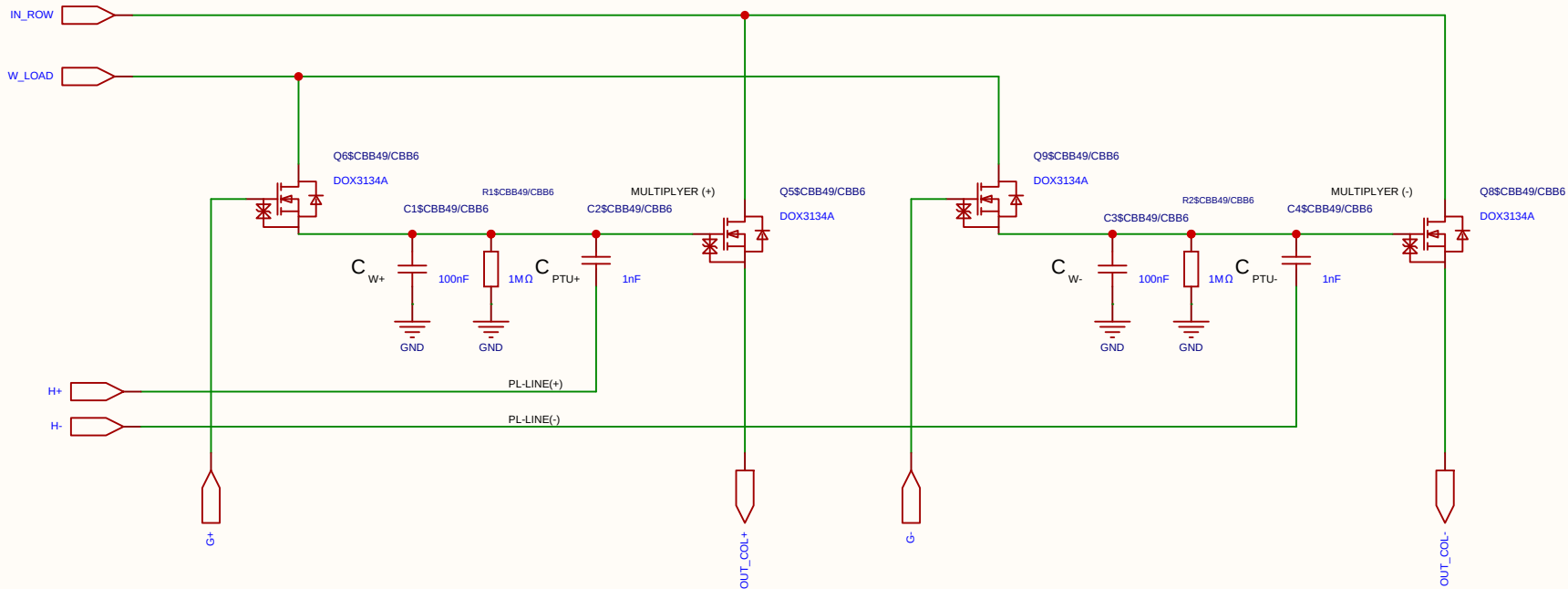
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

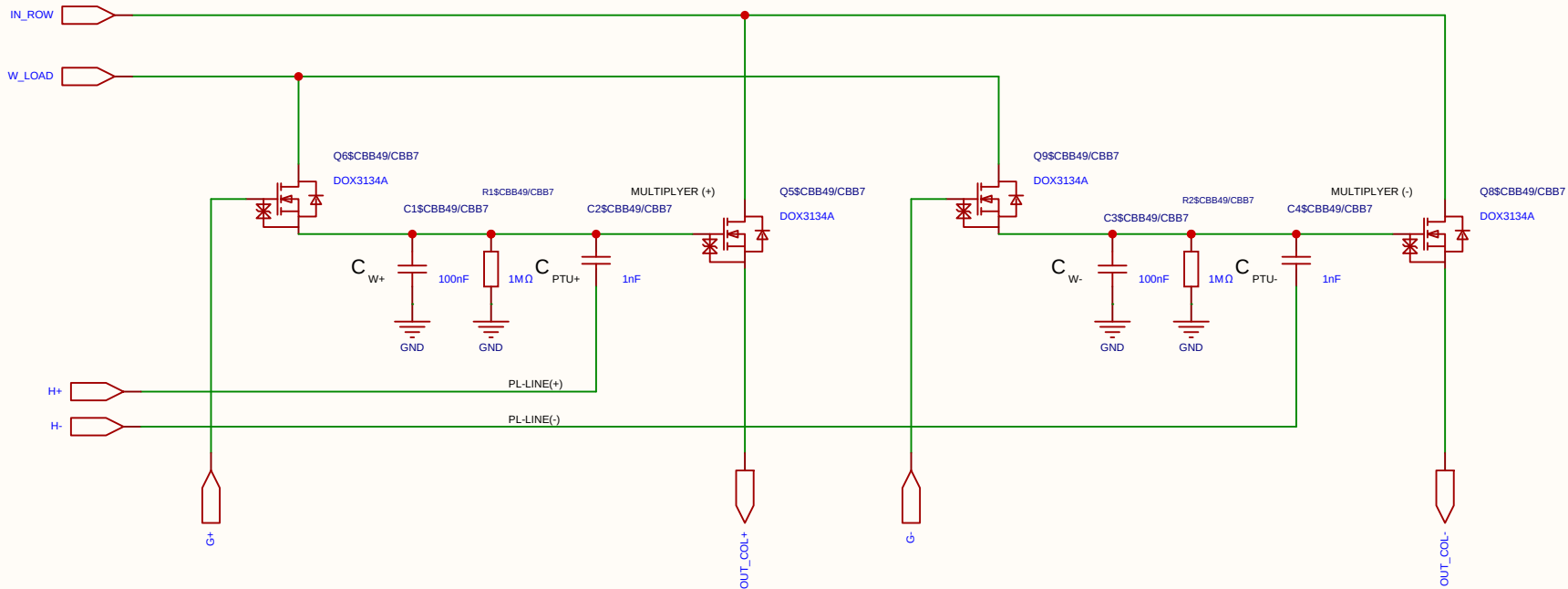
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

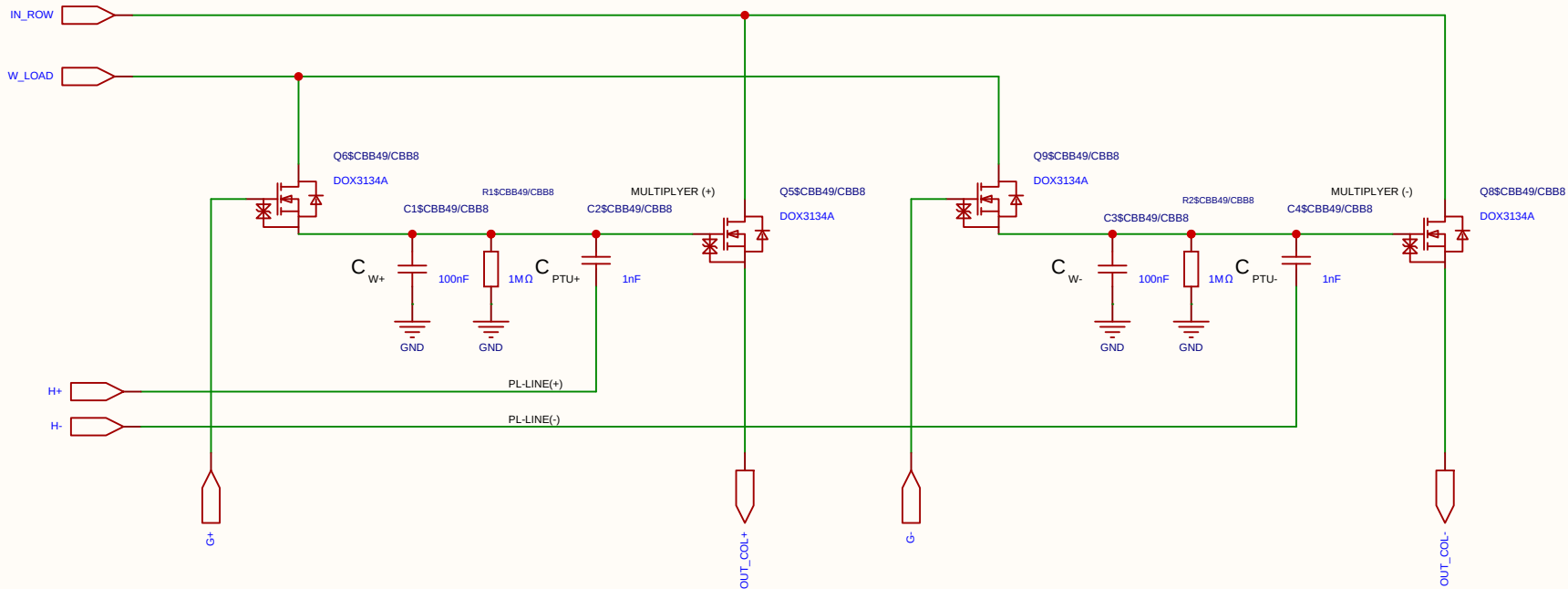
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

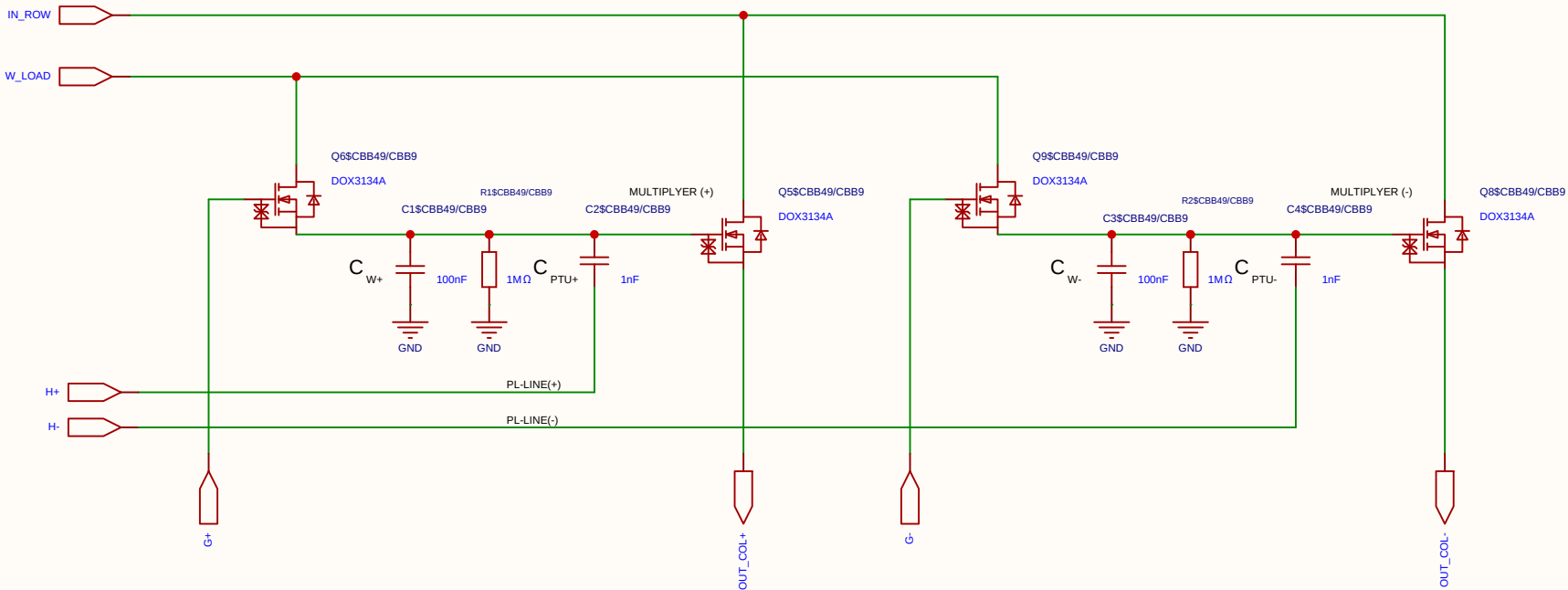
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

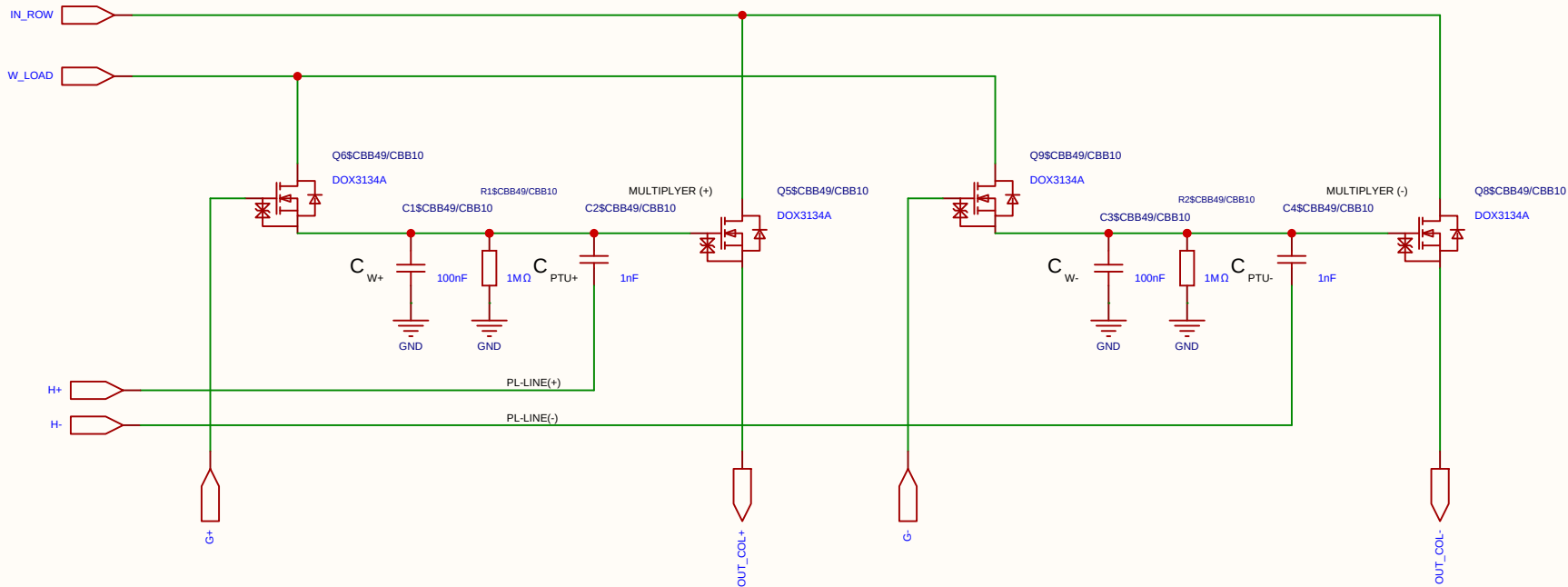
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

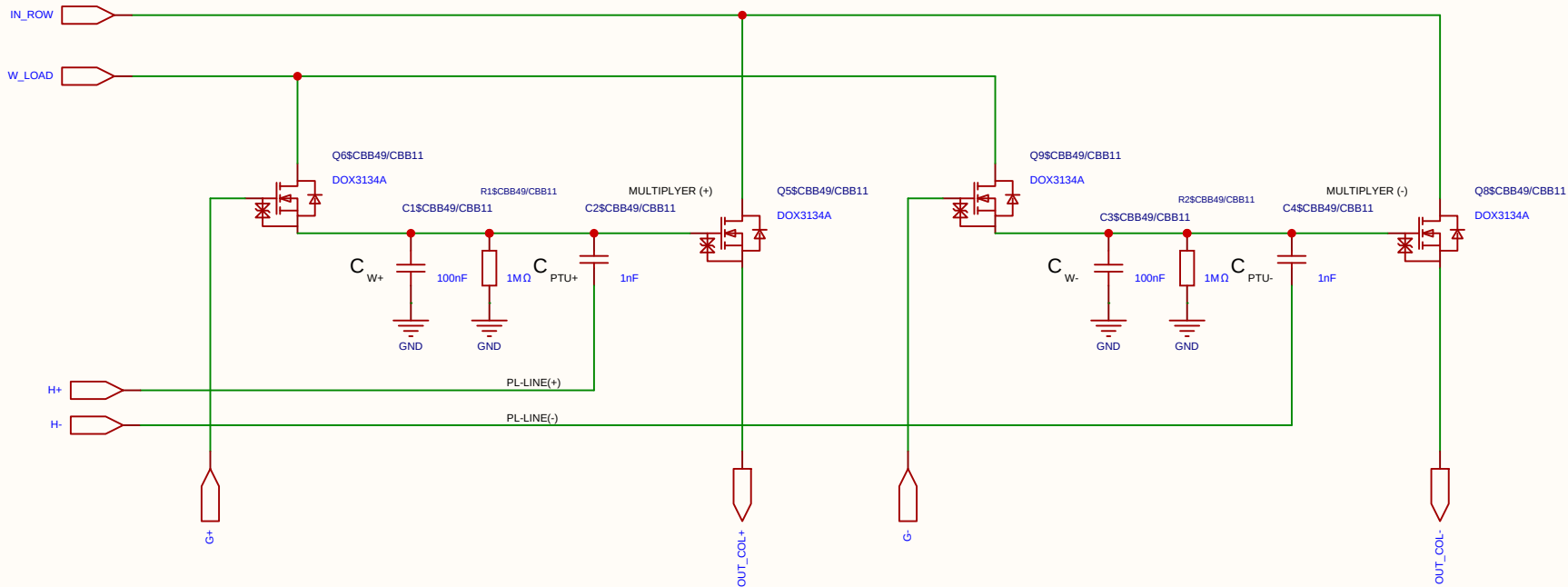
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

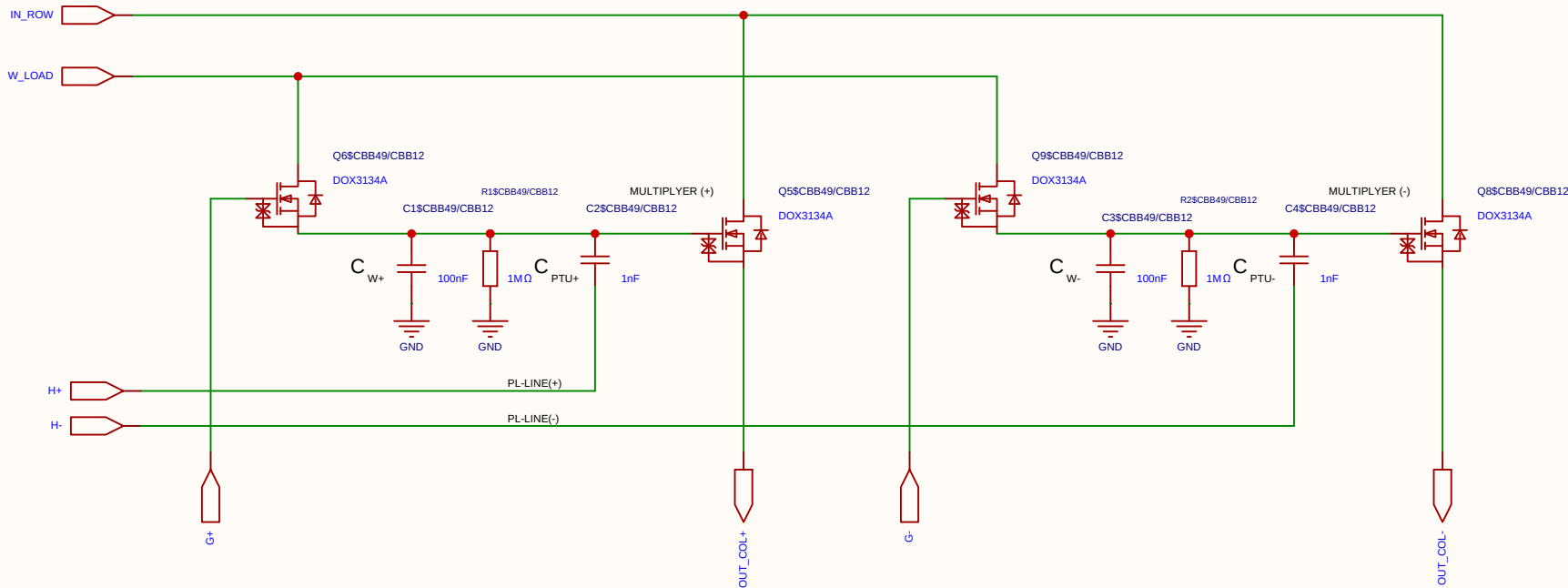
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

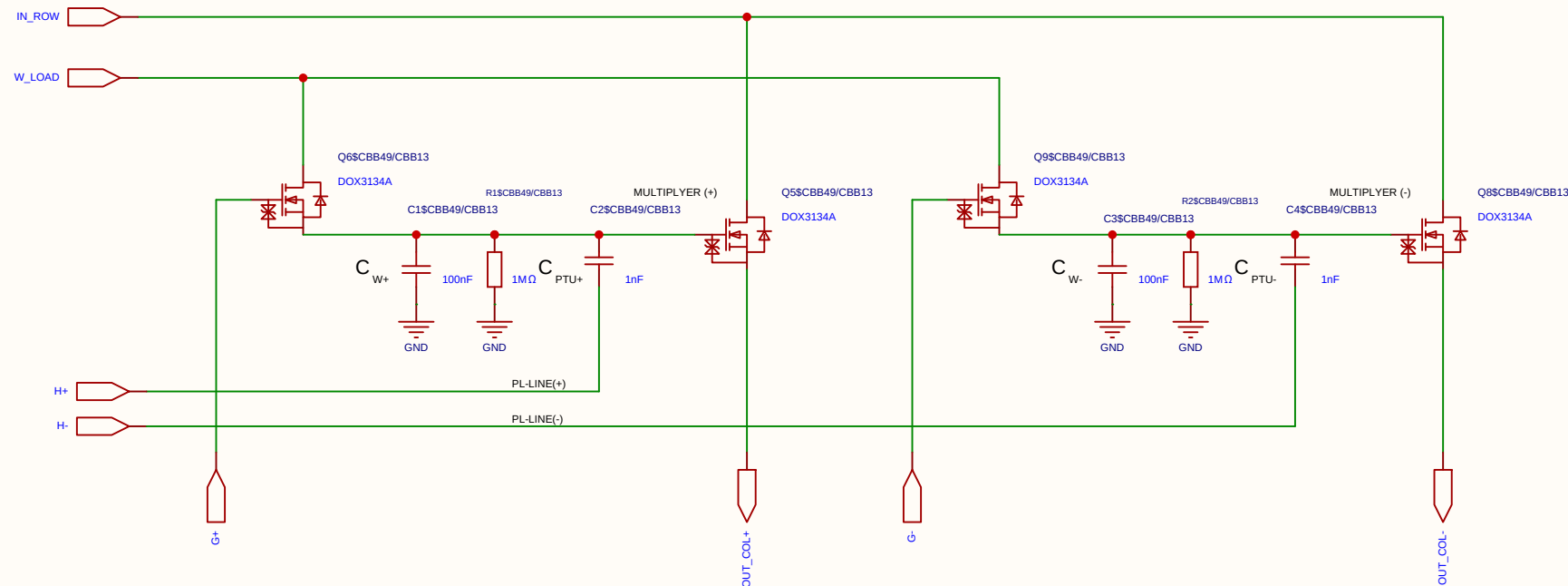
This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel

Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

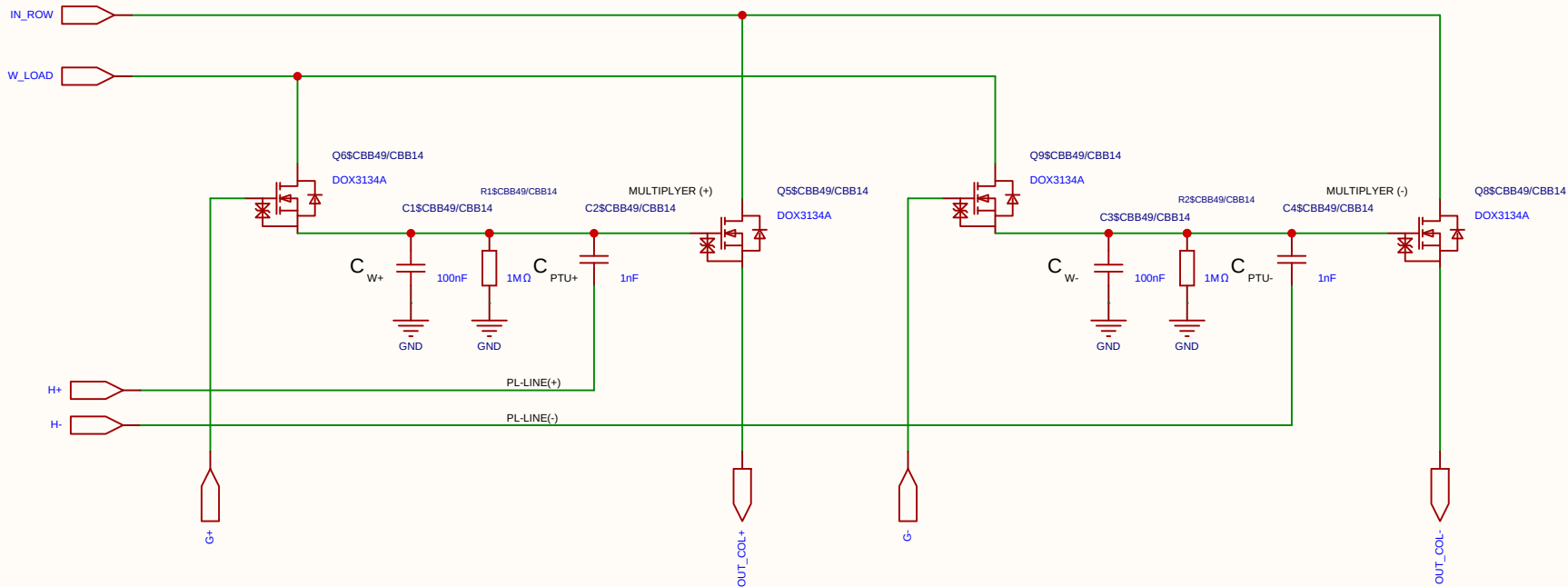
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_I row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_S) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

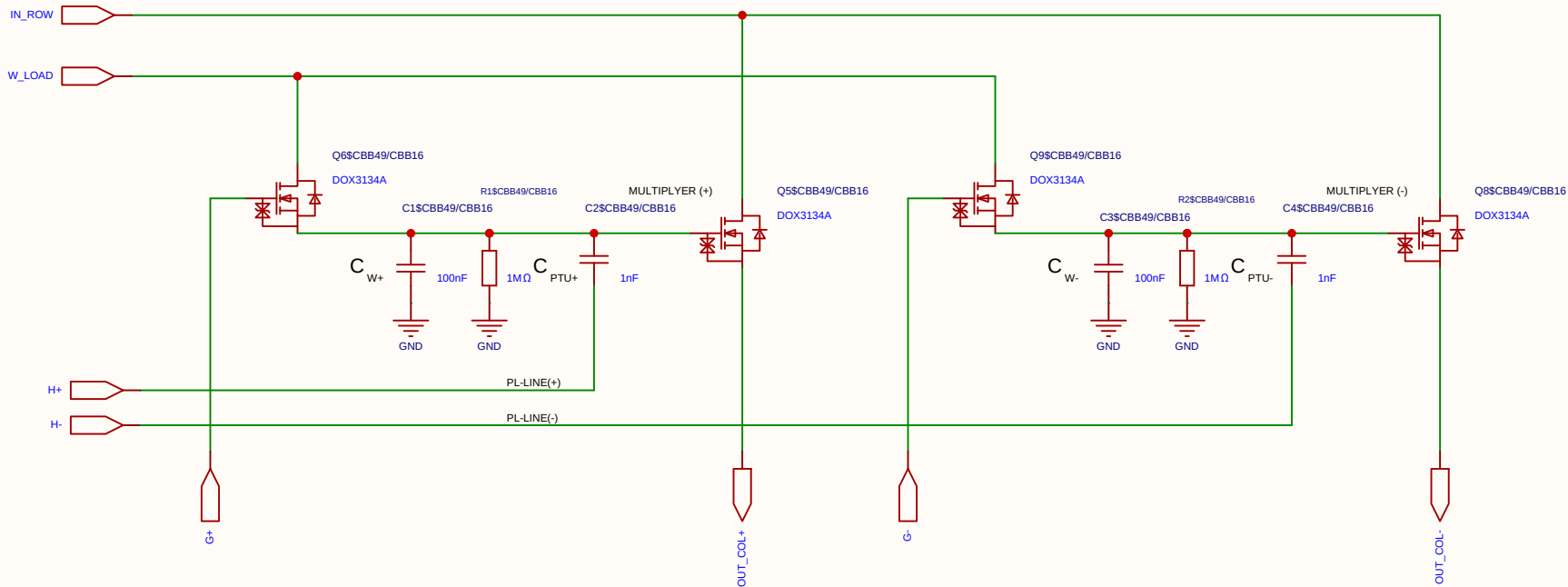
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.



Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

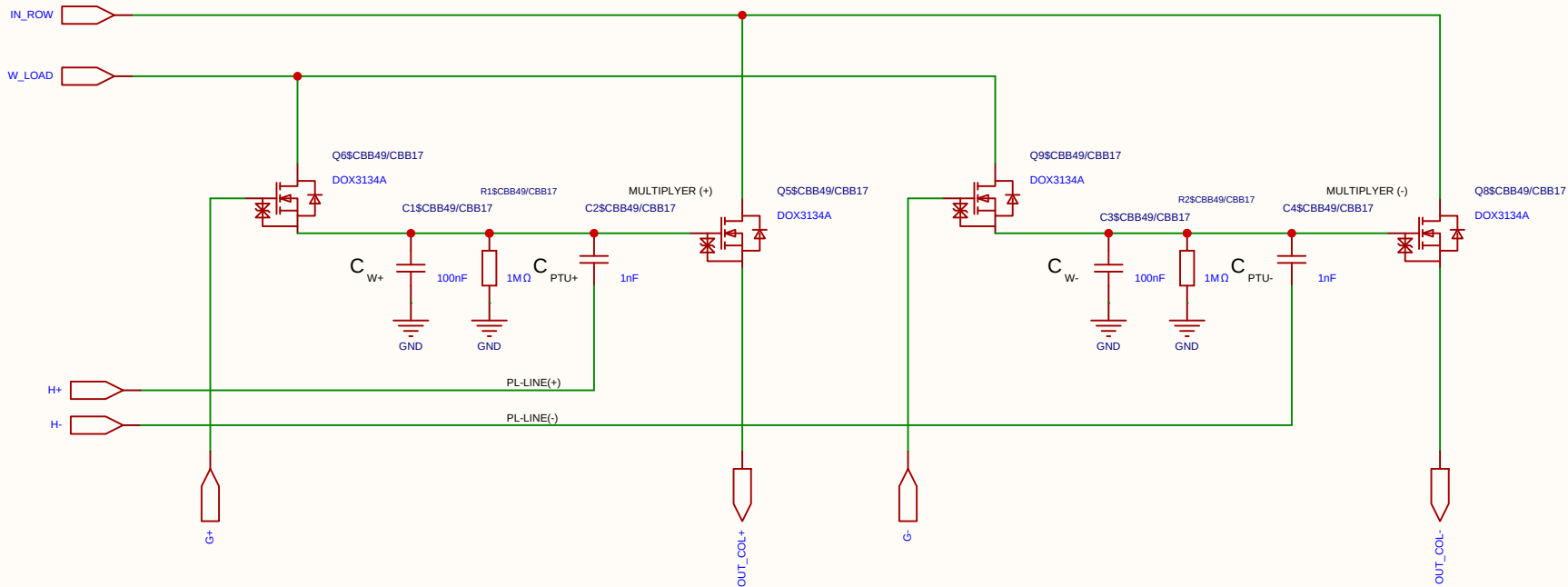
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

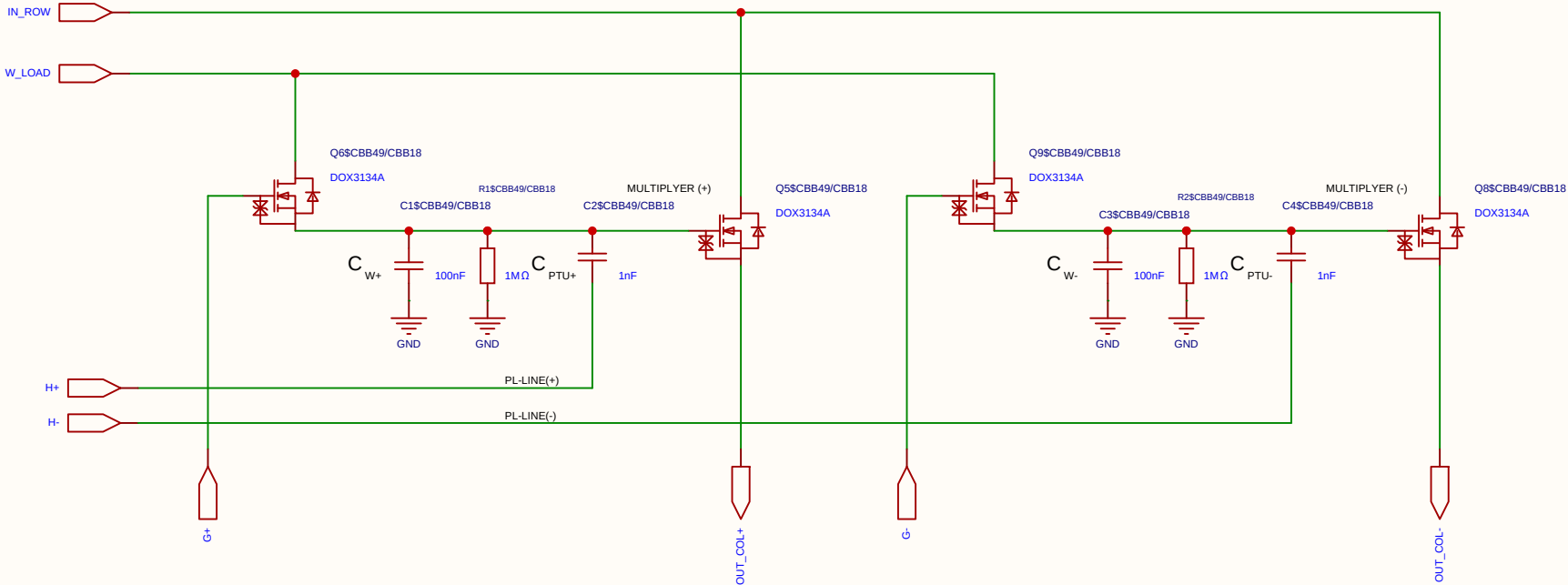
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

The schematic diagram illustrates a 2-4 bit multiplier circuit. It features two main multiplier blocks, MULTIPLIER (+) and MULTIPLIER (-), which are connected to the input signals IN_ROW and W_LOAD. The circuit includes several components: resistors (R1, R2), capacitors (C1, C2, C3, C4), and transistors (Q6, Q5, Q9, Q8). The output signals are OUT_COL+ and OUT_COL-.

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

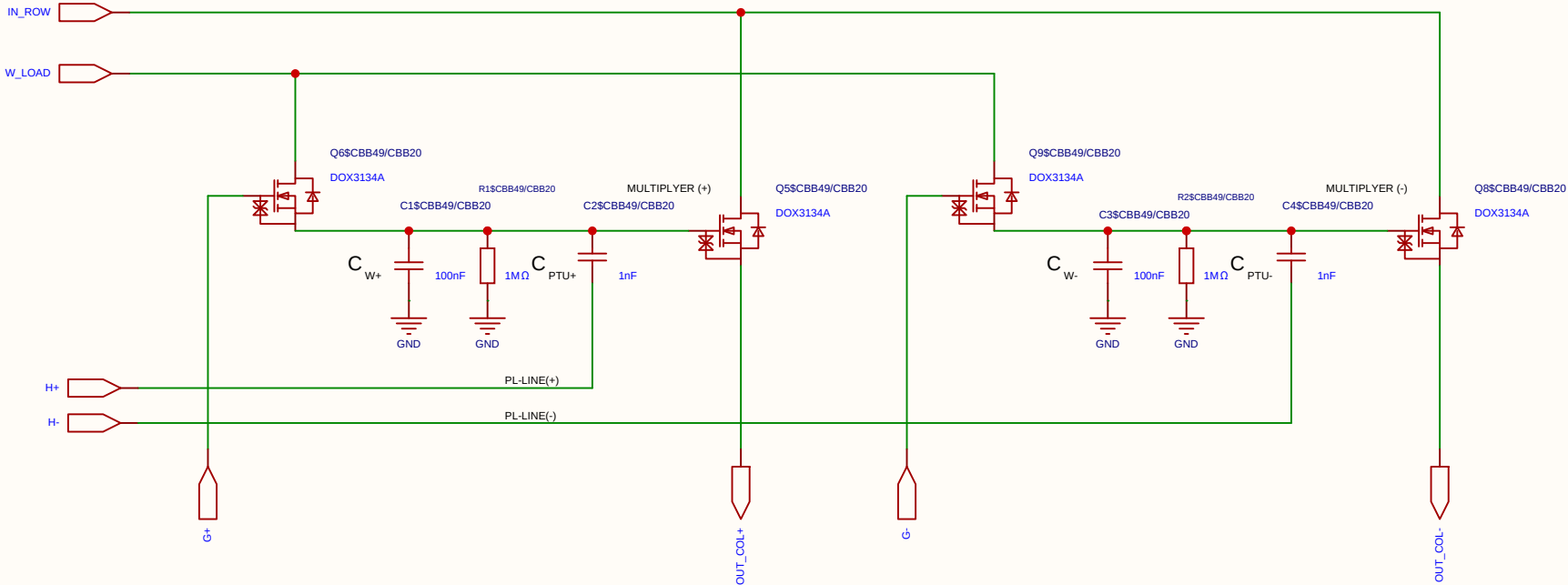
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.



Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

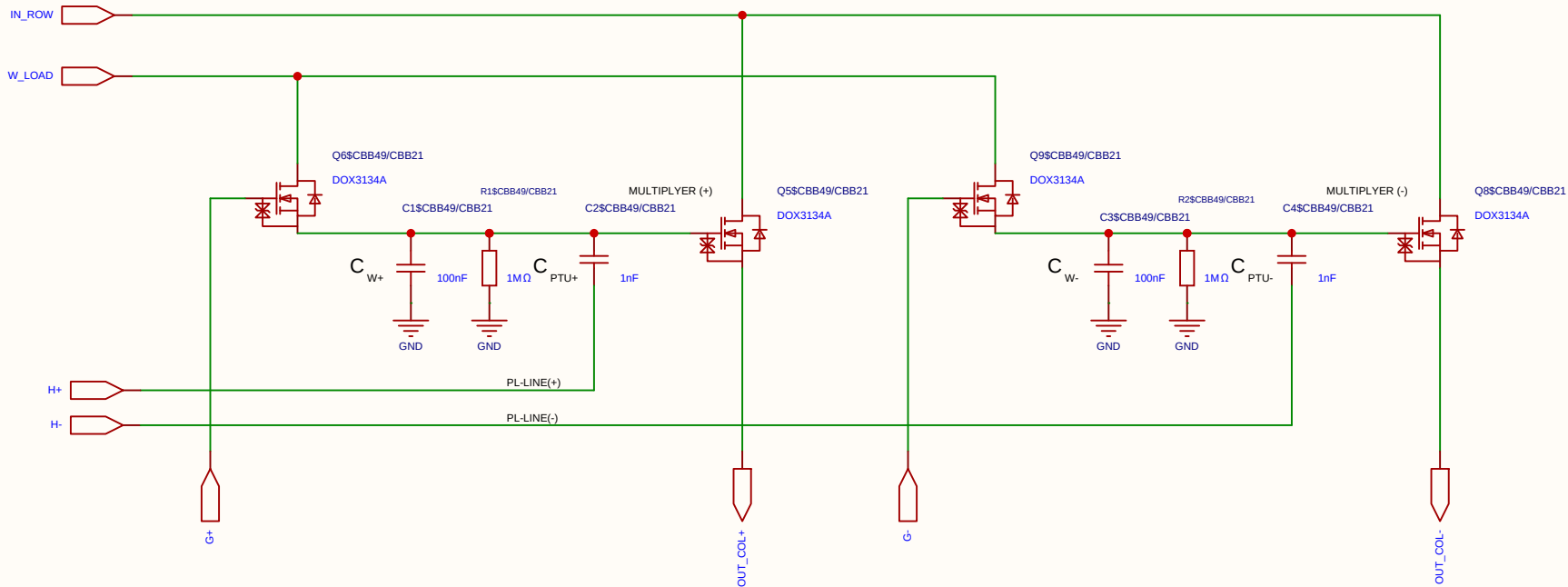
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

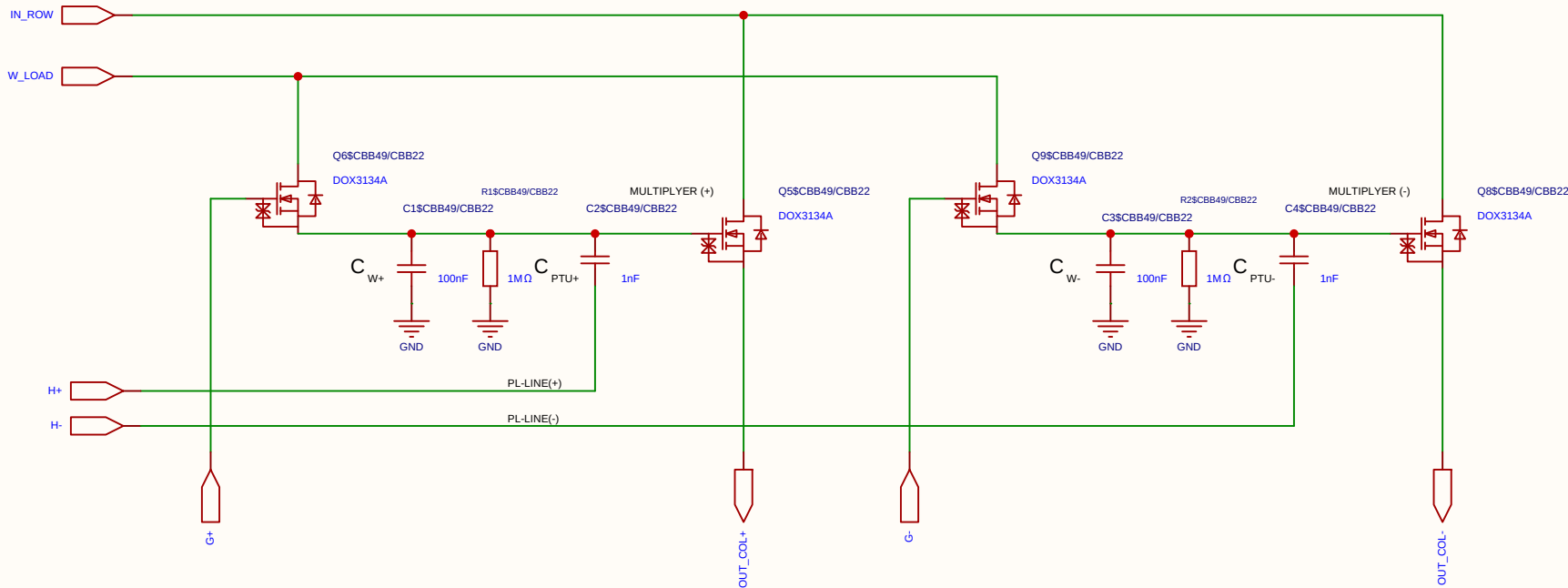
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

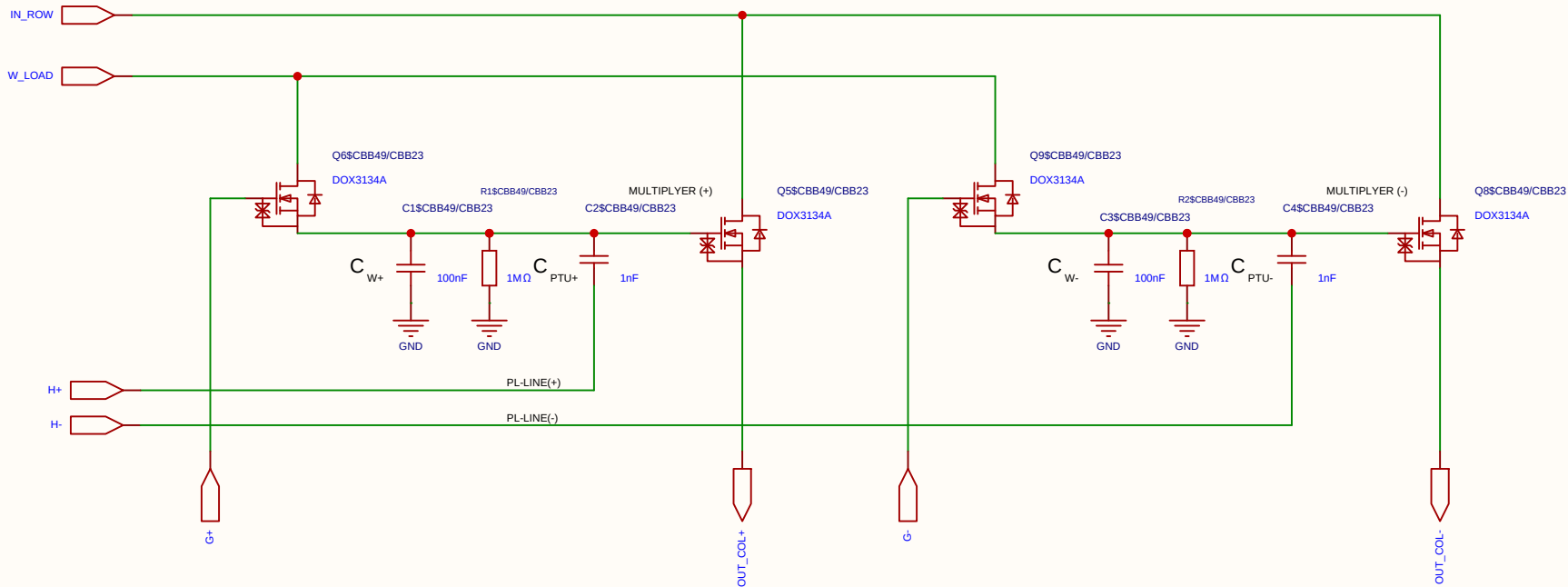
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

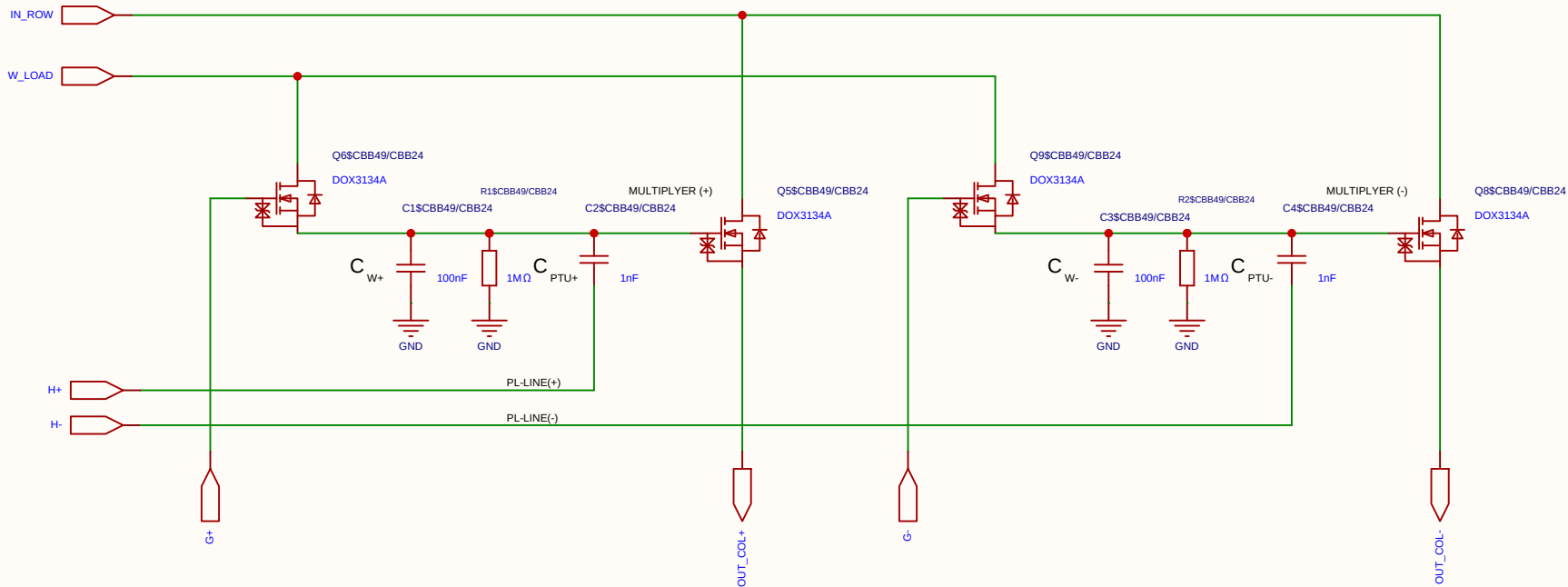
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

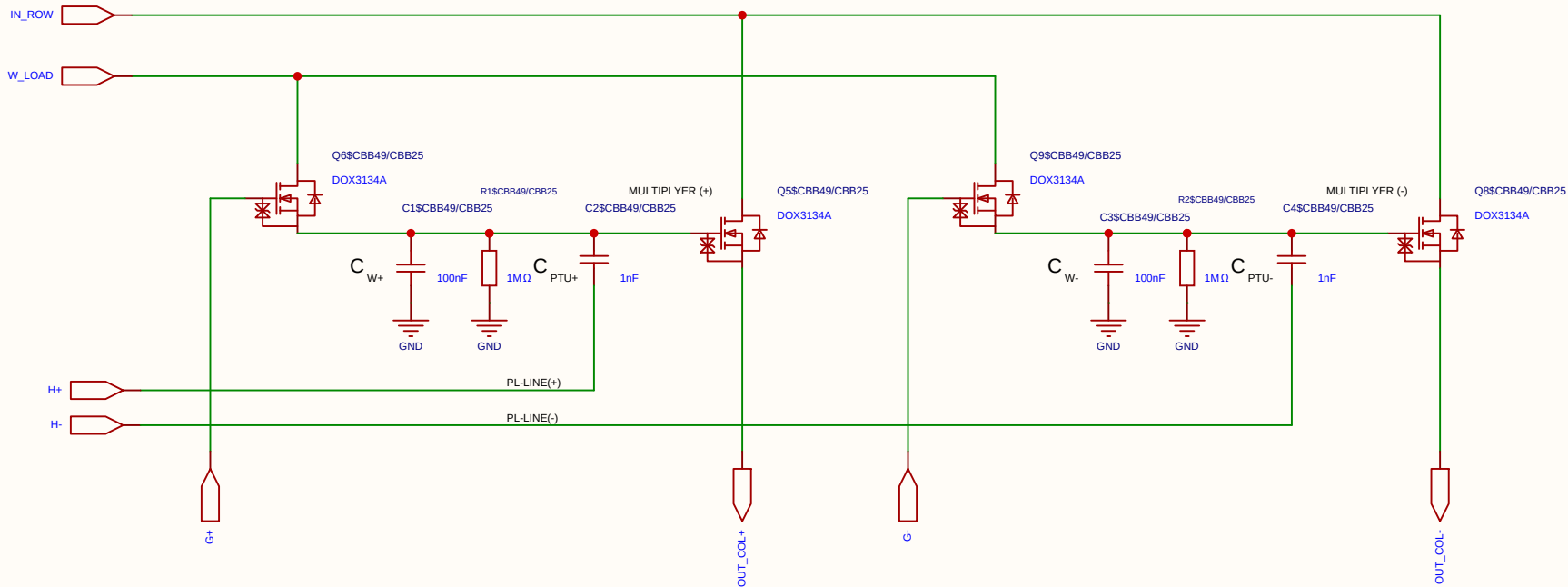
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

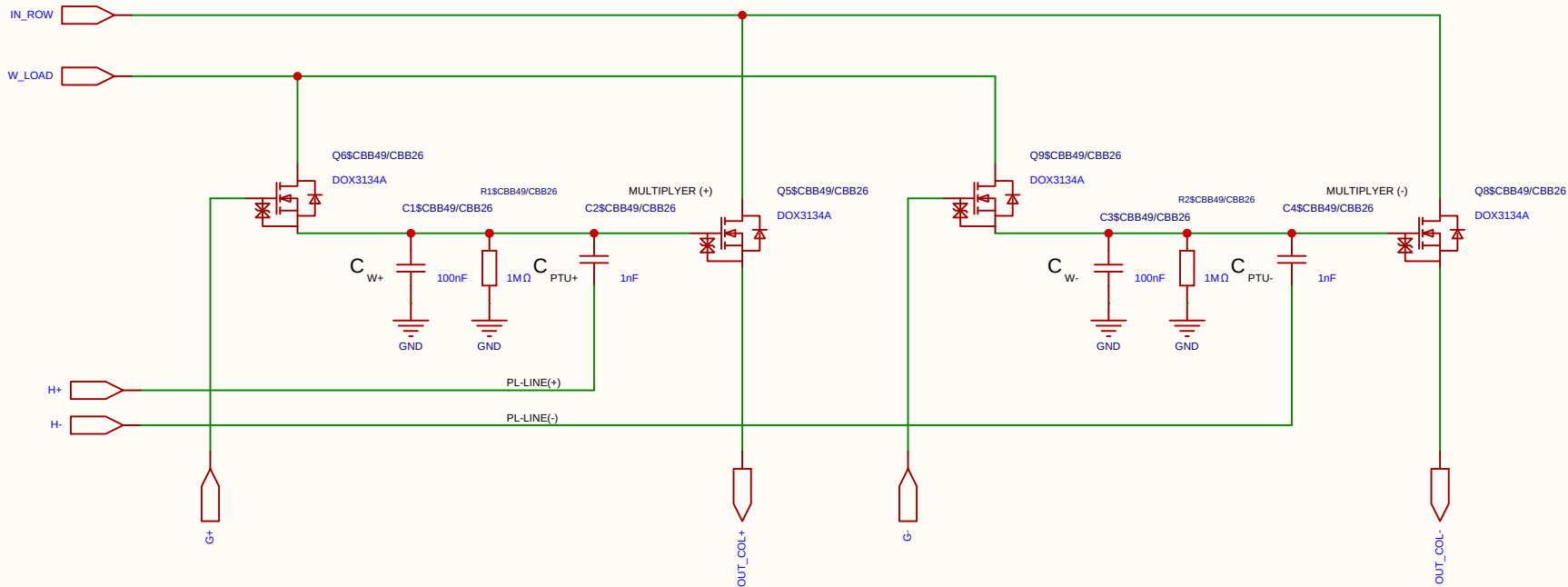
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

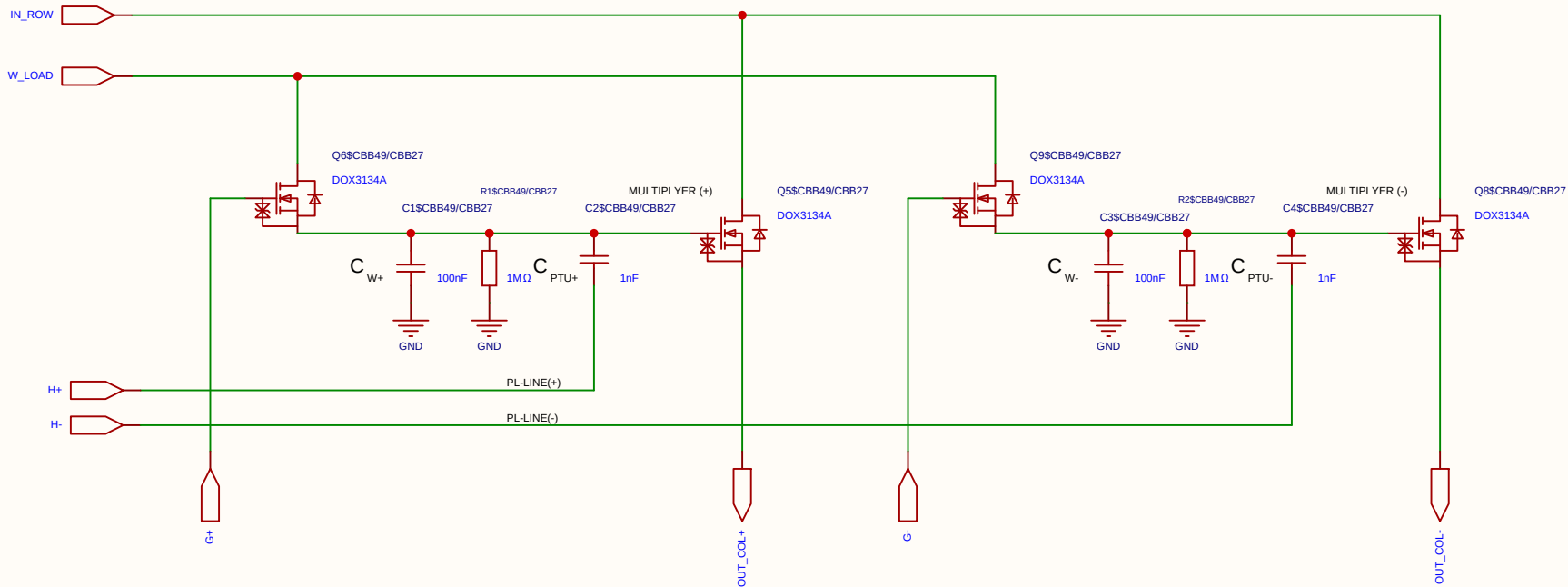
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

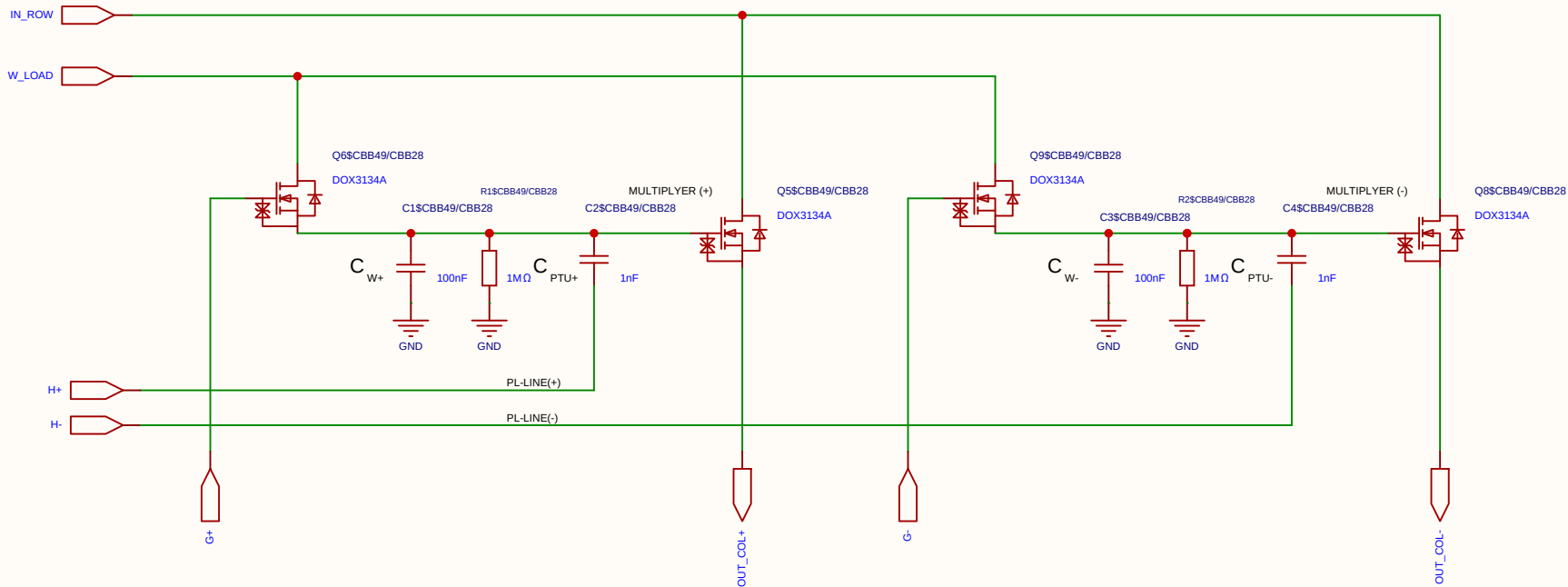
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

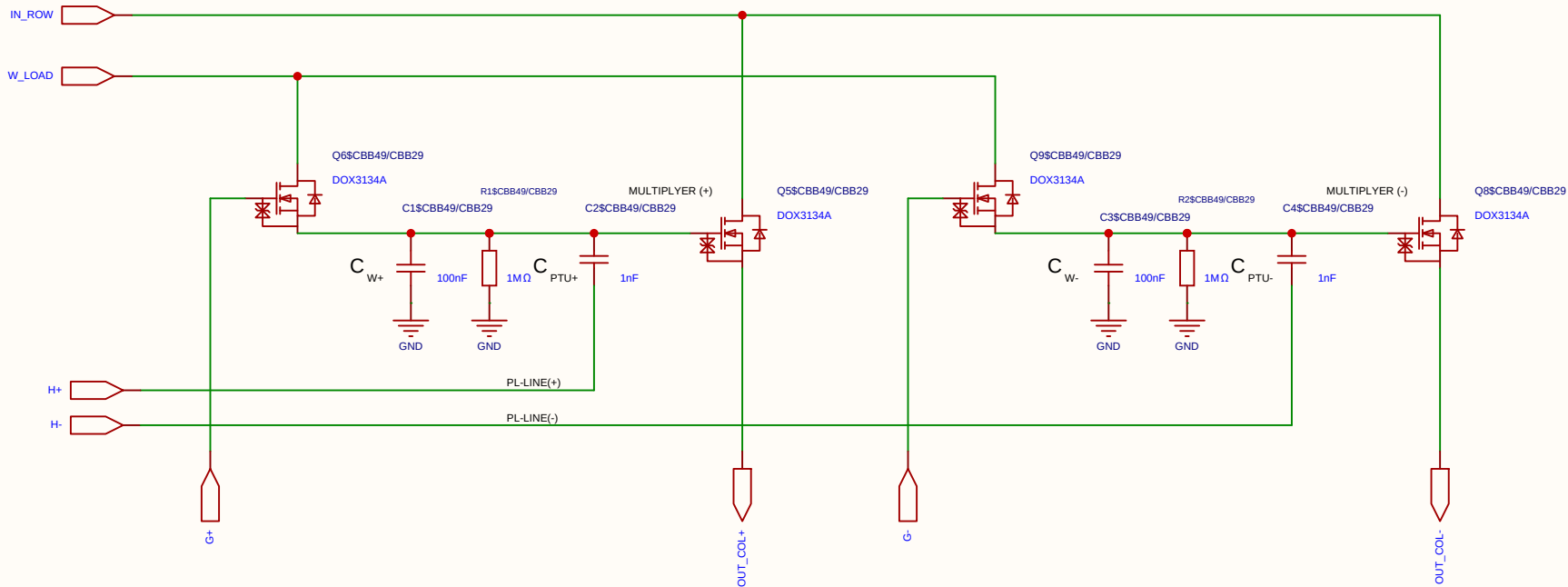
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

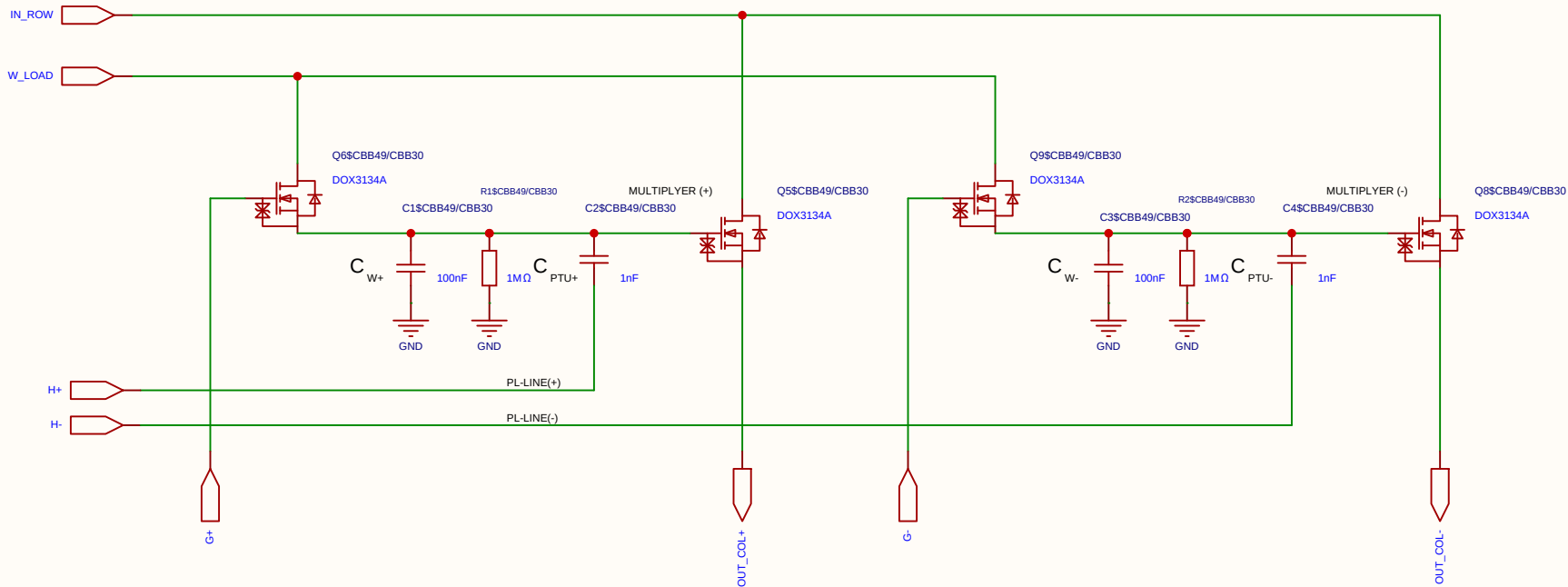
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

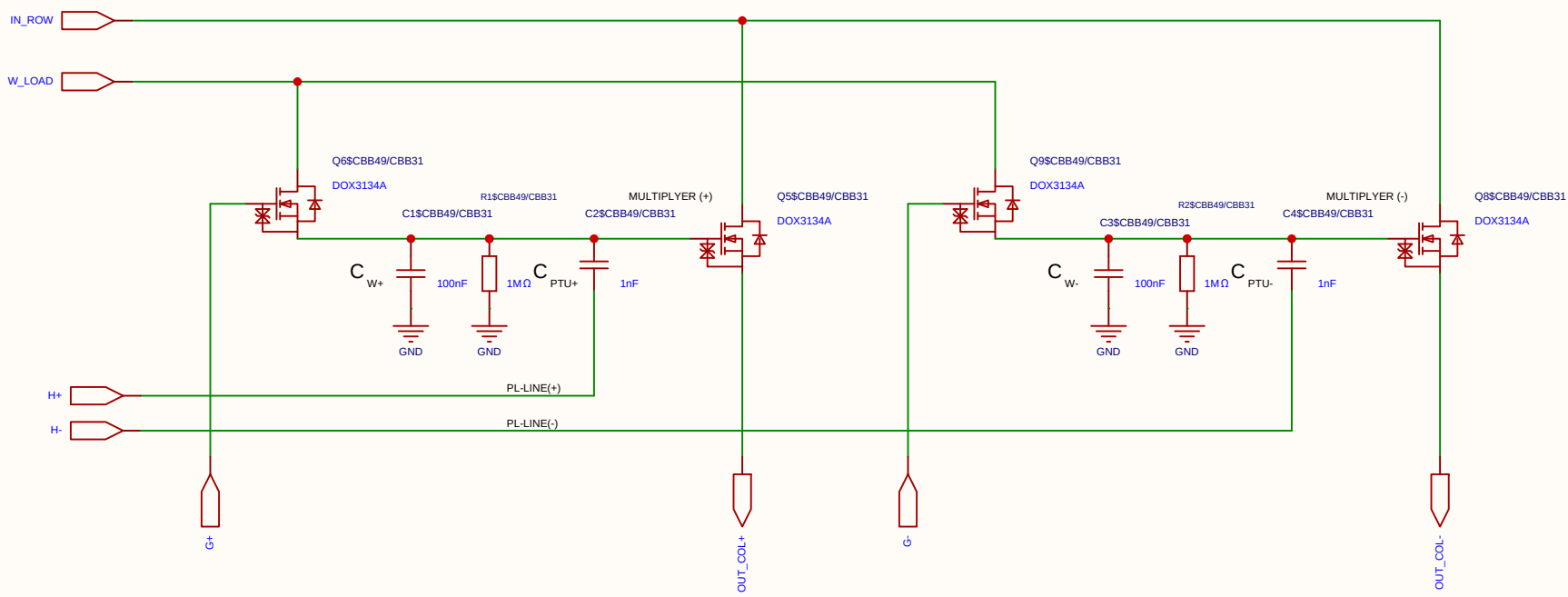
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

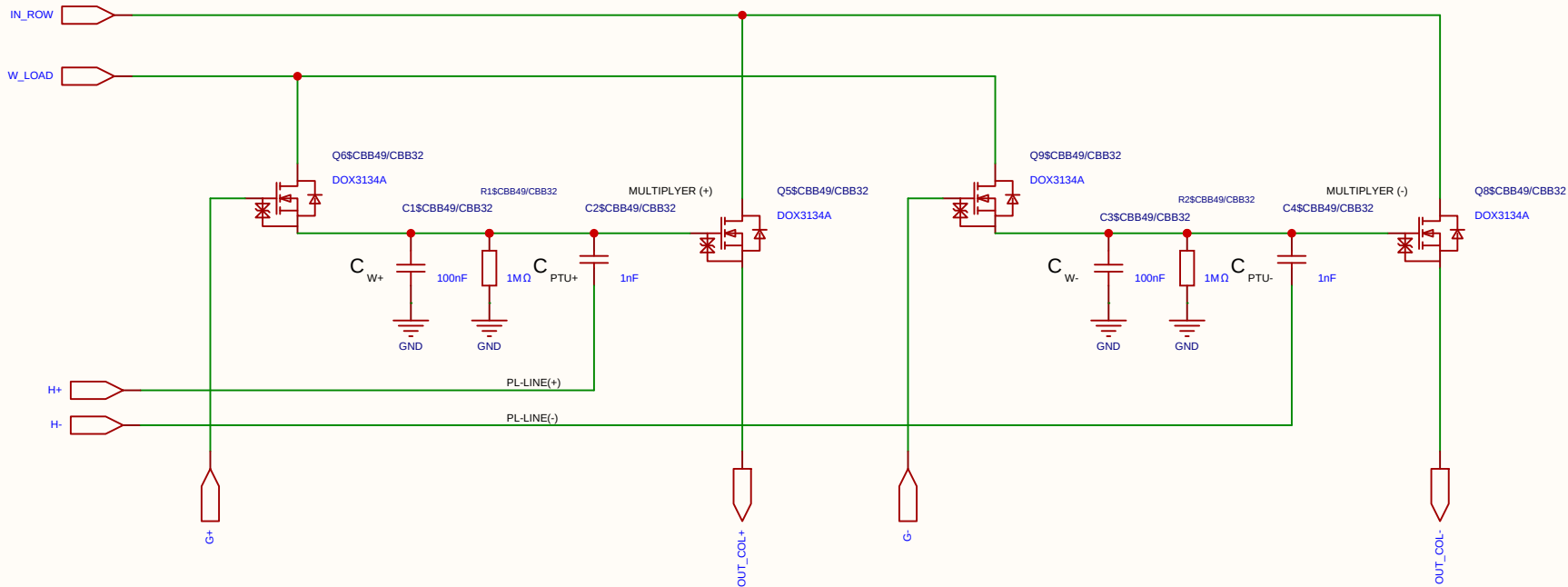
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

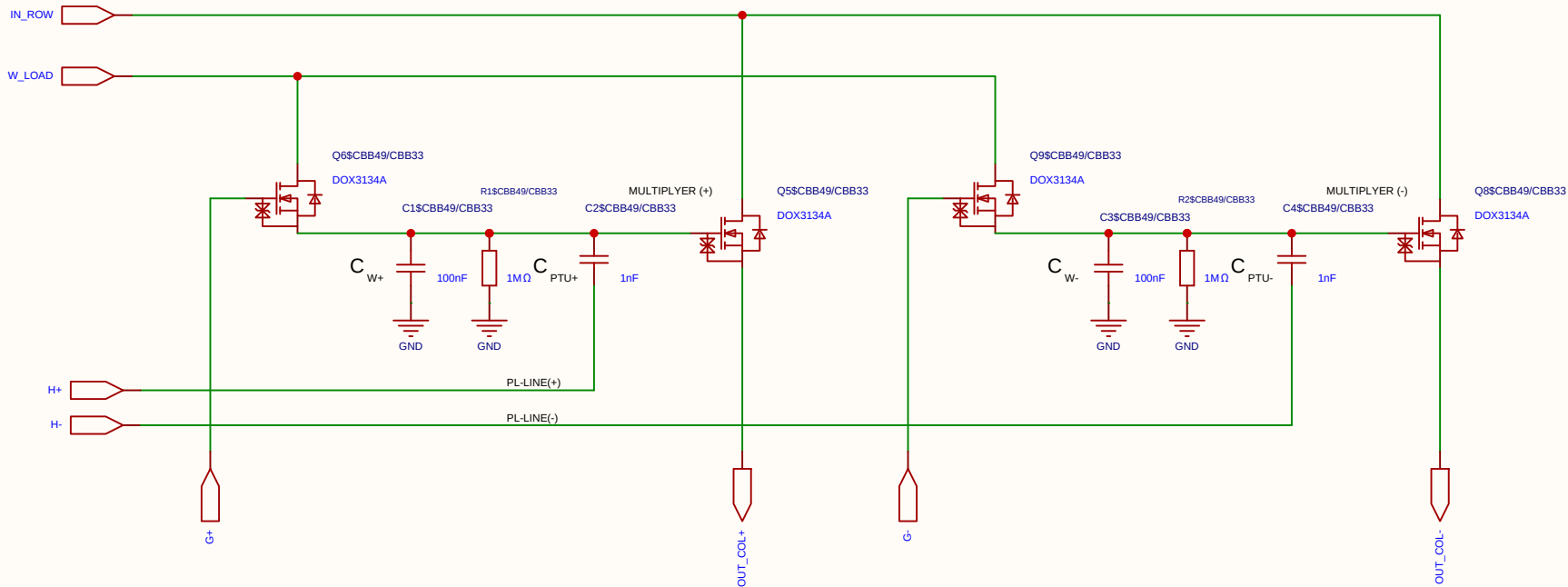
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

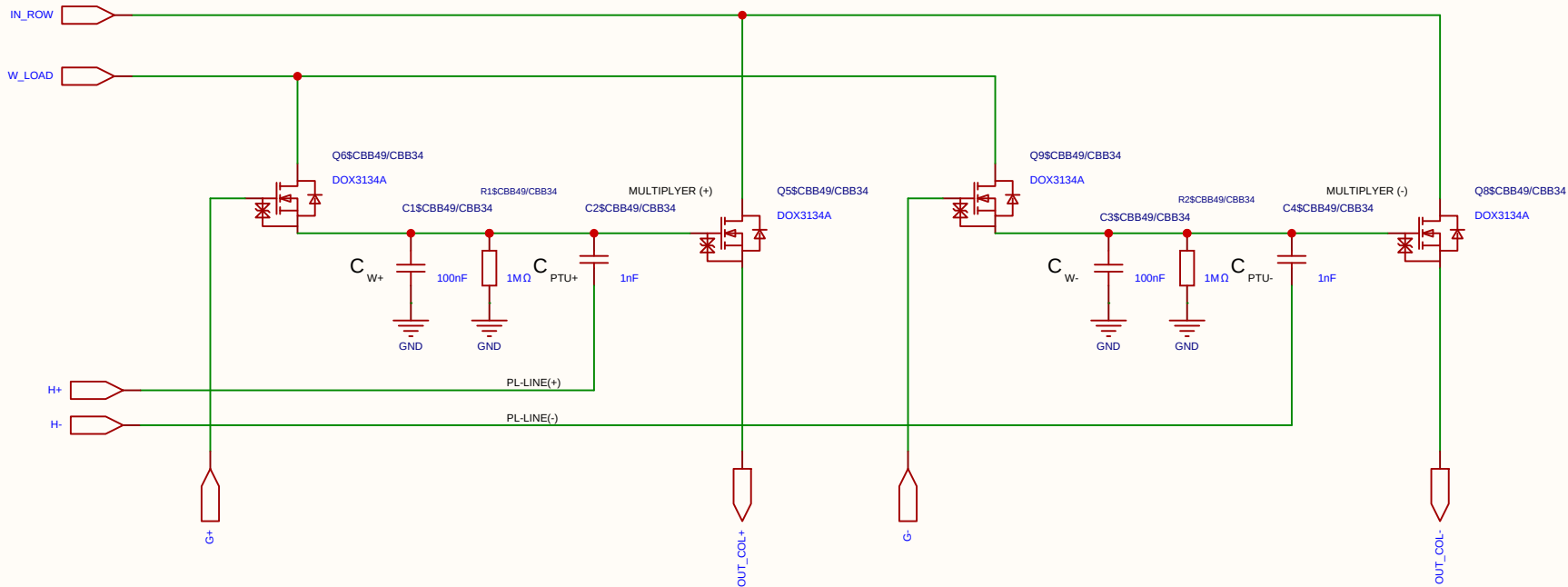
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_3_8_MUL_CELL
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

State	PL Voltage	Perturbation at Gate	Comment
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

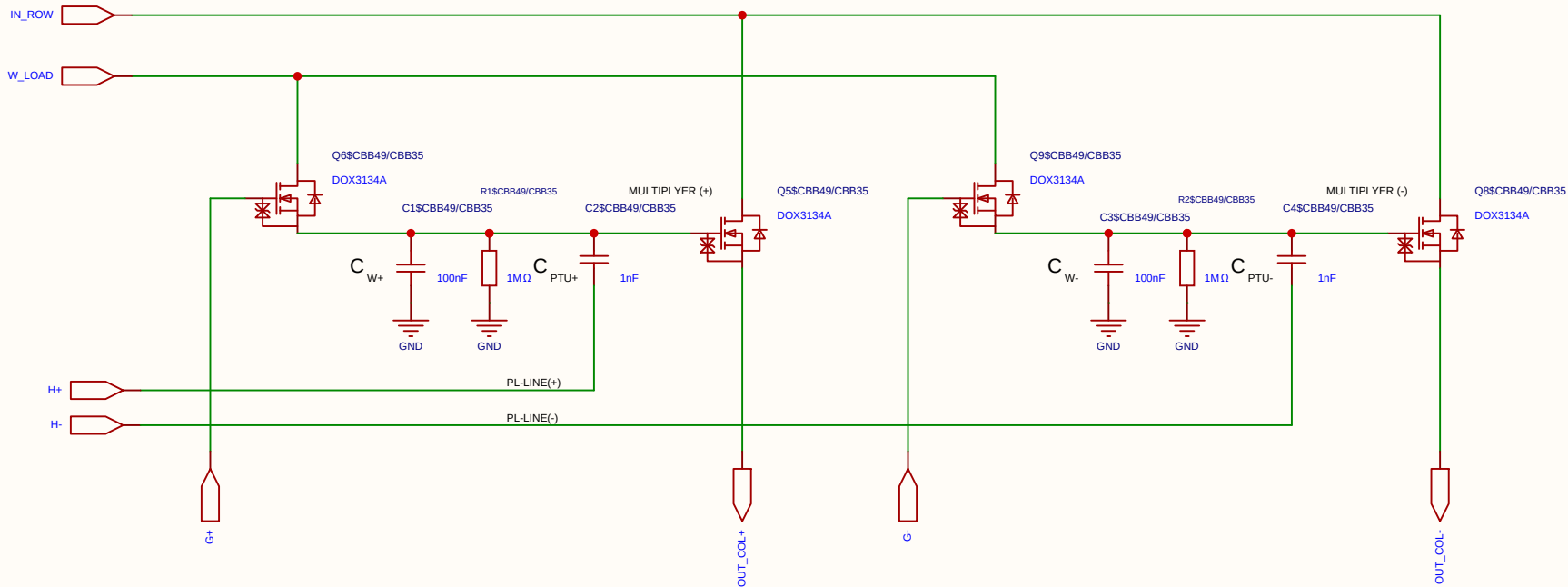
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

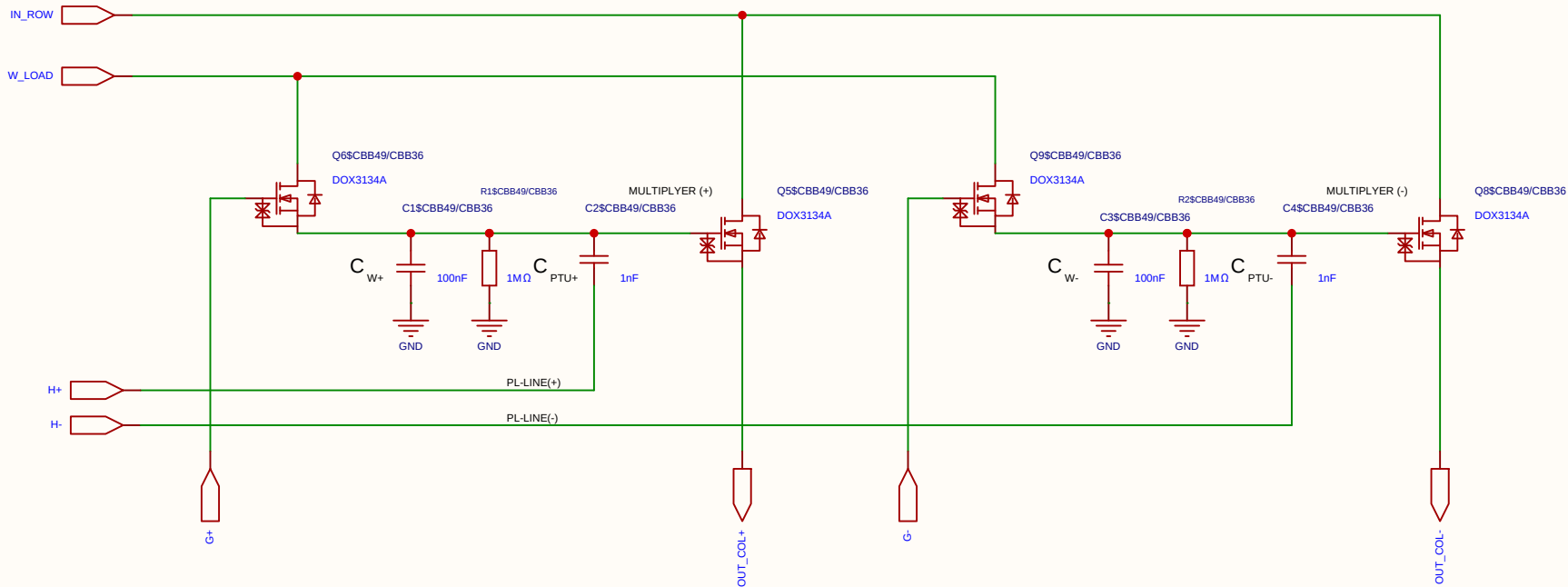
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

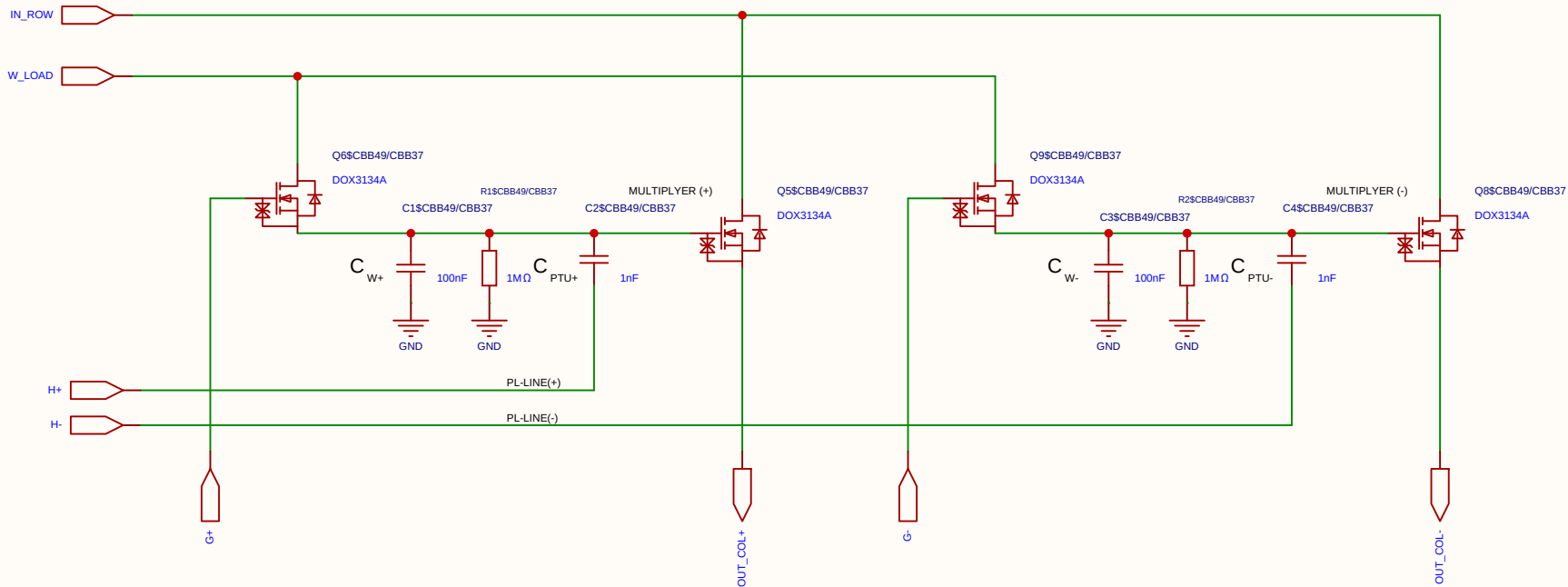
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

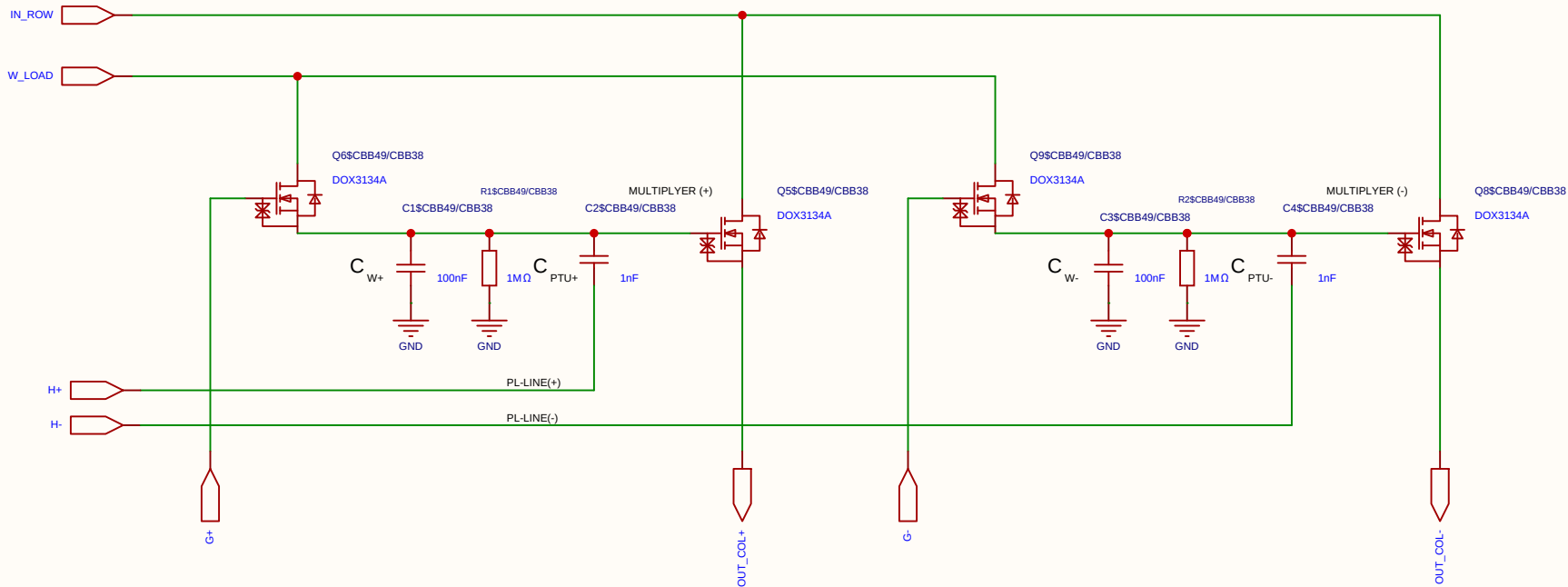
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

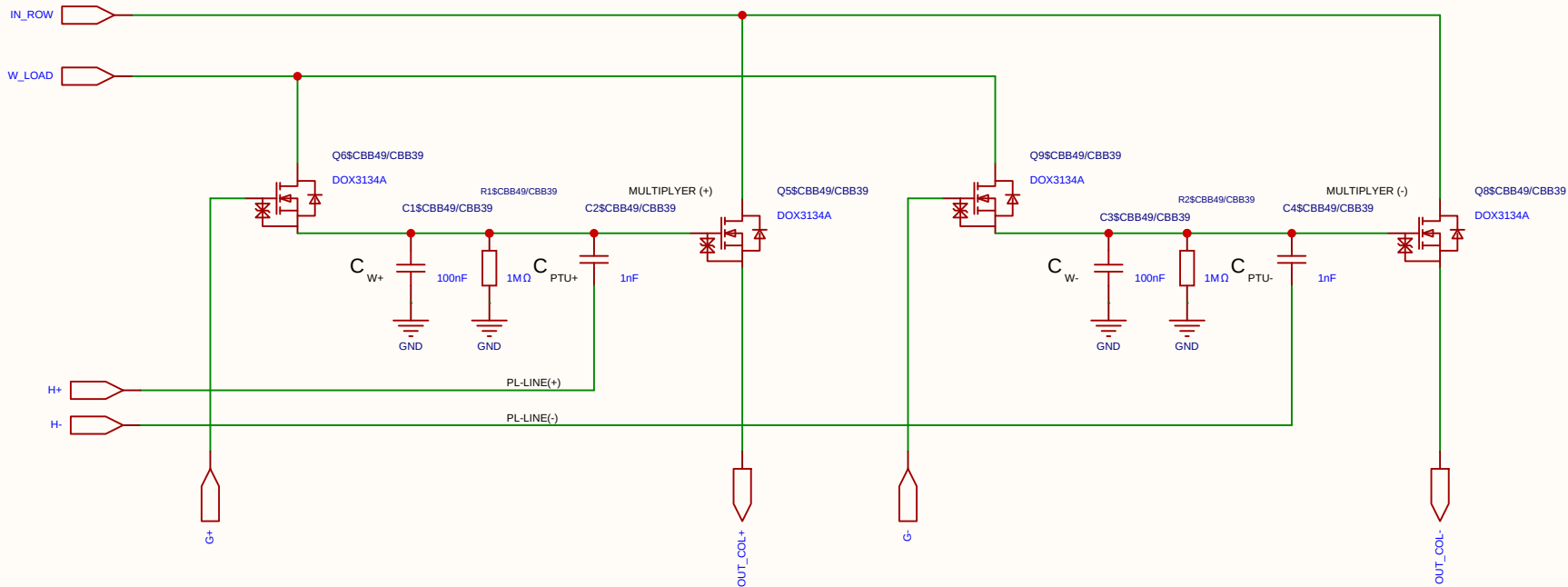
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

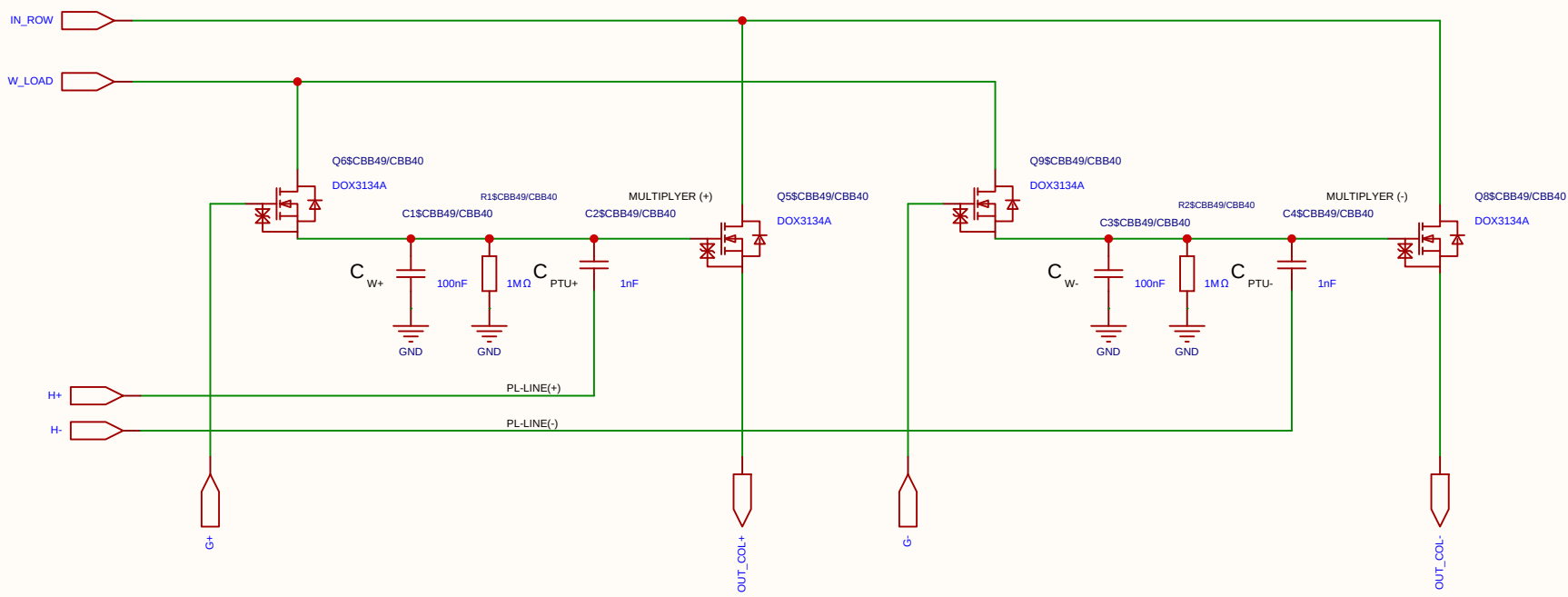
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

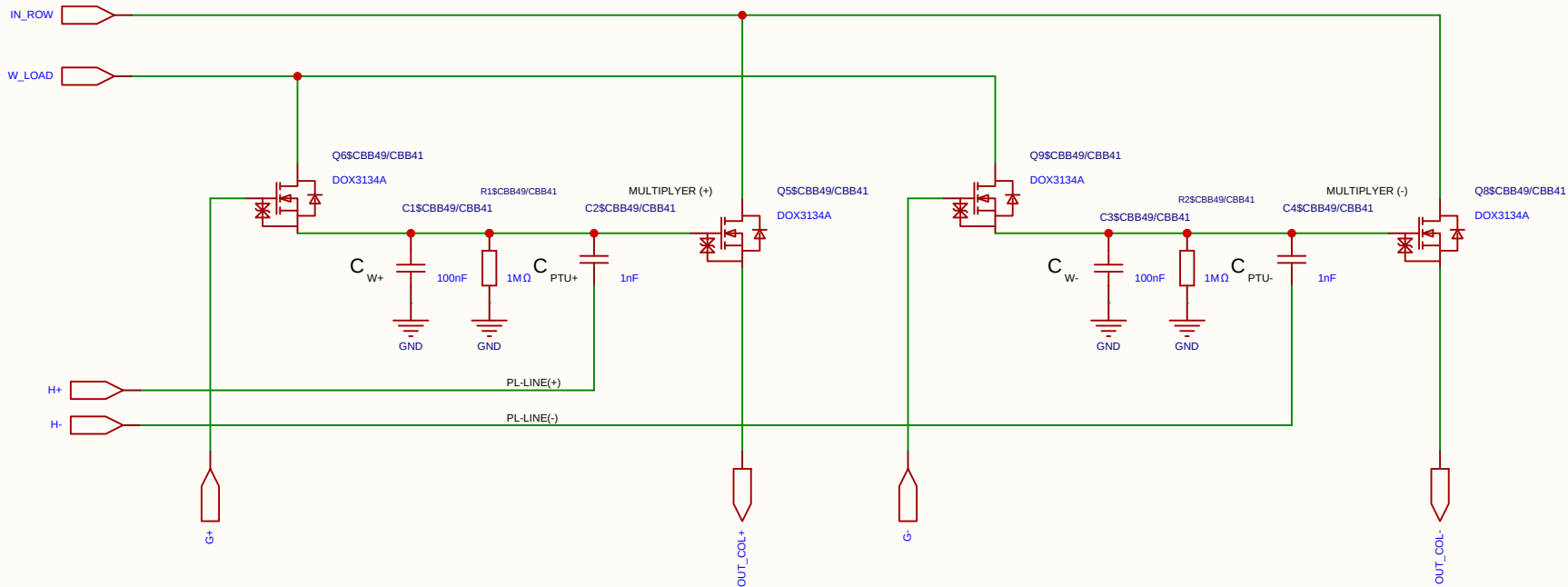
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

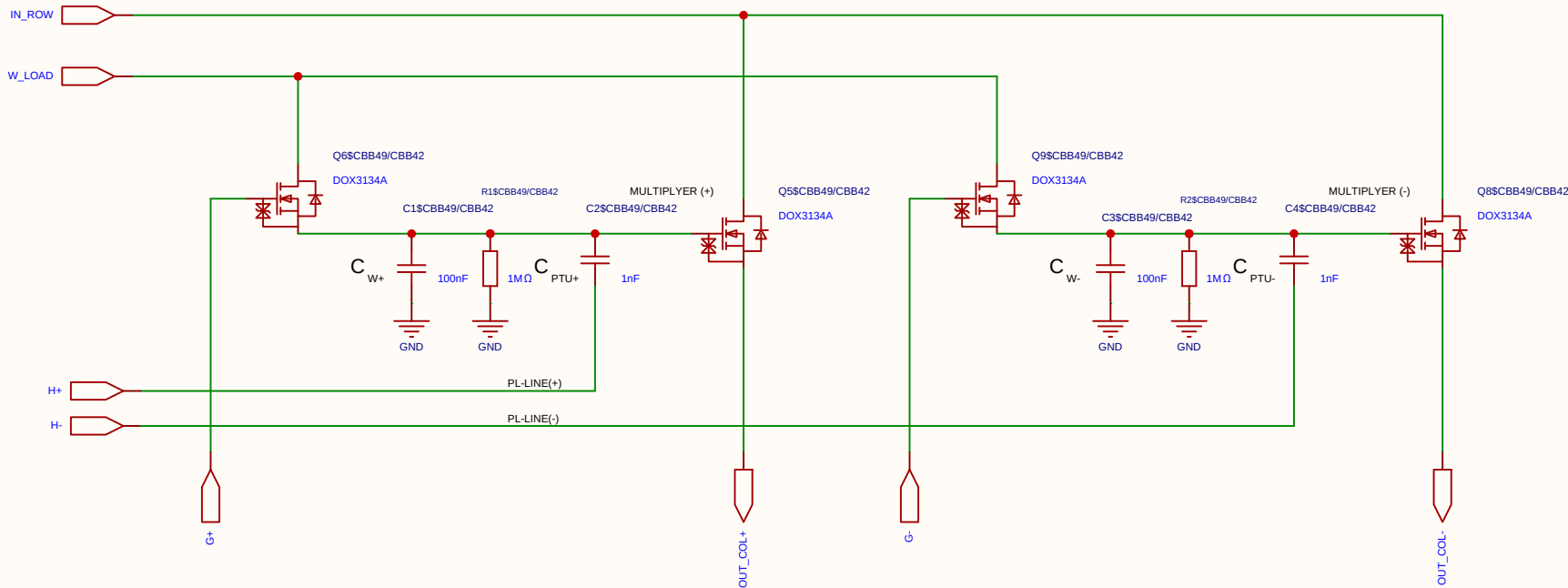
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

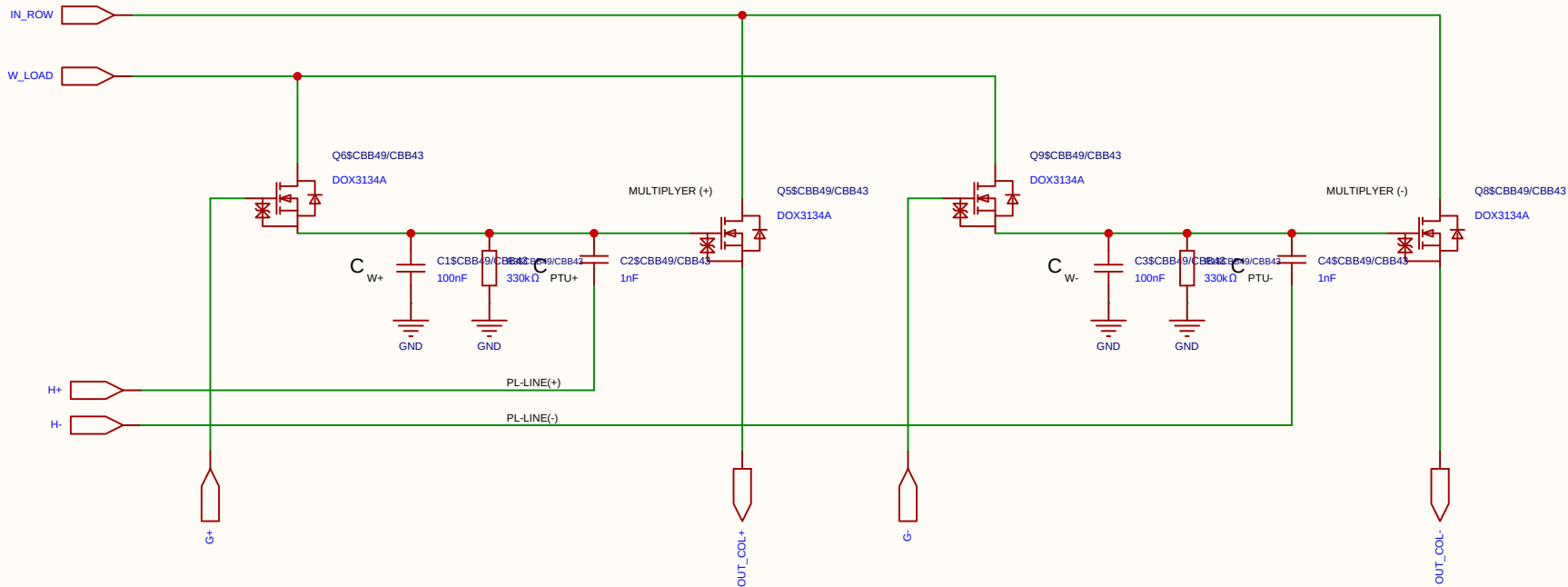
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

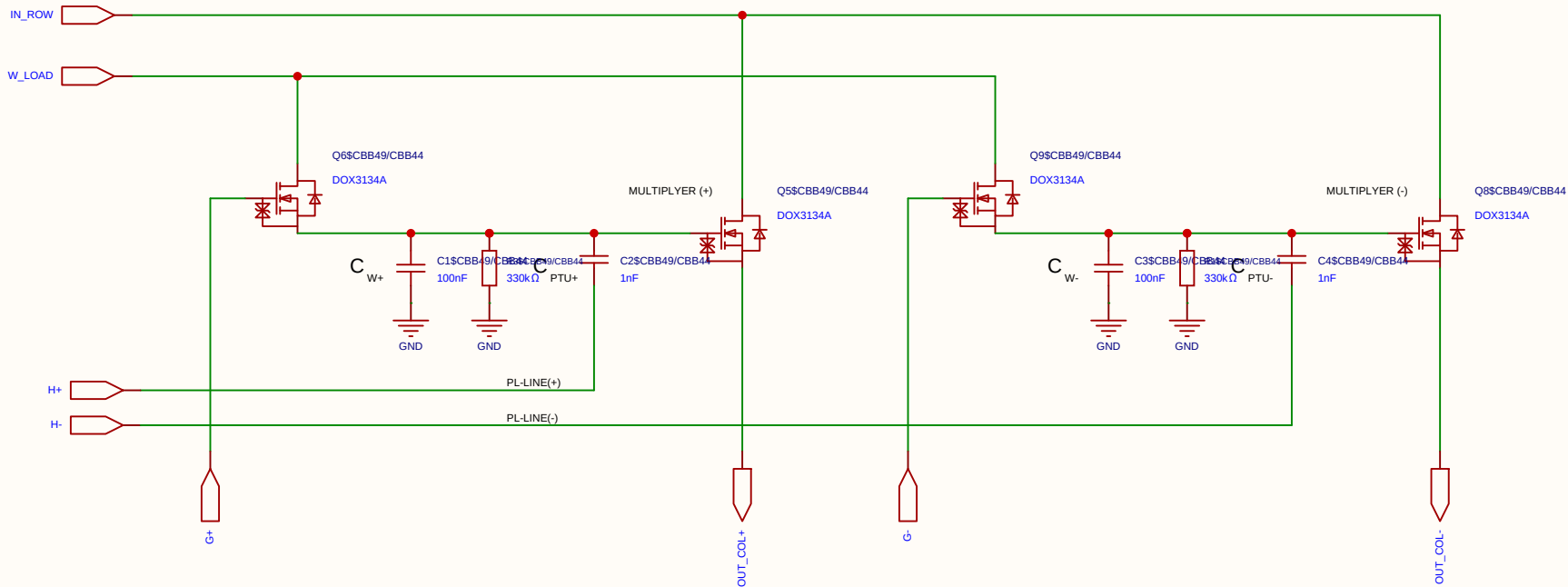
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

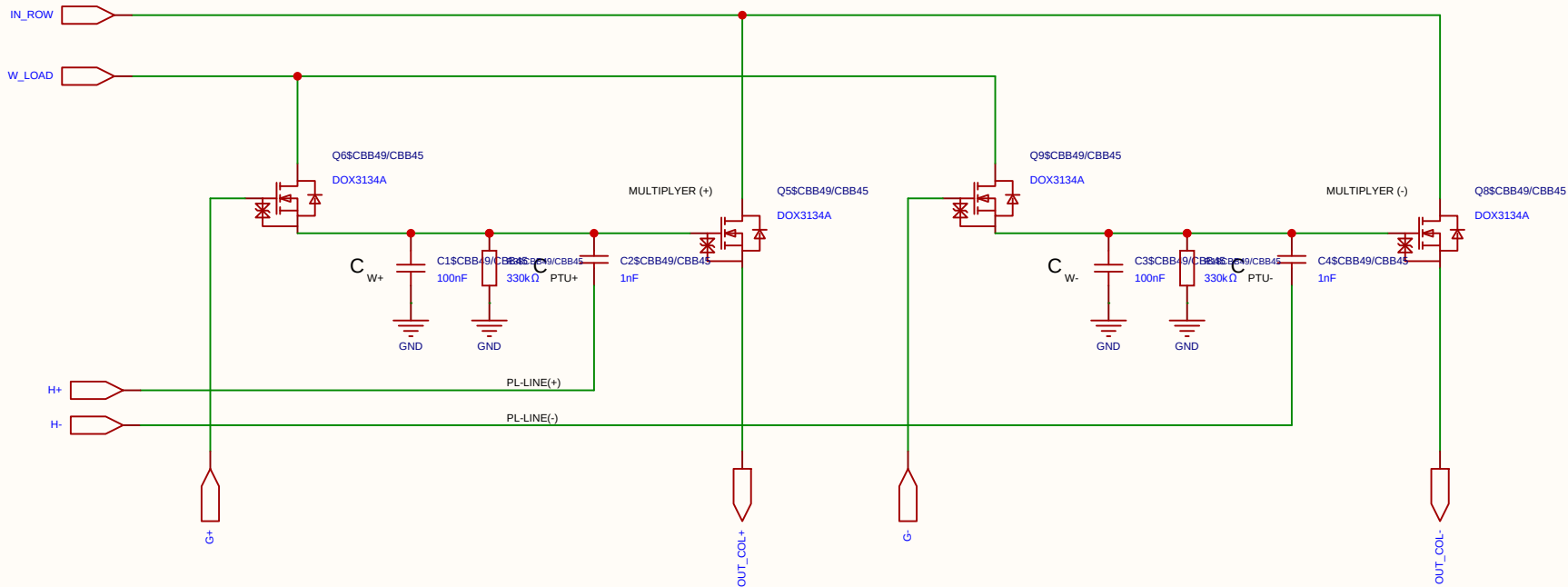
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

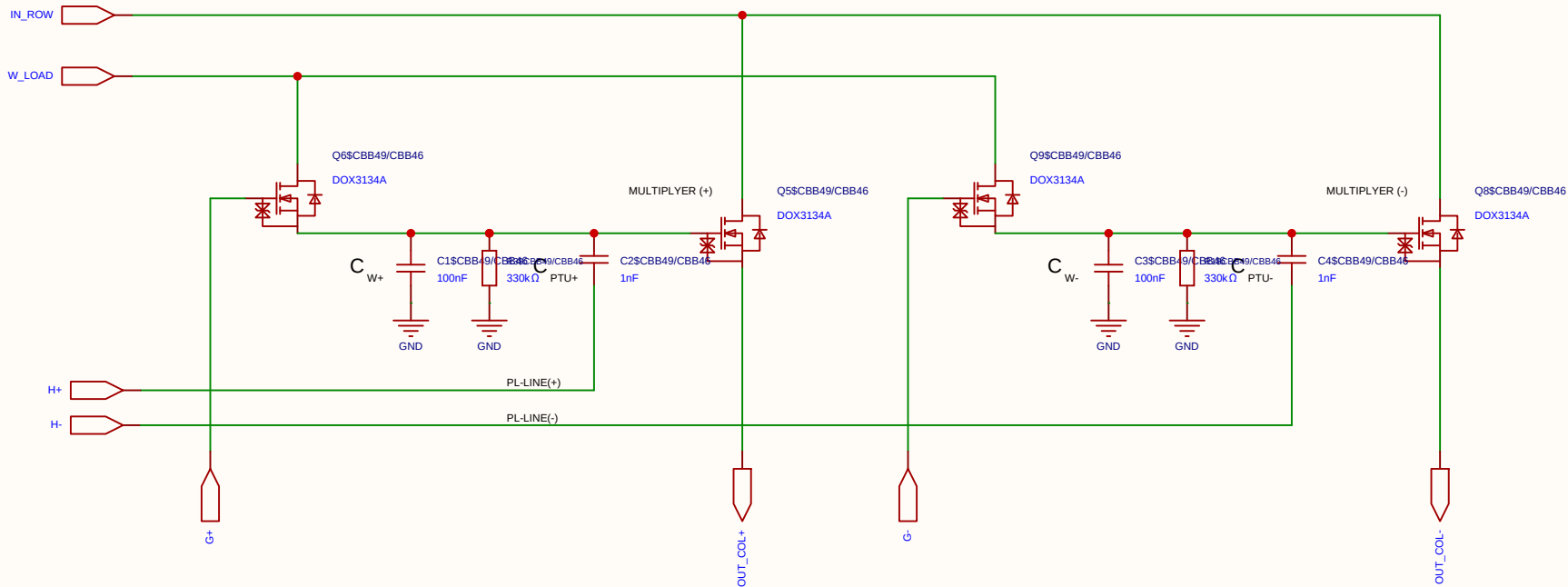
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

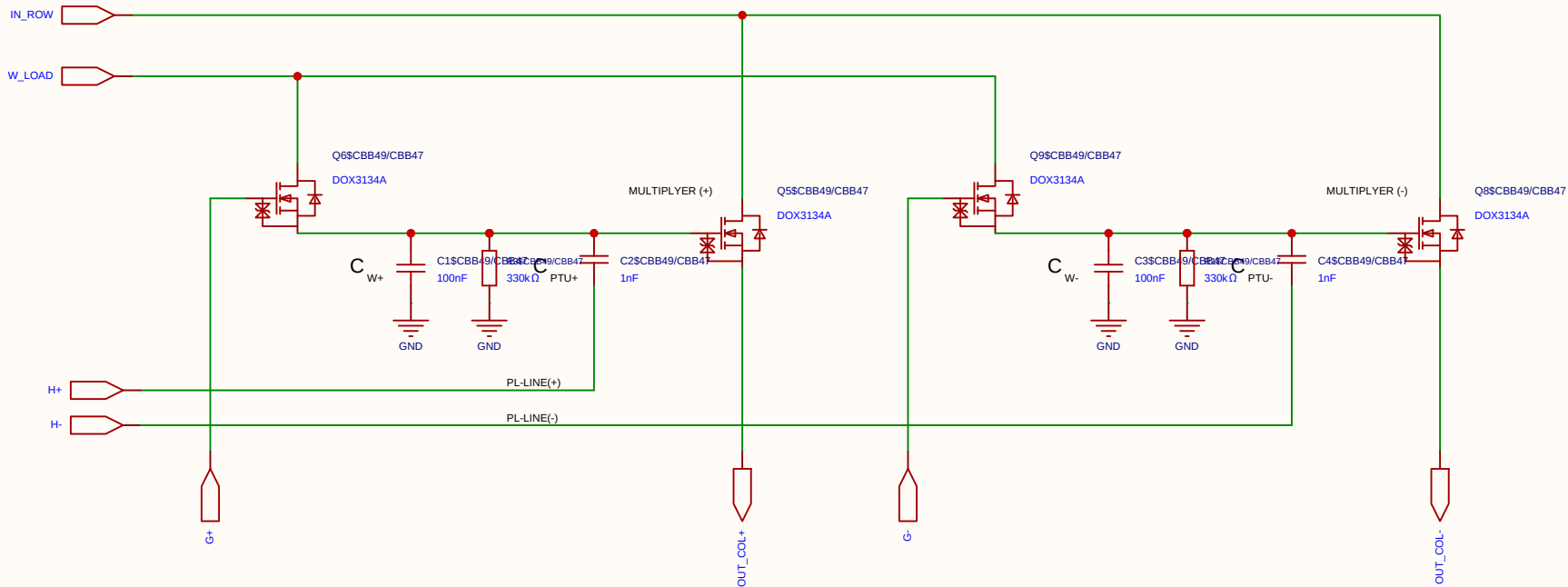
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C		Create at	2026-06-19
			Update at	2026-06-19
Board			Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

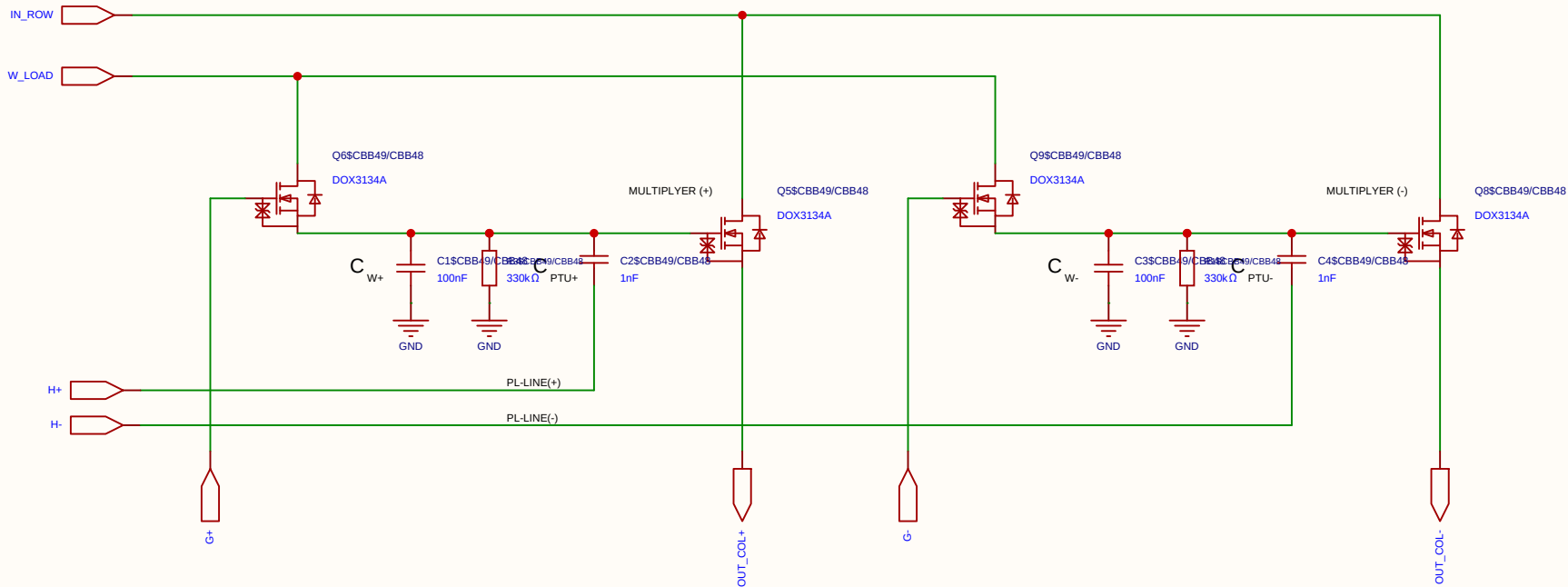
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

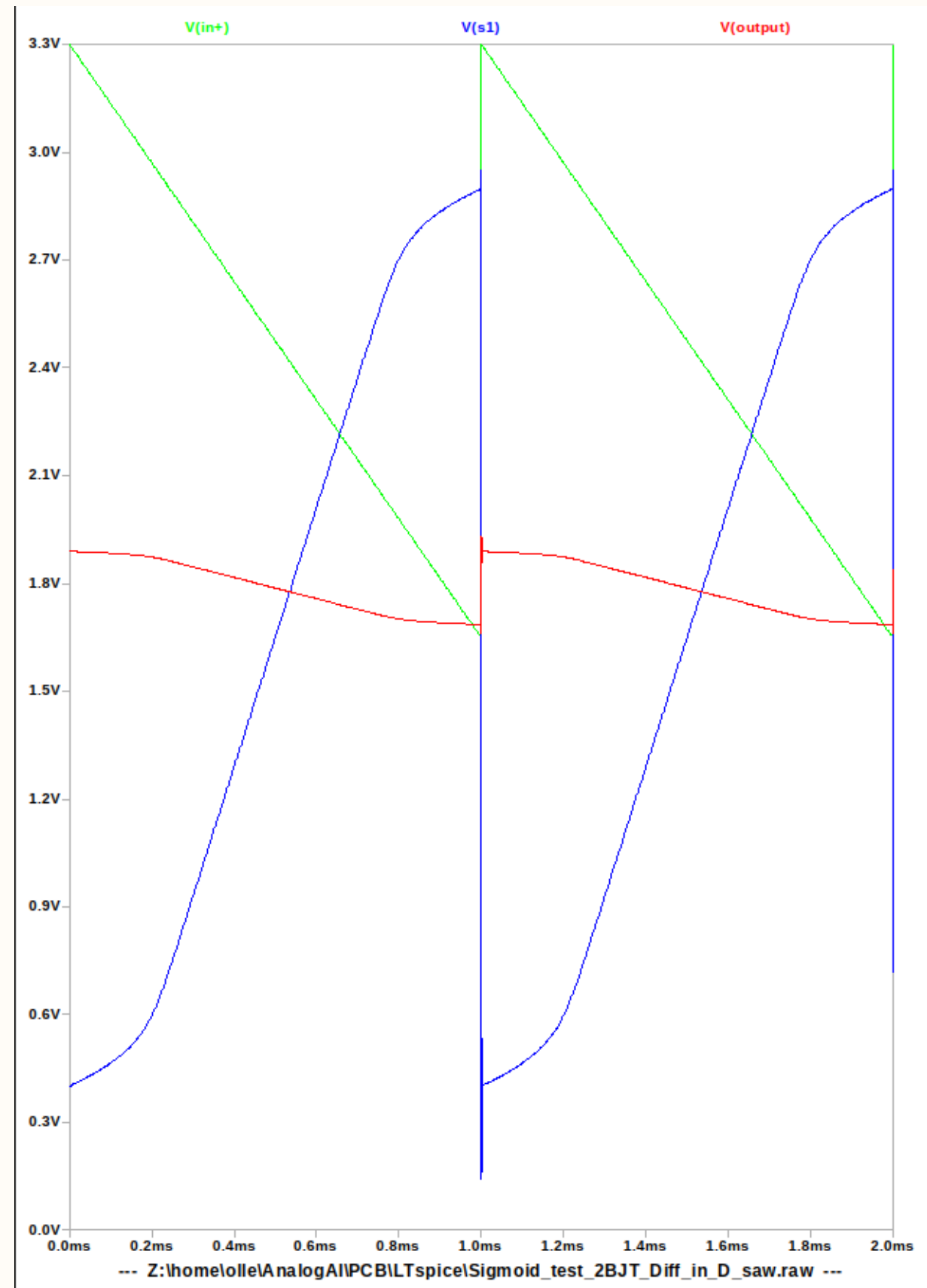
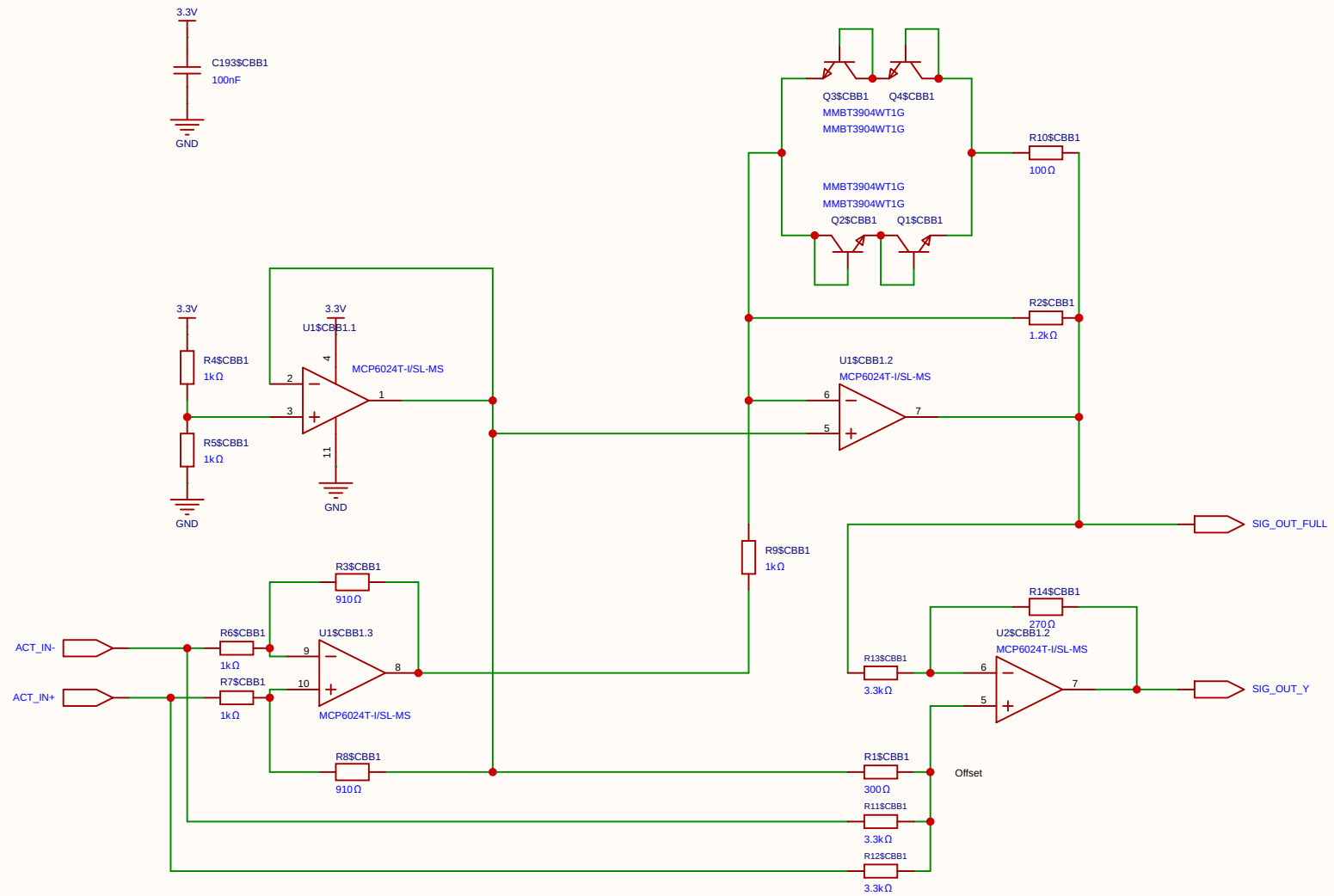
STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

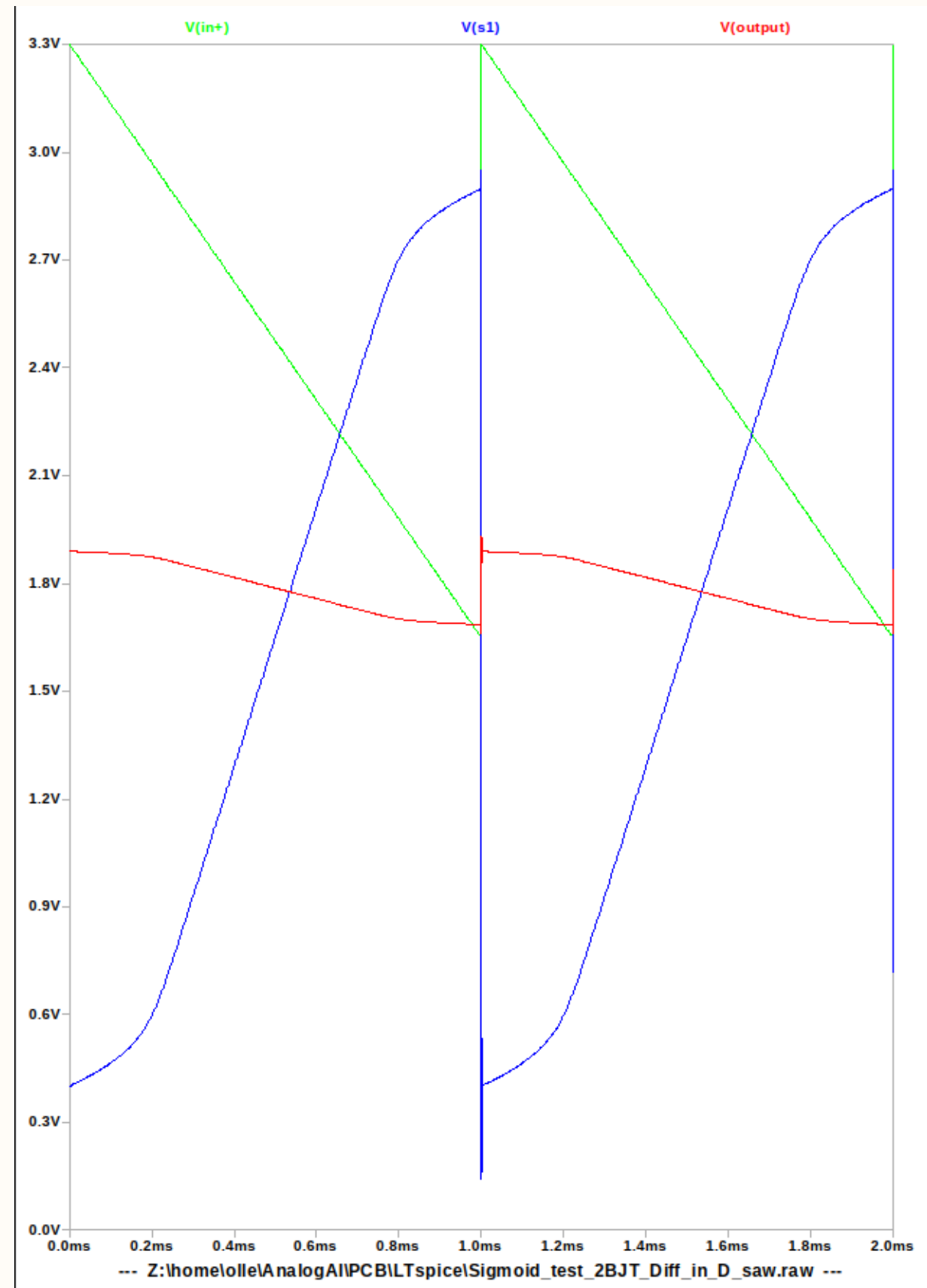
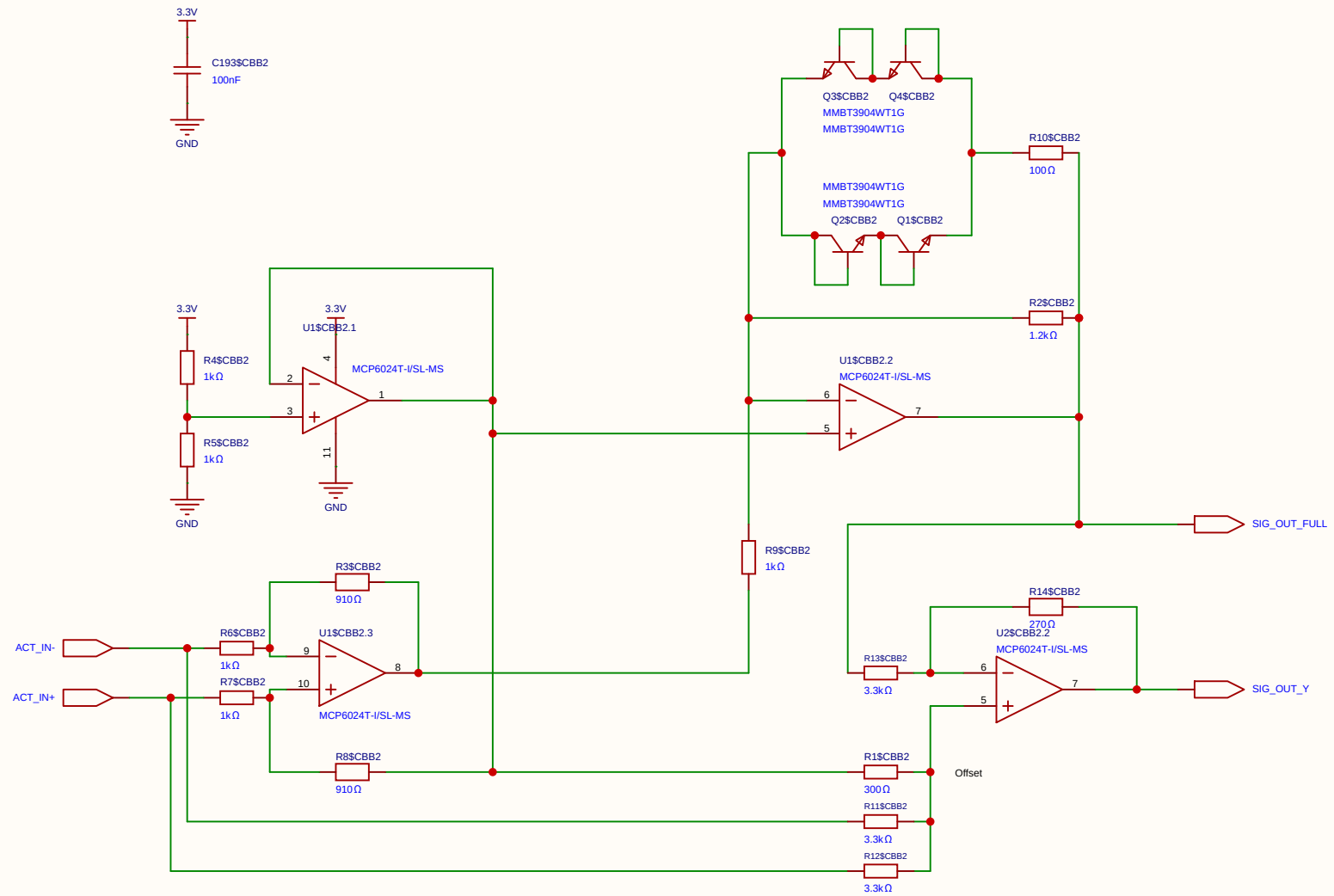
This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

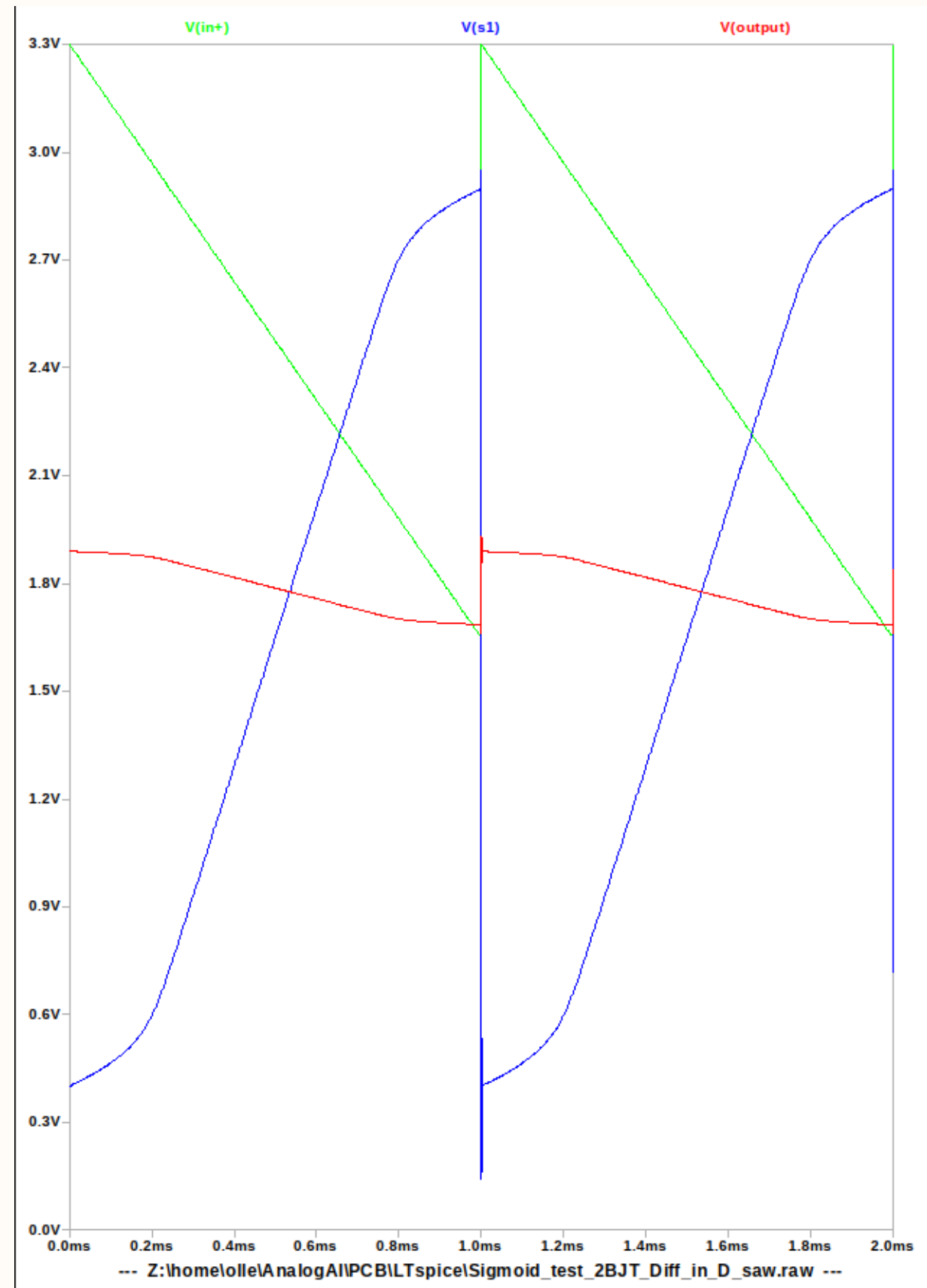
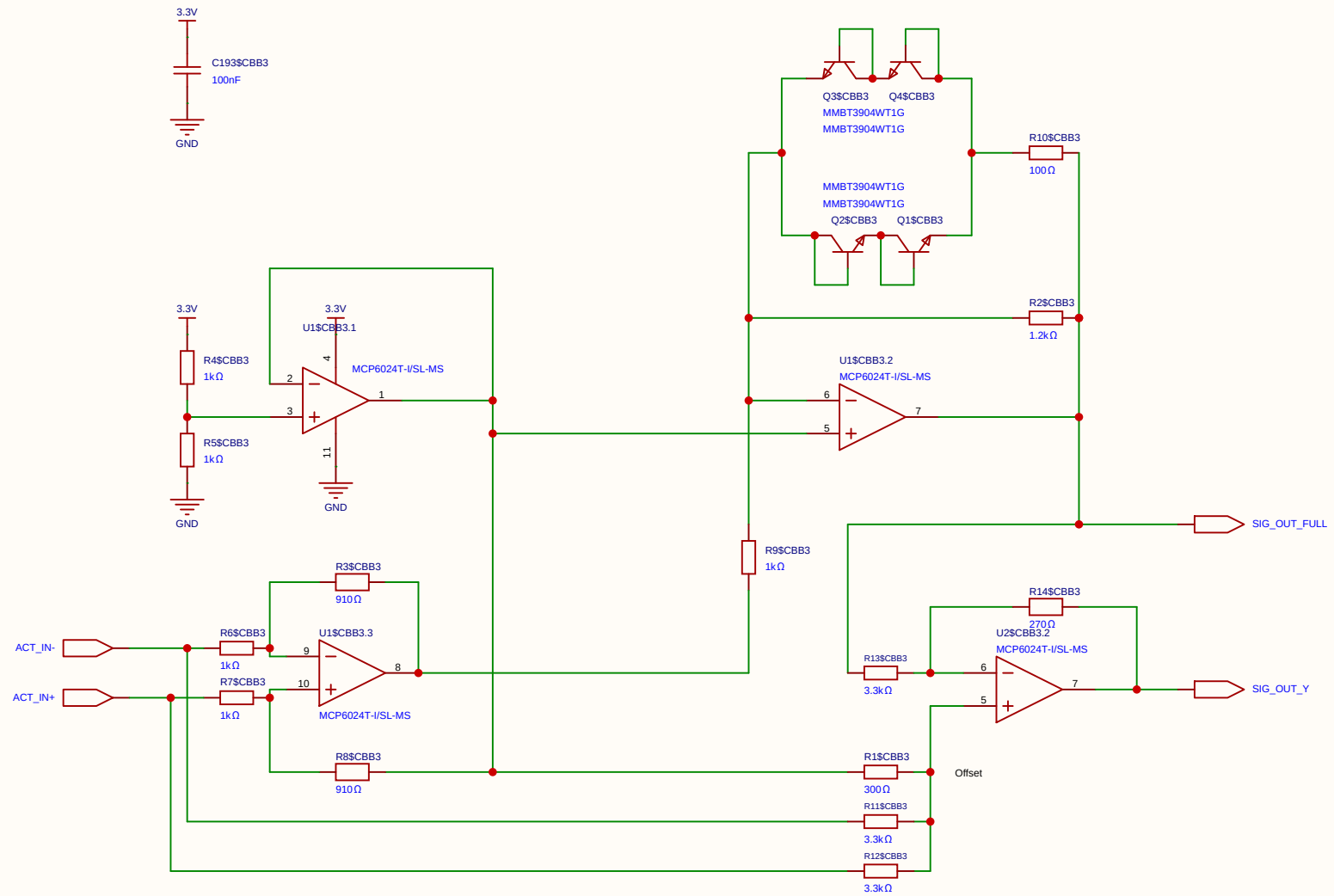
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EasyEDA		V1.0	A4	EasyEDA.com	



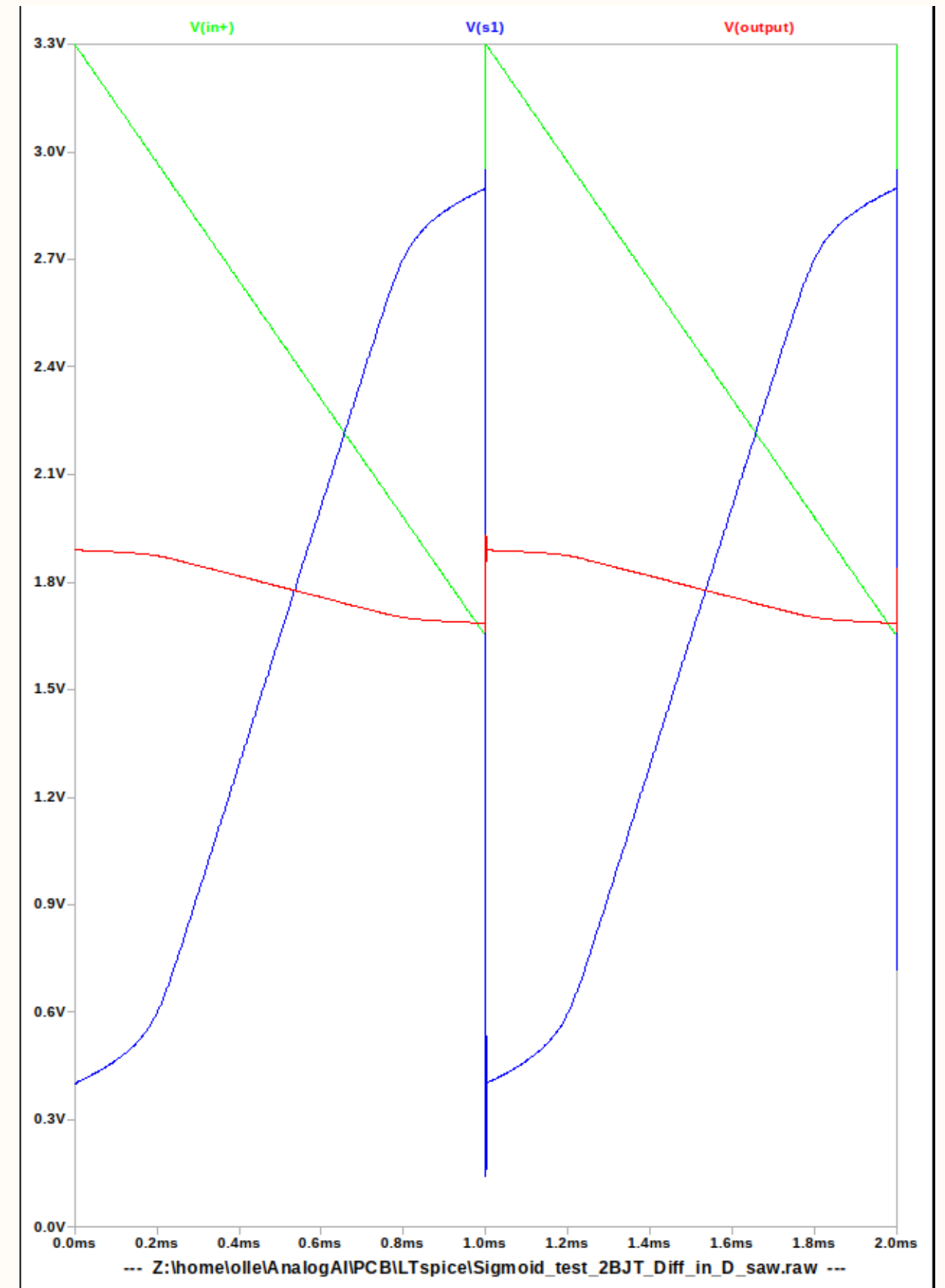
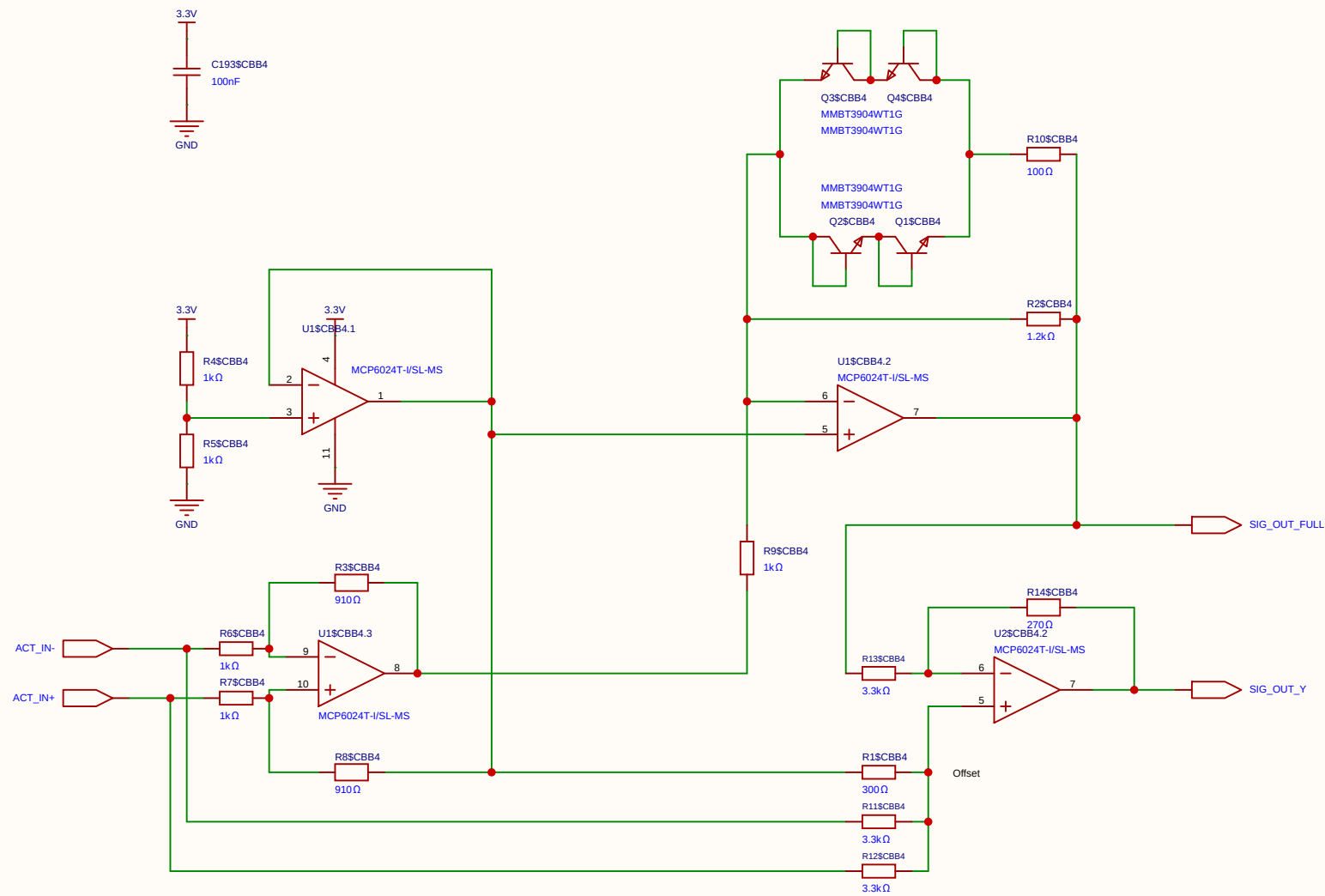
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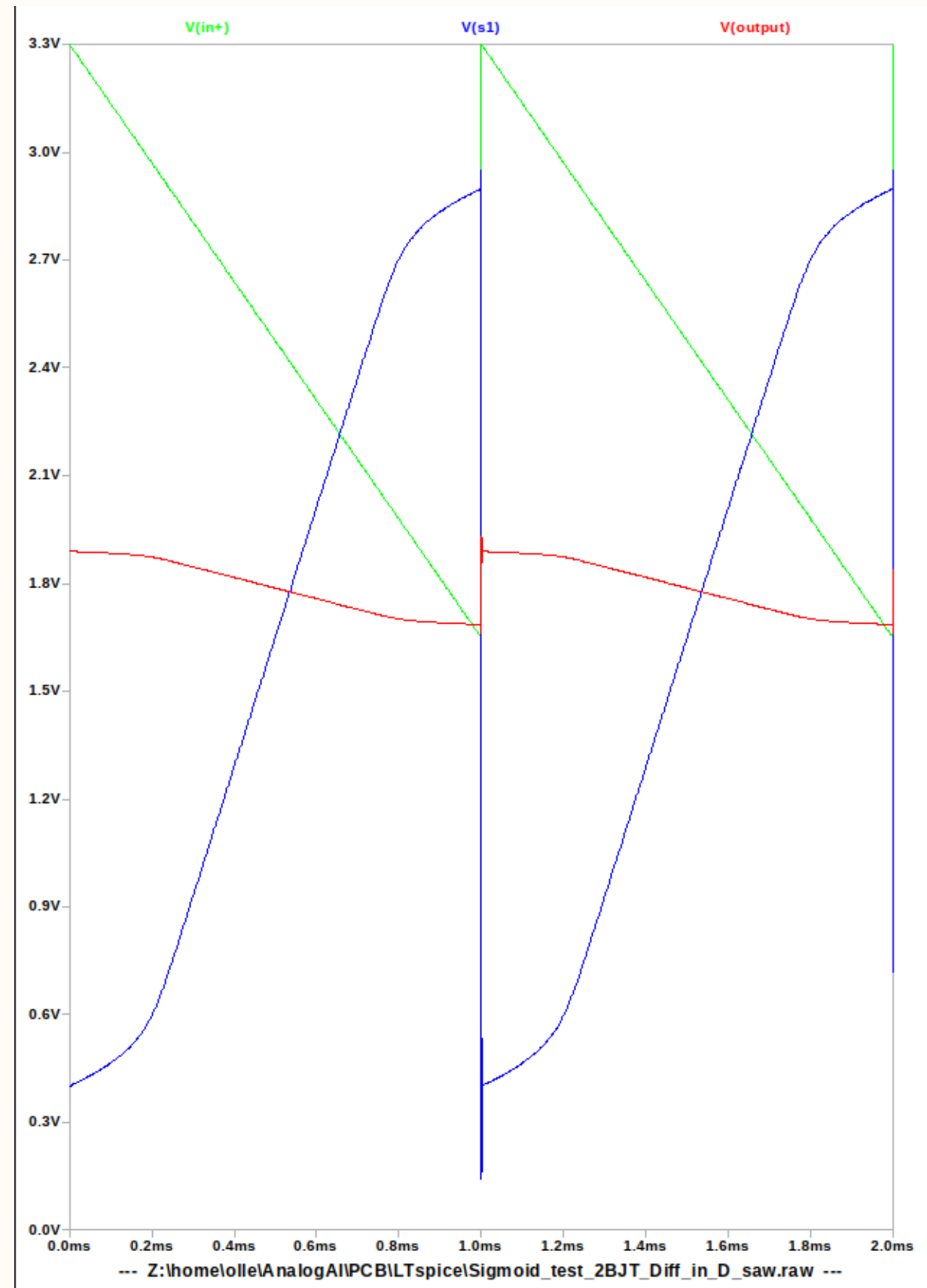
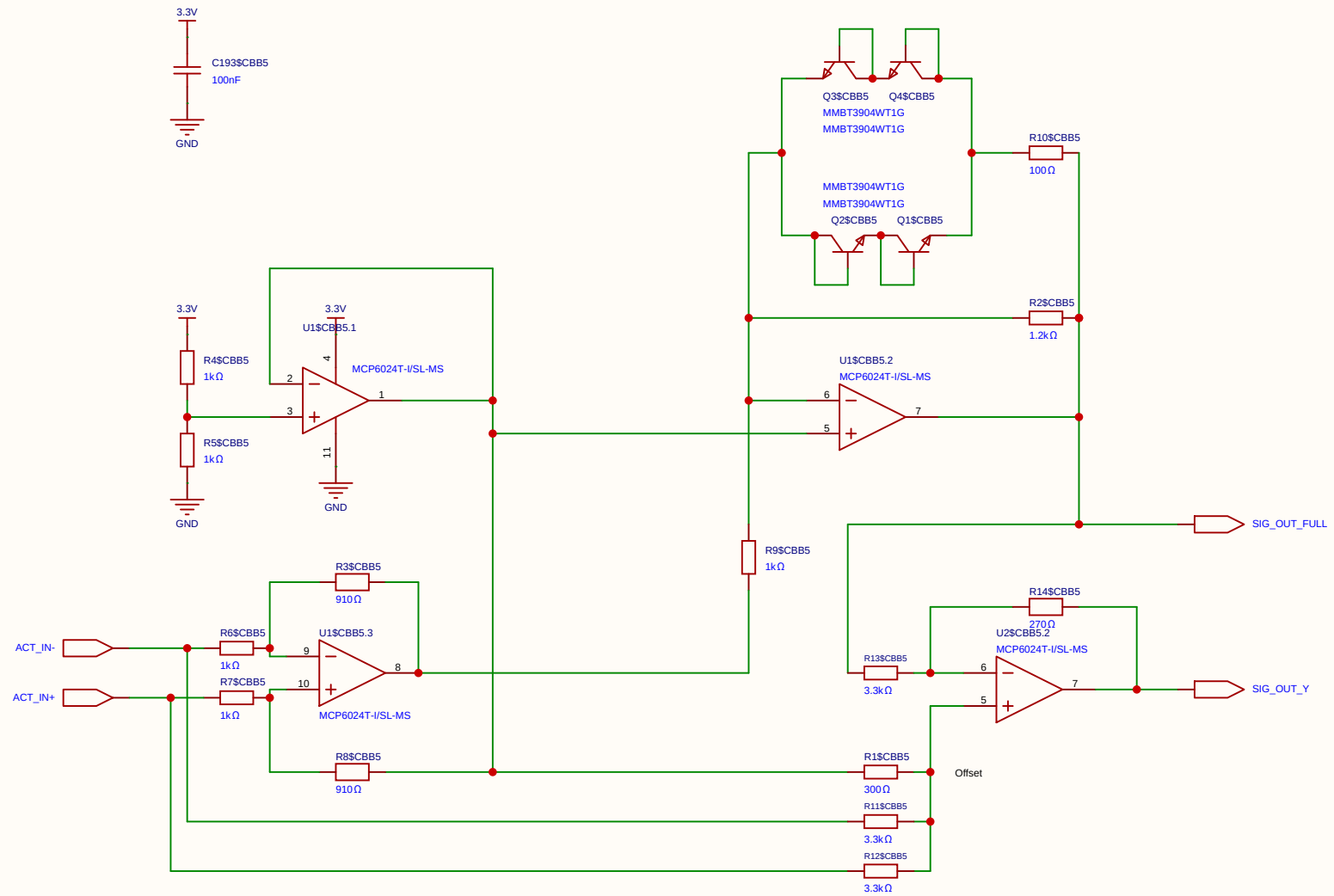
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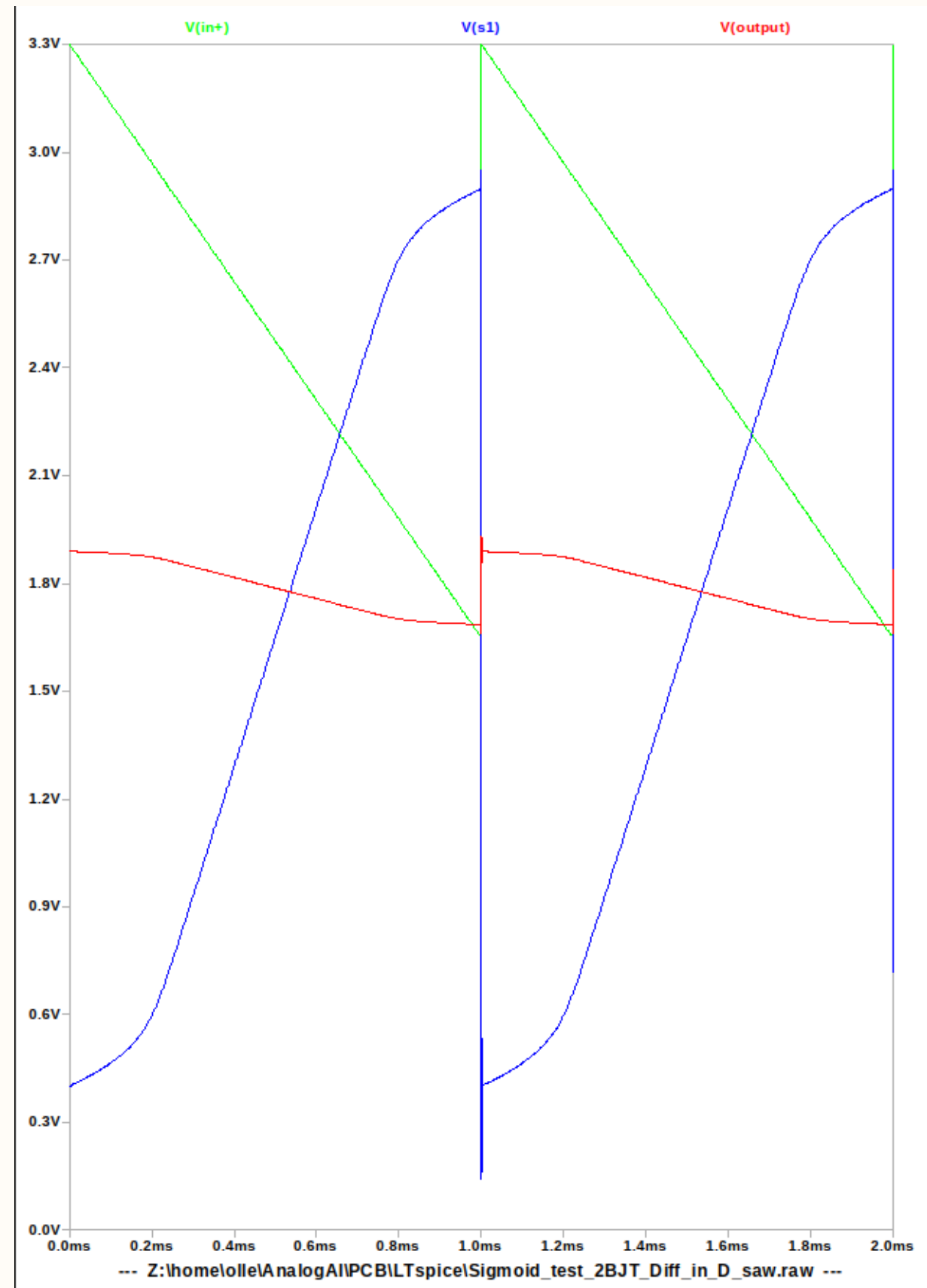
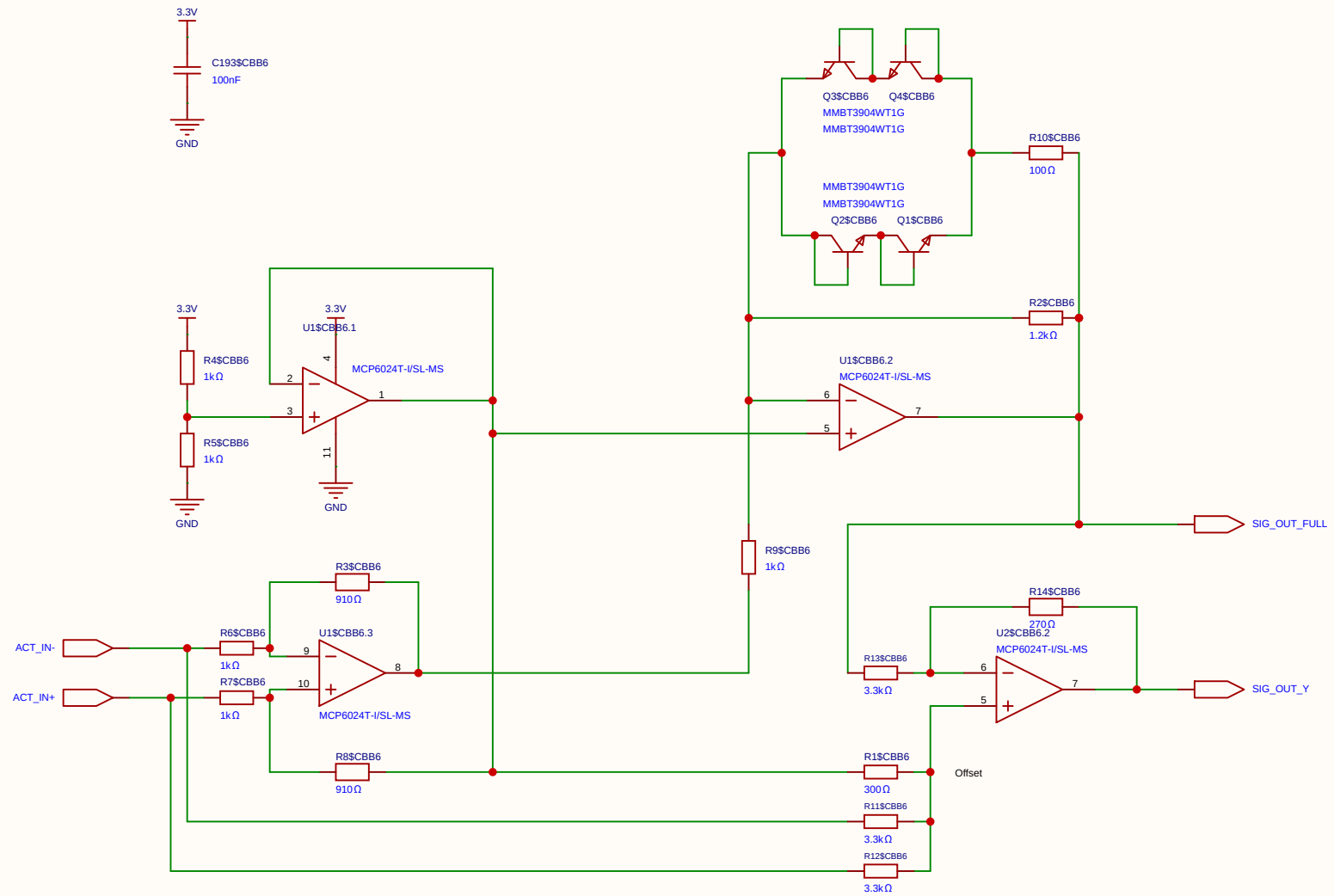
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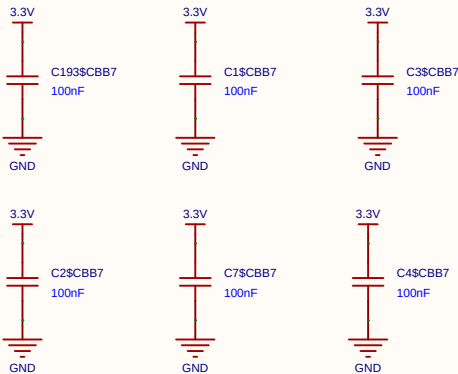
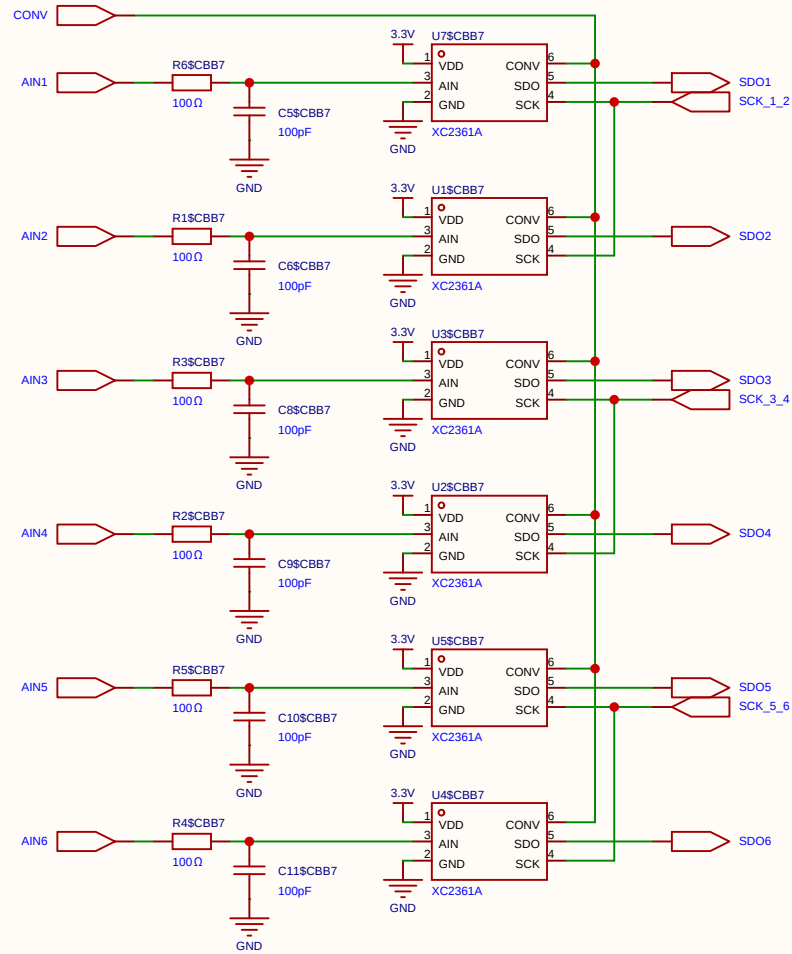
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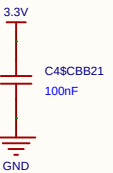
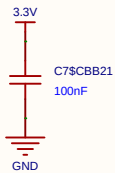
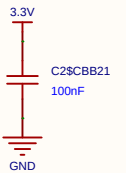
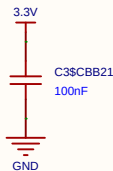
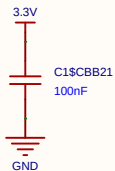
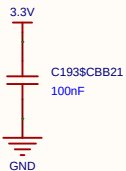
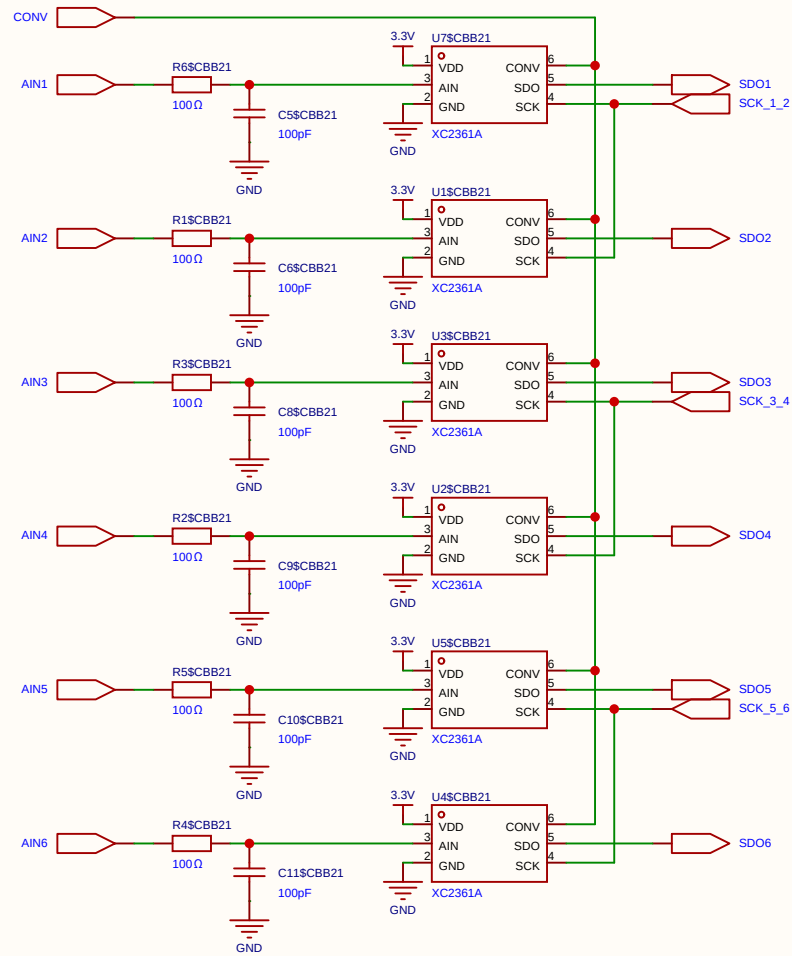
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EasyEDA		V1.0	A4	EasyEDA.com	



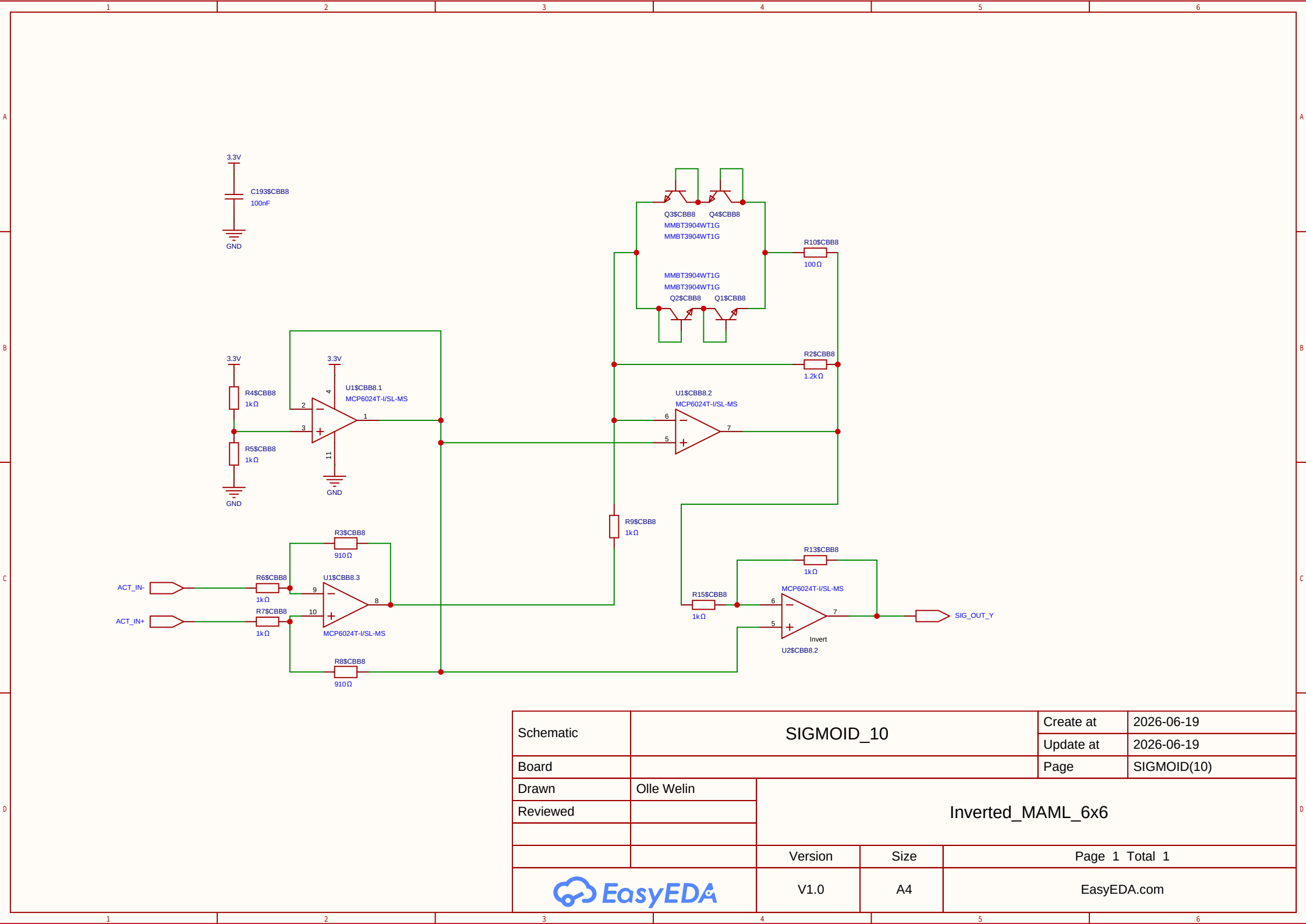
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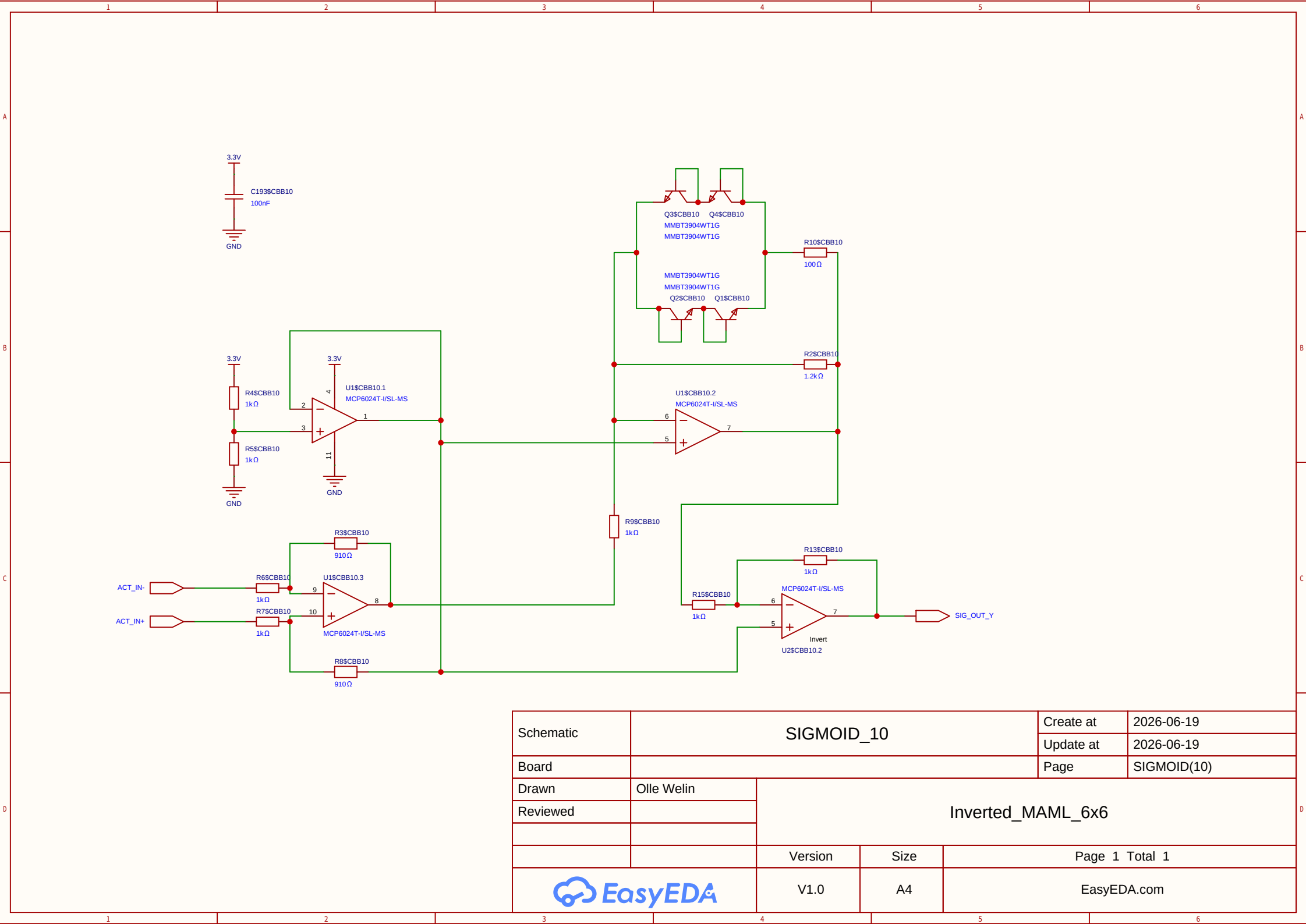
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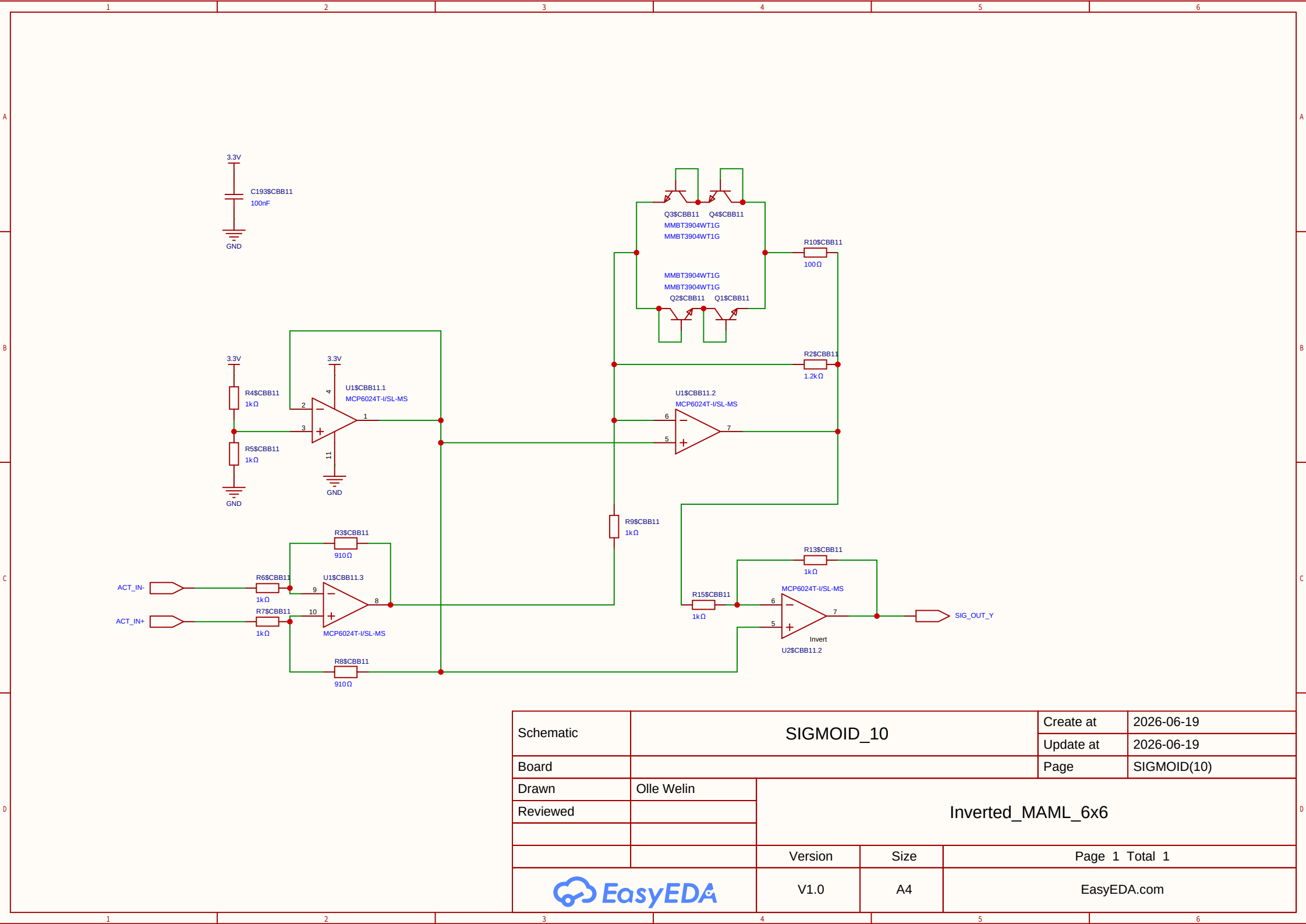
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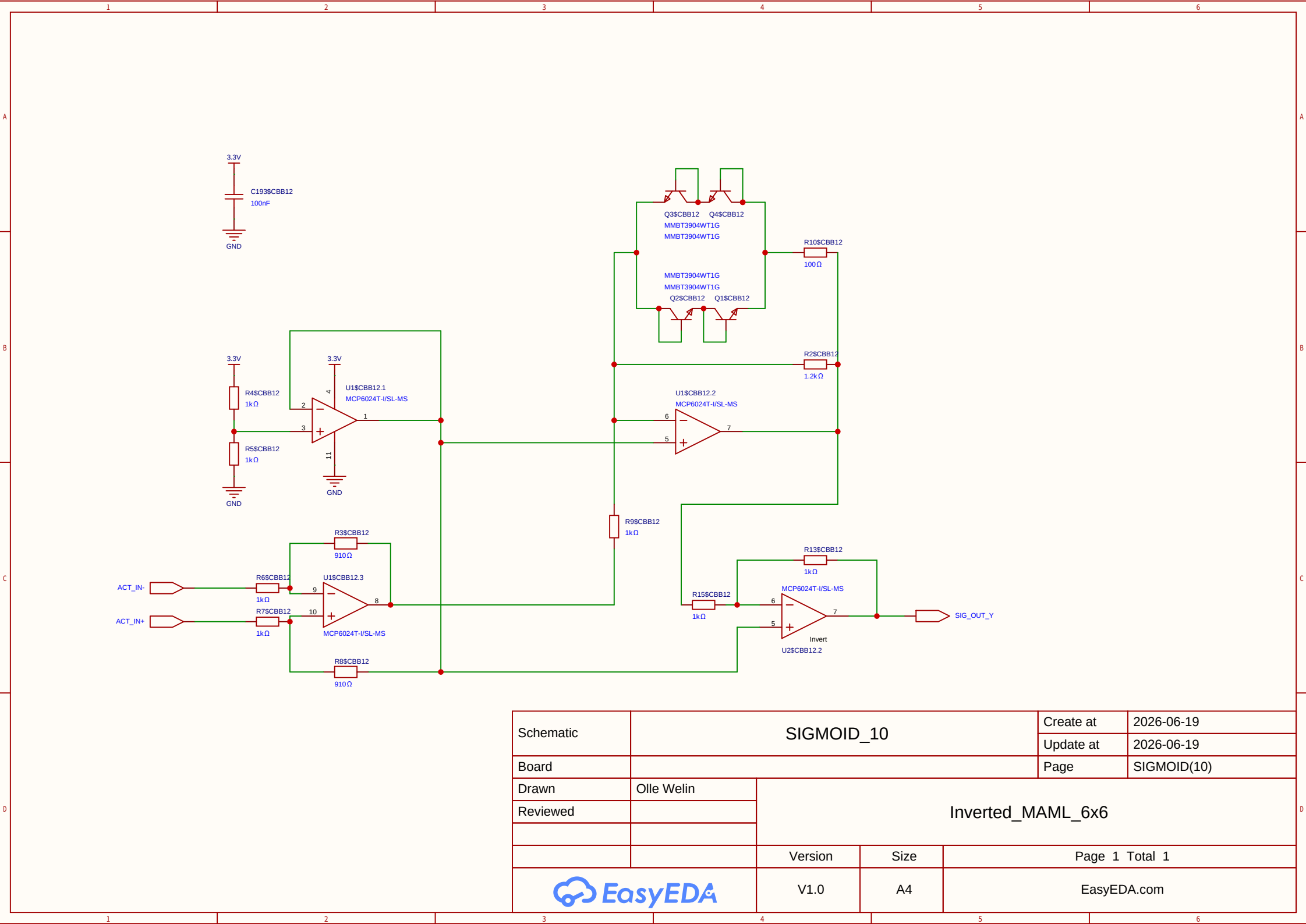
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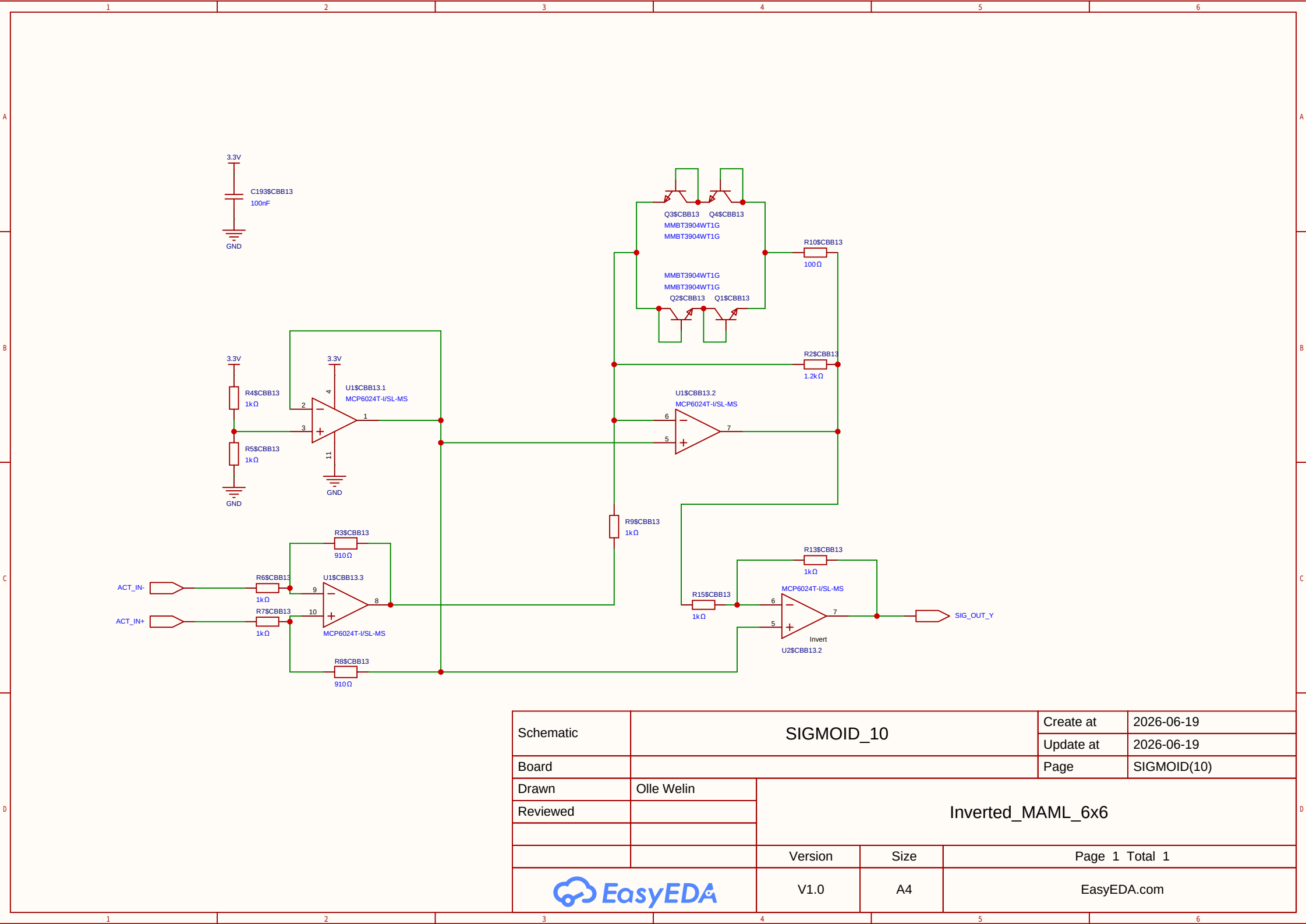
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Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	



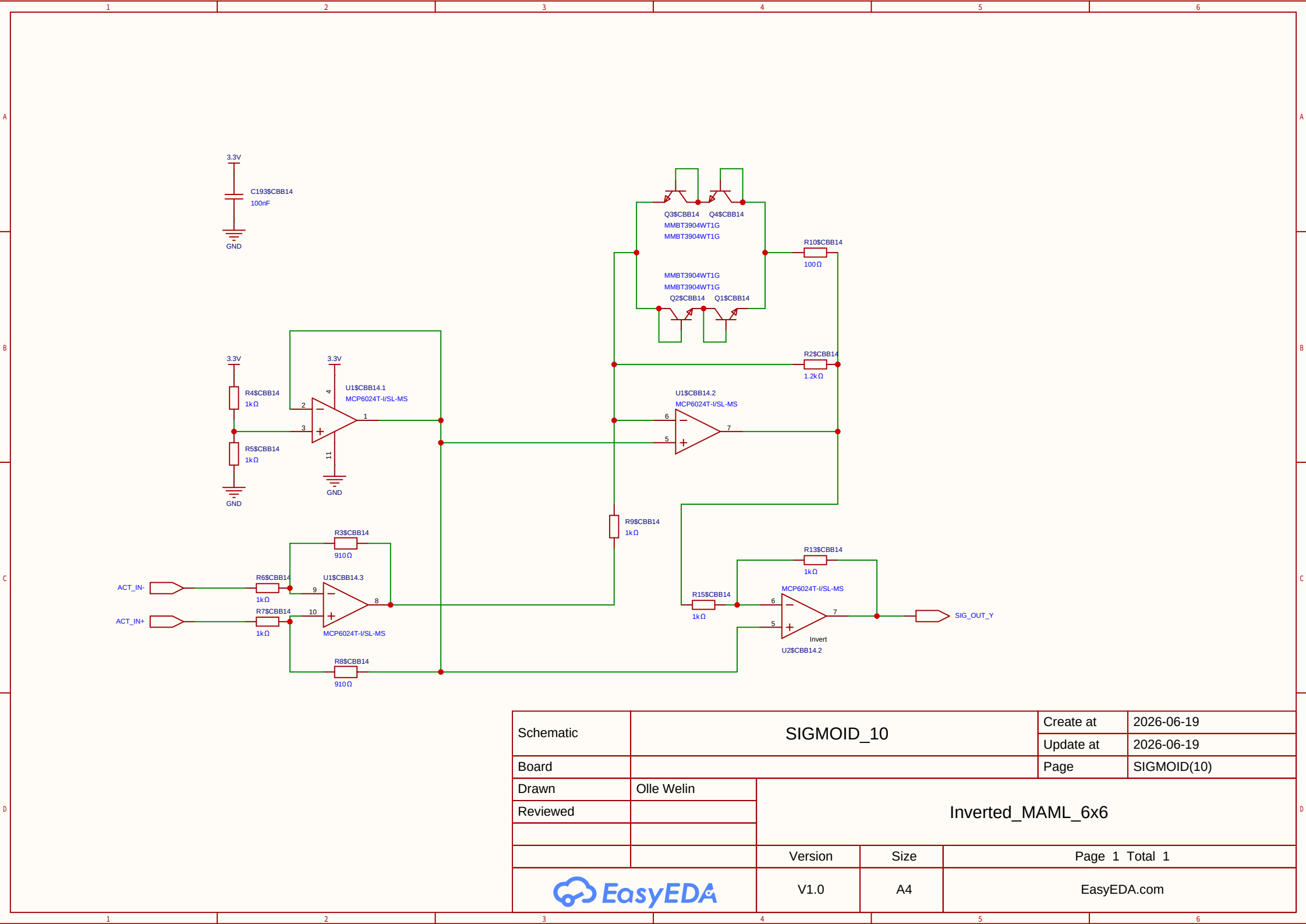
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Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	



Schematic	SIGMOID_10			Create at	2026-06-19
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Board				Page	SIGMOID(10)
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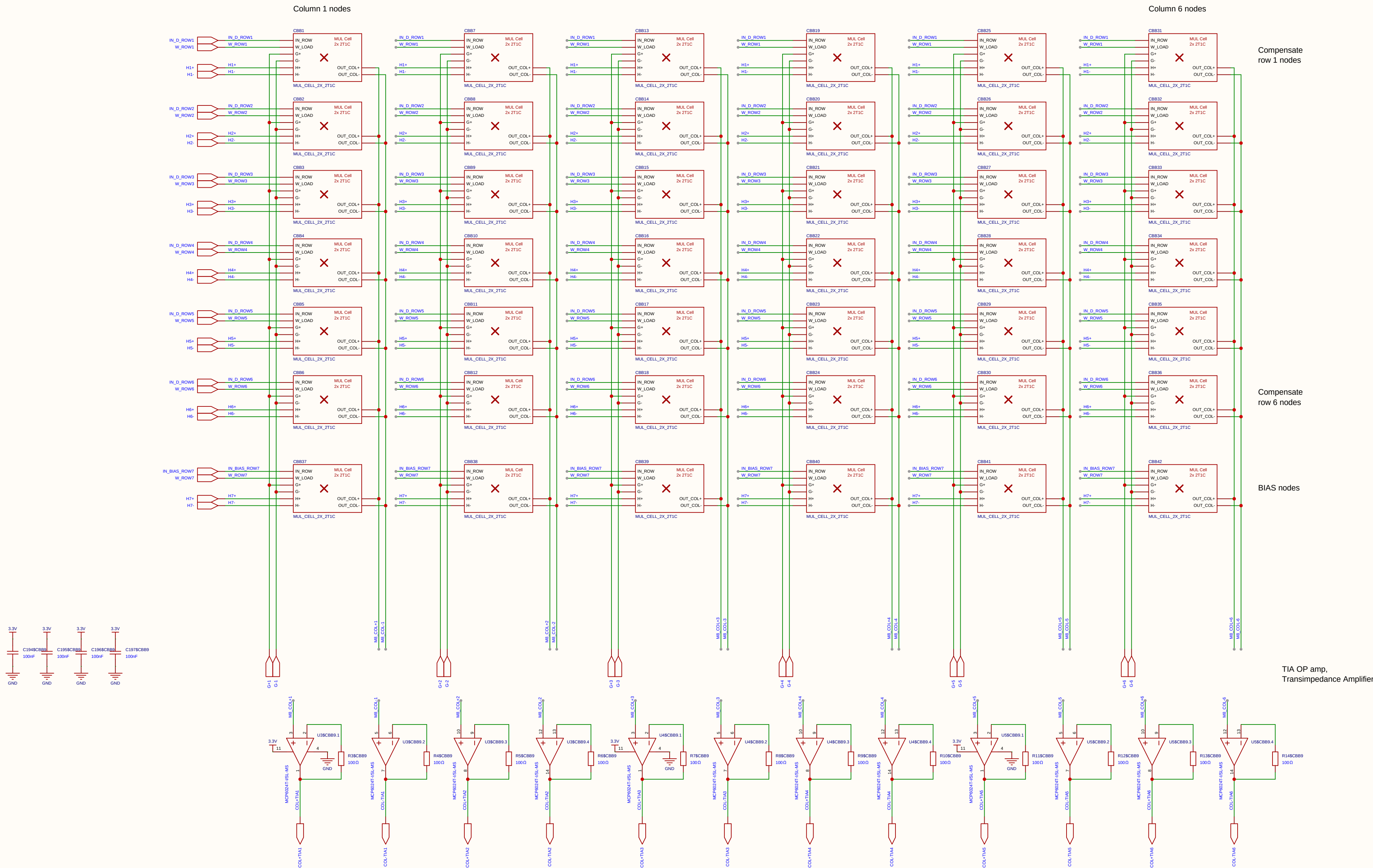


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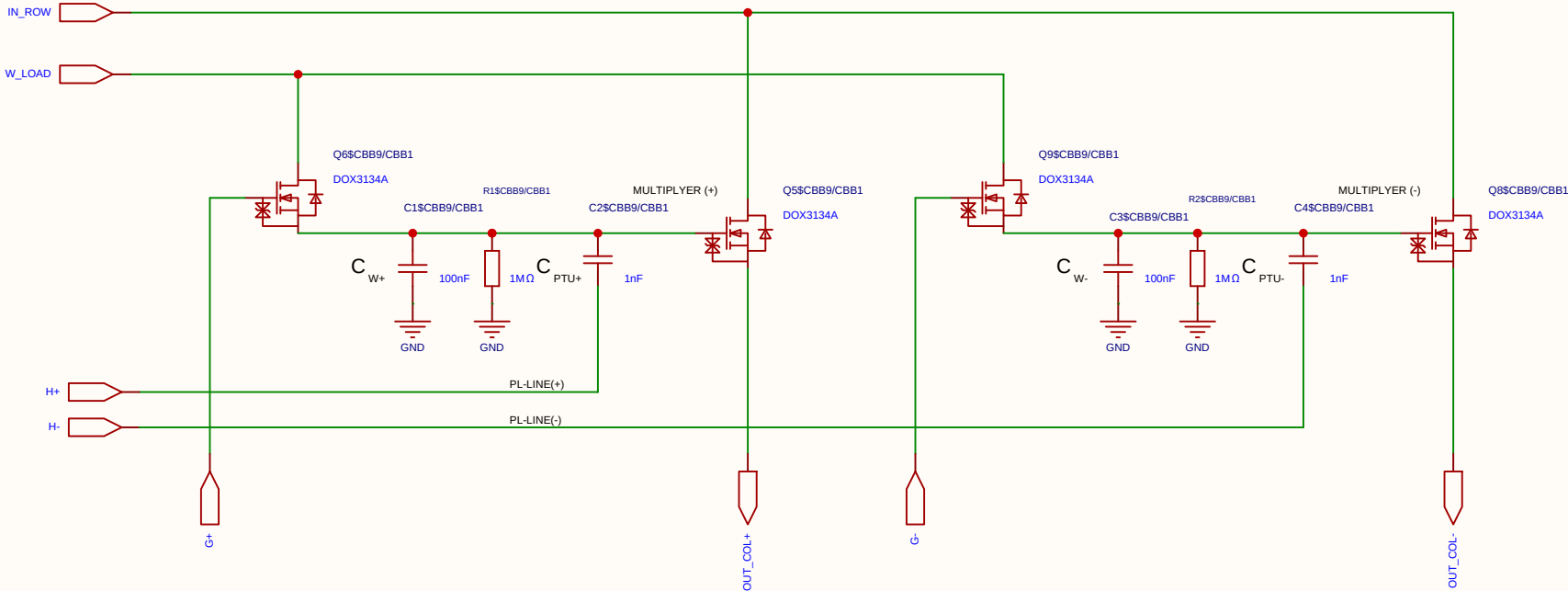
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Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$



Schematic	matrix8			Create at	2026-06-25
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Board				Page	matx8
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

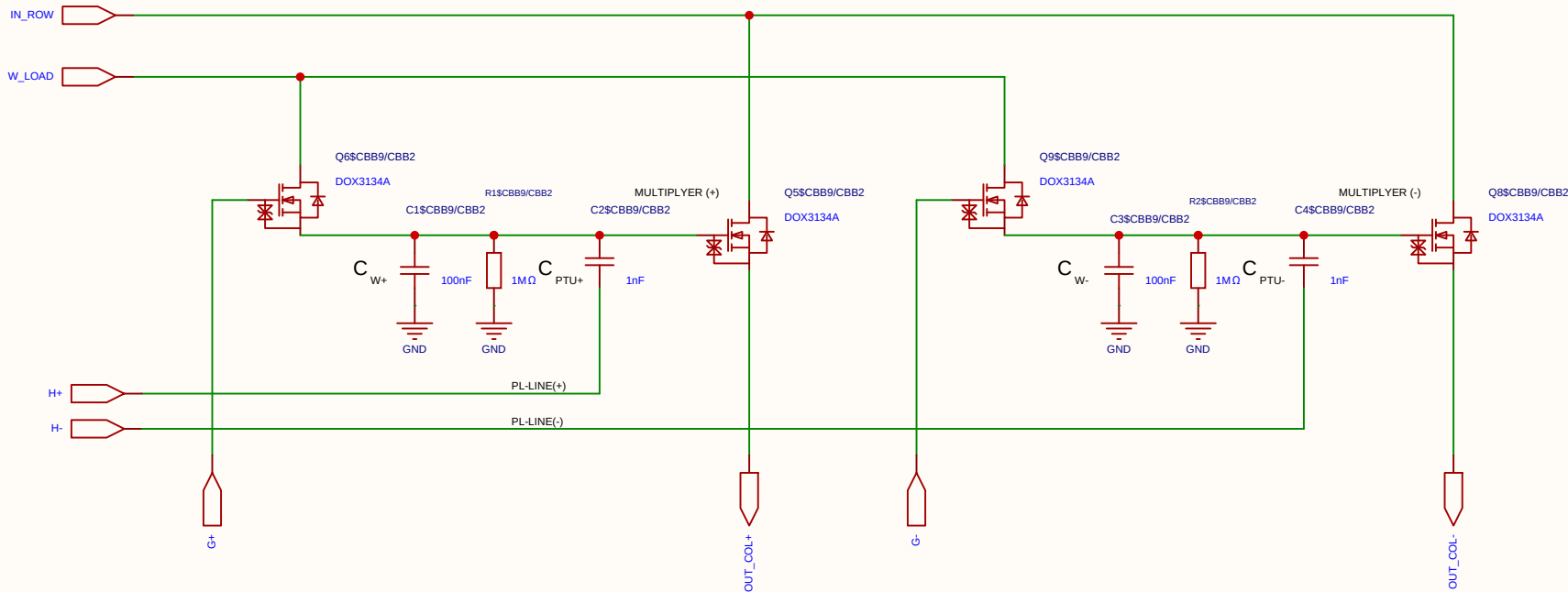
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

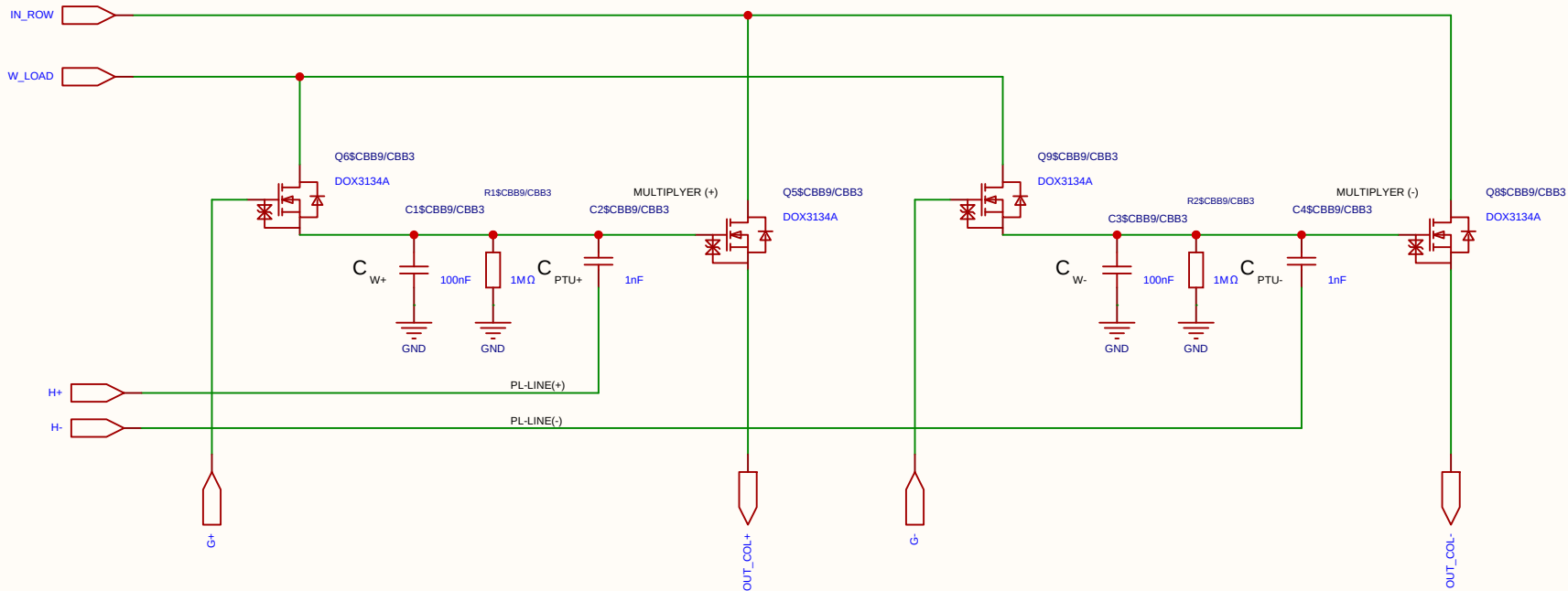
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

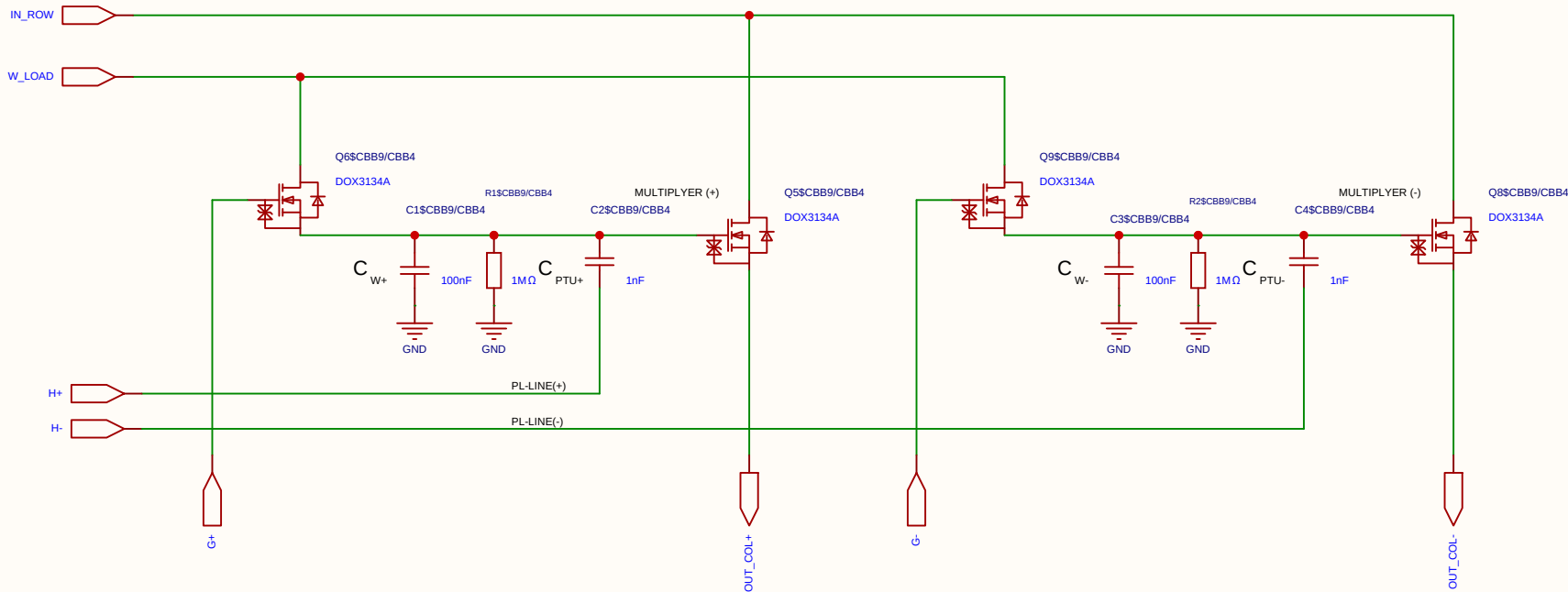
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
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Reviewed					
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

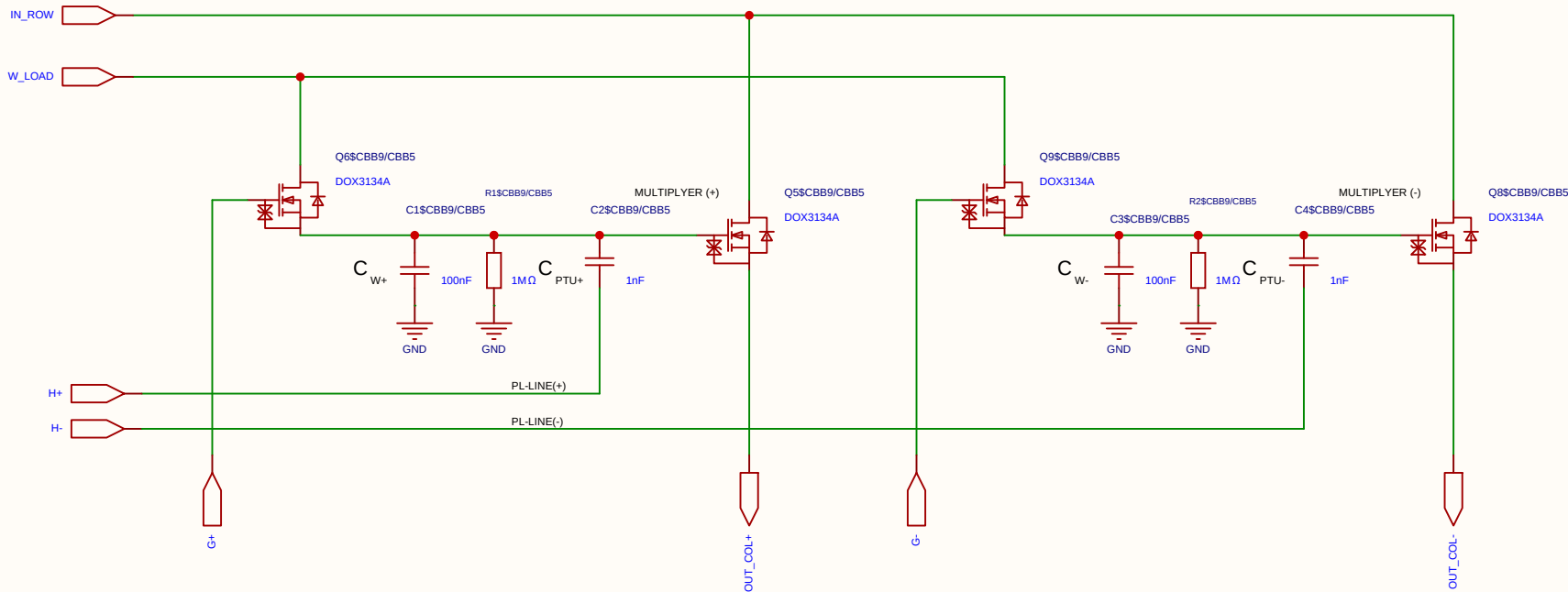
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

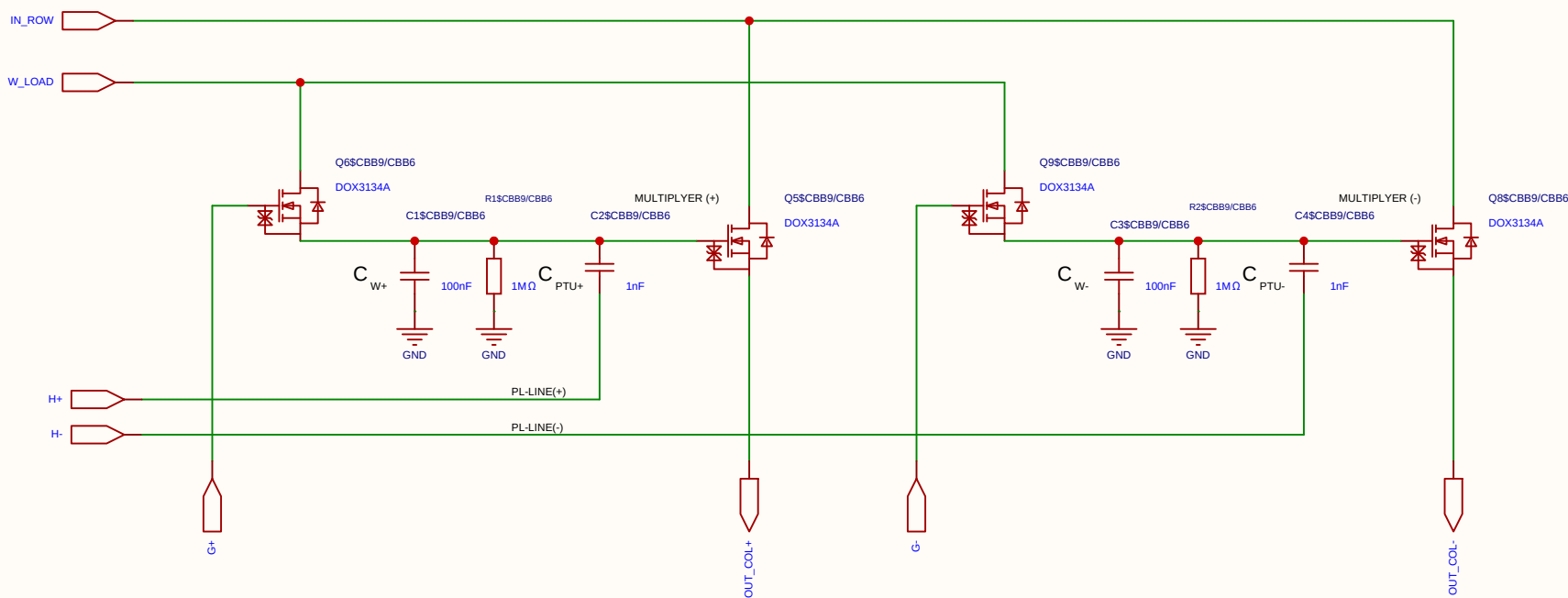
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

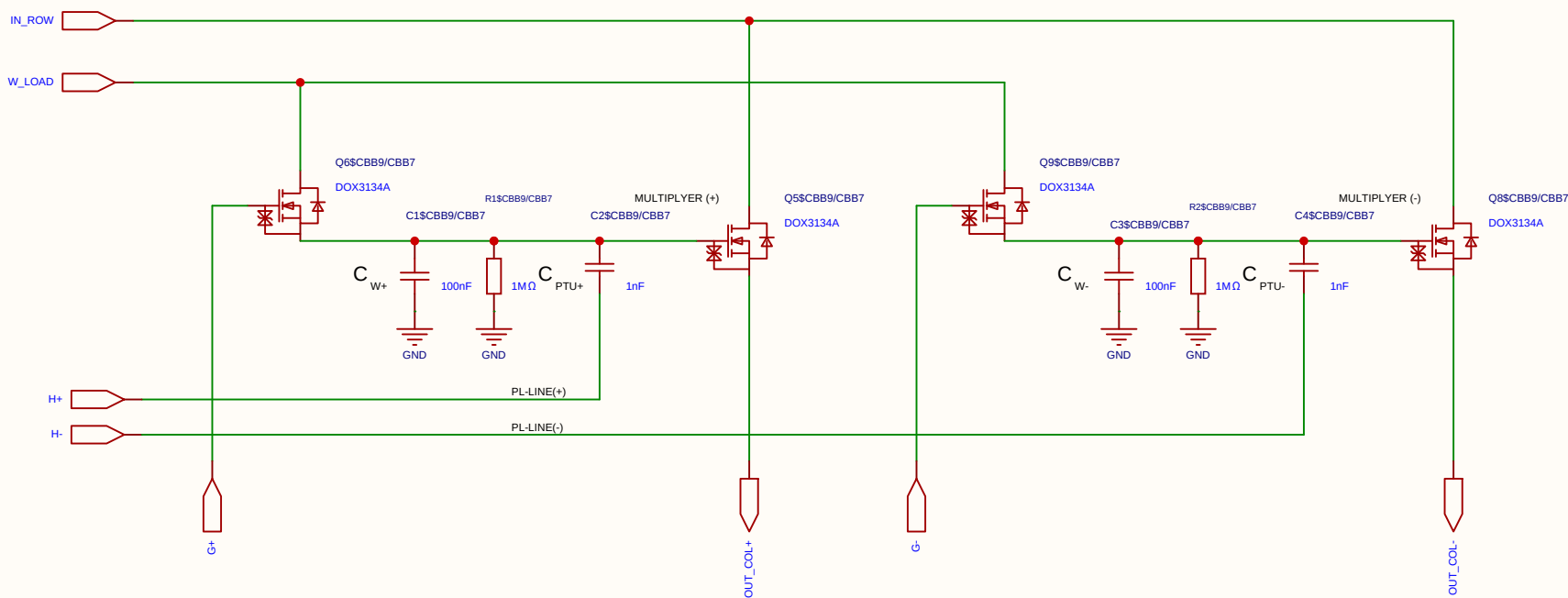
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

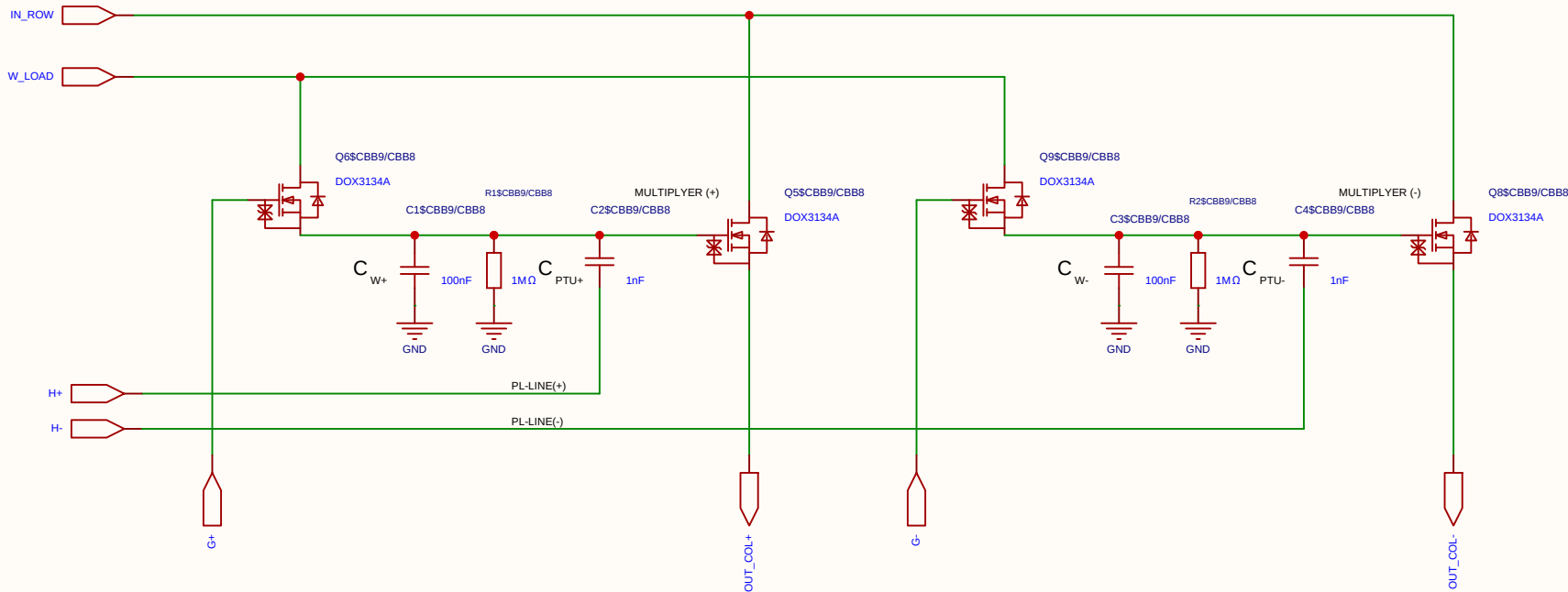
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

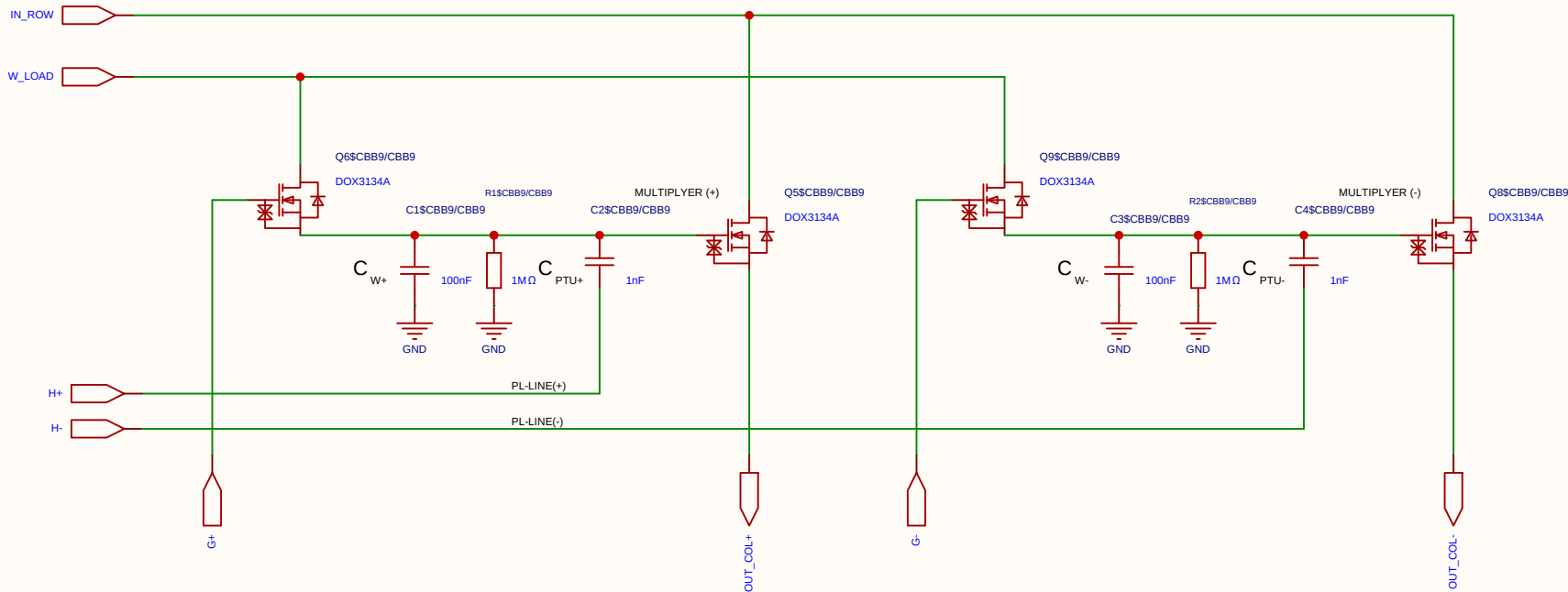
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
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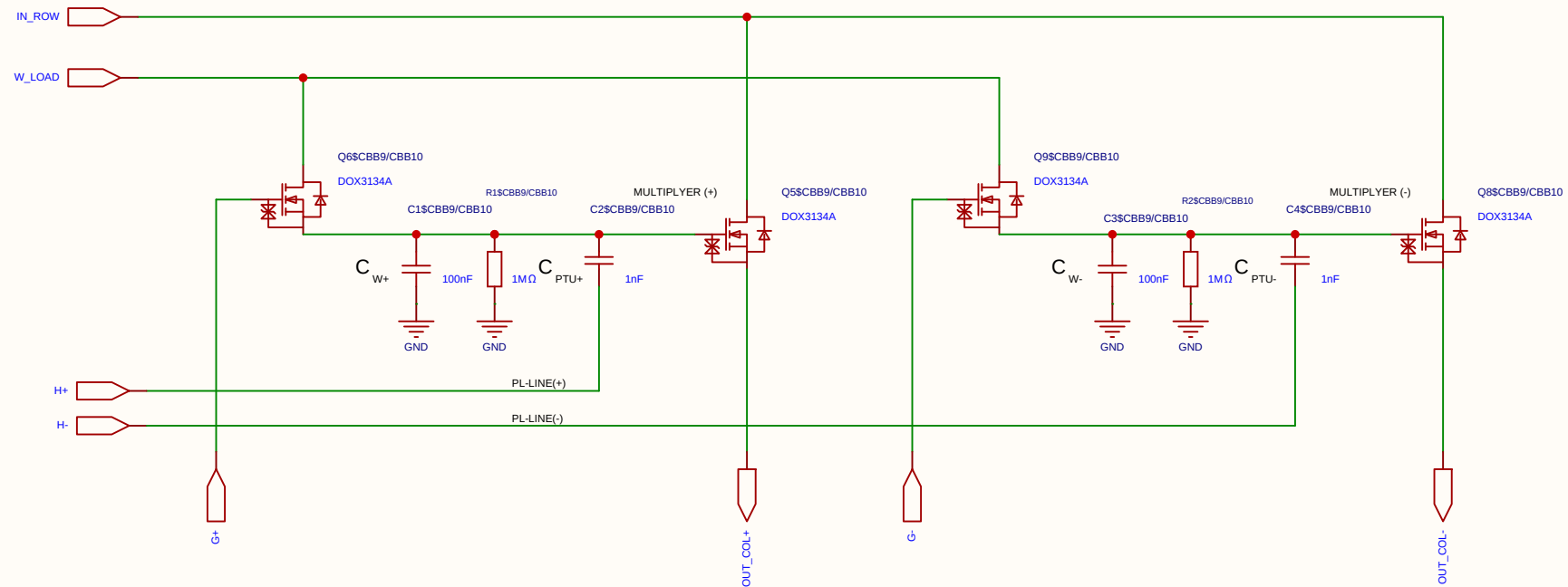
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
Resistor Network	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

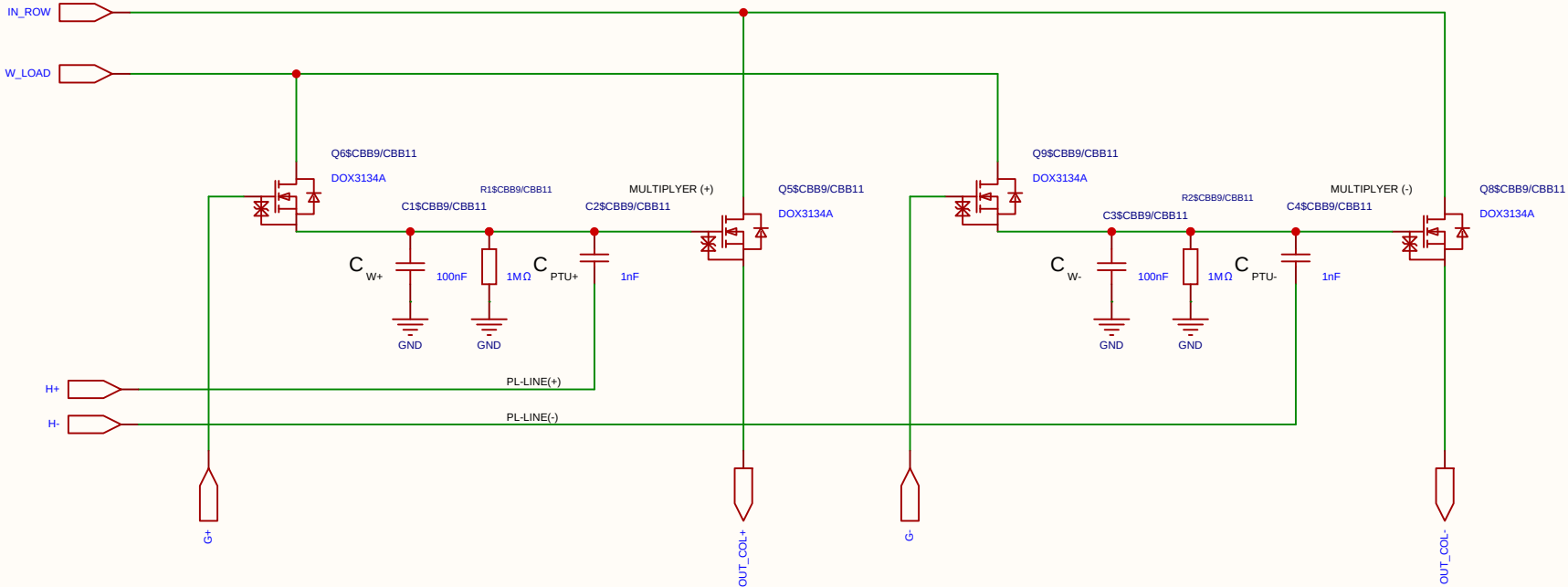
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

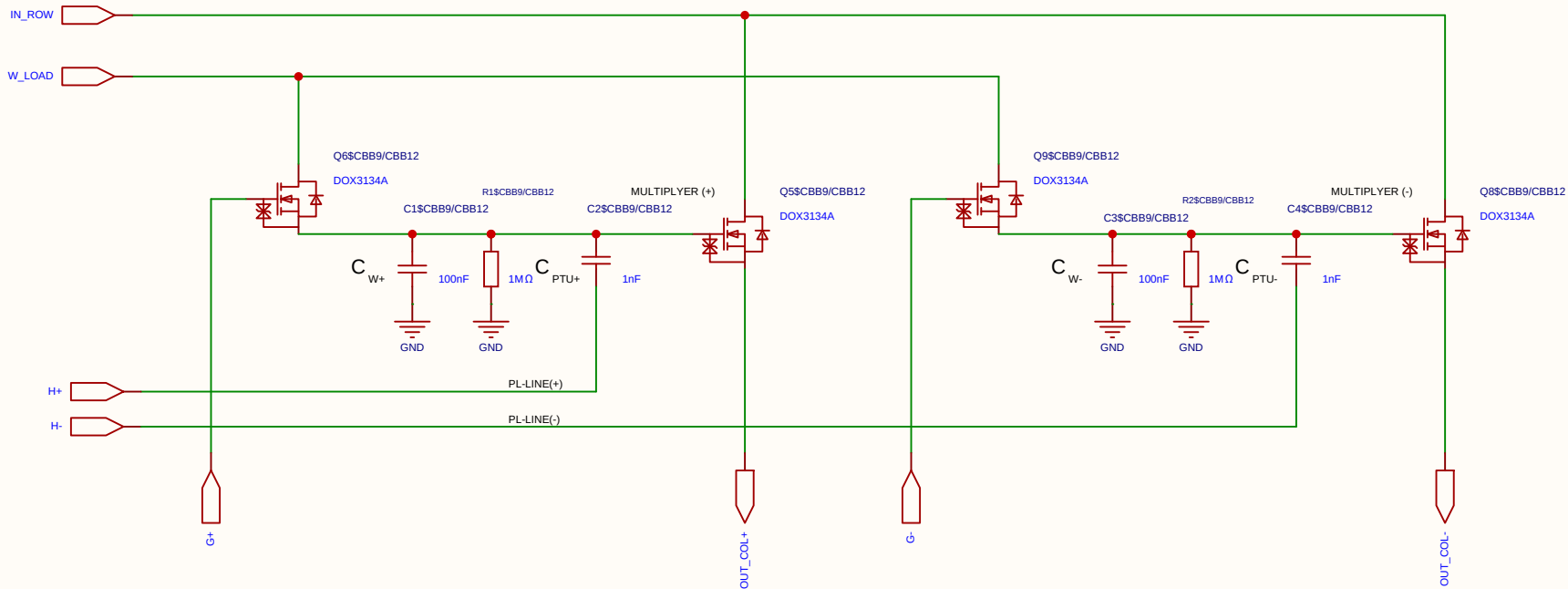
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

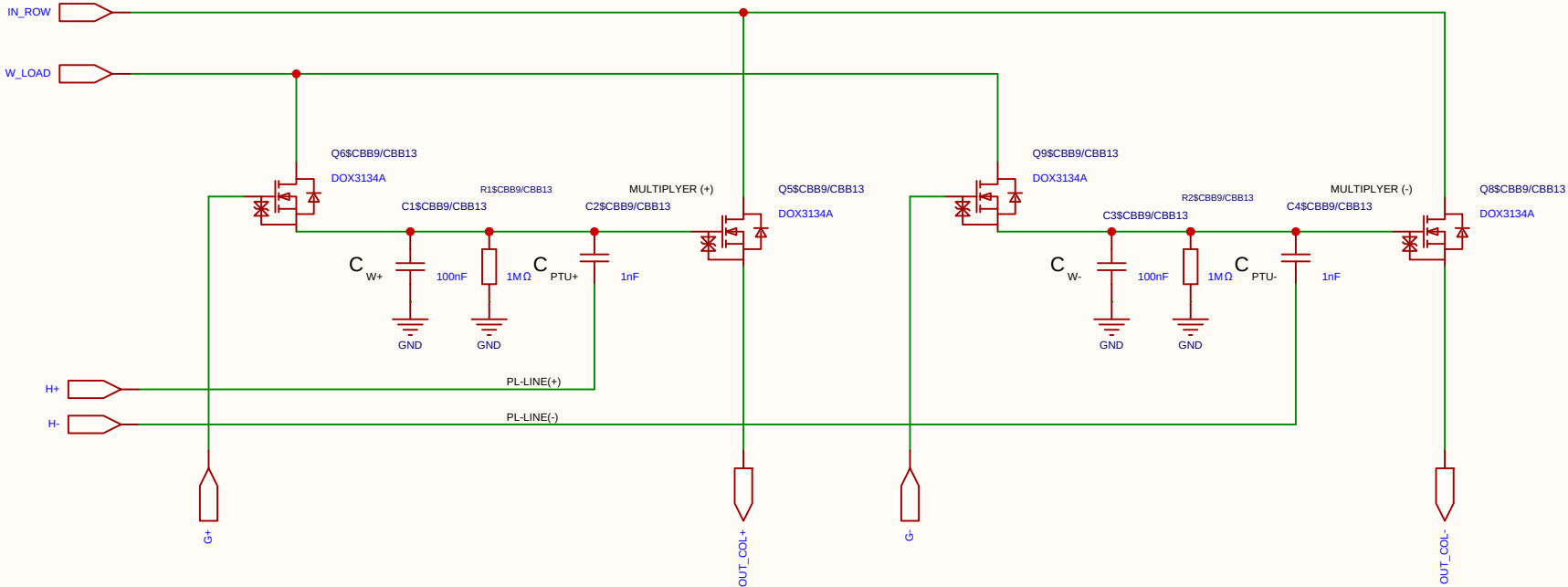
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

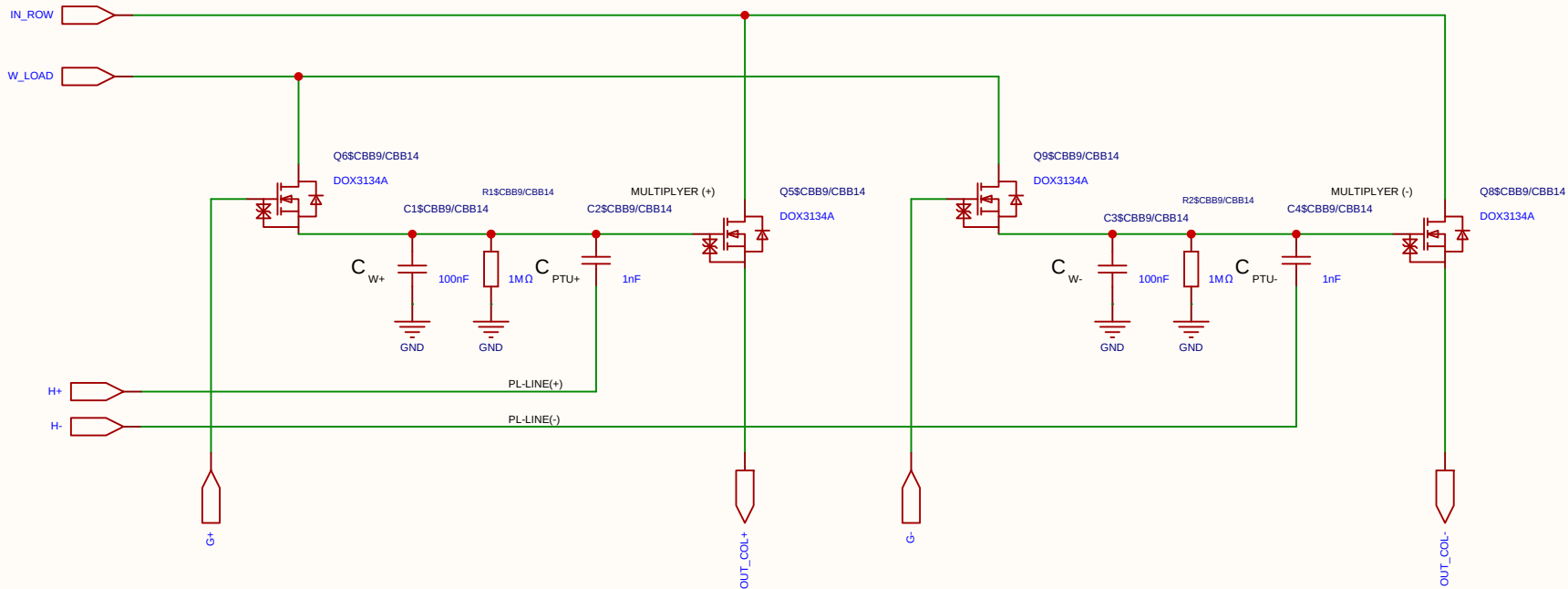
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

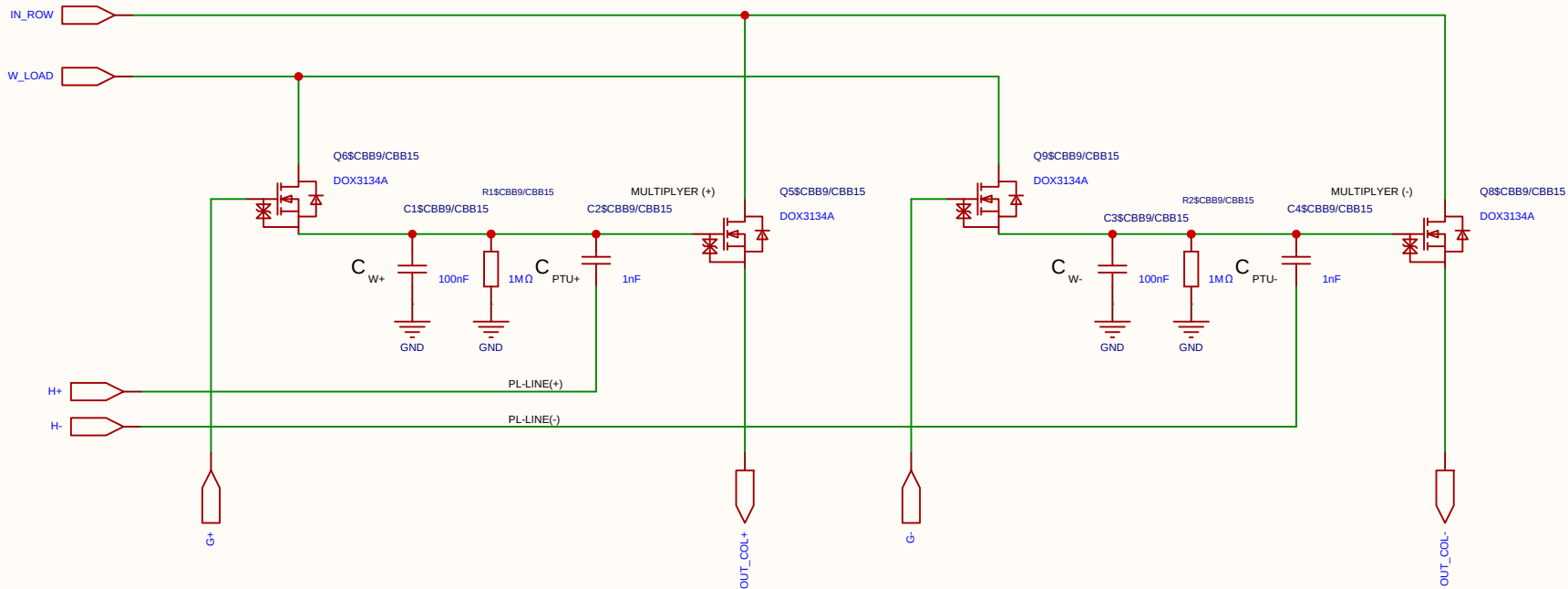
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

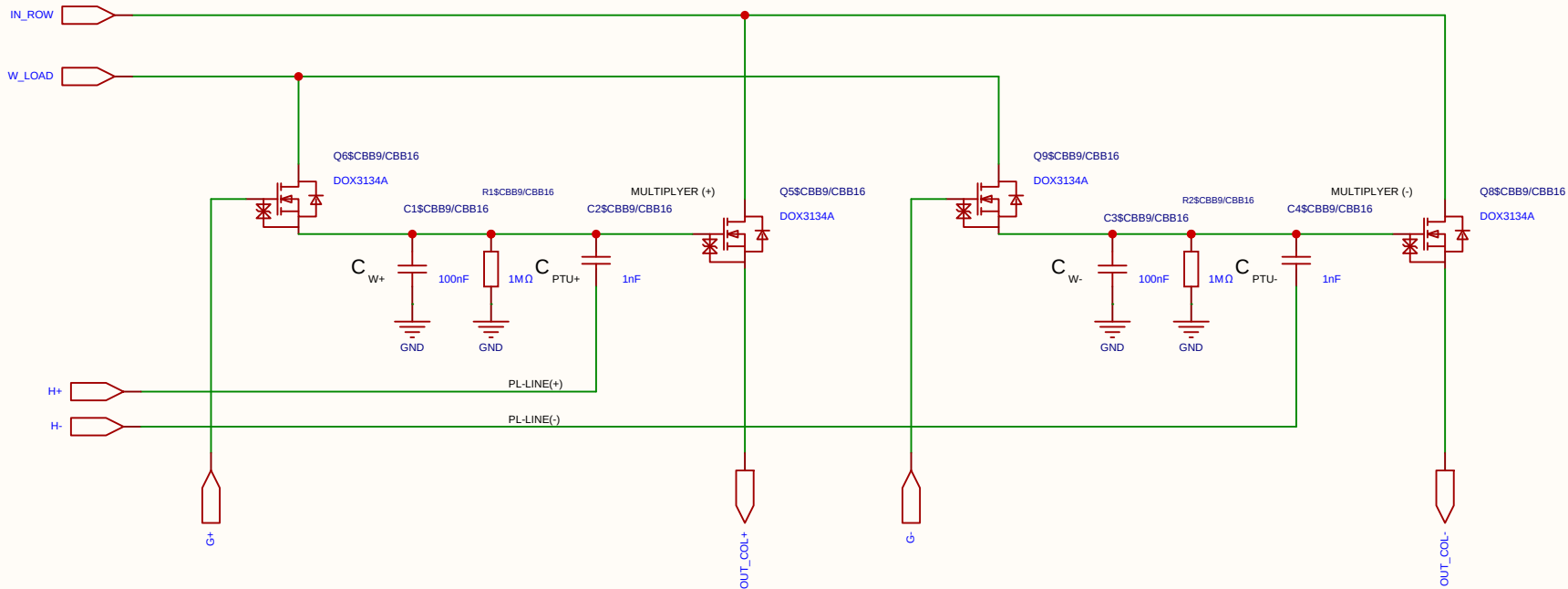
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
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Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

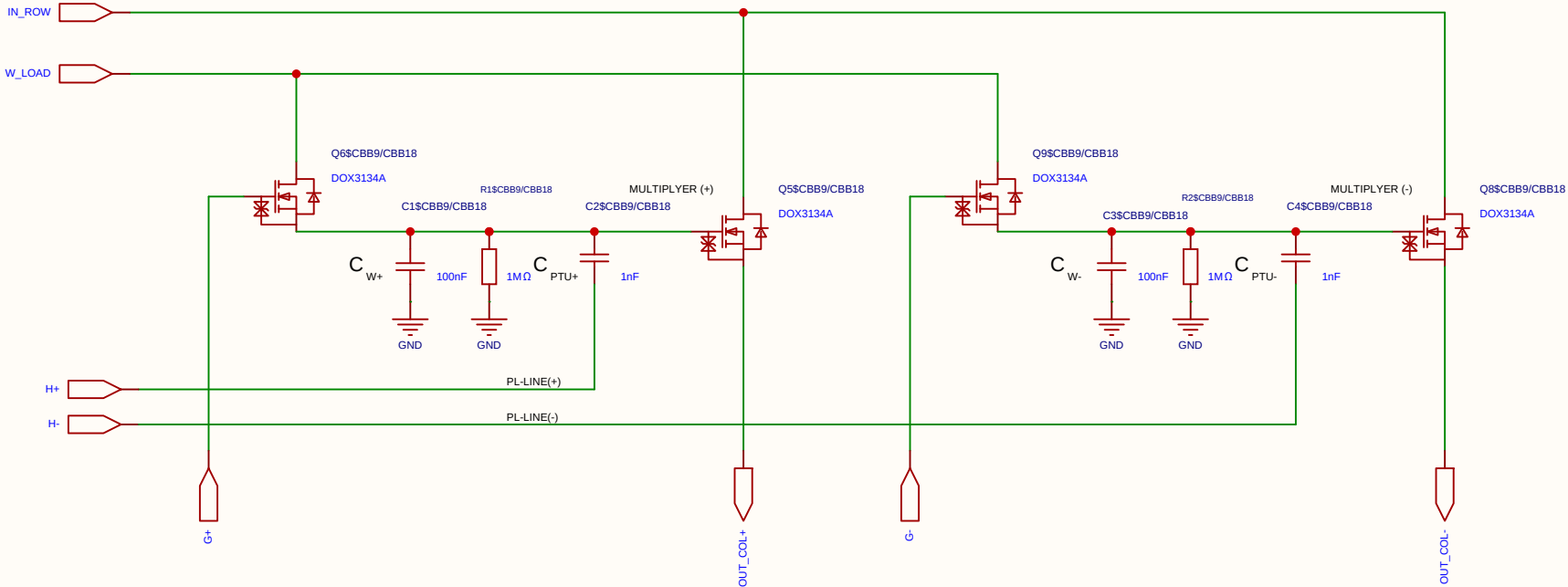
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

State	PL Voltage	Perturbation at Gate	Comment
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_S) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.



Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

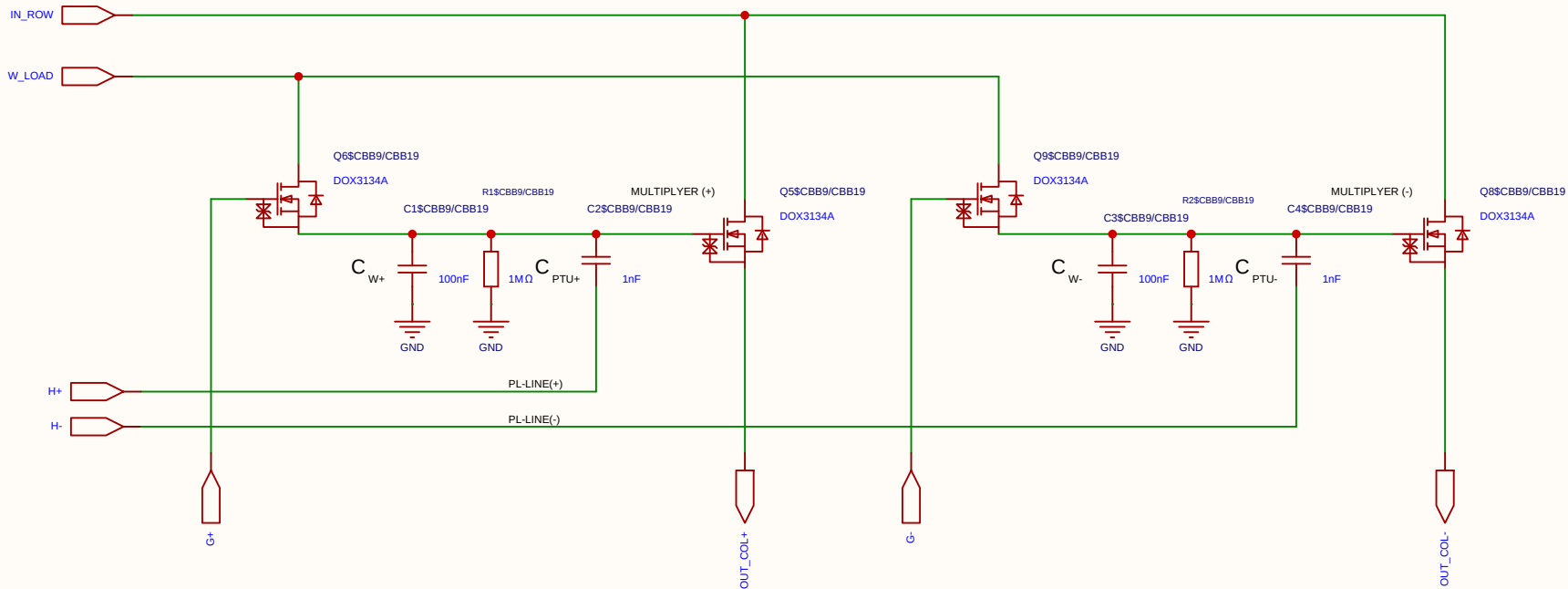
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

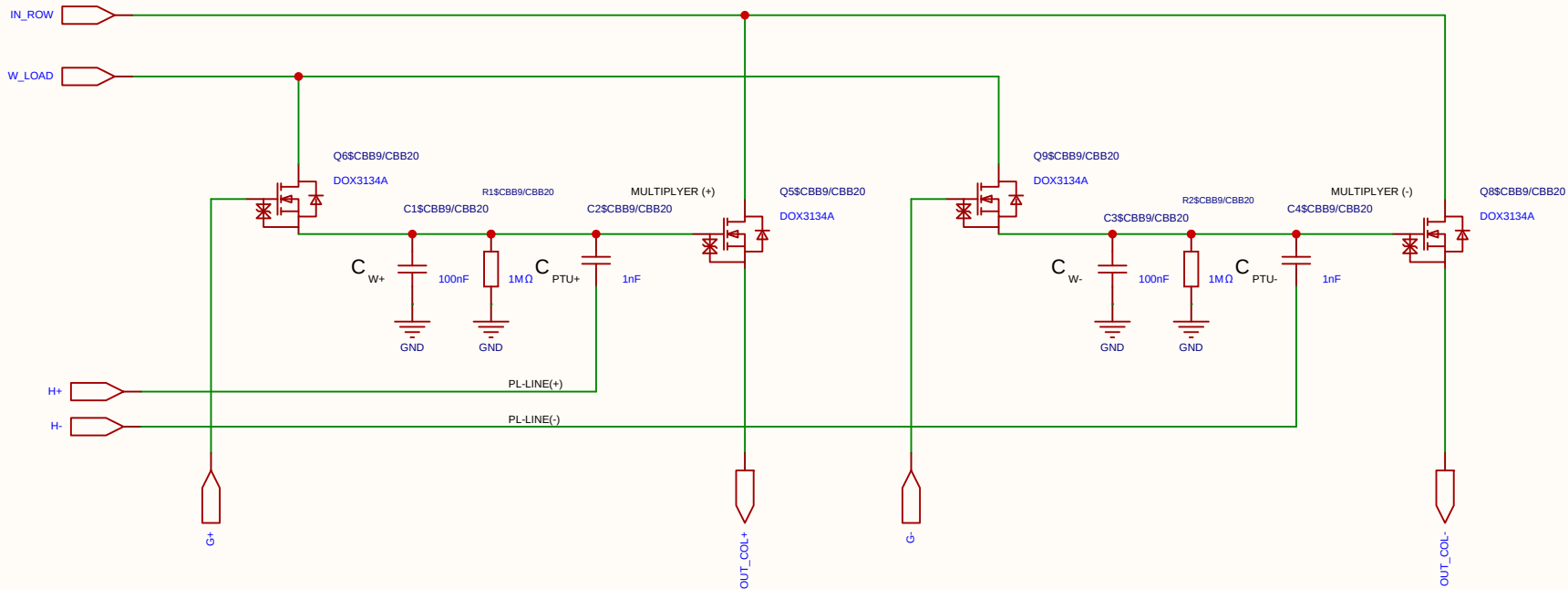
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

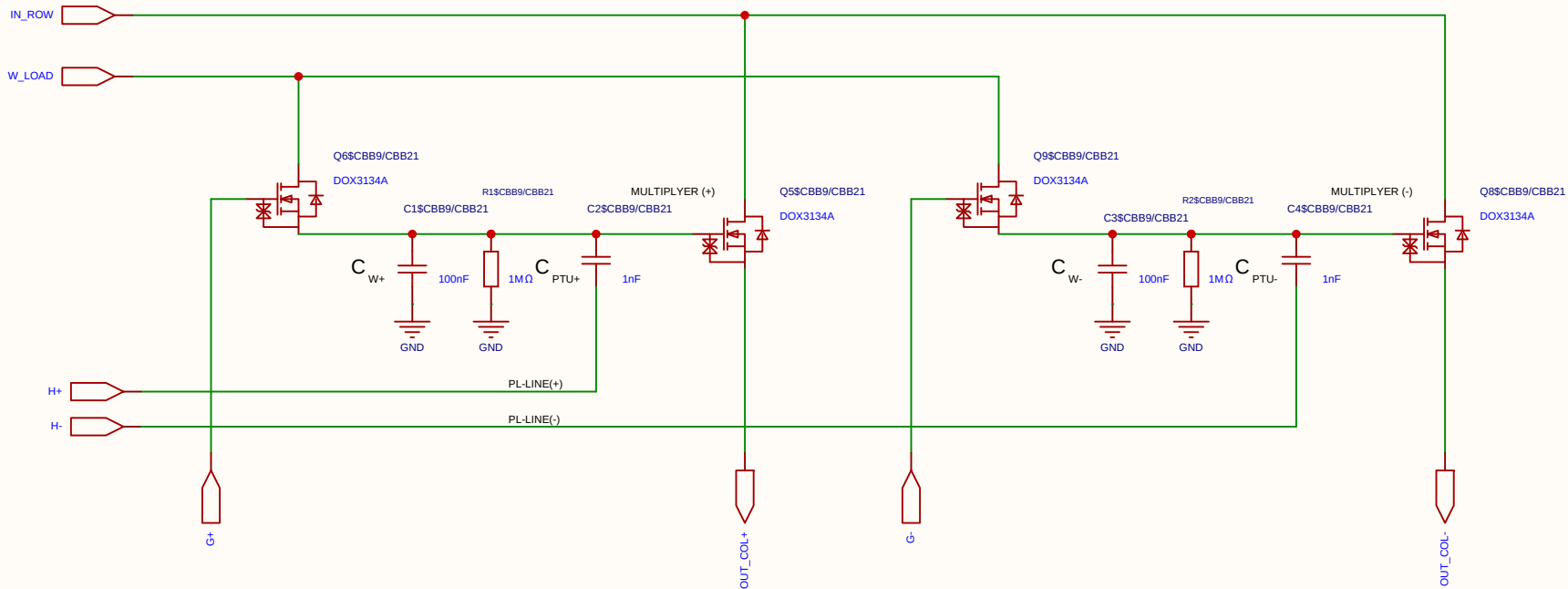
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_3_8_MUL_CELL
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

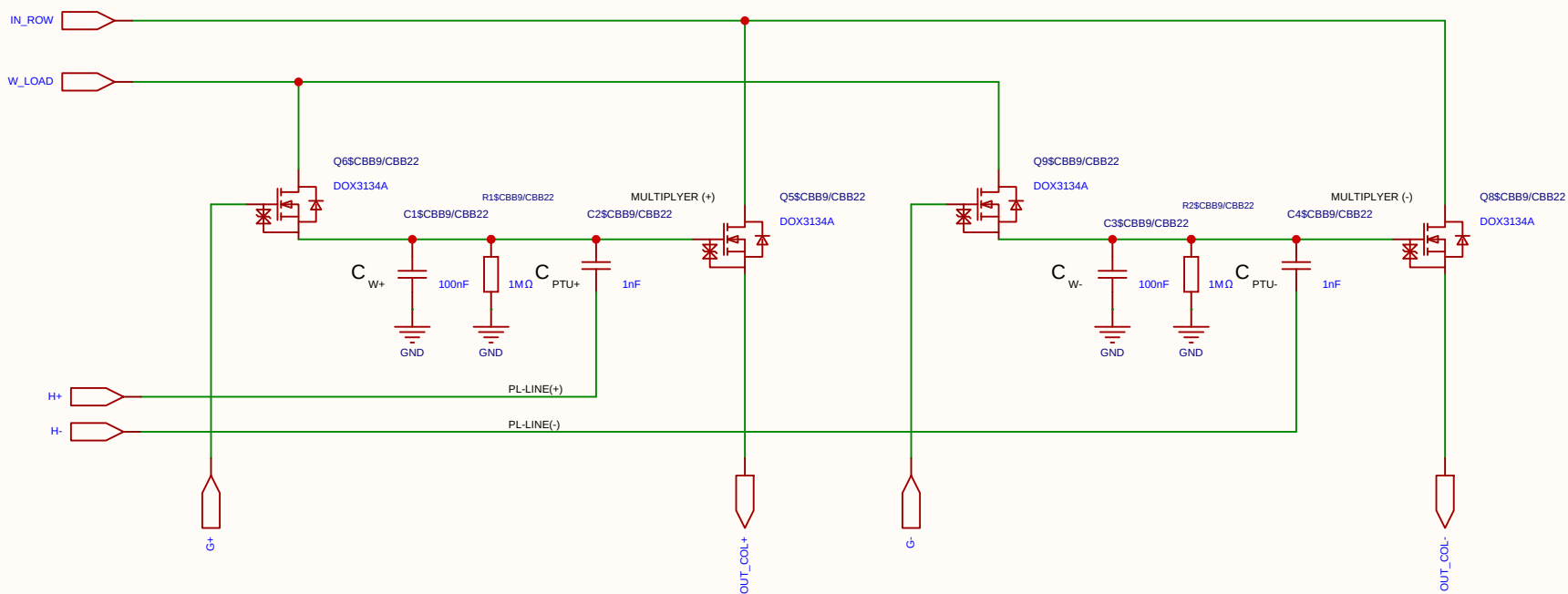
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

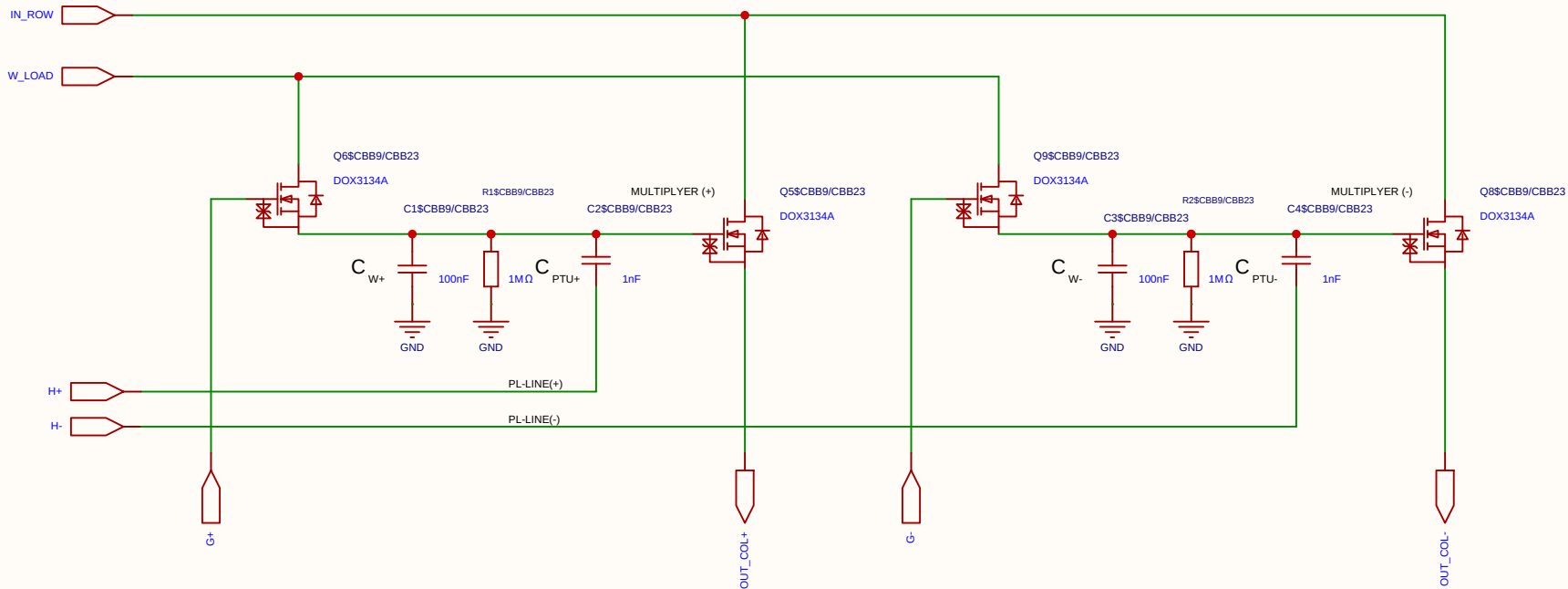
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

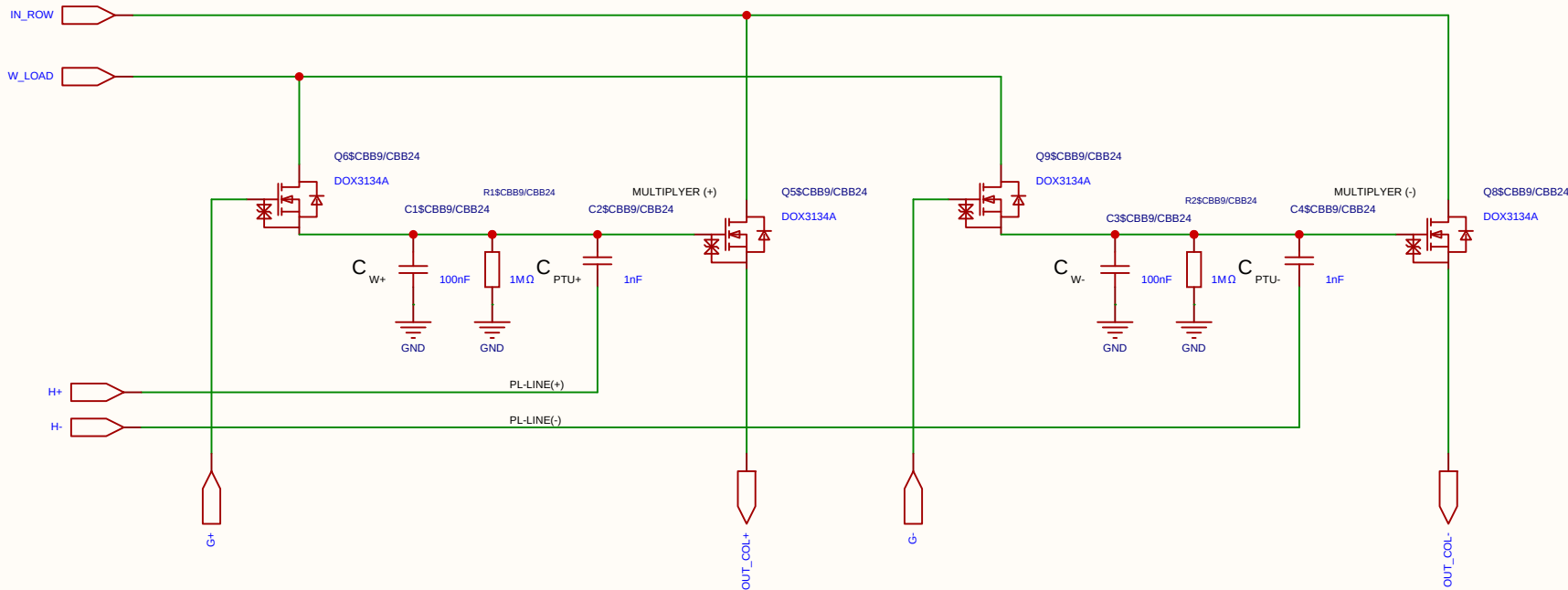
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_3_8_MUL_CELL
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

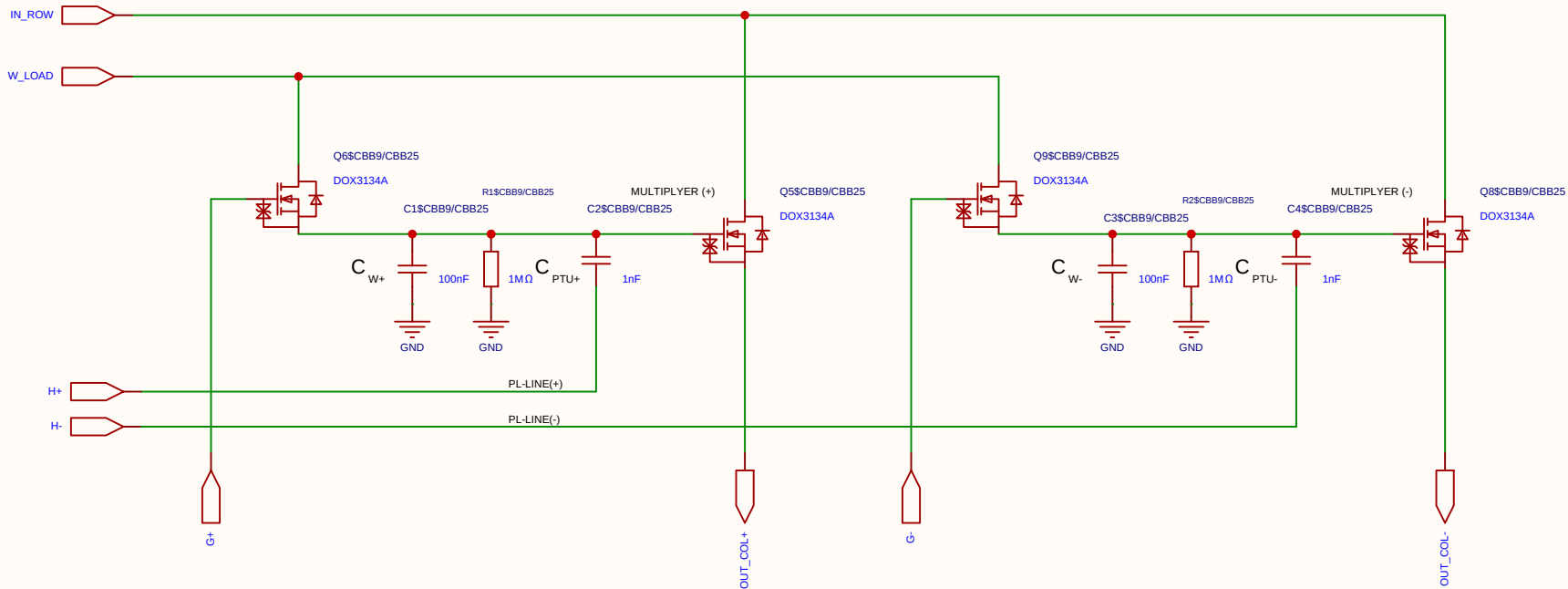
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

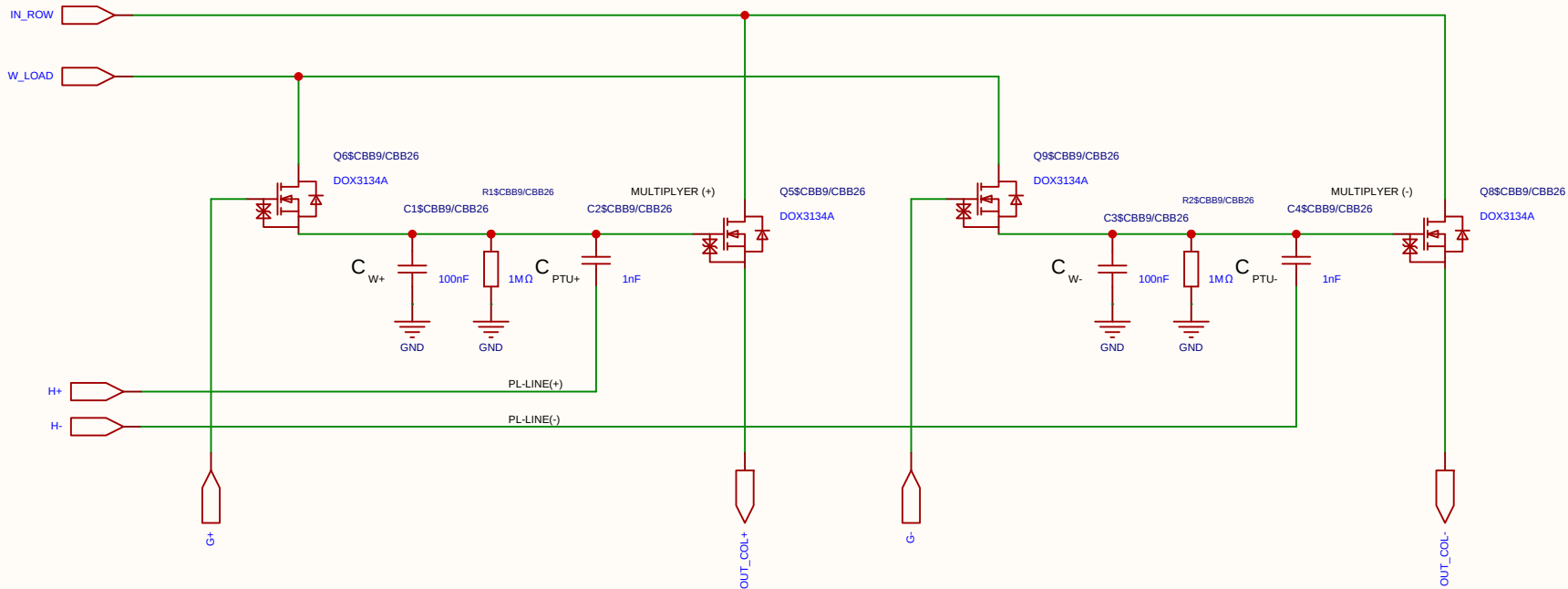
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

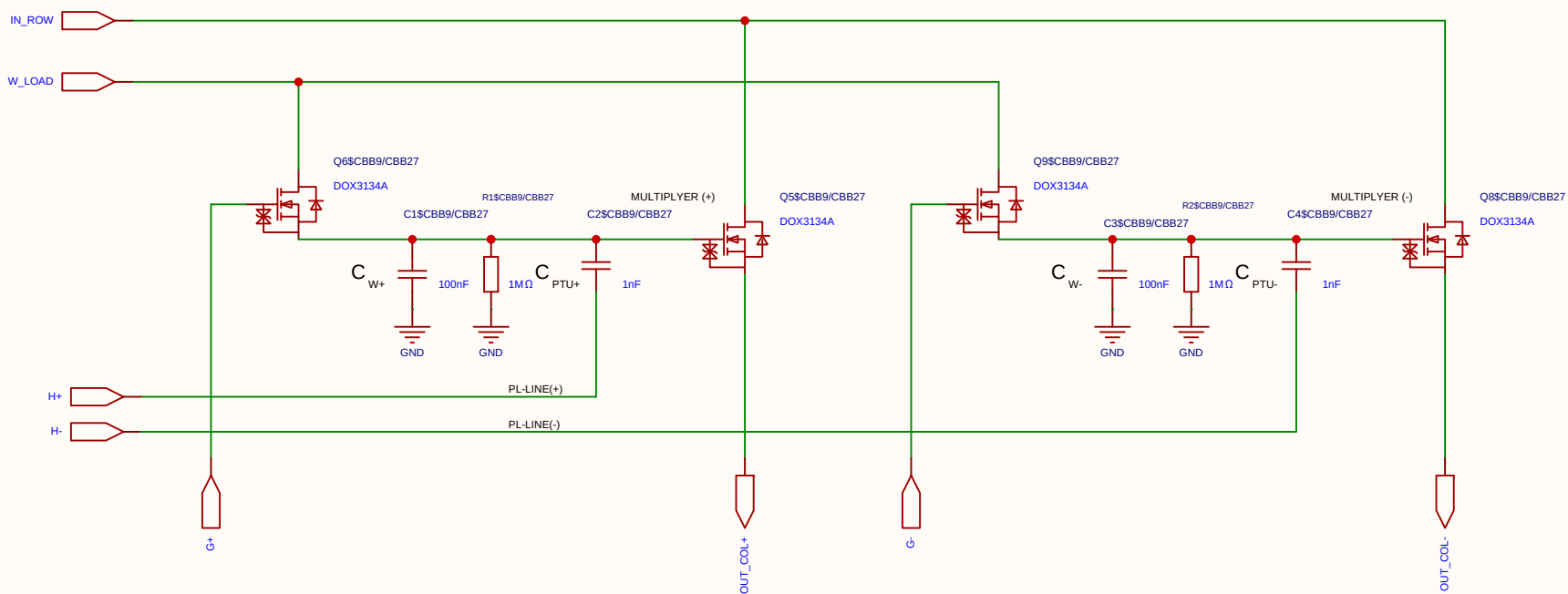
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

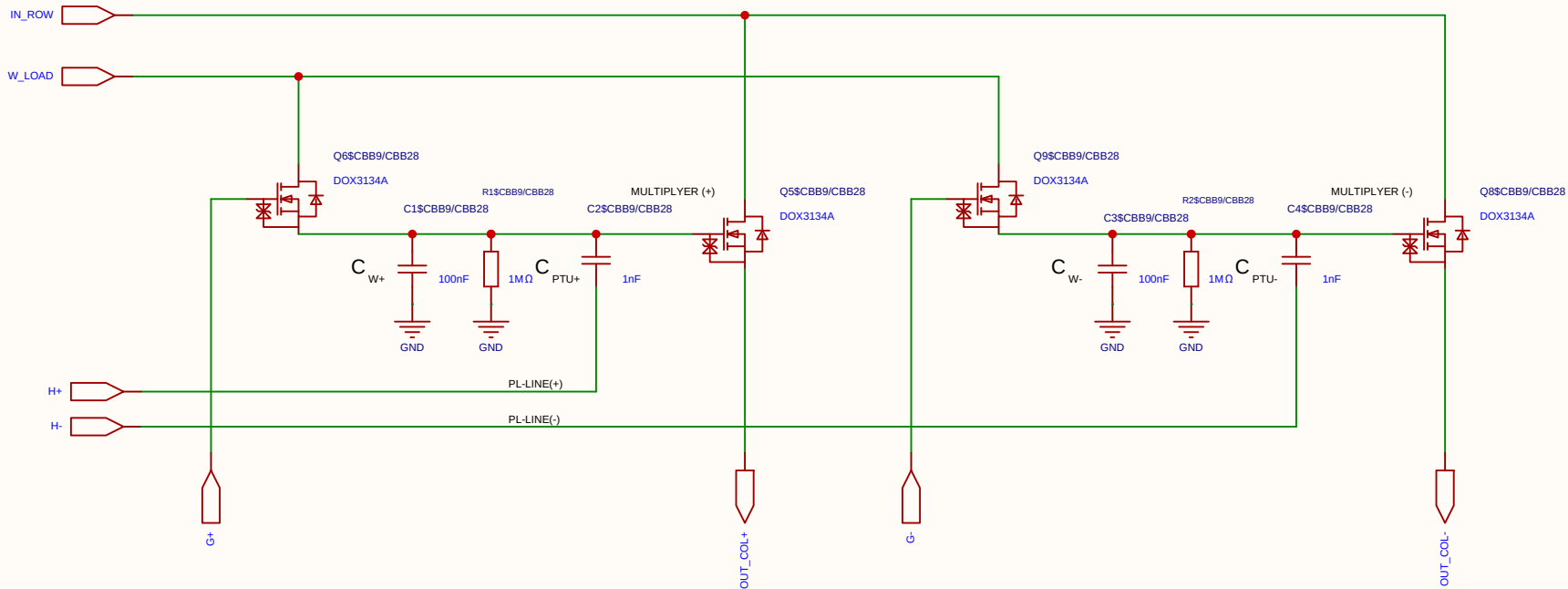
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

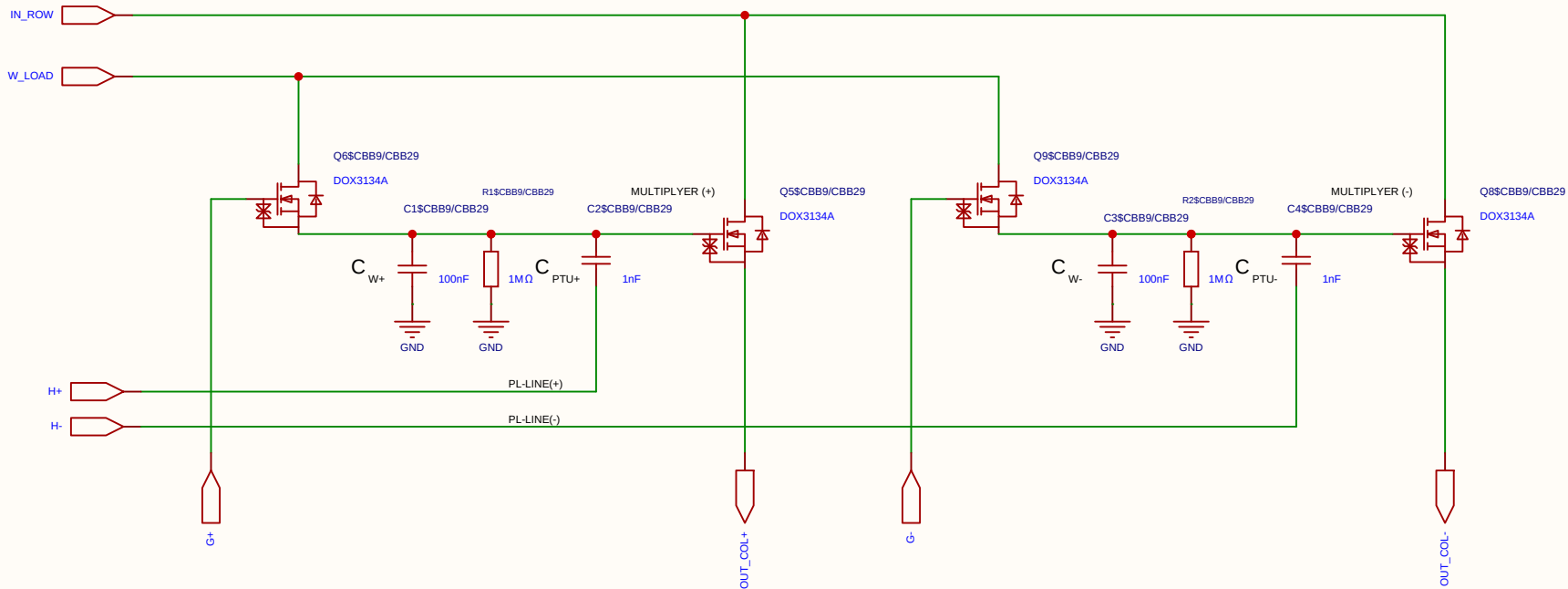
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

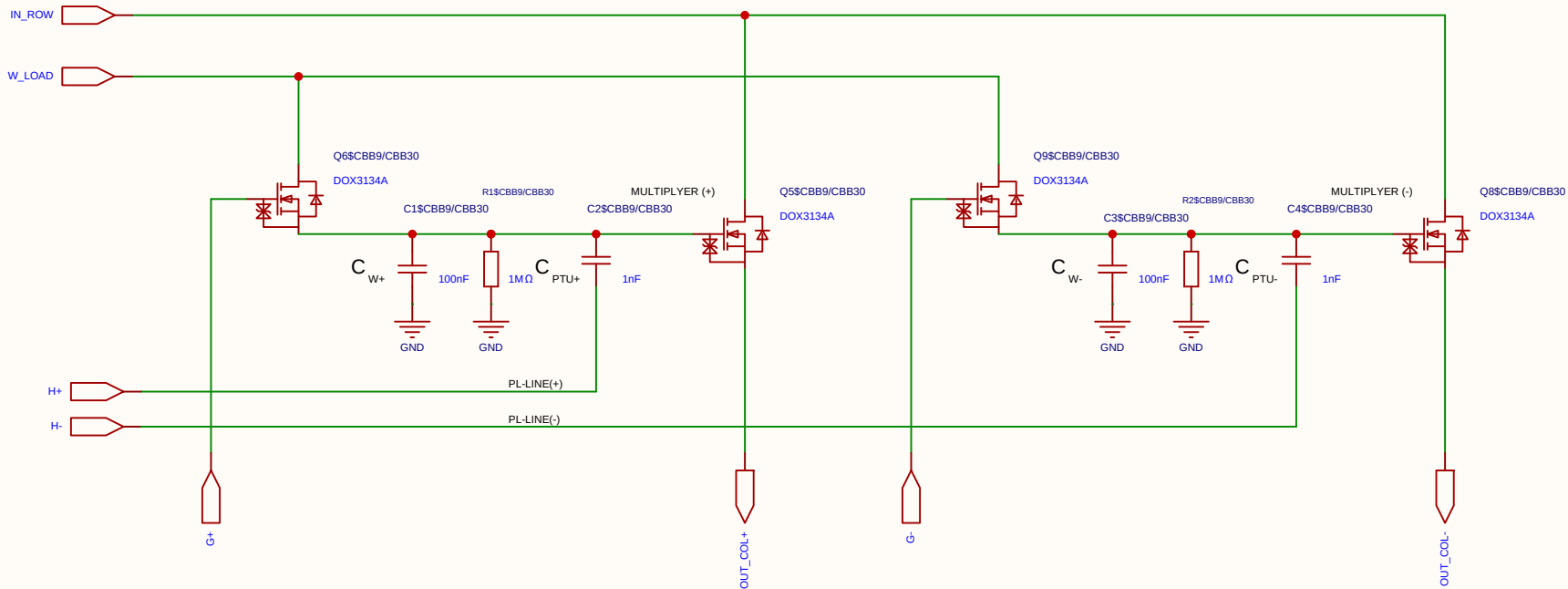
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

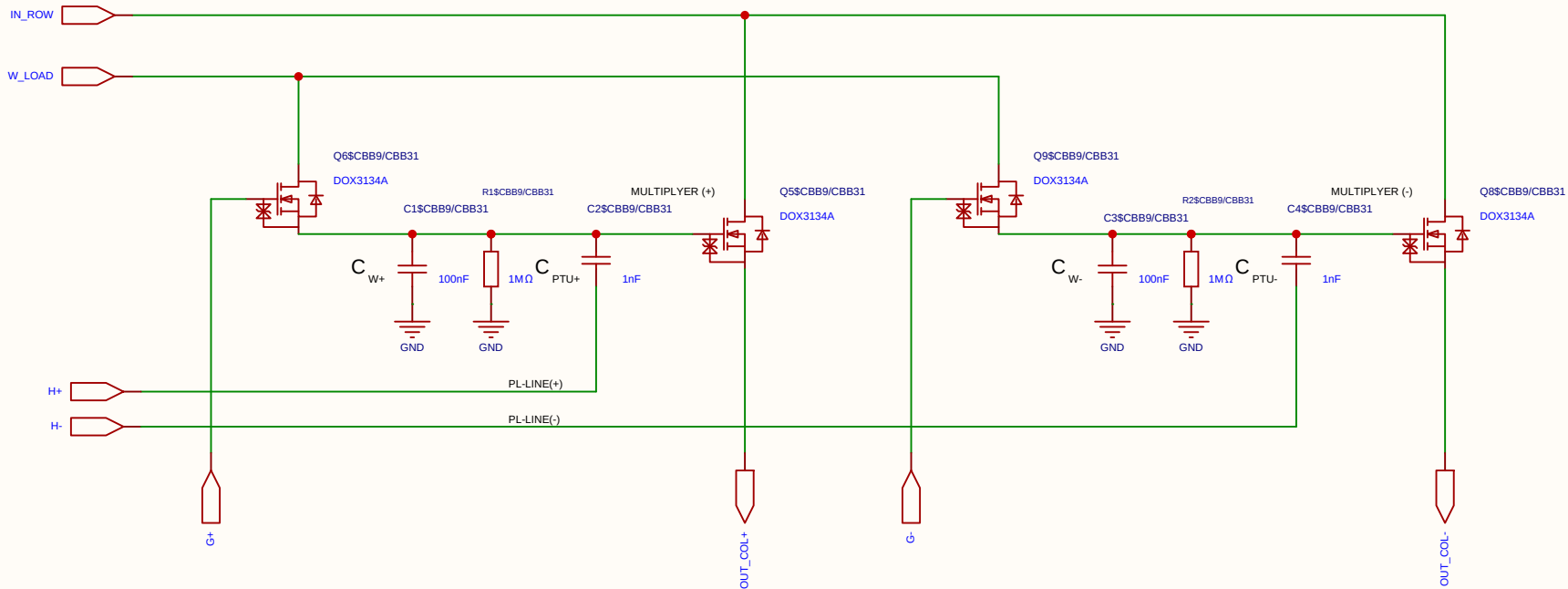
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

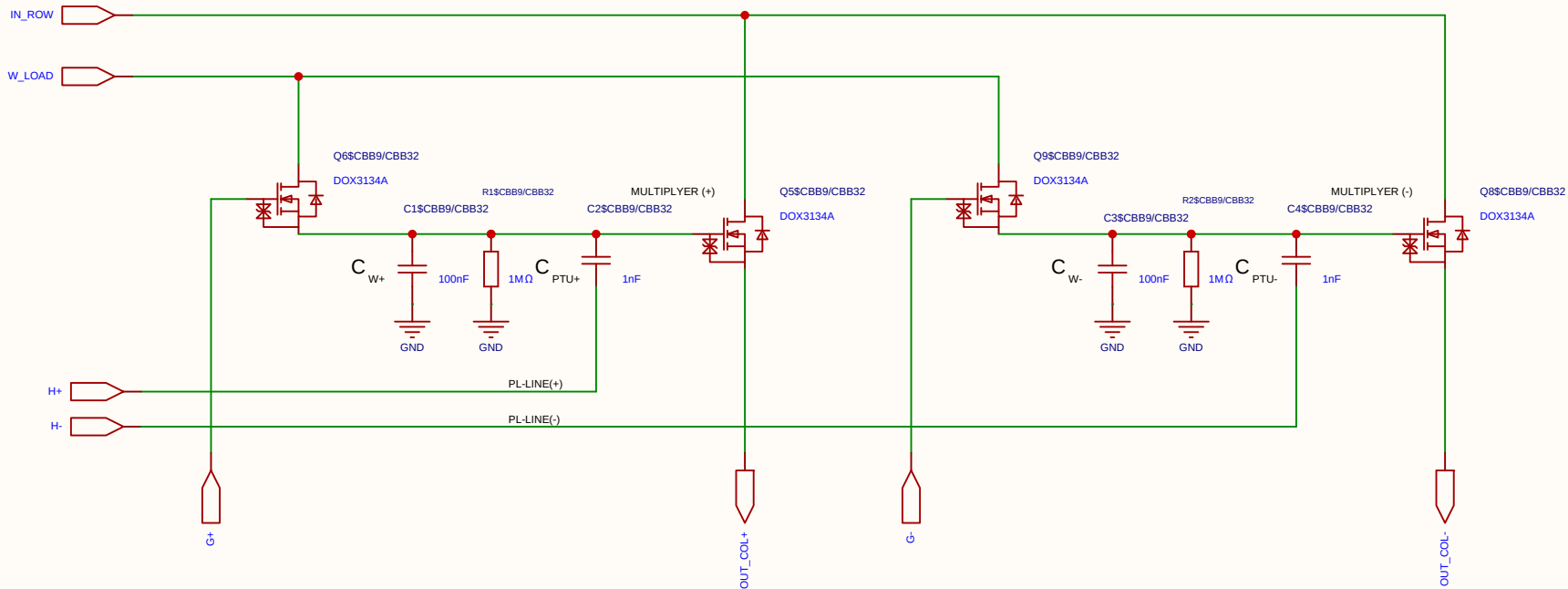
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

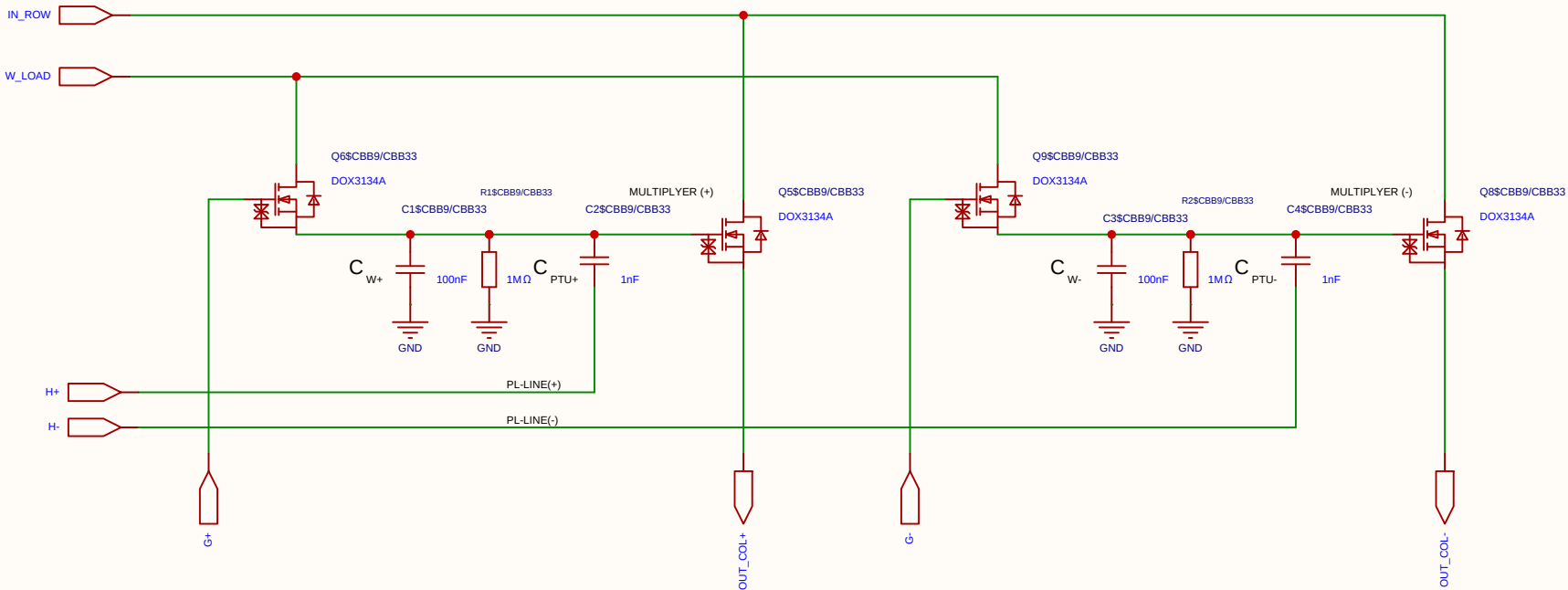
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

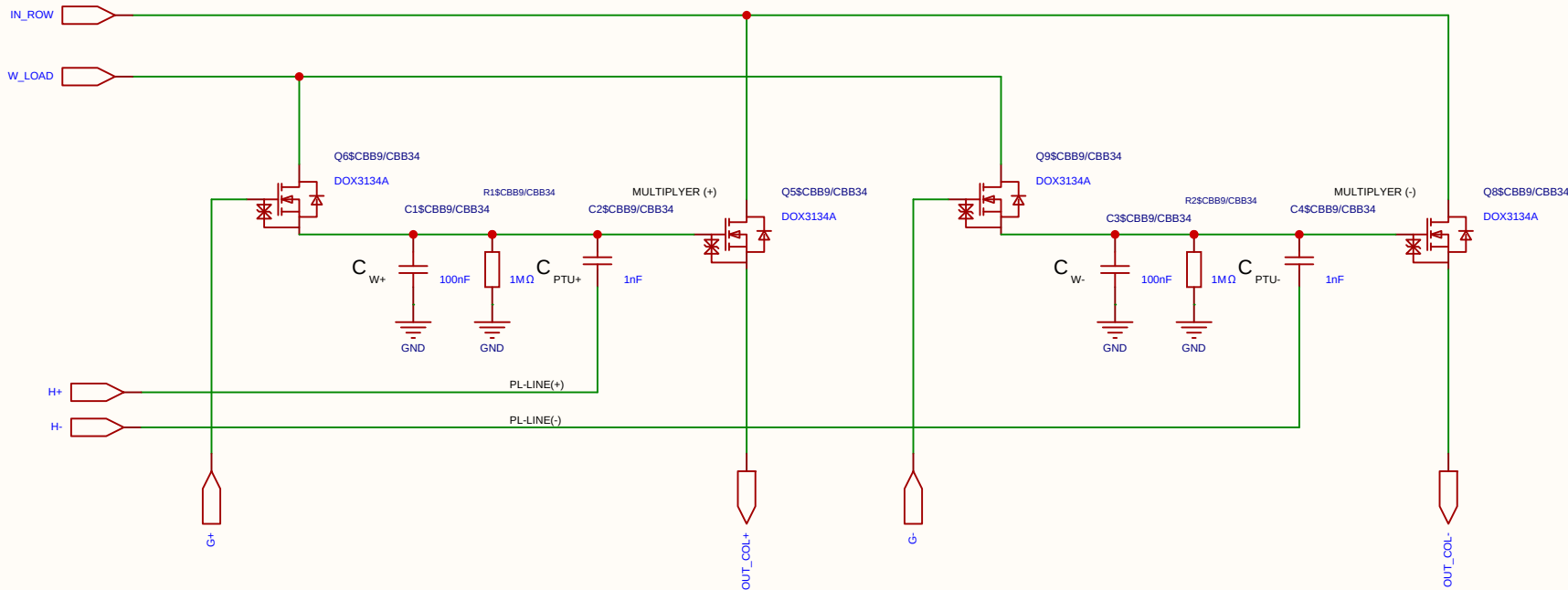
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

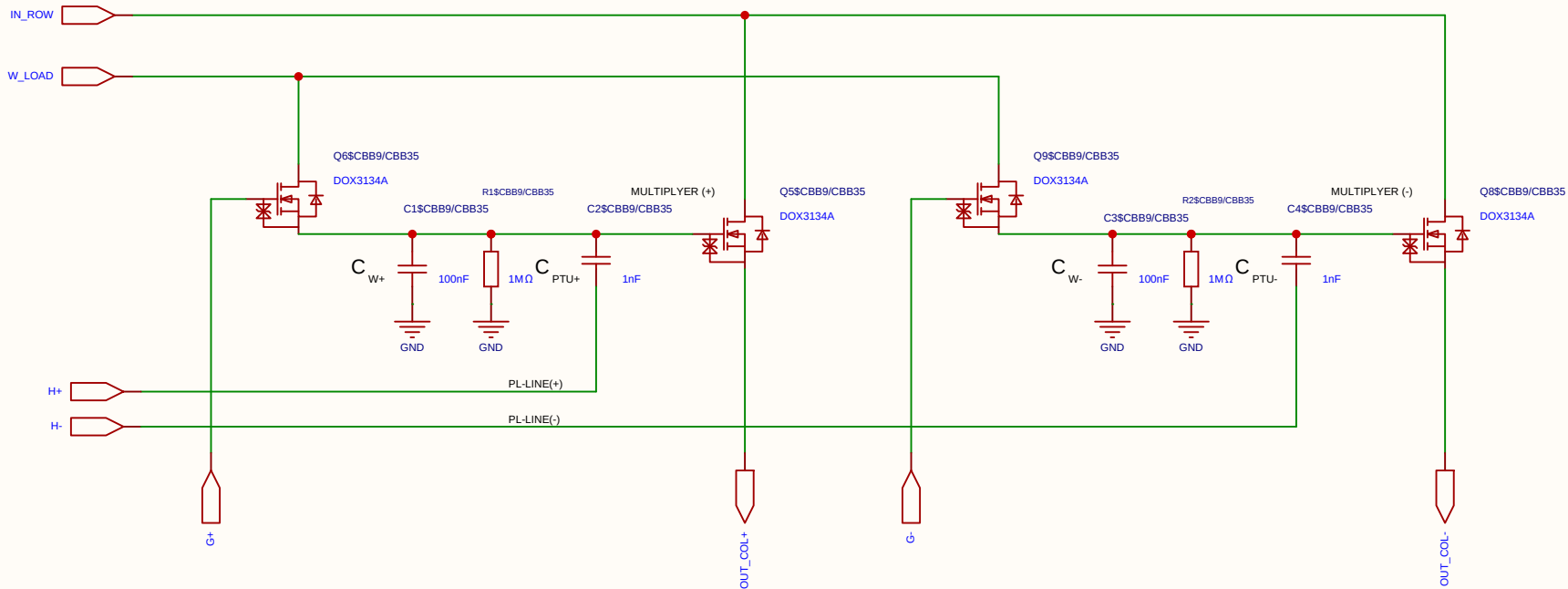
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

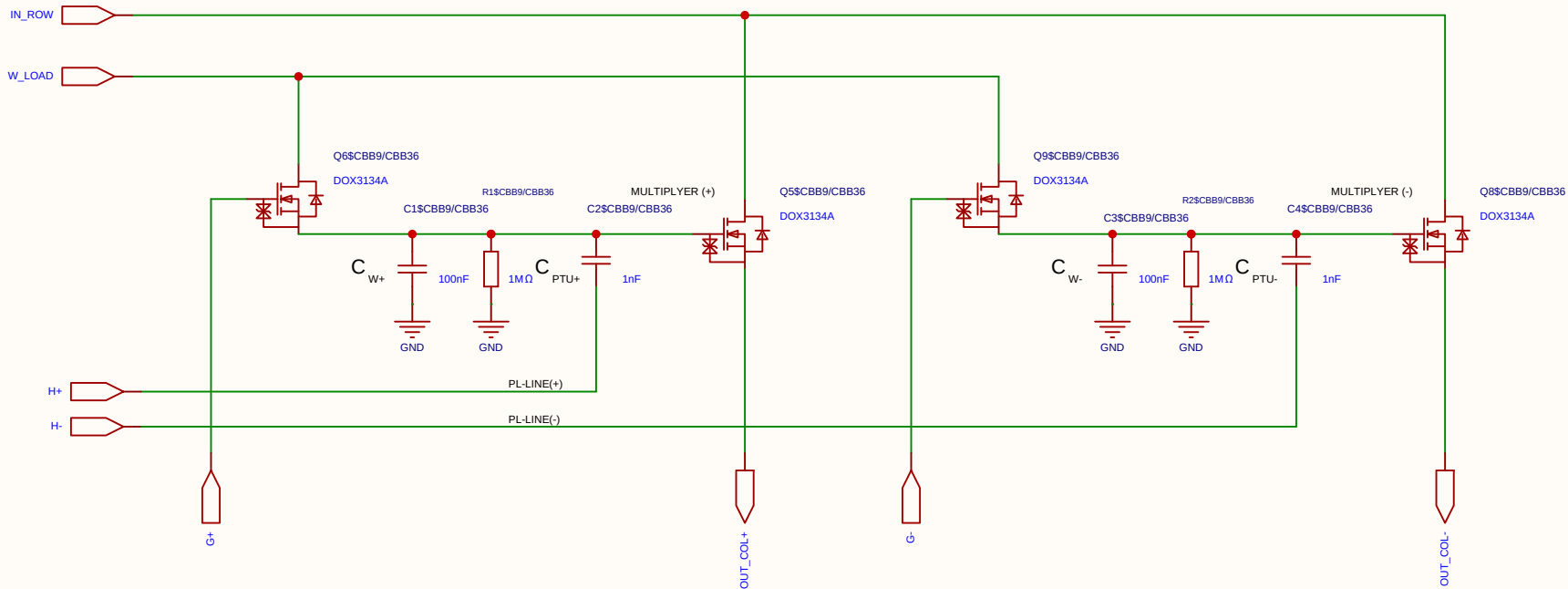
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

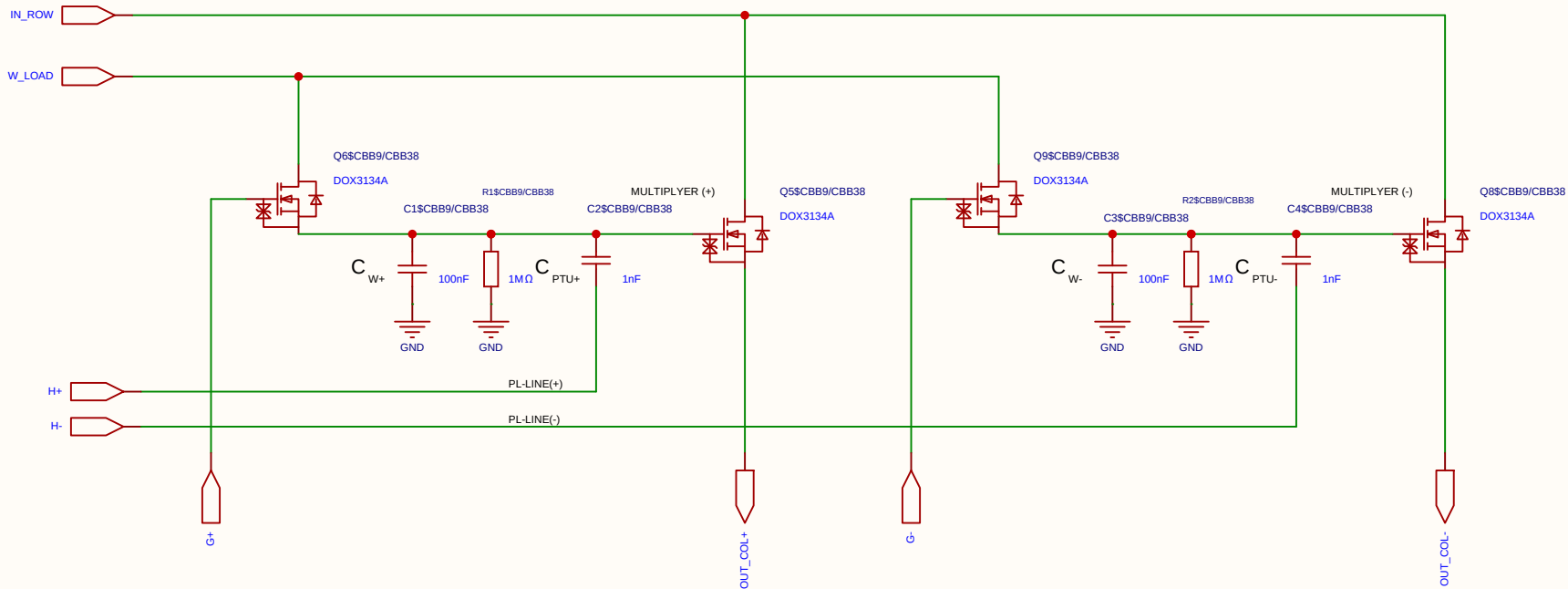
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

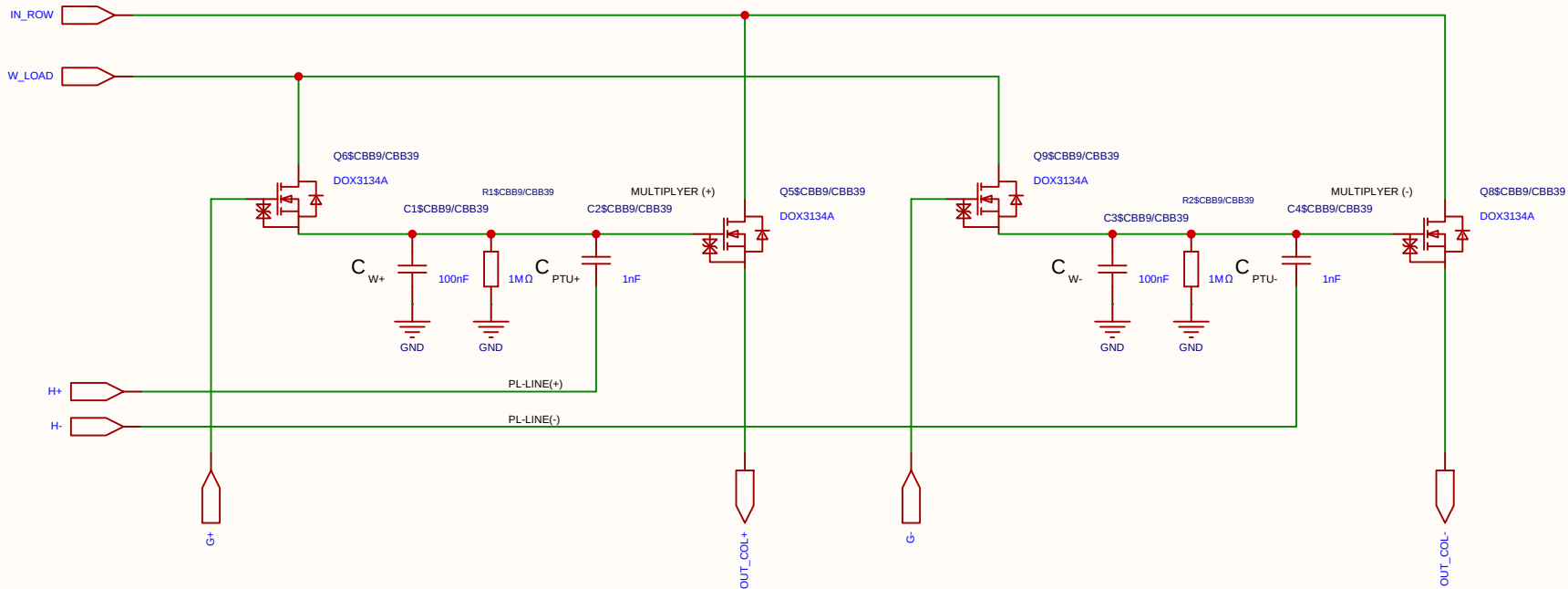
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

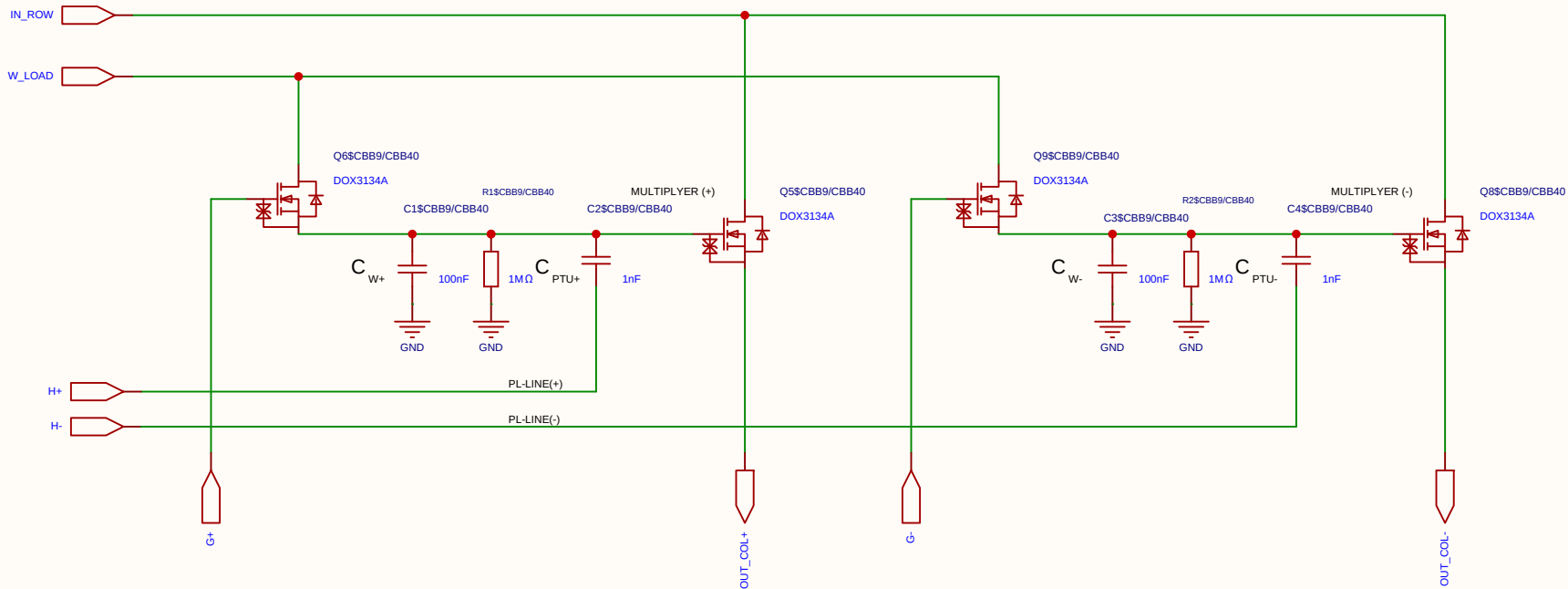
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

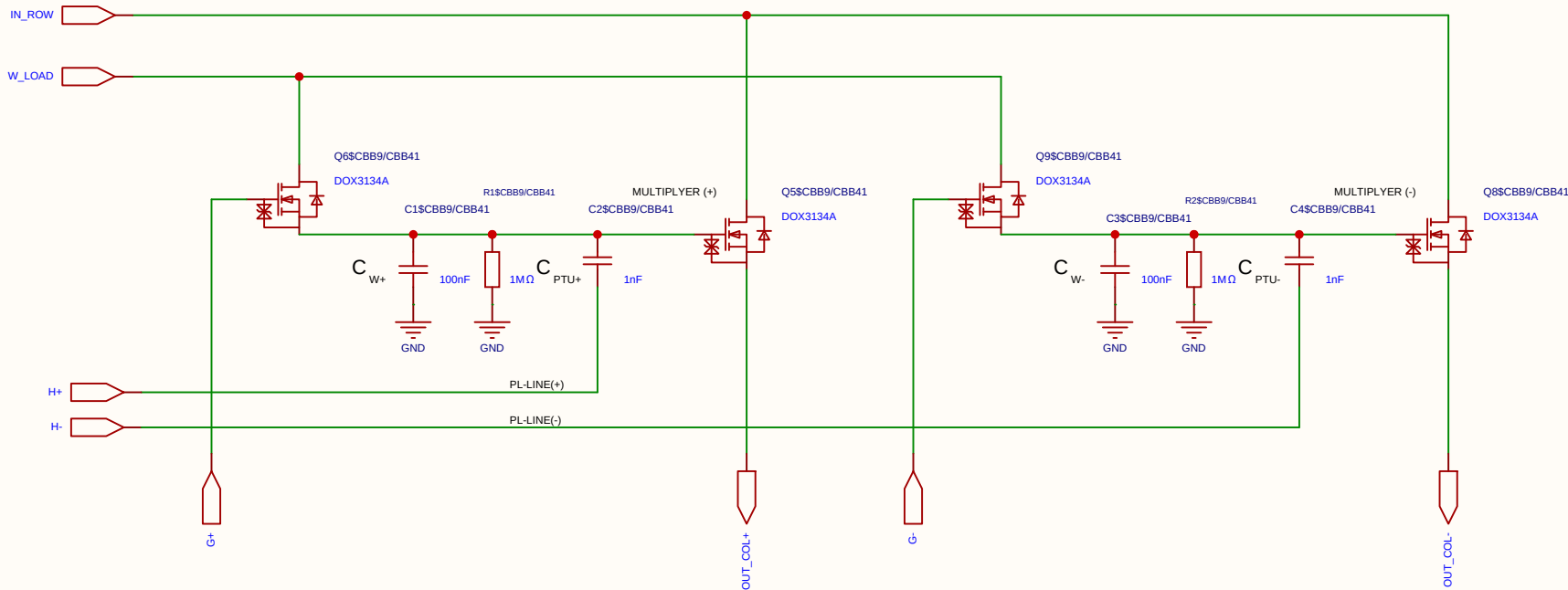
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

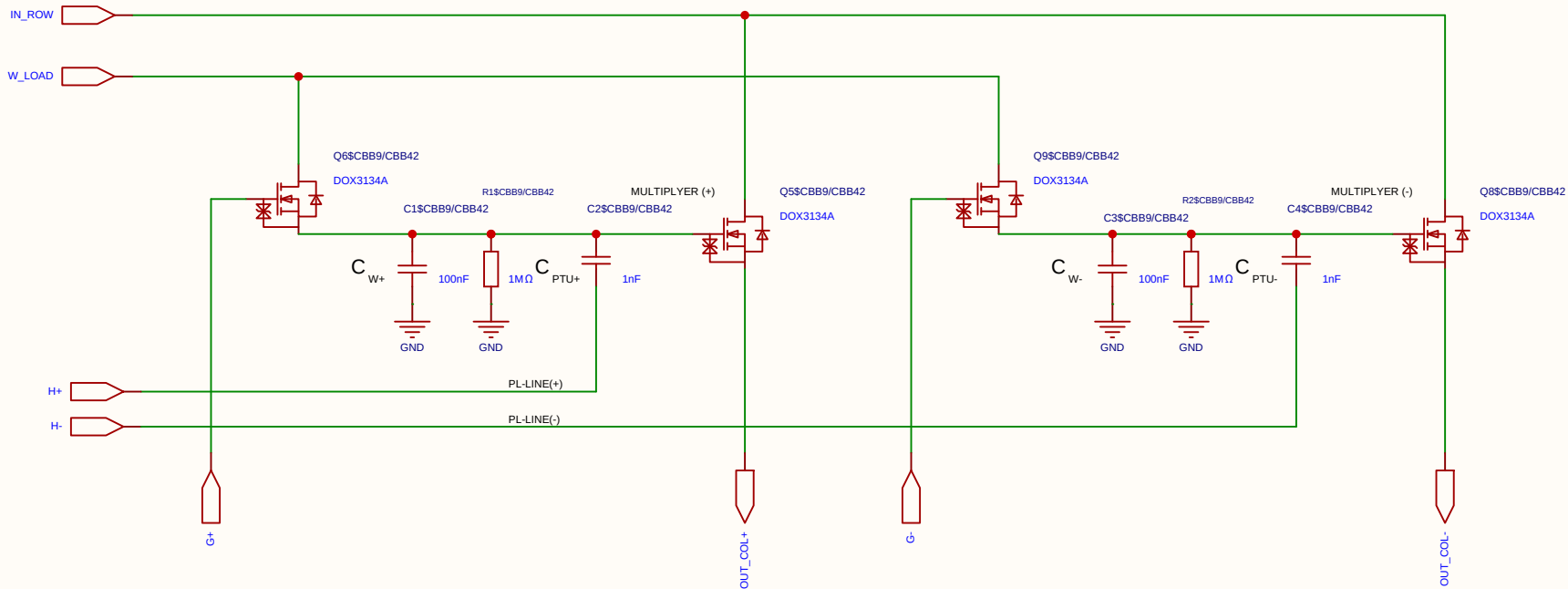
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

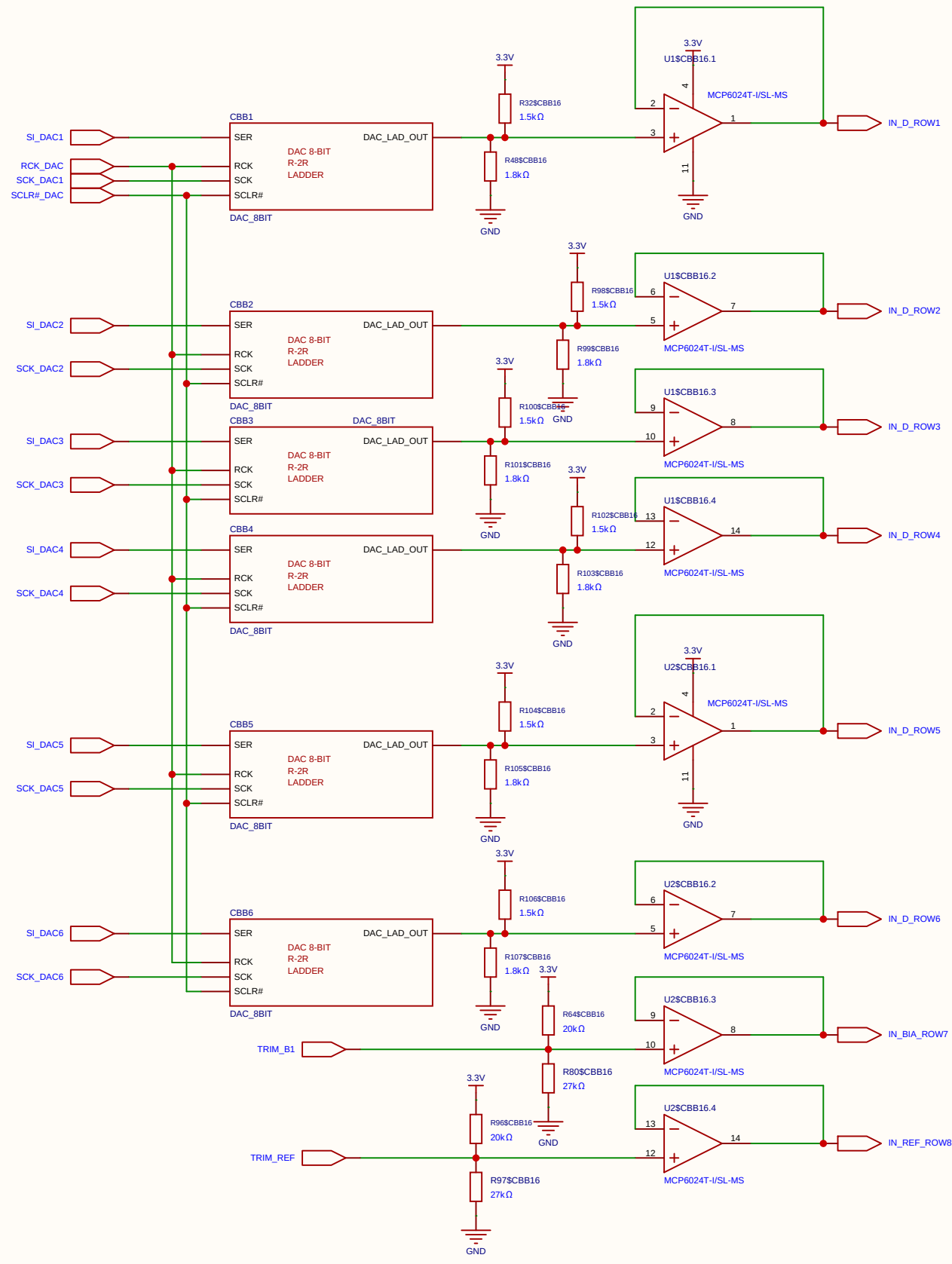
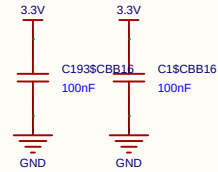
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

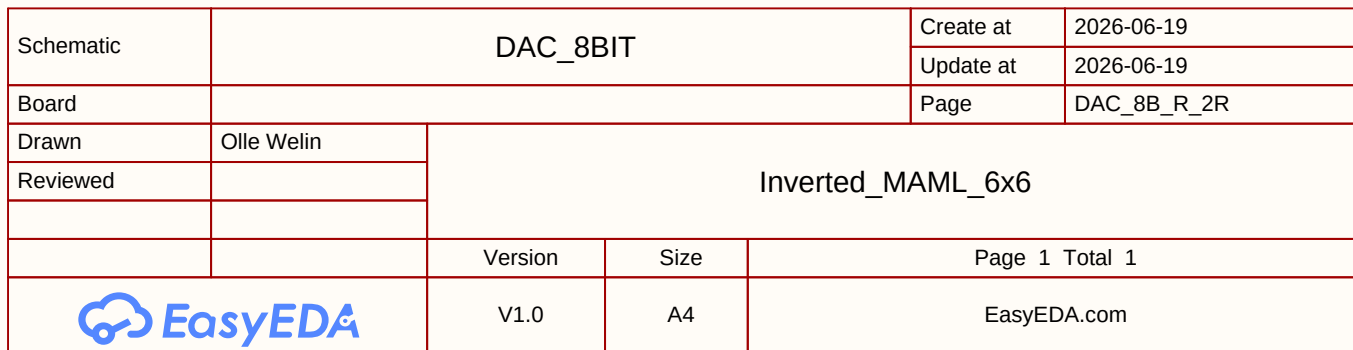
COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

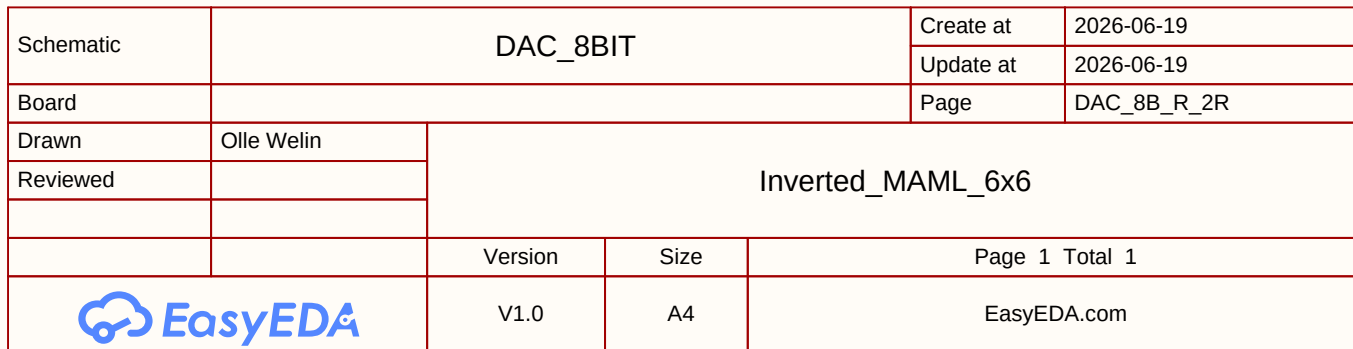
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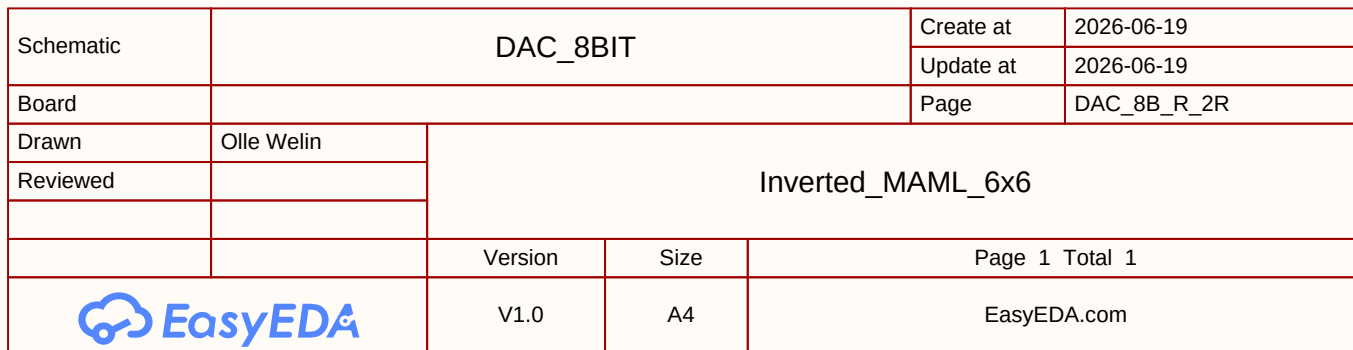
Voltages Data swing (Row / V_{ds}) 250 mV Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.

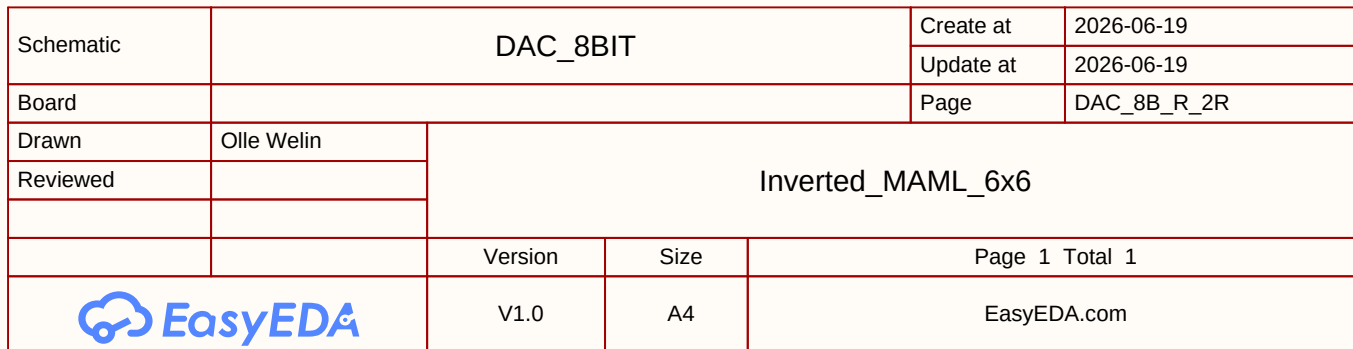


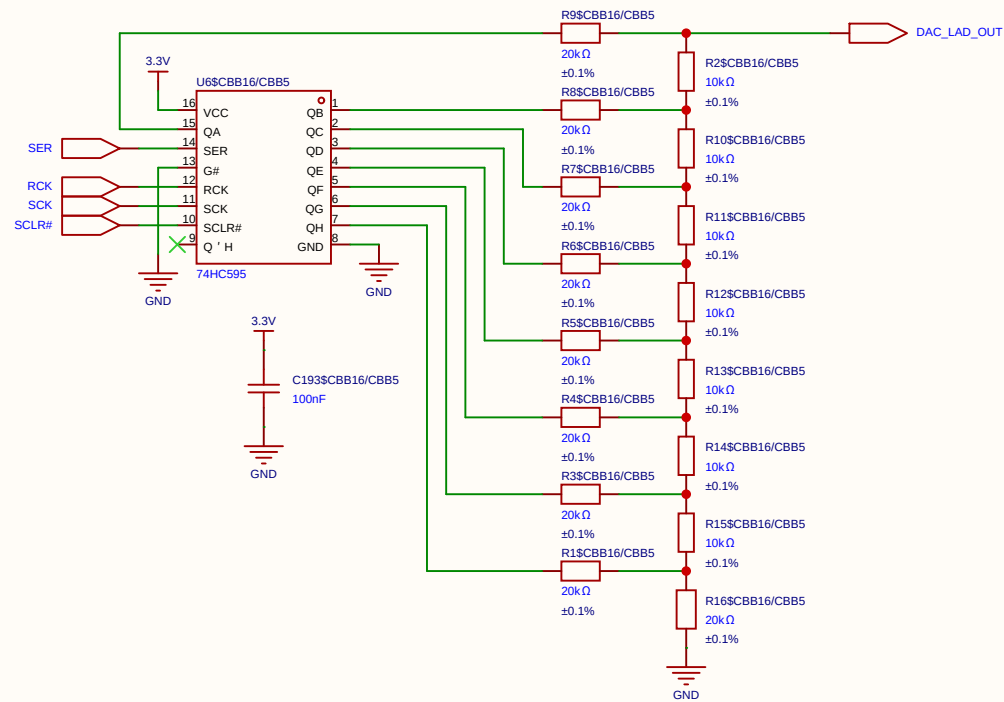
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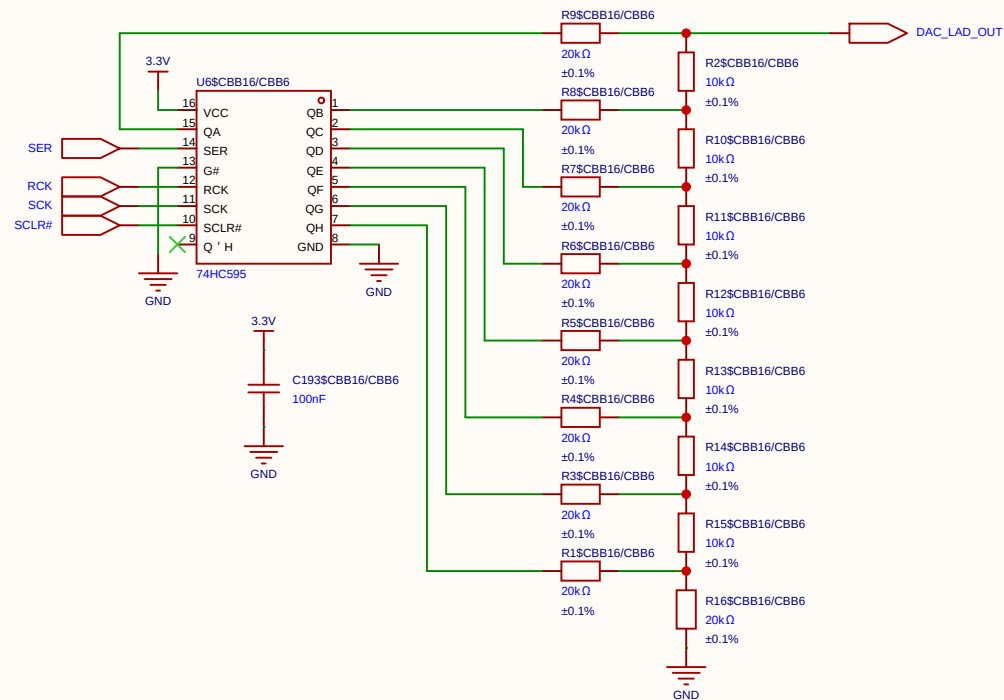






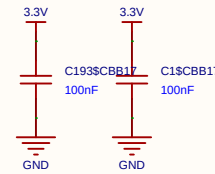
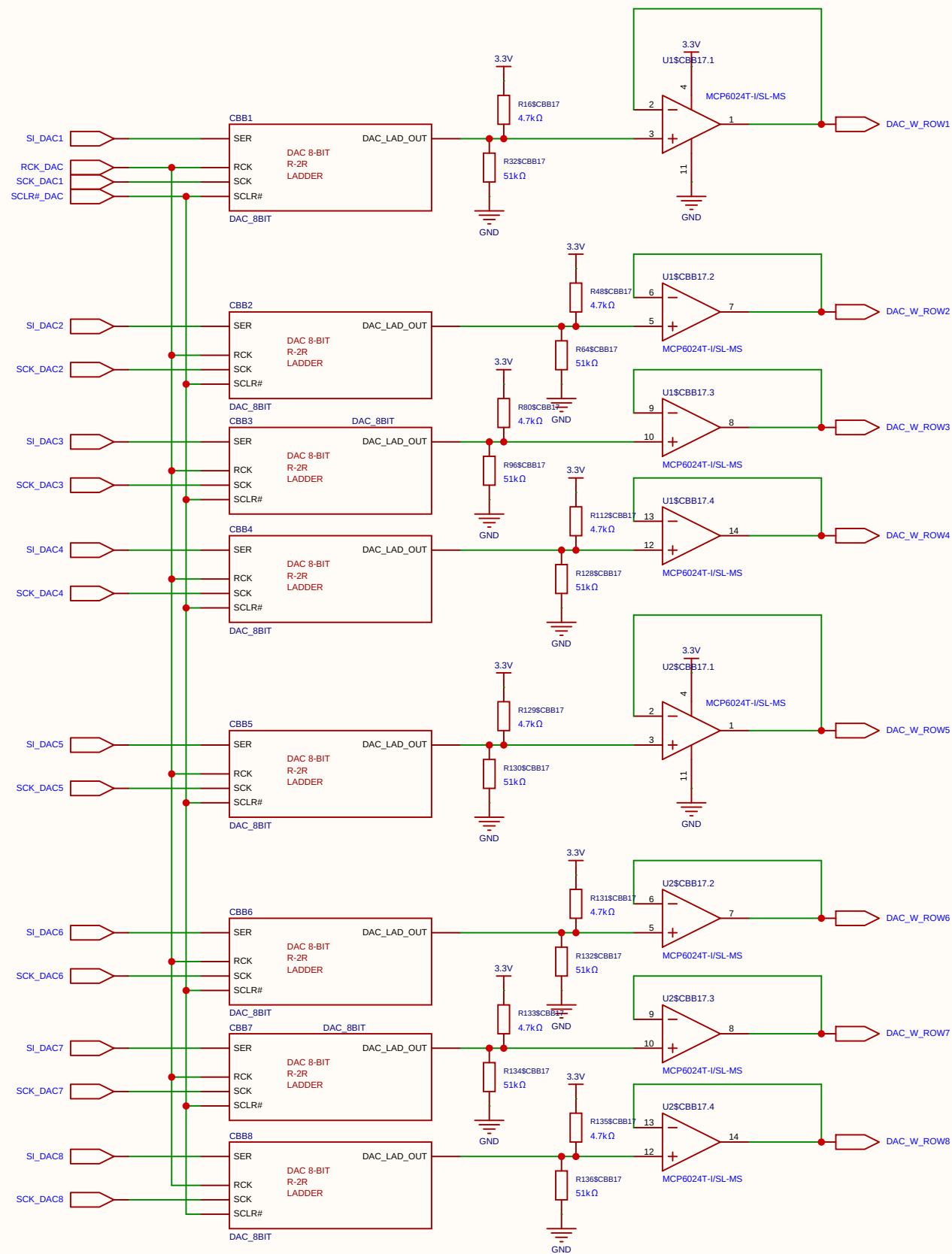


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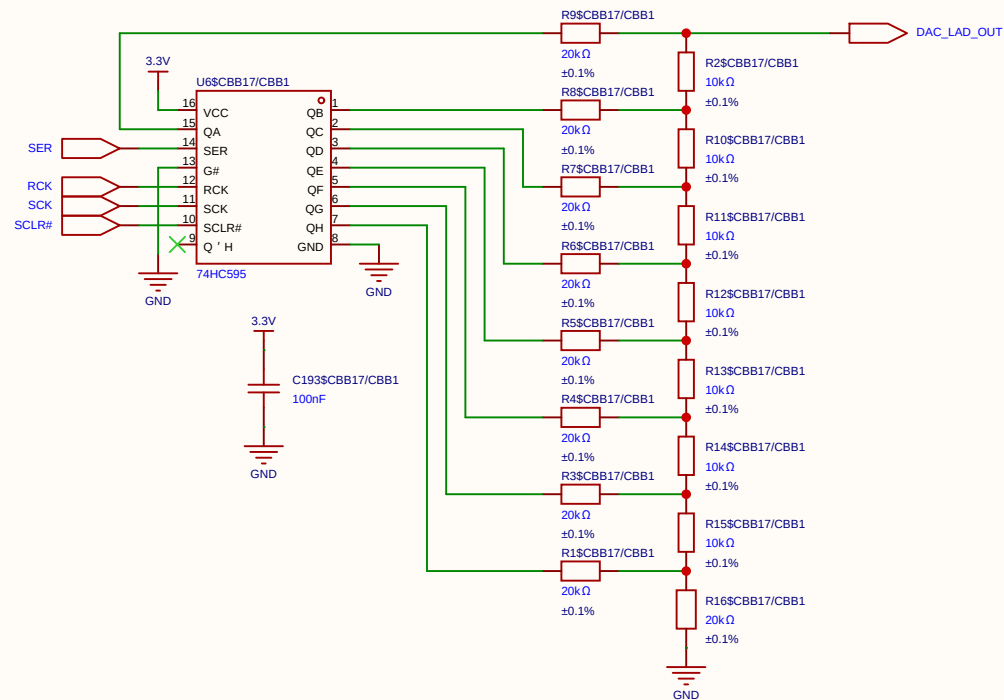


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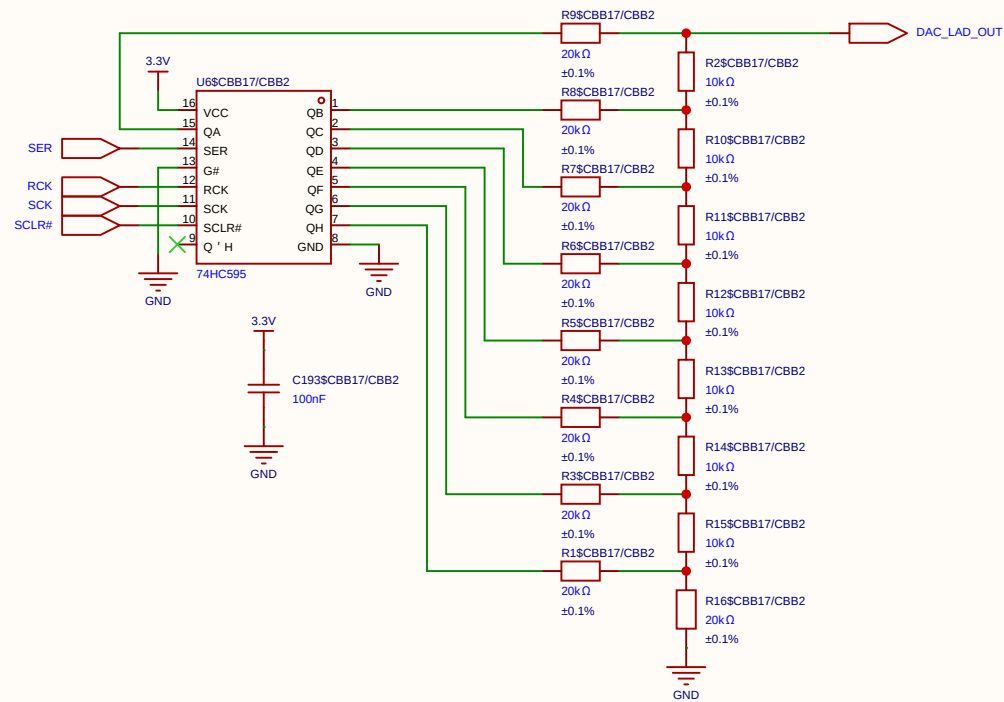
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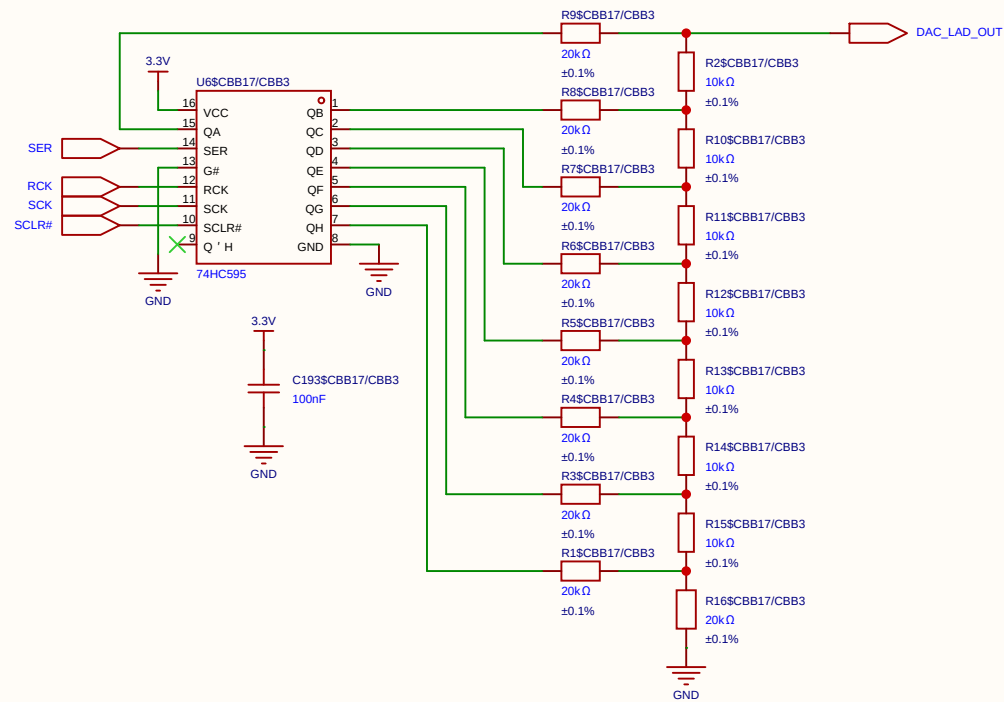
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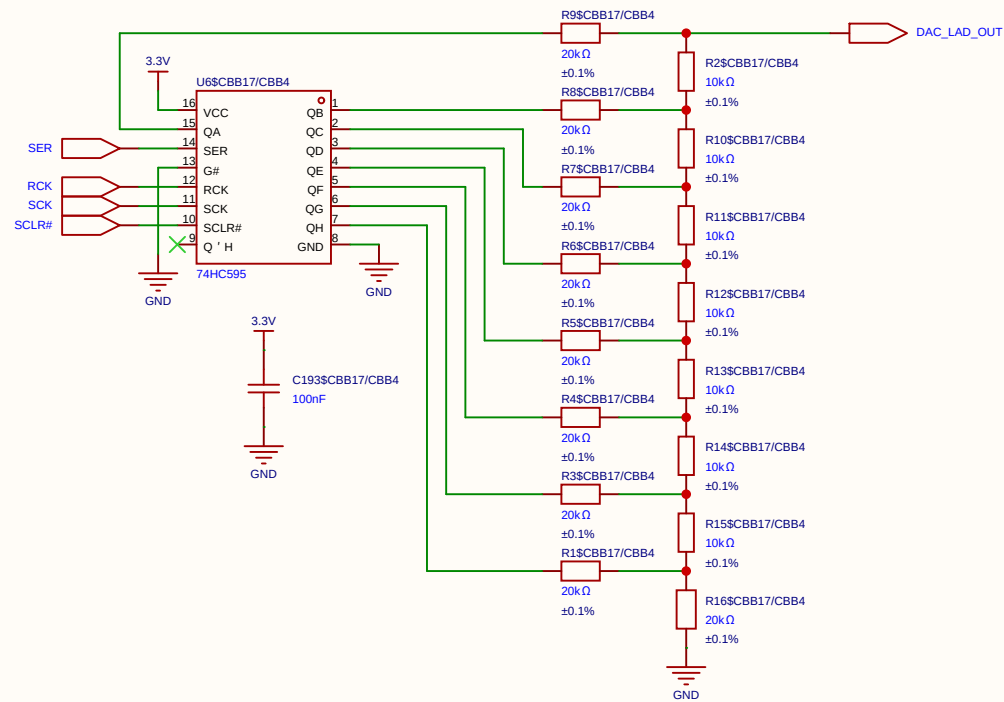
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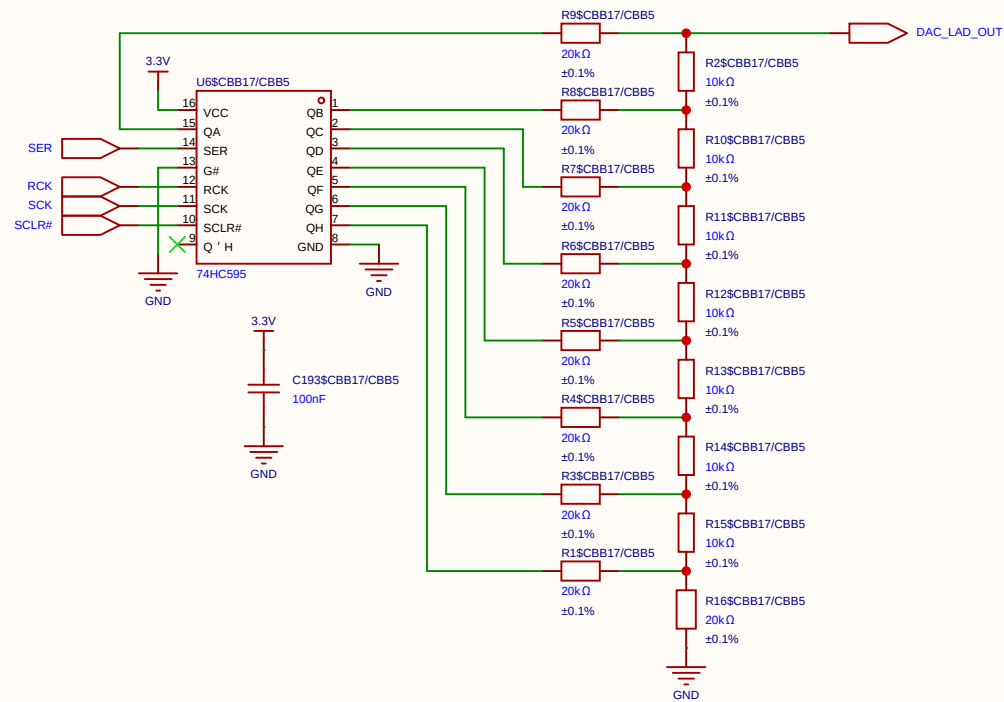
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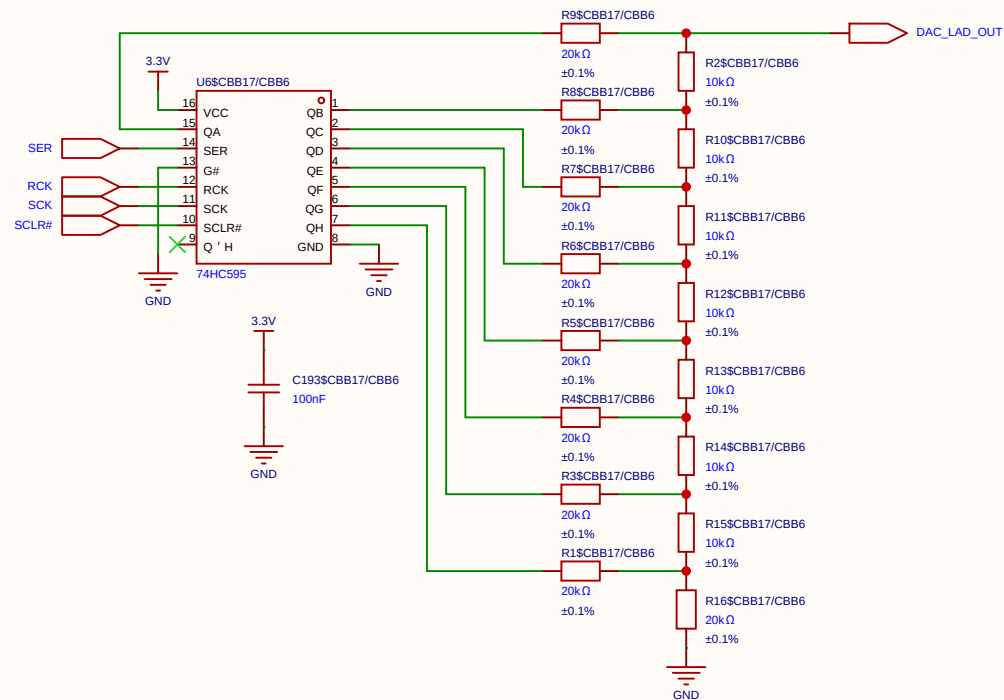
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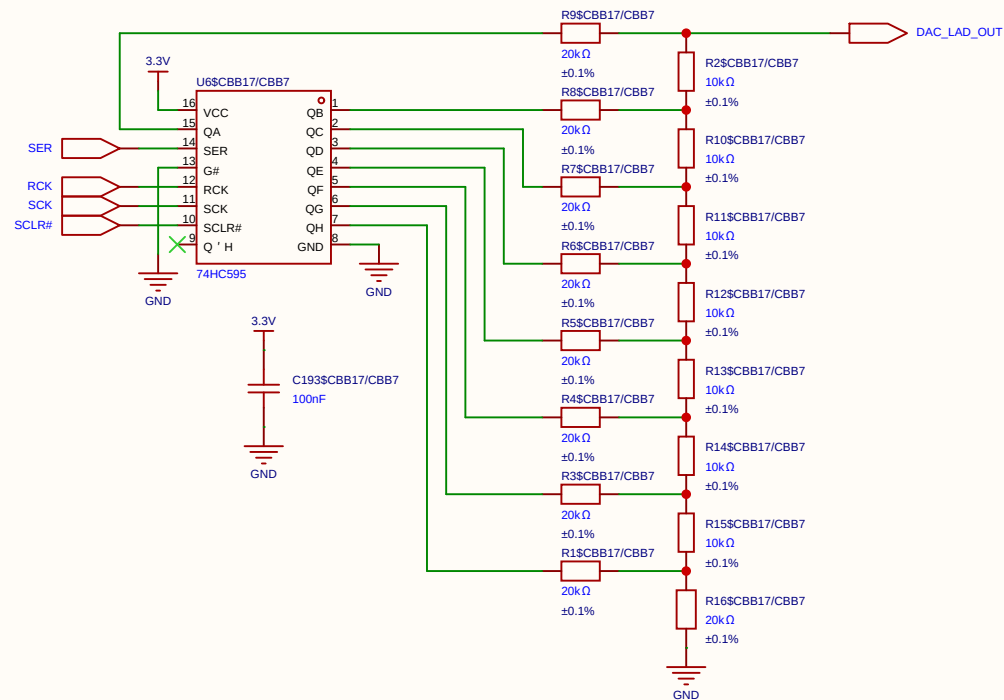
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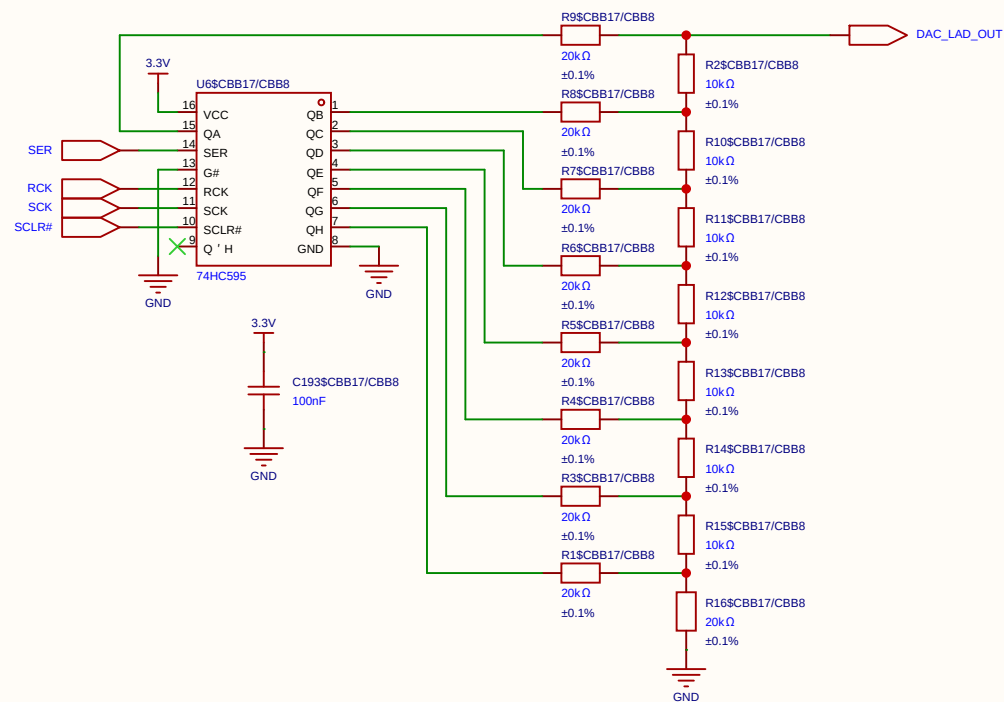
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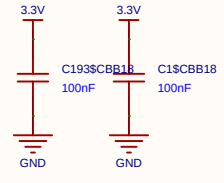
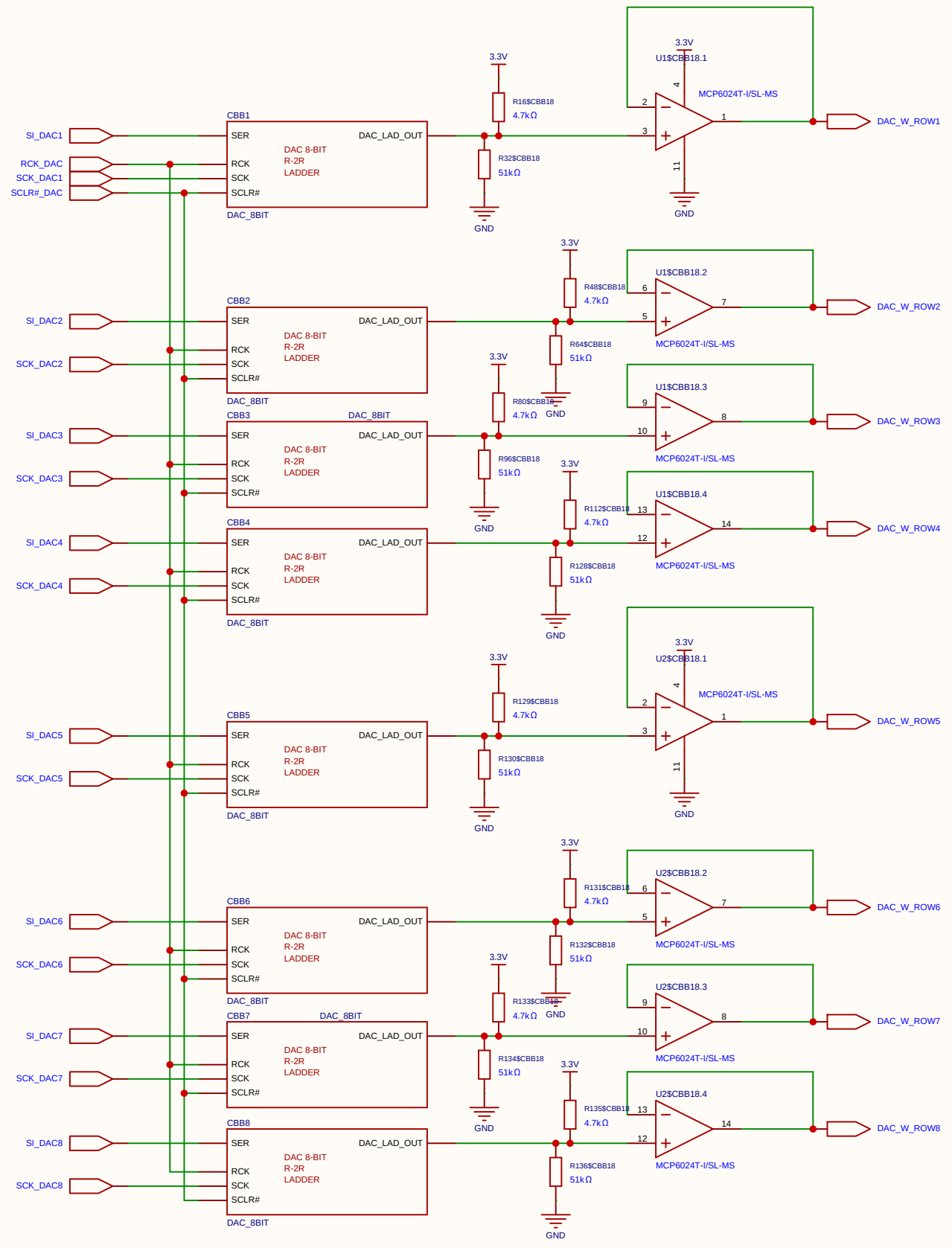


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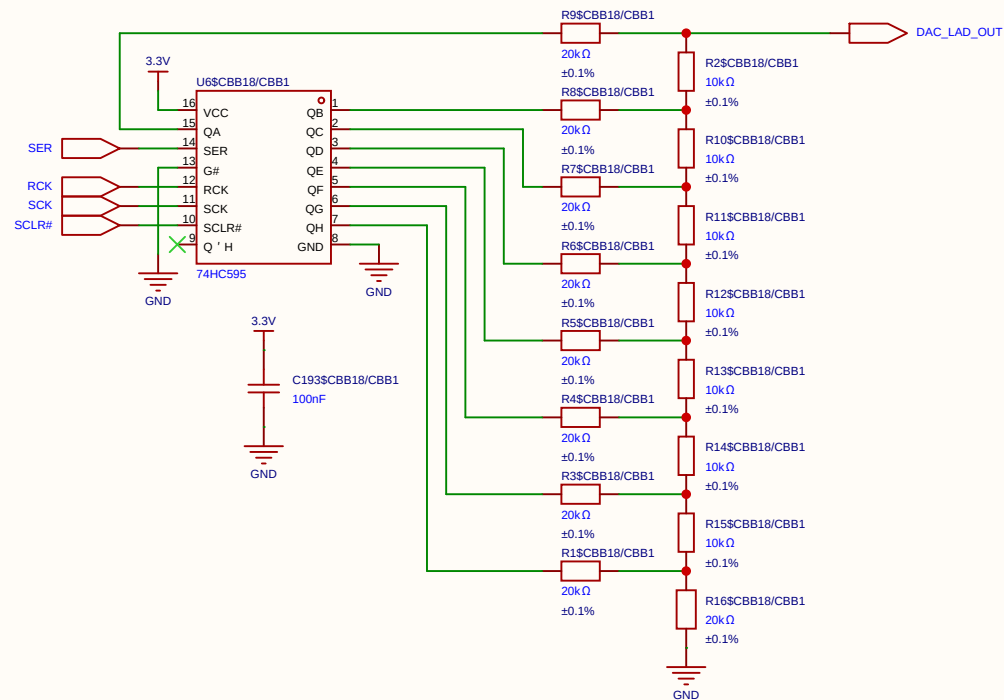


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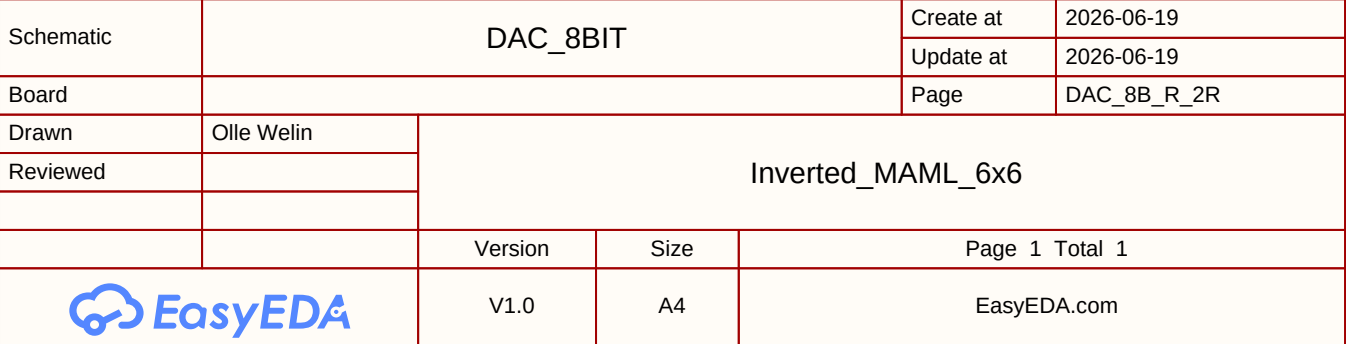
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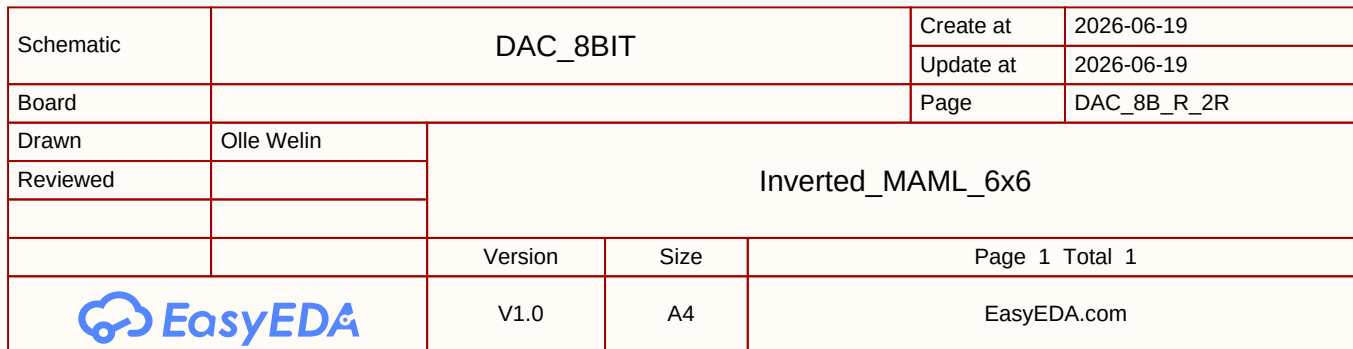


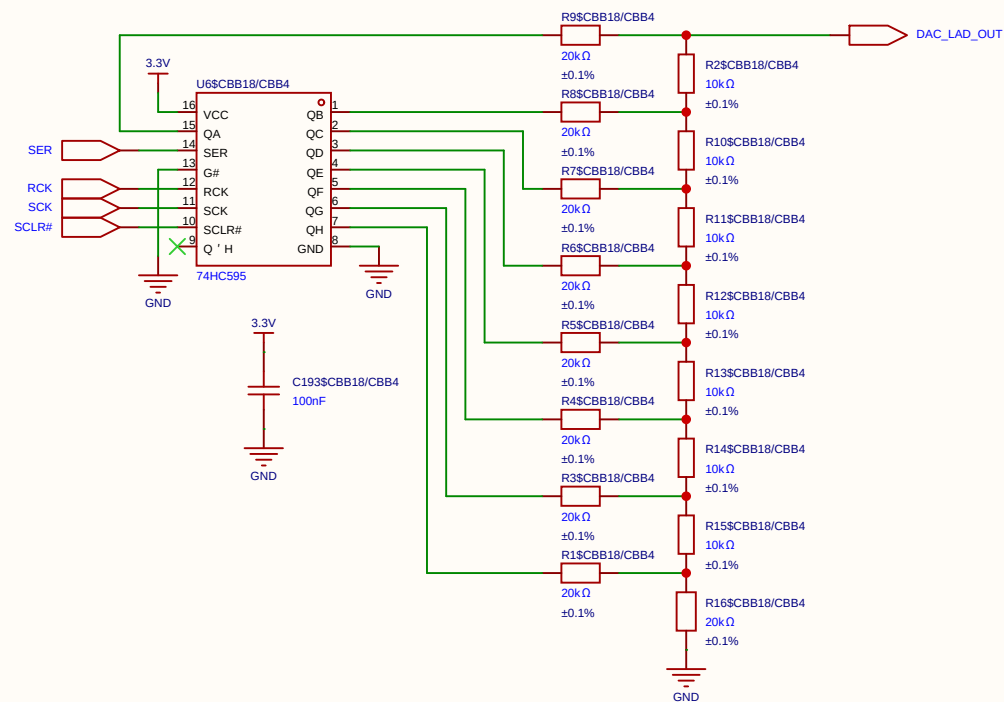
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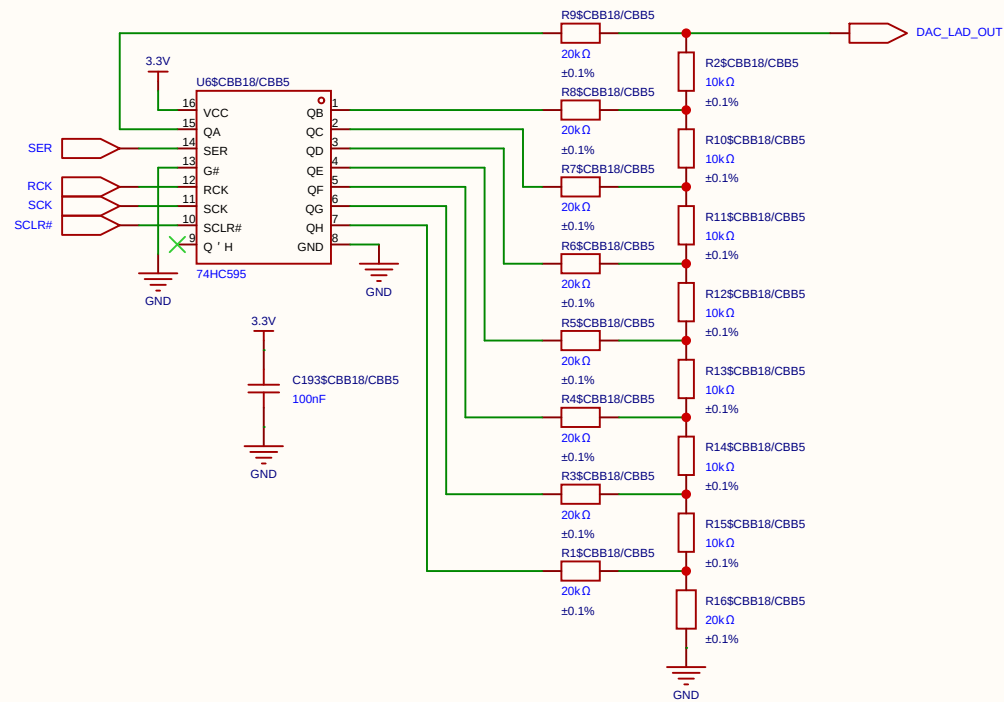
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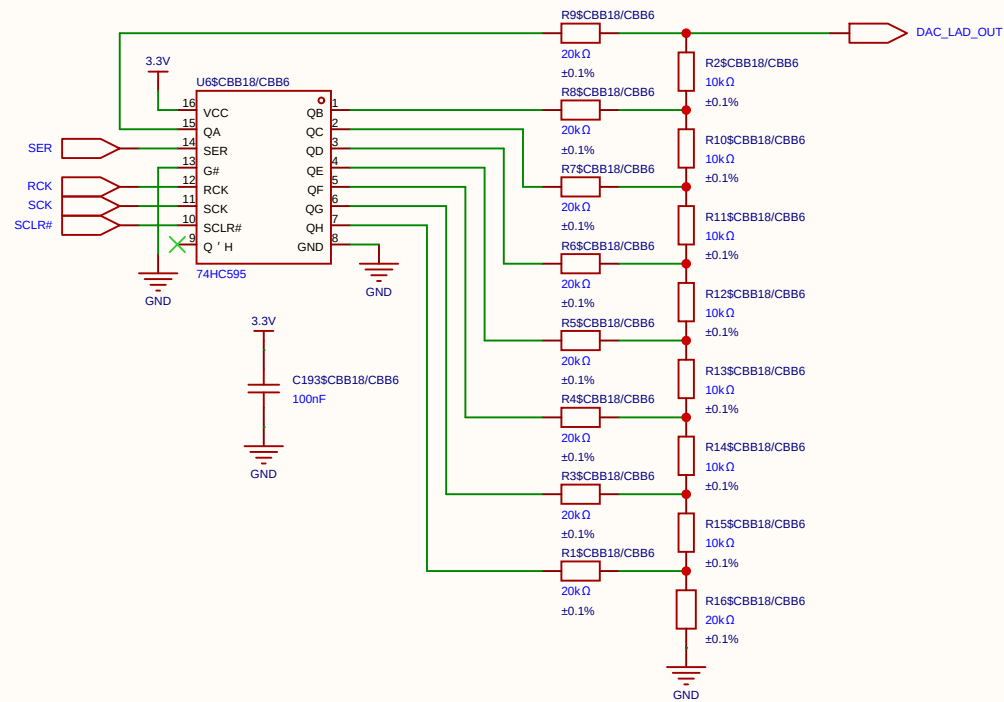




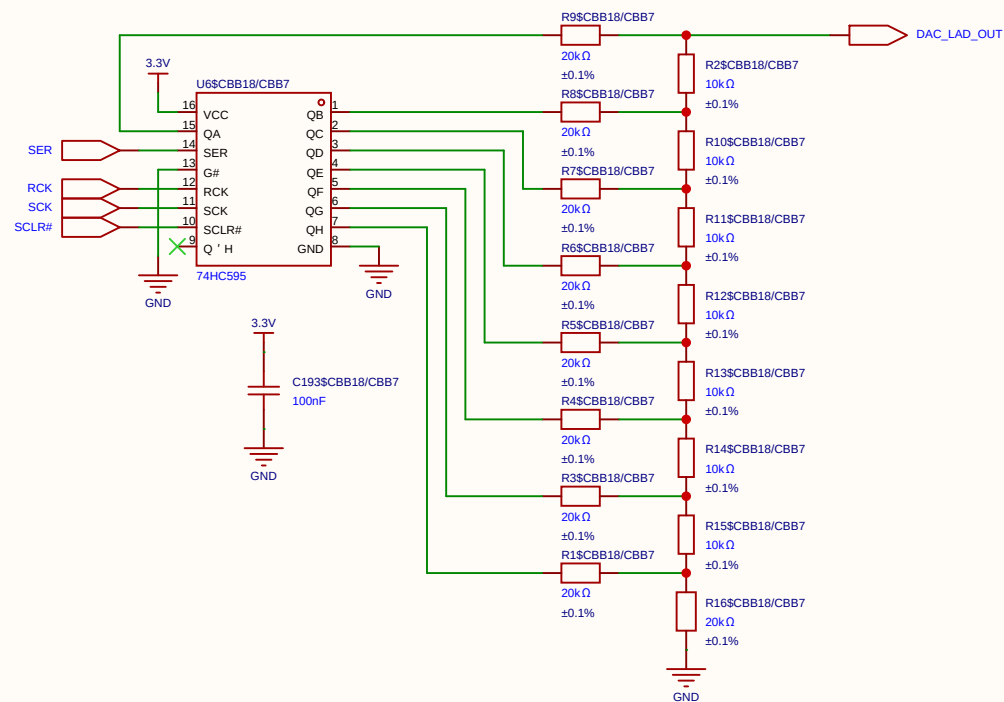
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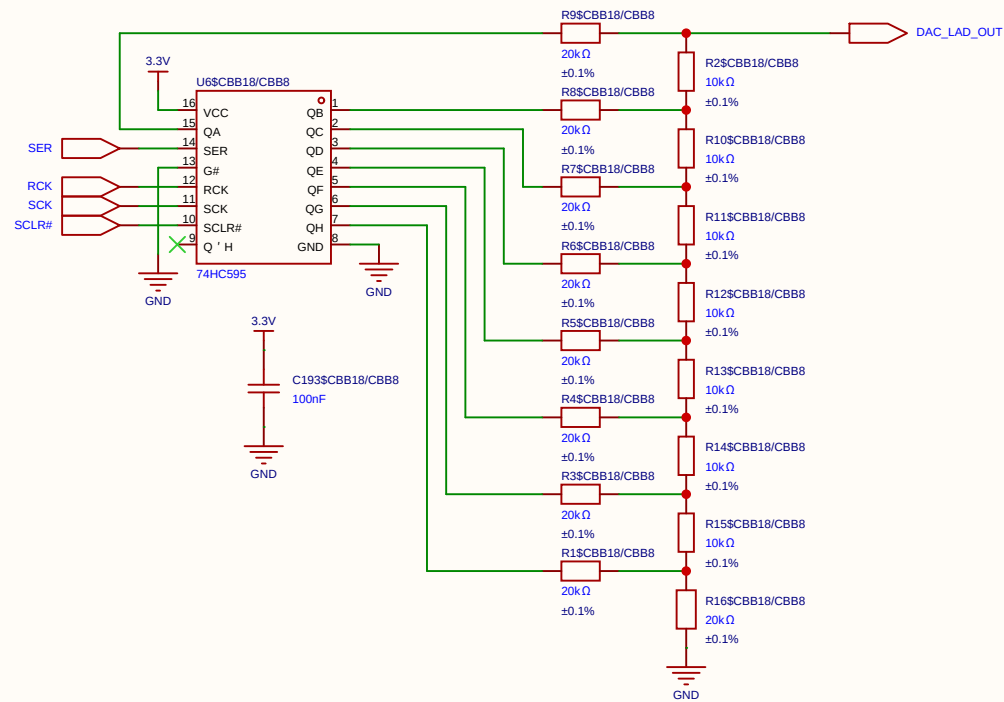
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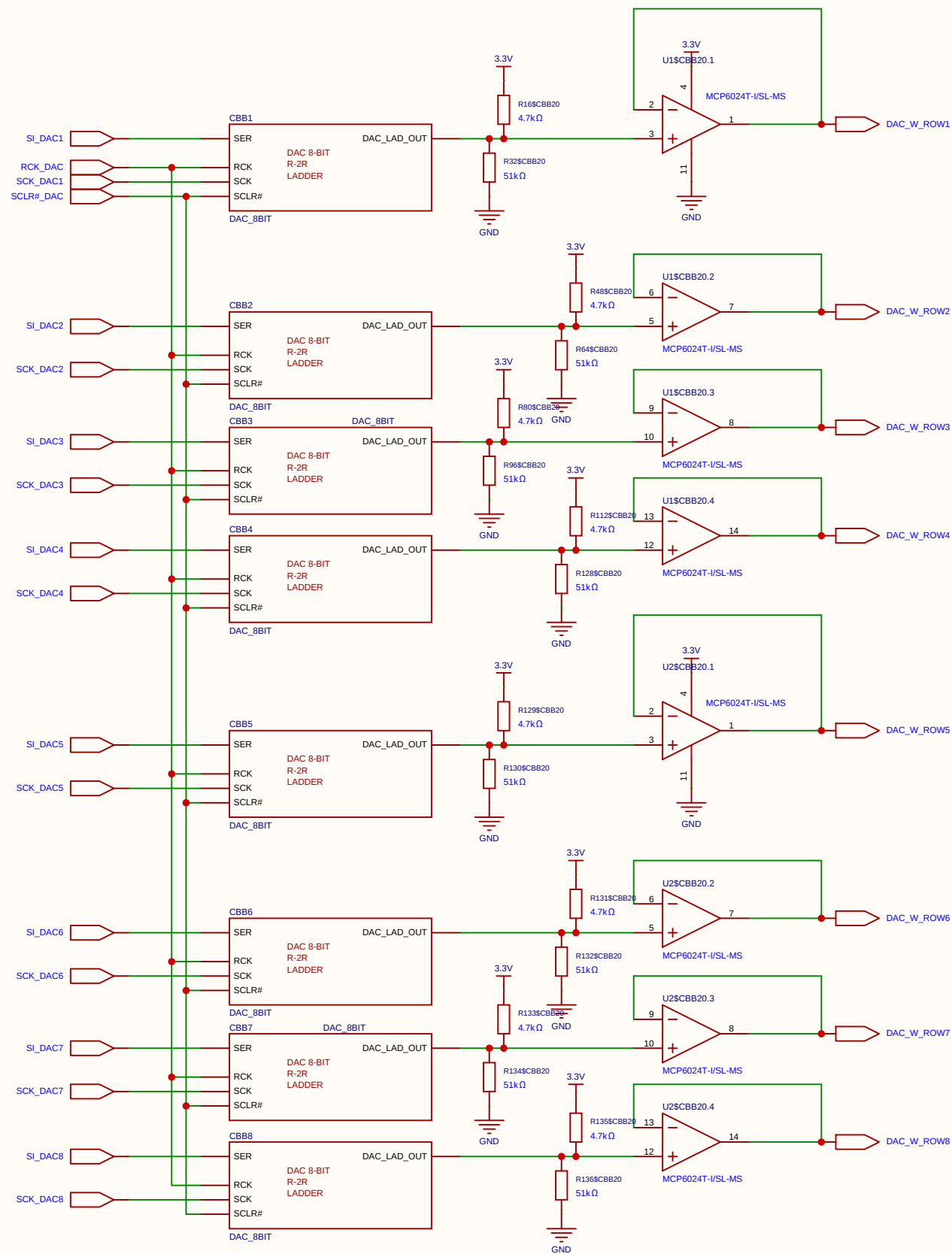


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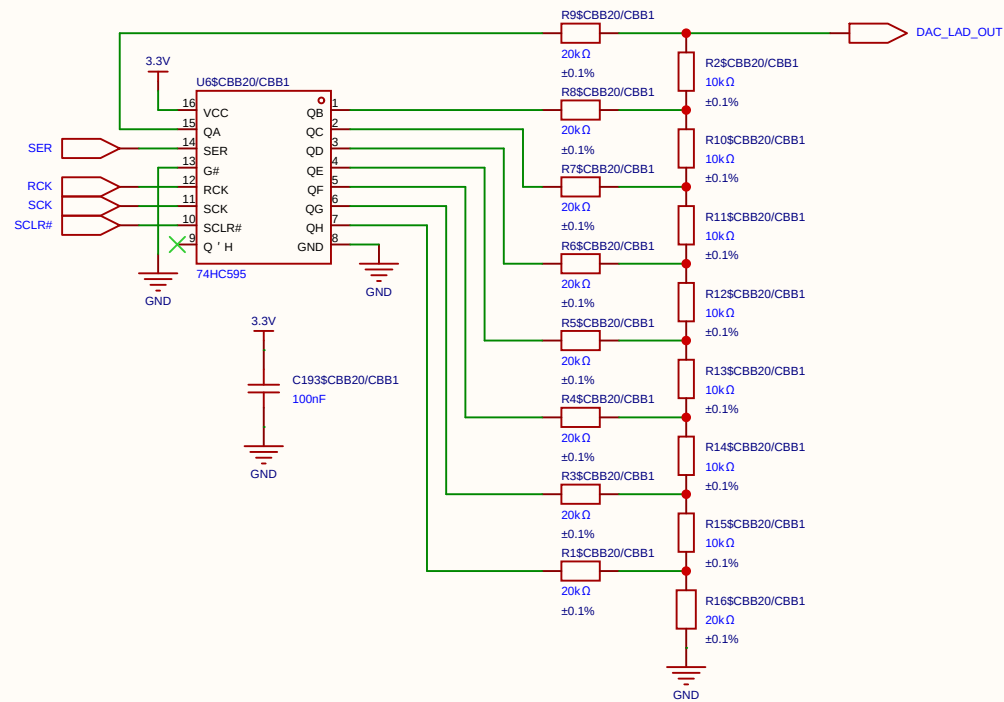


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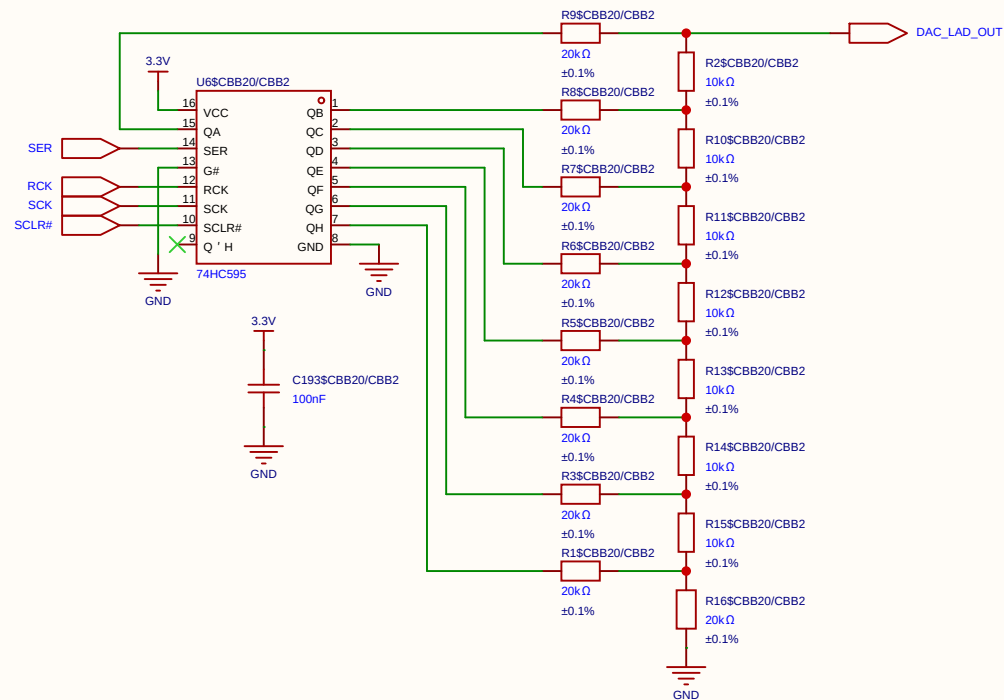
Weight swing (Gate / V_{gs}) 1.0 V Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).



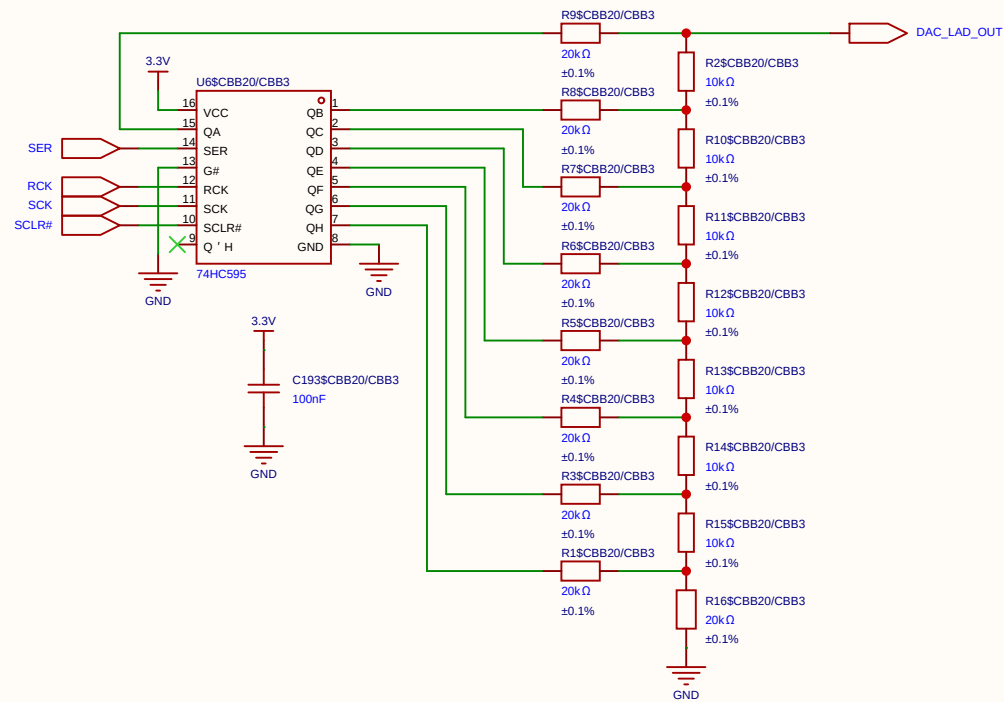
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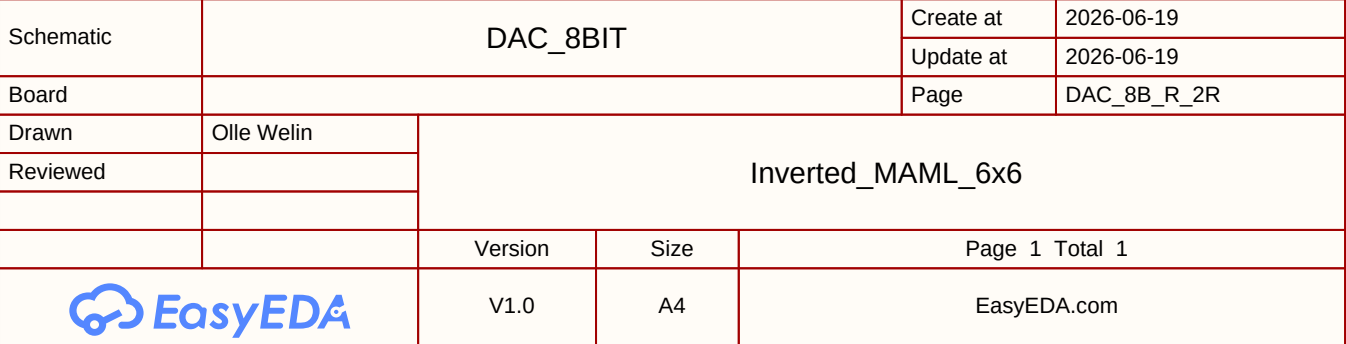
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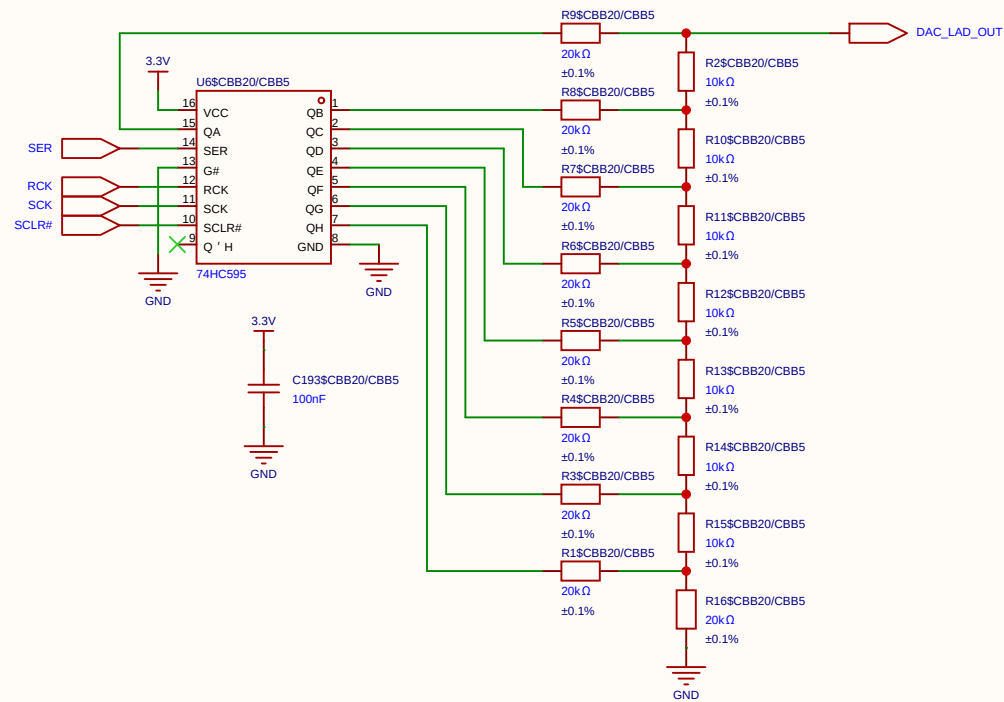


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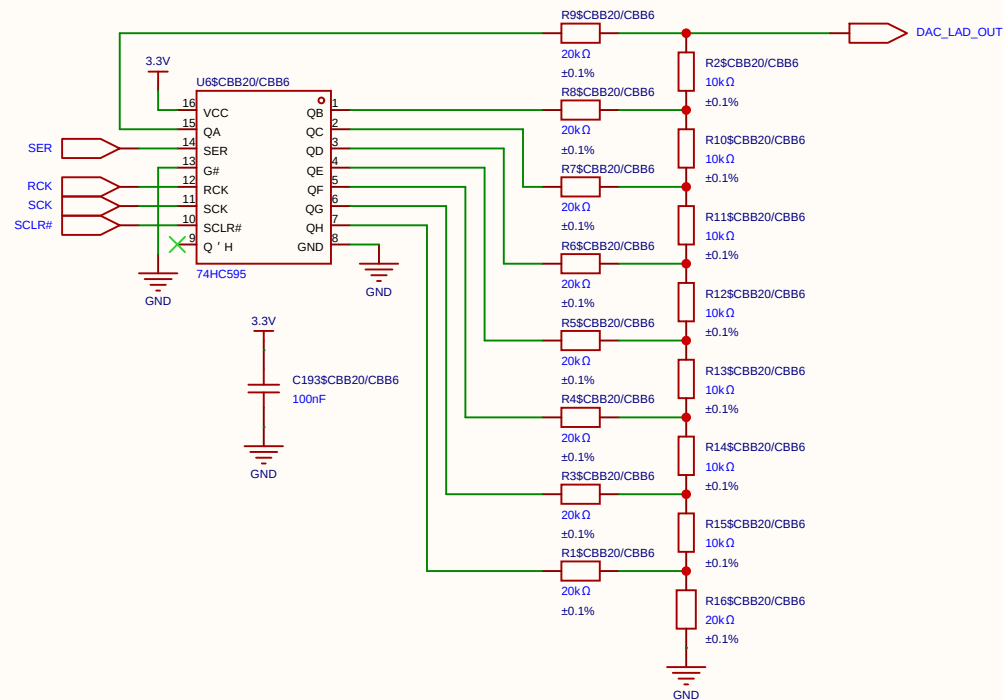


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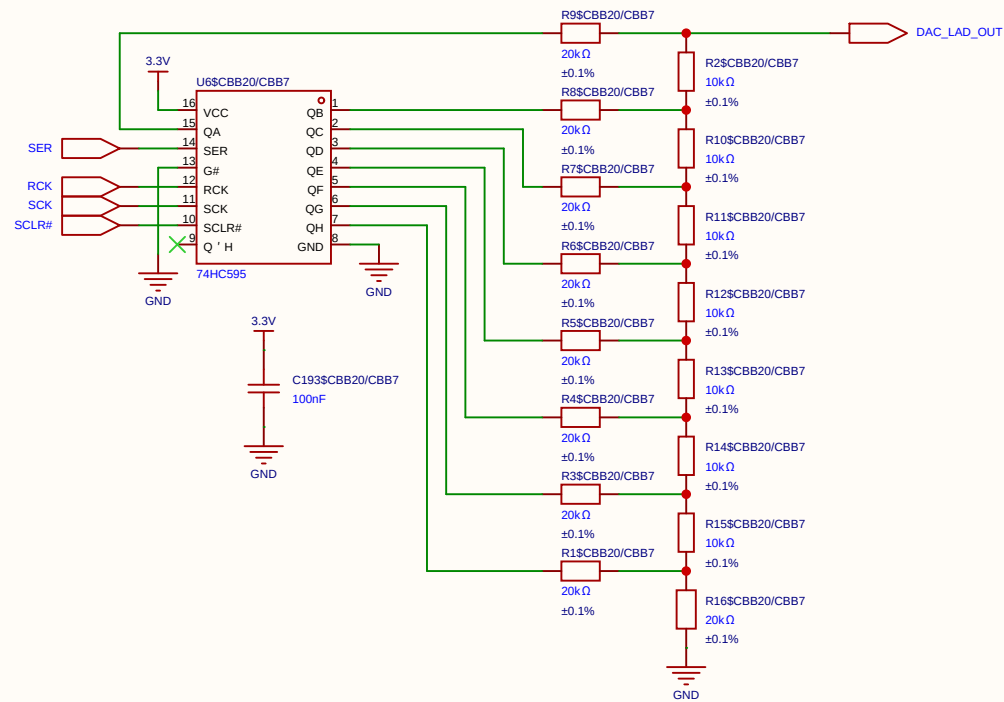




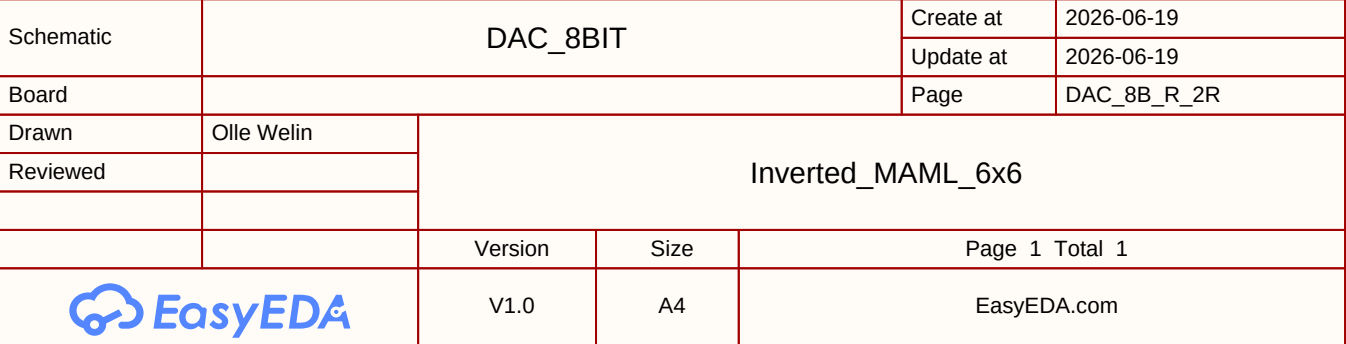
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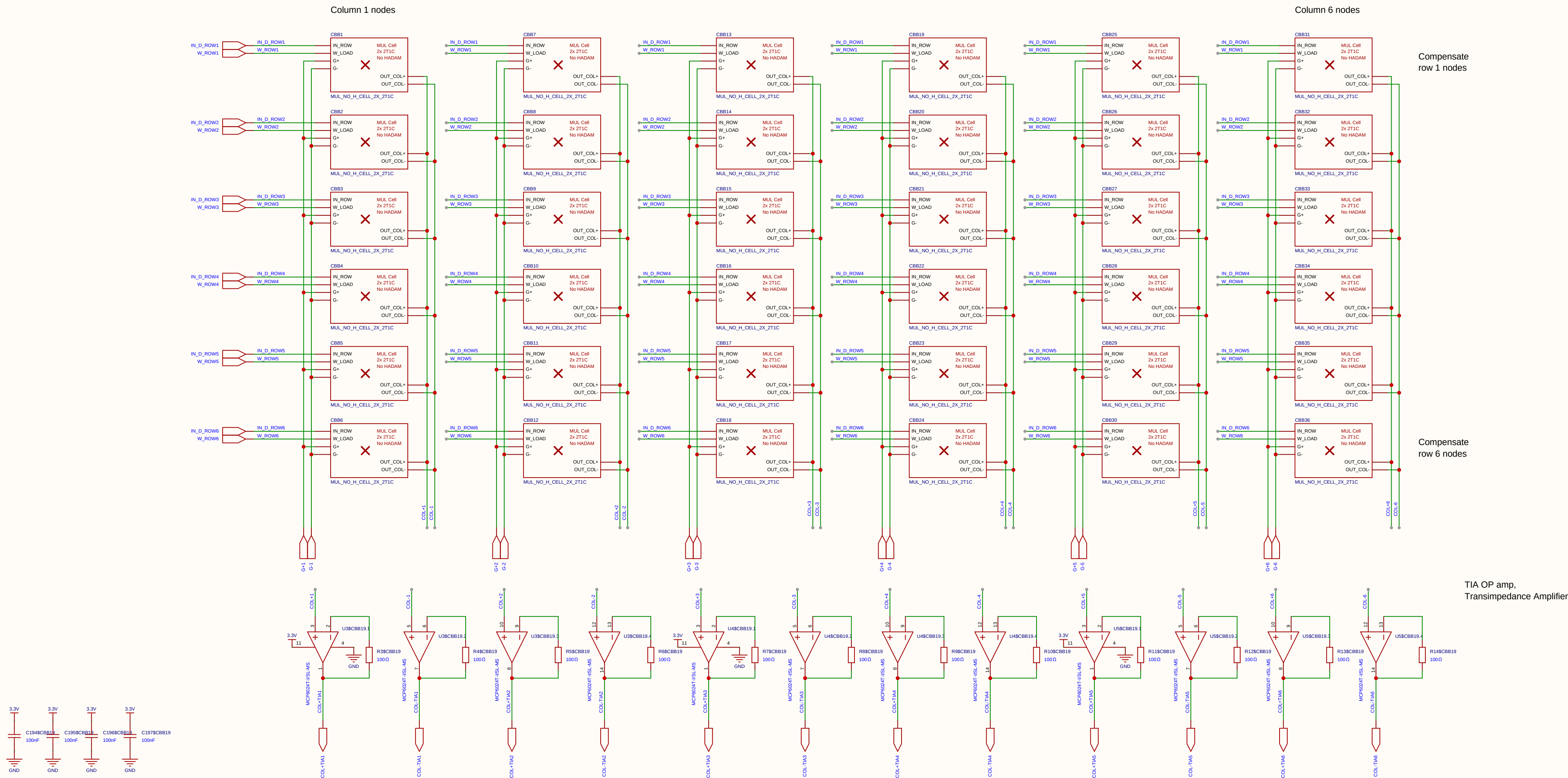
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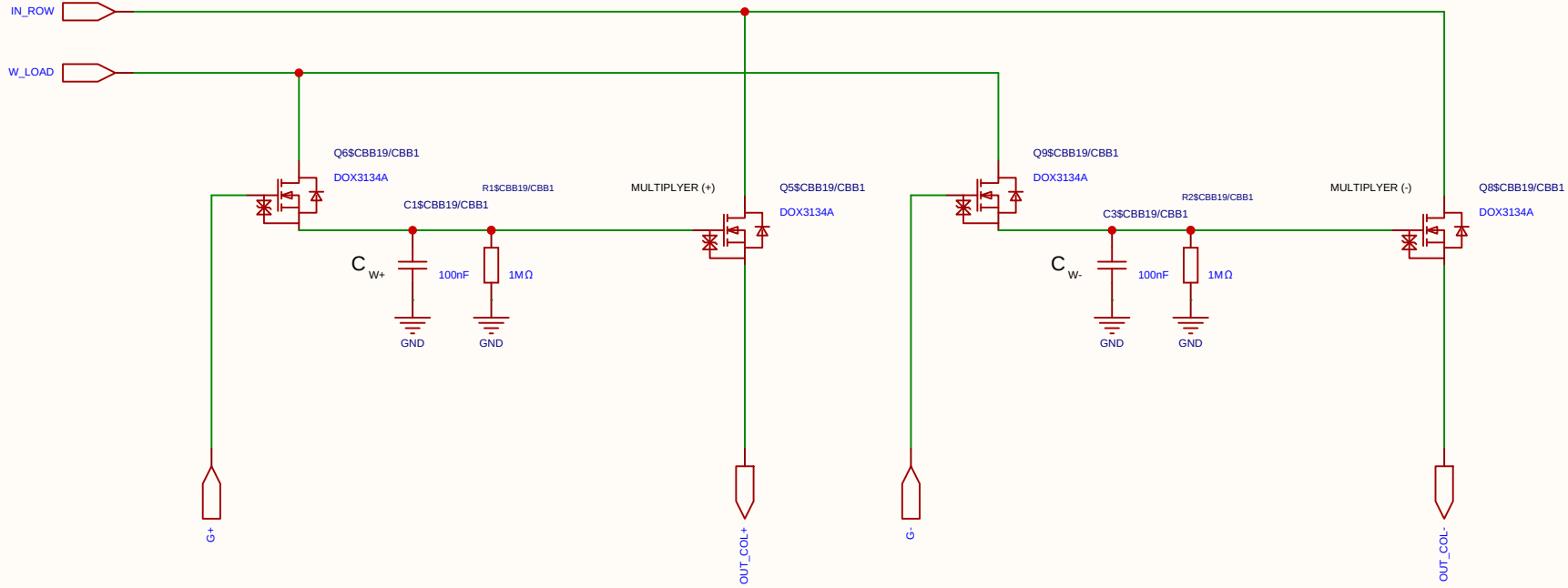
$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$



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Analog Multiplication CELL Kernel

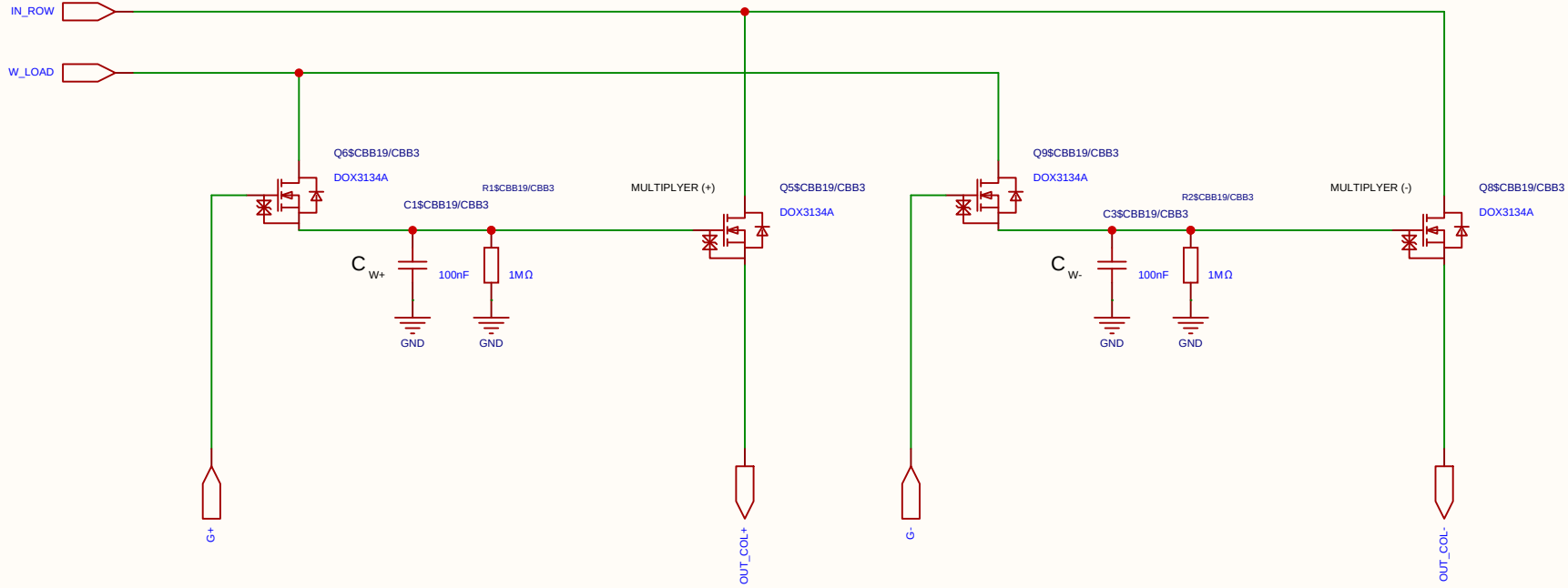
Differential 2x 2T1C



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Analog Multiplication CELL Kernel

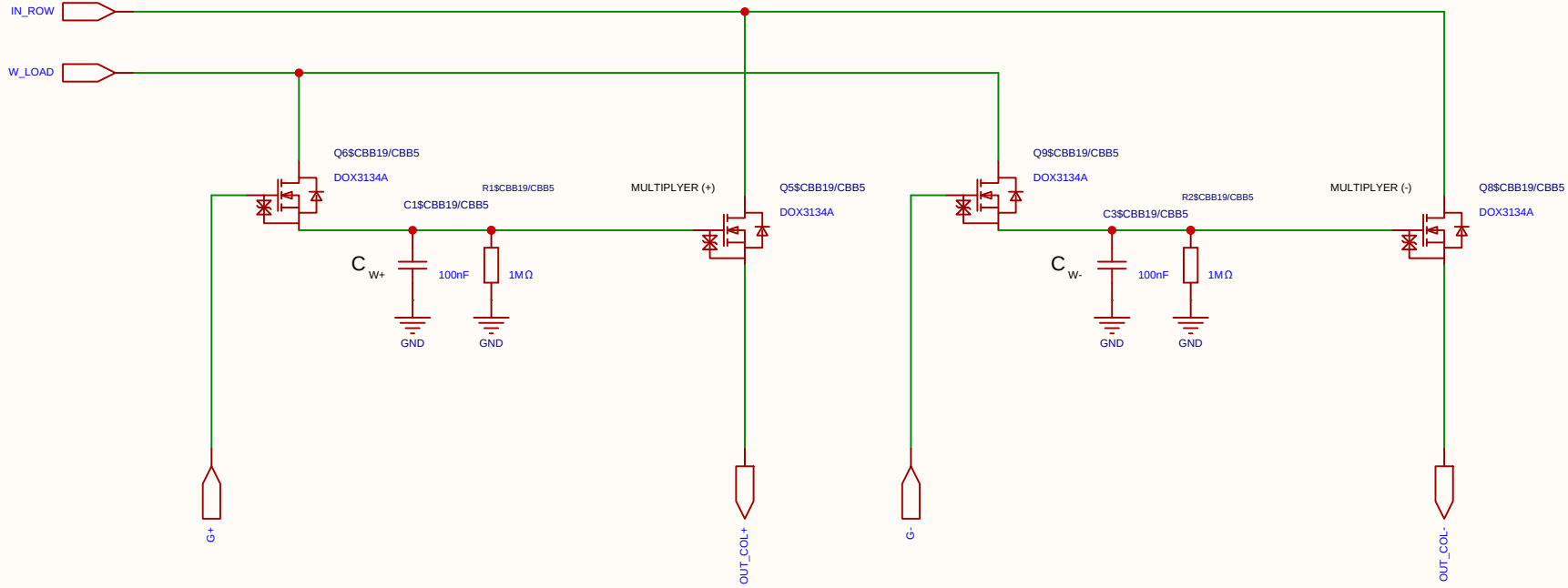
Differential 2x 2T1C



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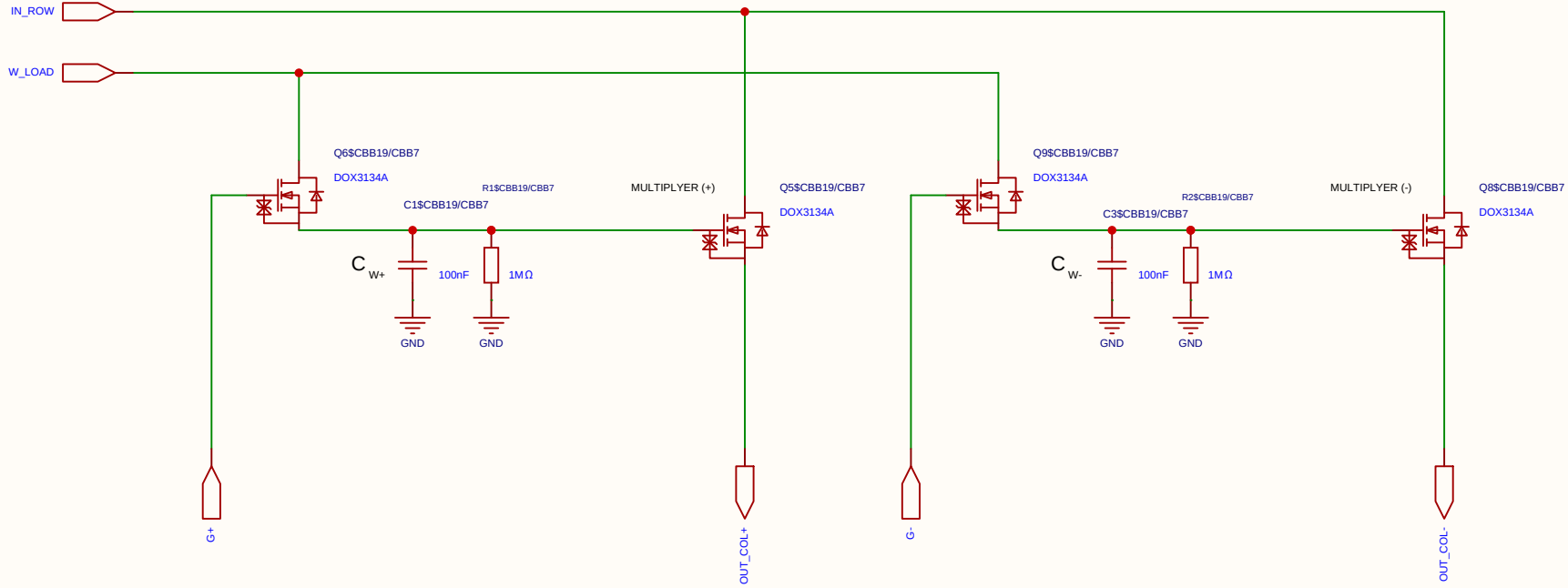
Differential 2x 2T1C



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Analog Multiplication CELL Kernel

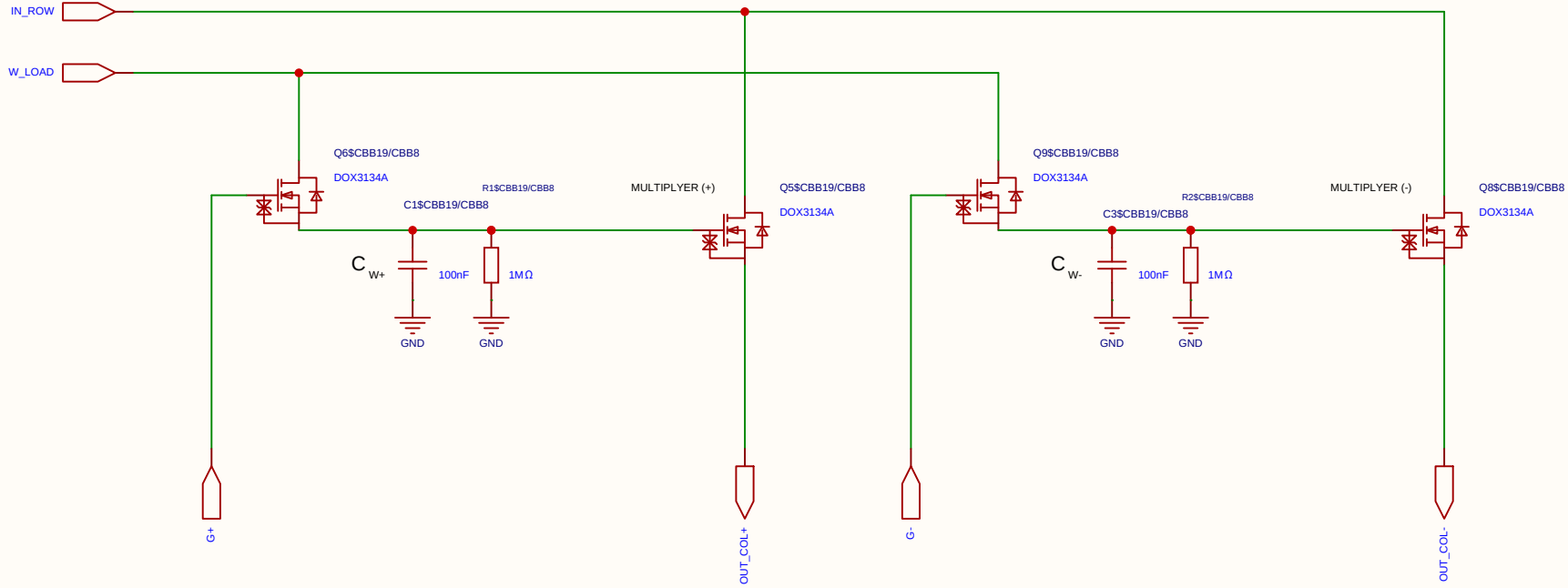
Differential 2x 2T1C



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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

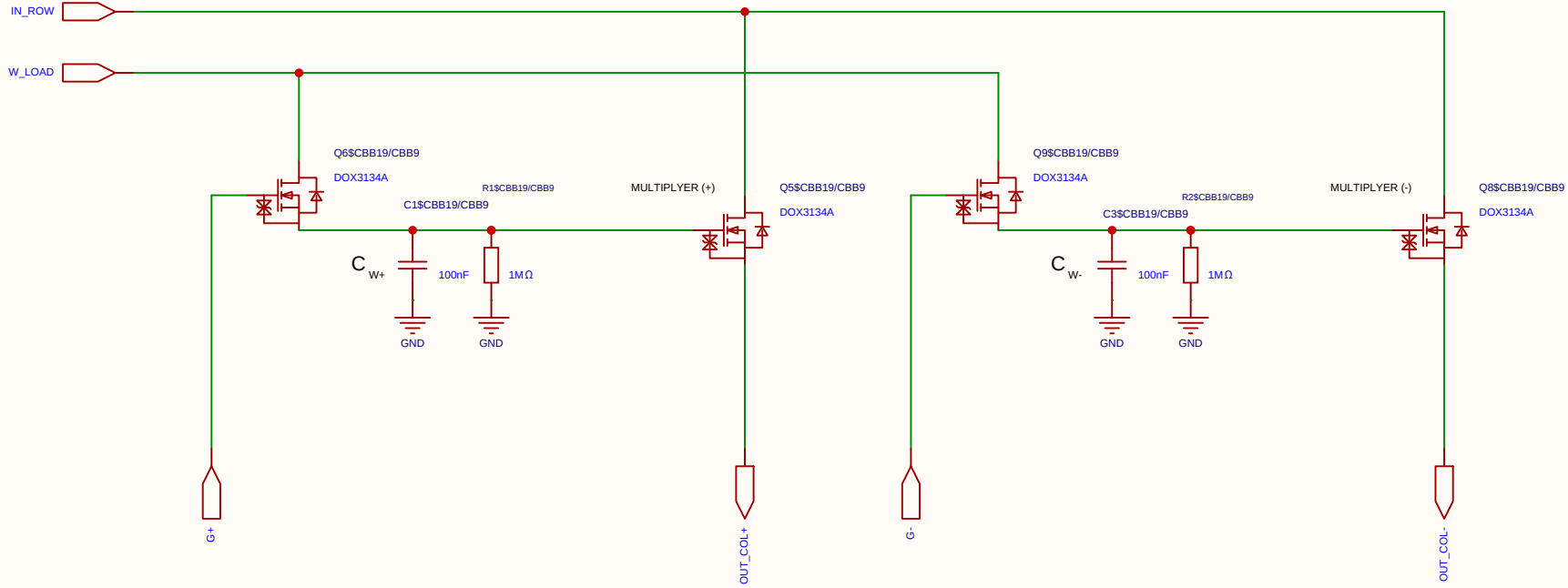
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

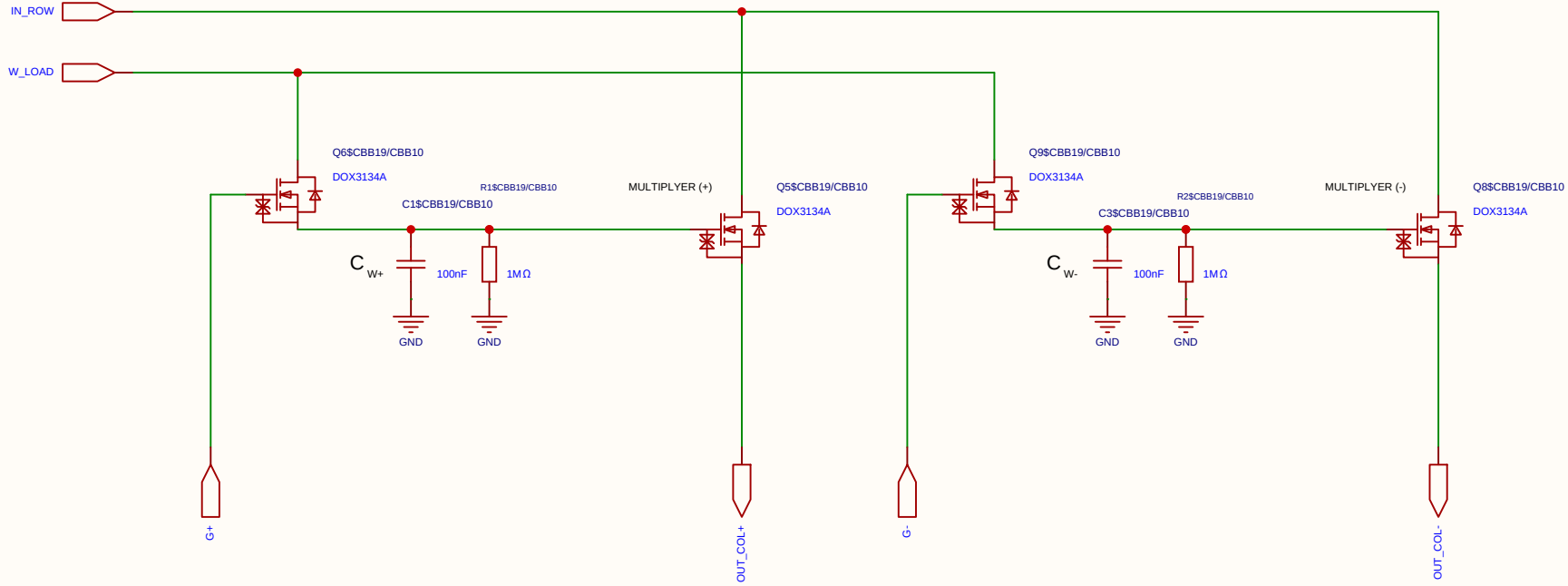
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C		Create at	2026-06-20
			Update at	2026-06-28
Board			Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel

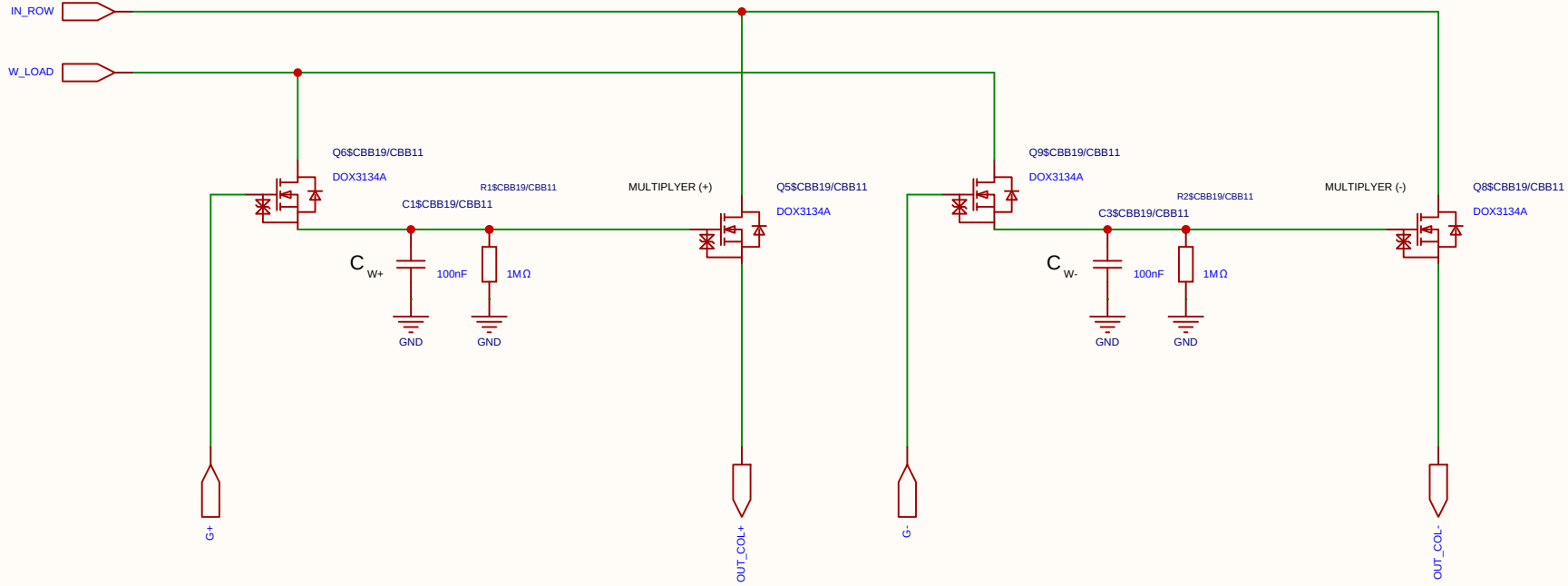
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

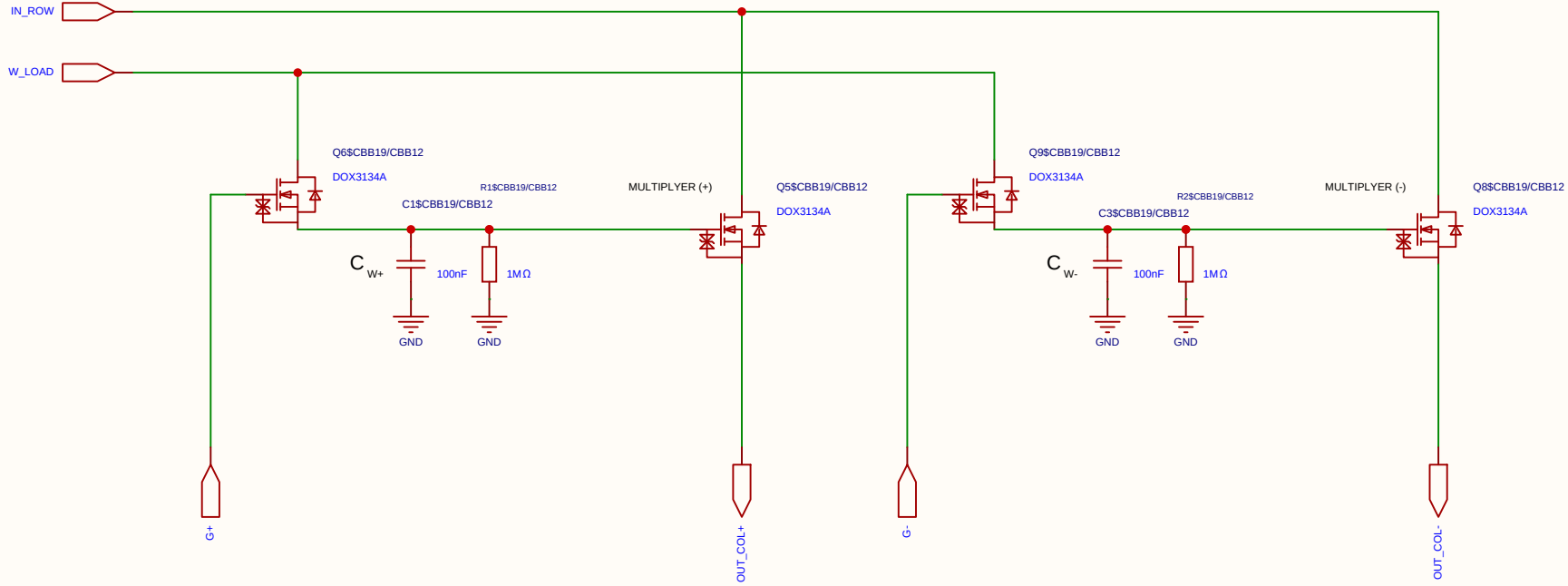
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

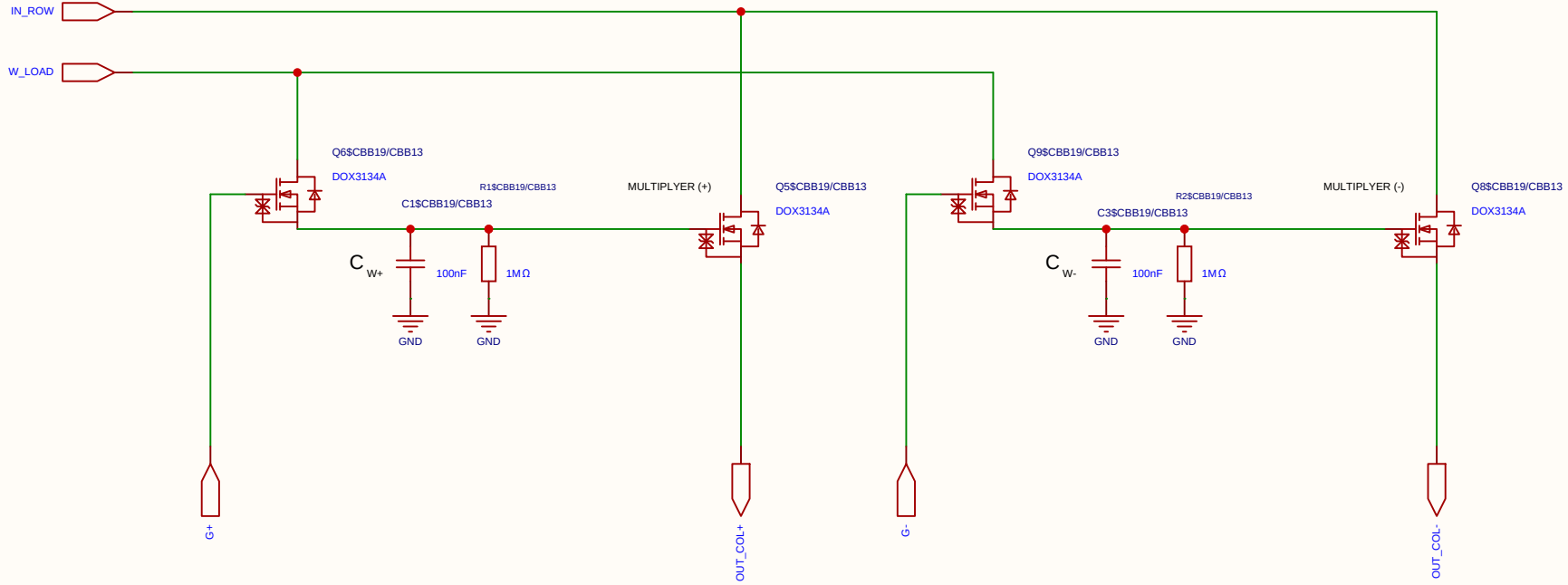
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

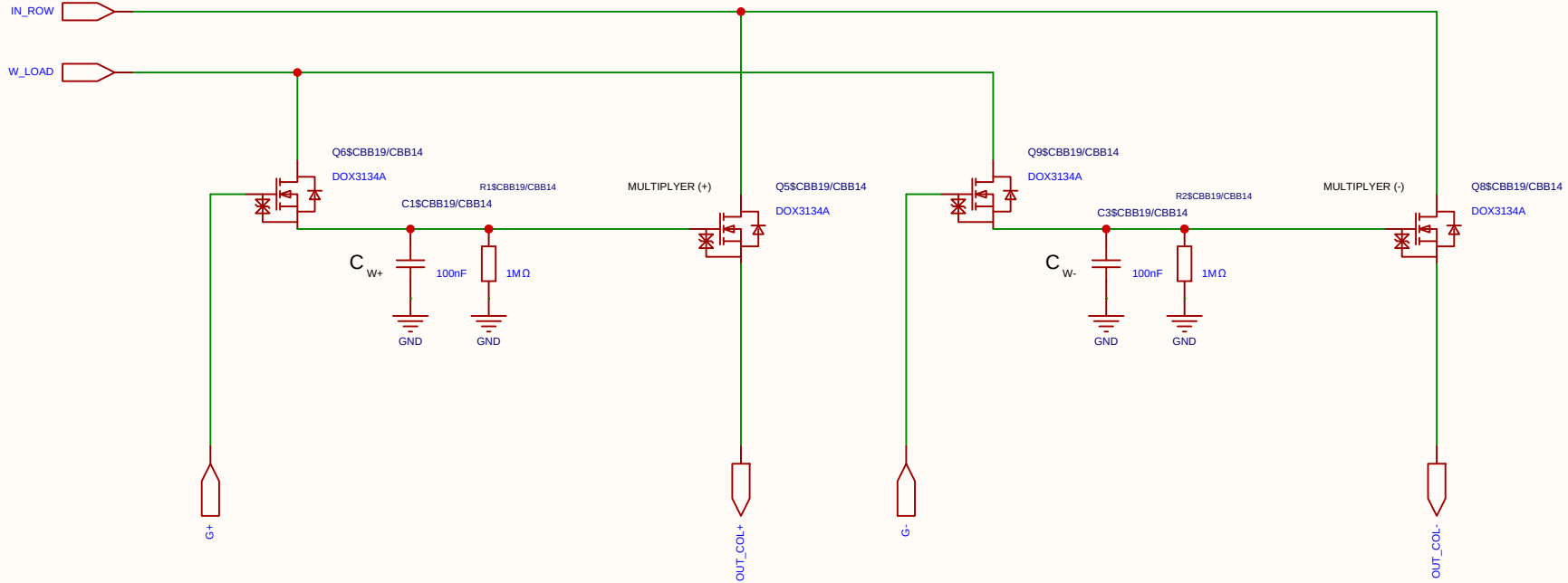
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C		Create at	2026-06-20
			Update at	2026-06-28
Board			Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel

Differential 2x 2T1C

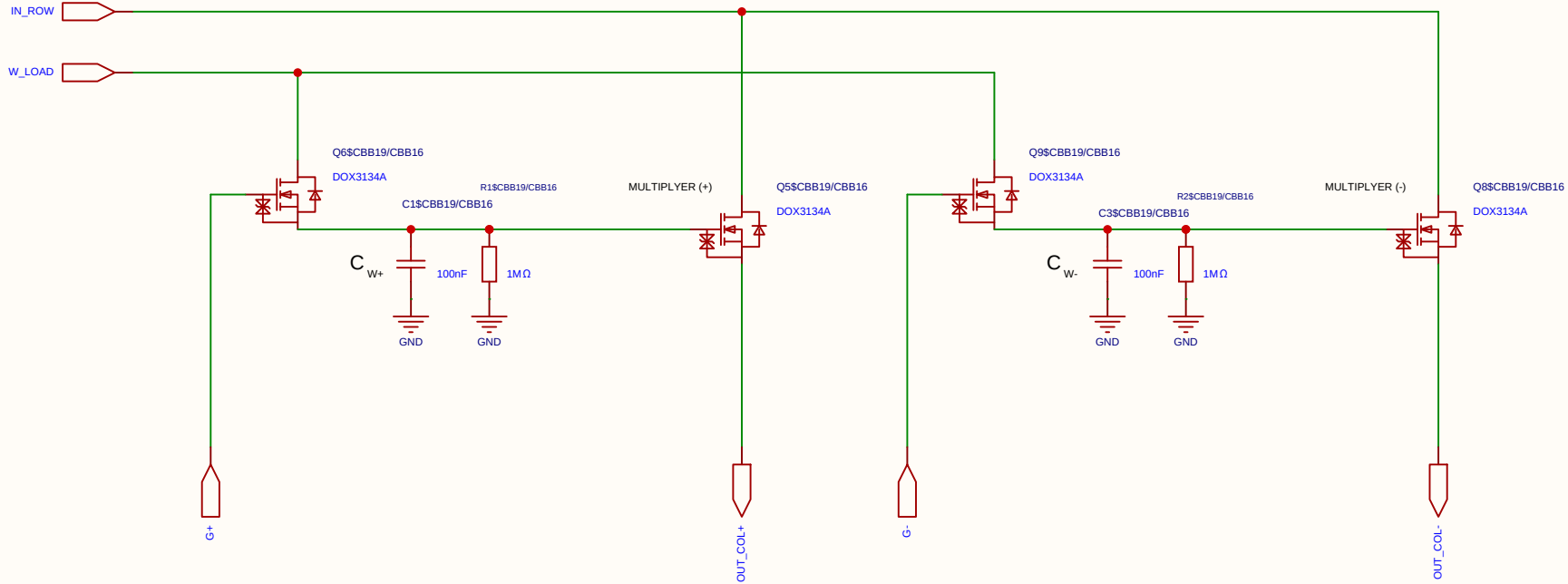


Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

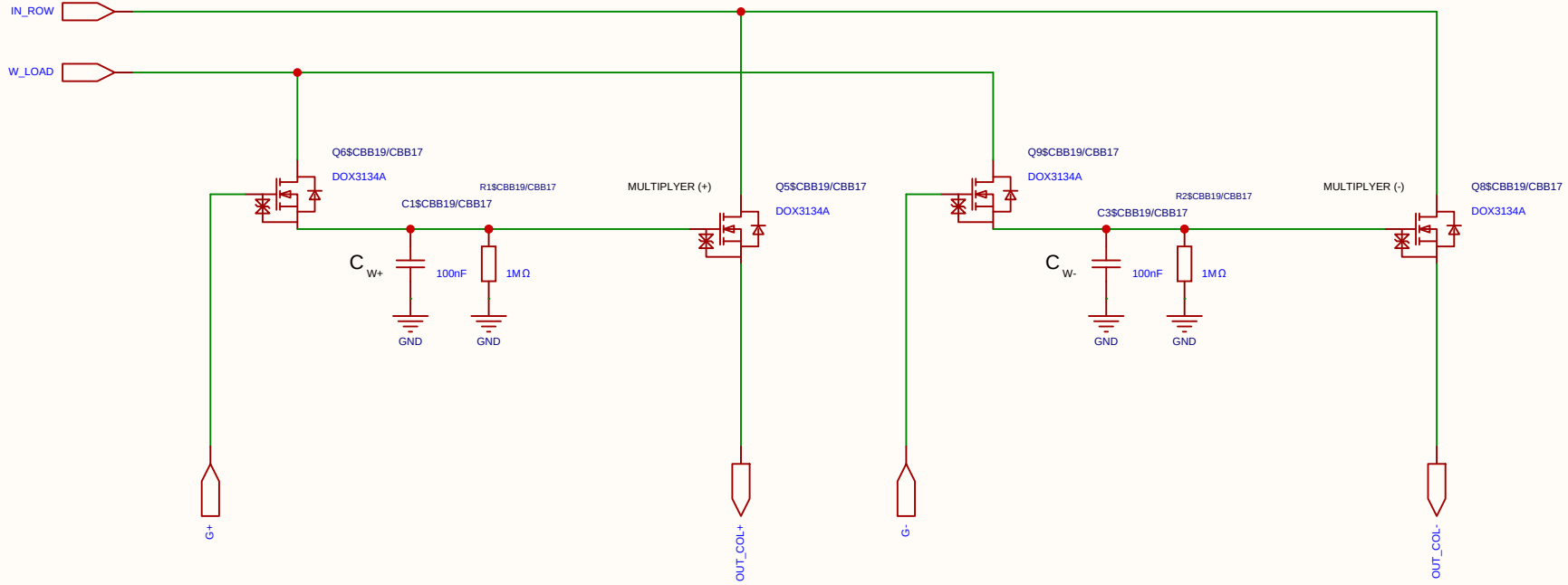
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

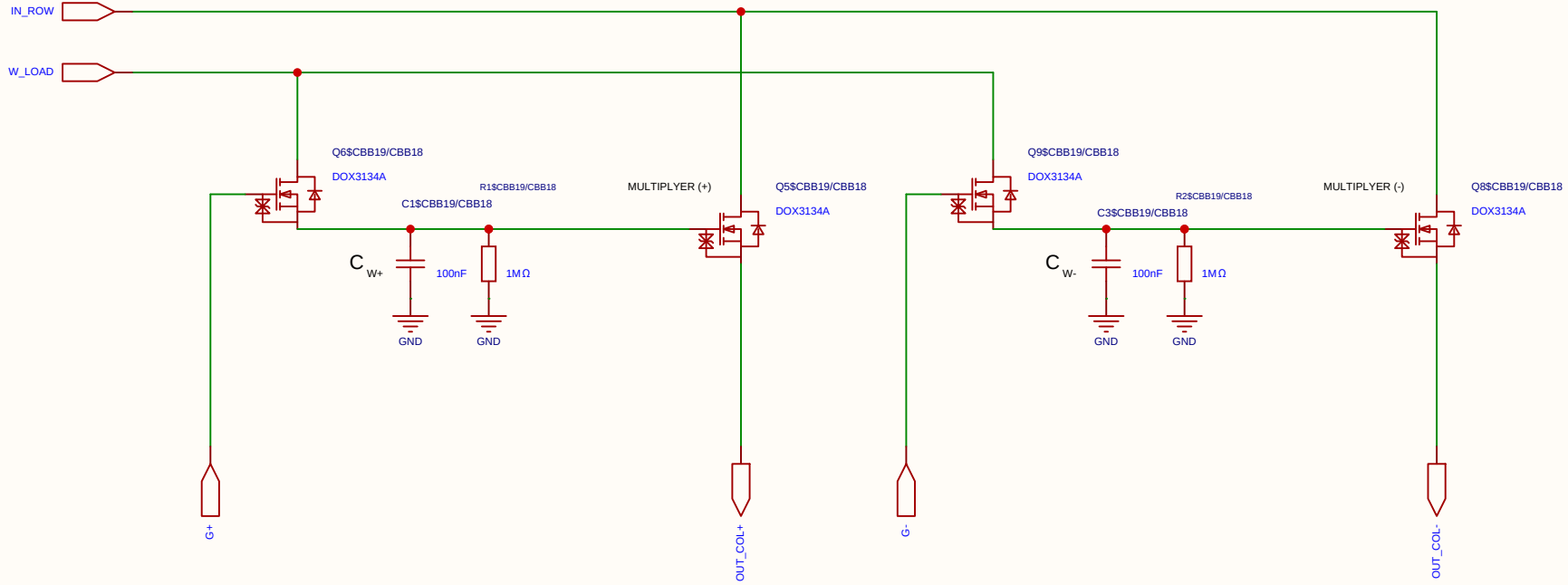
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

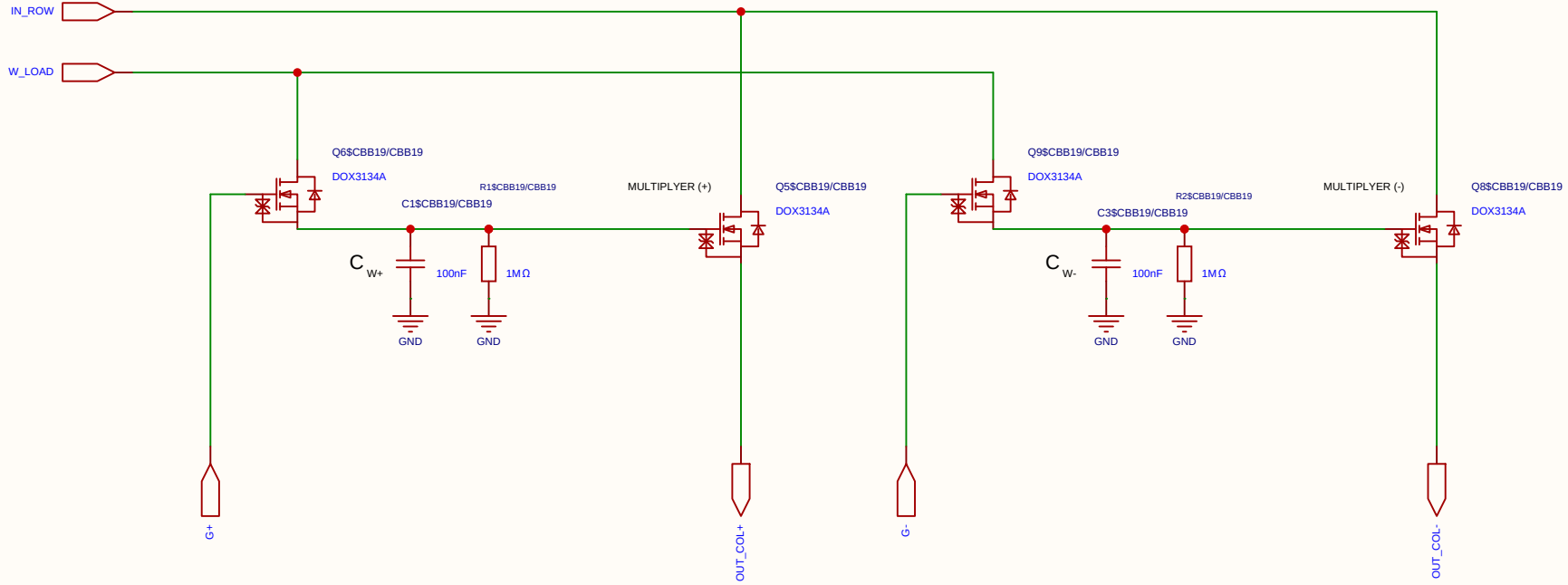
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

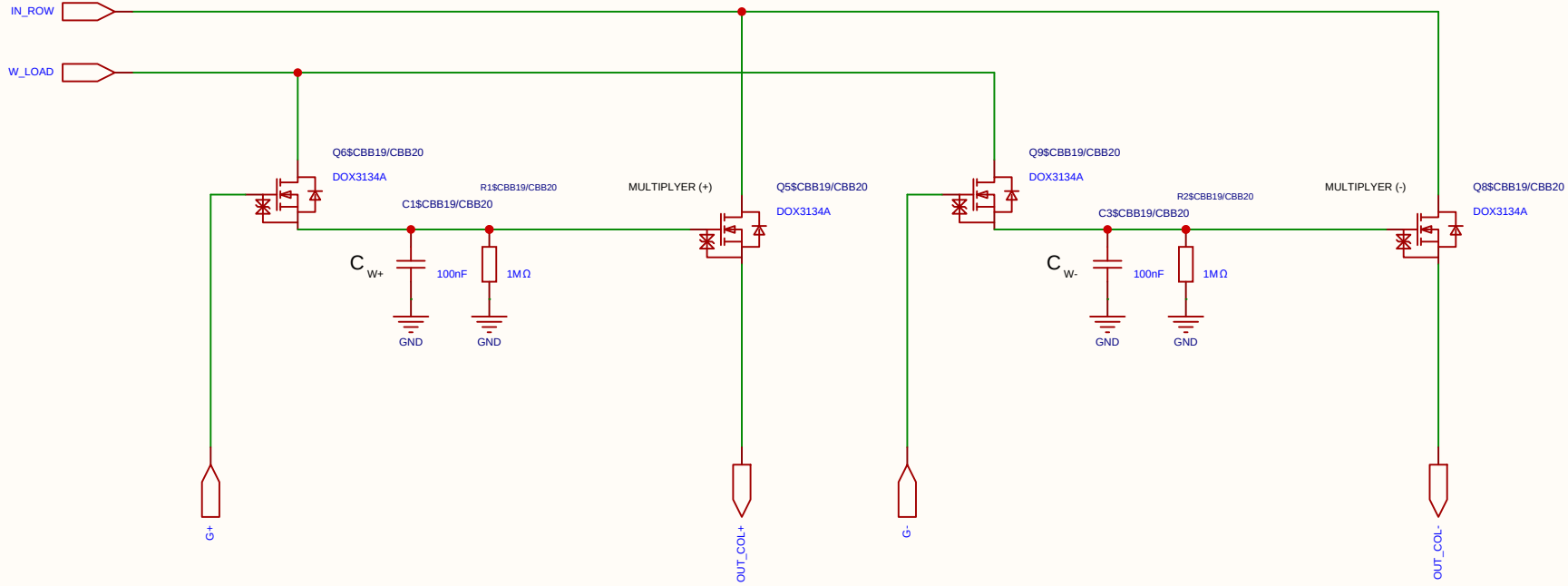
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

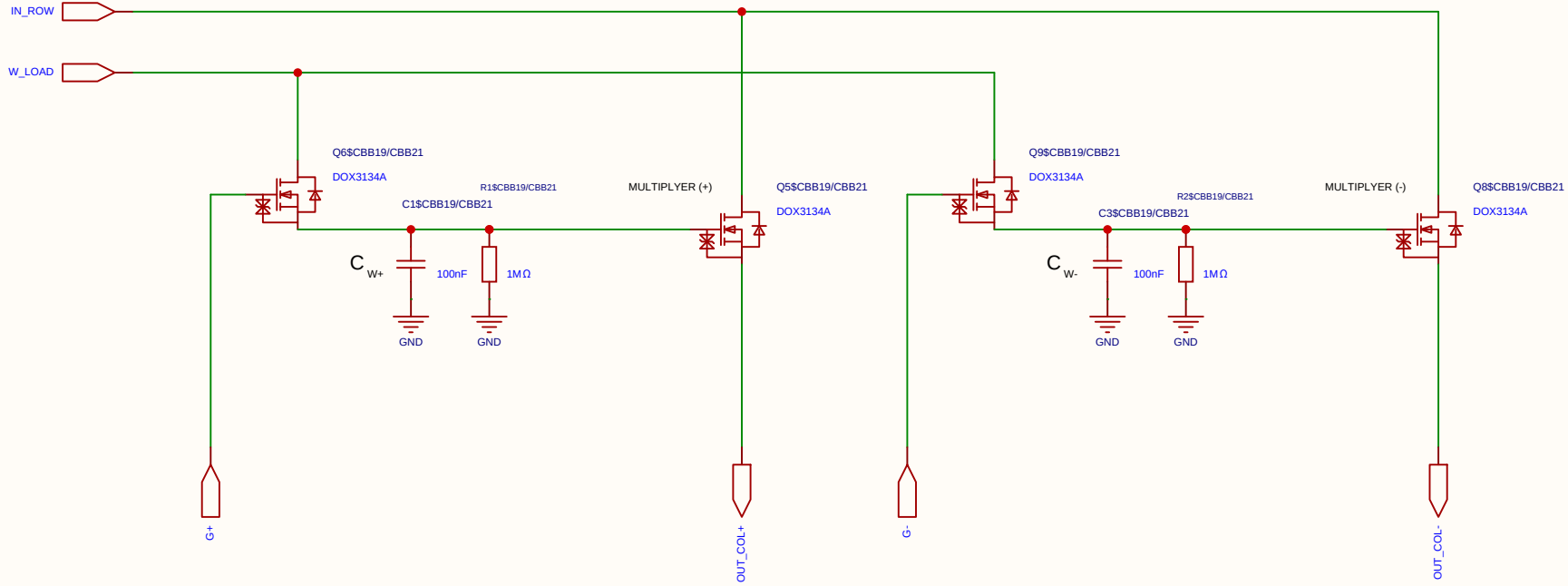
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

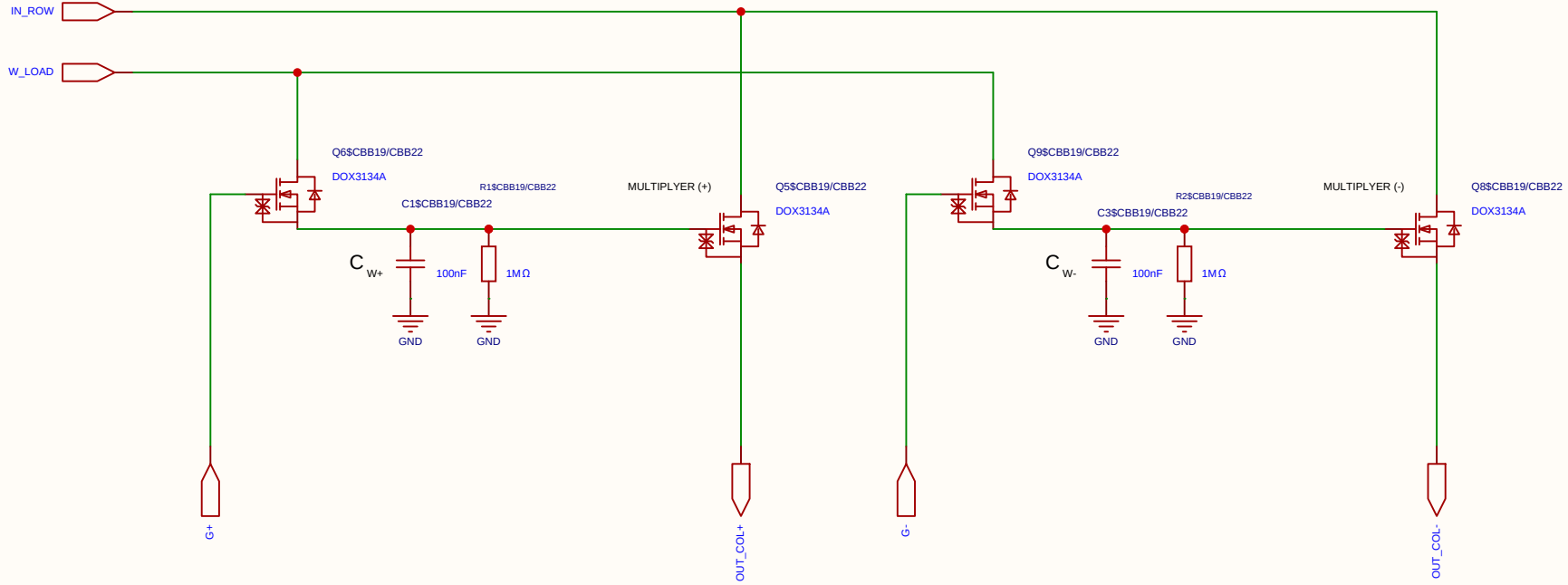
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

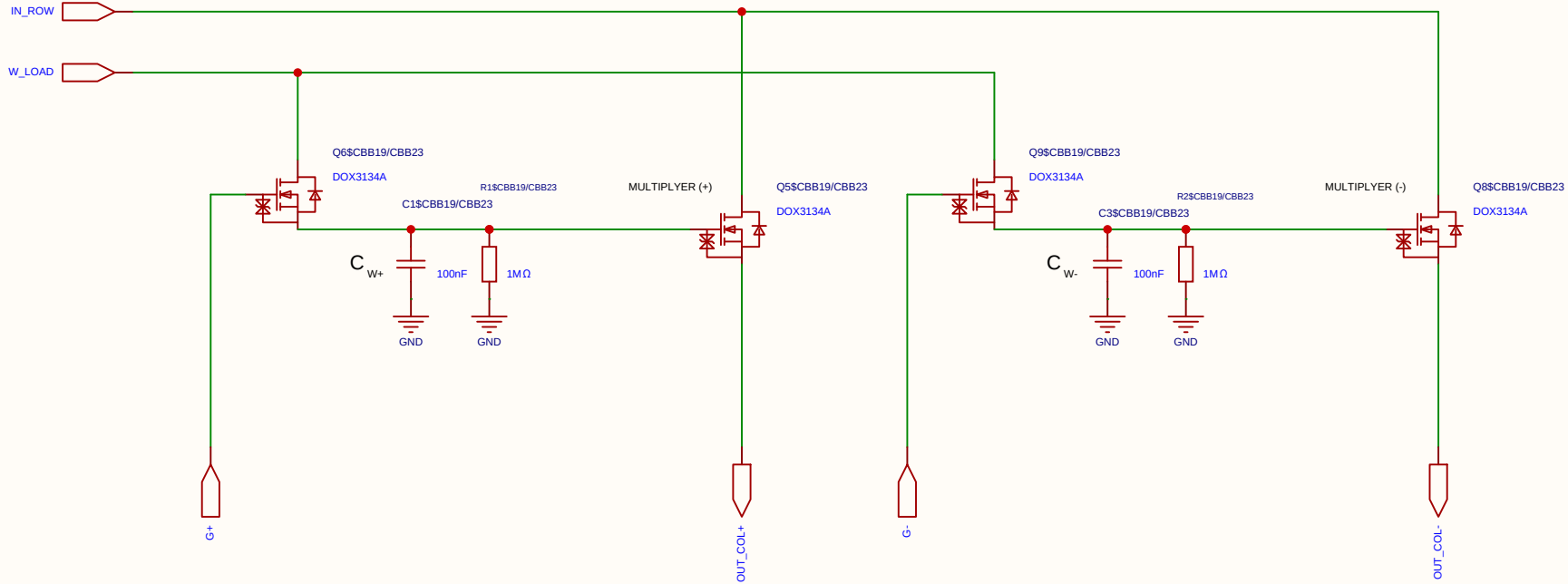
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

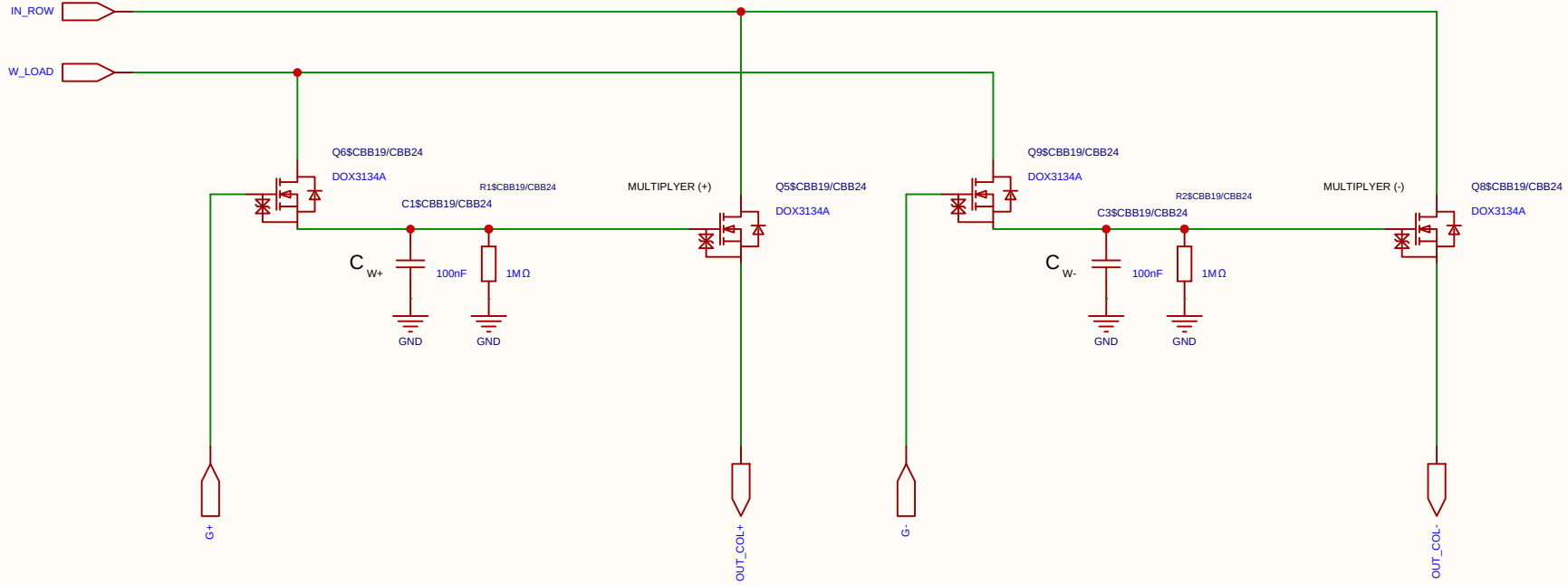
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

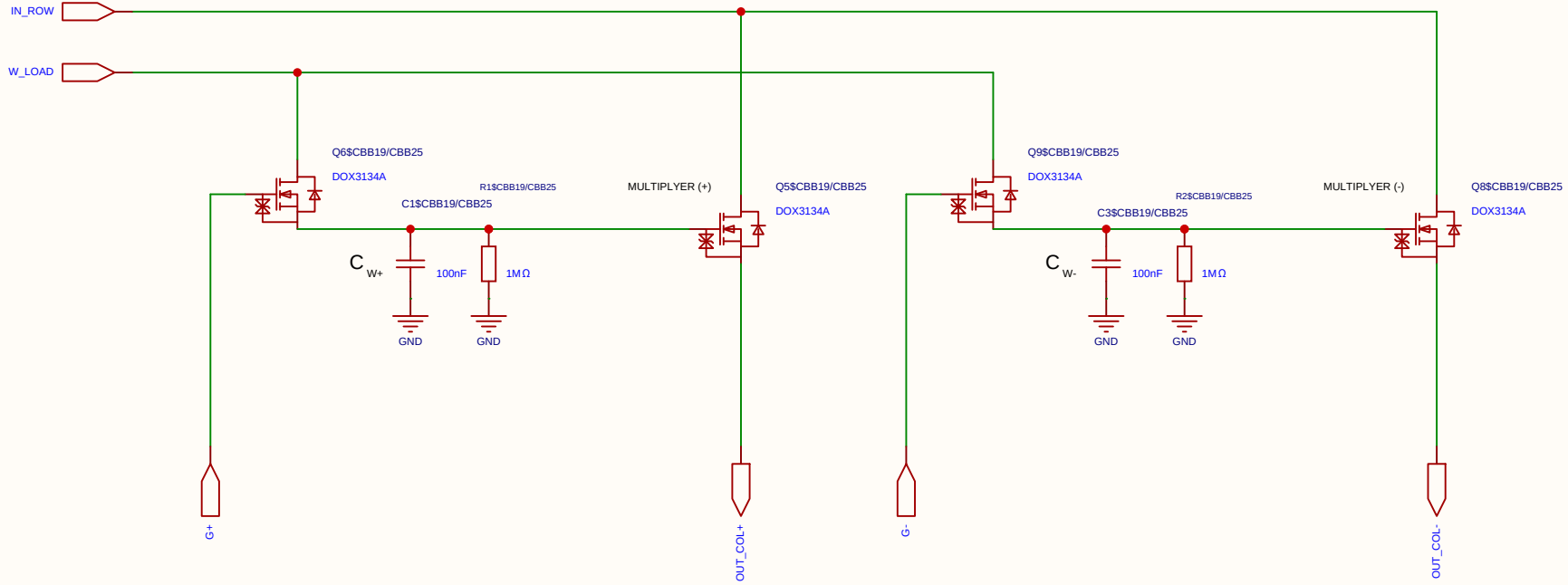
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

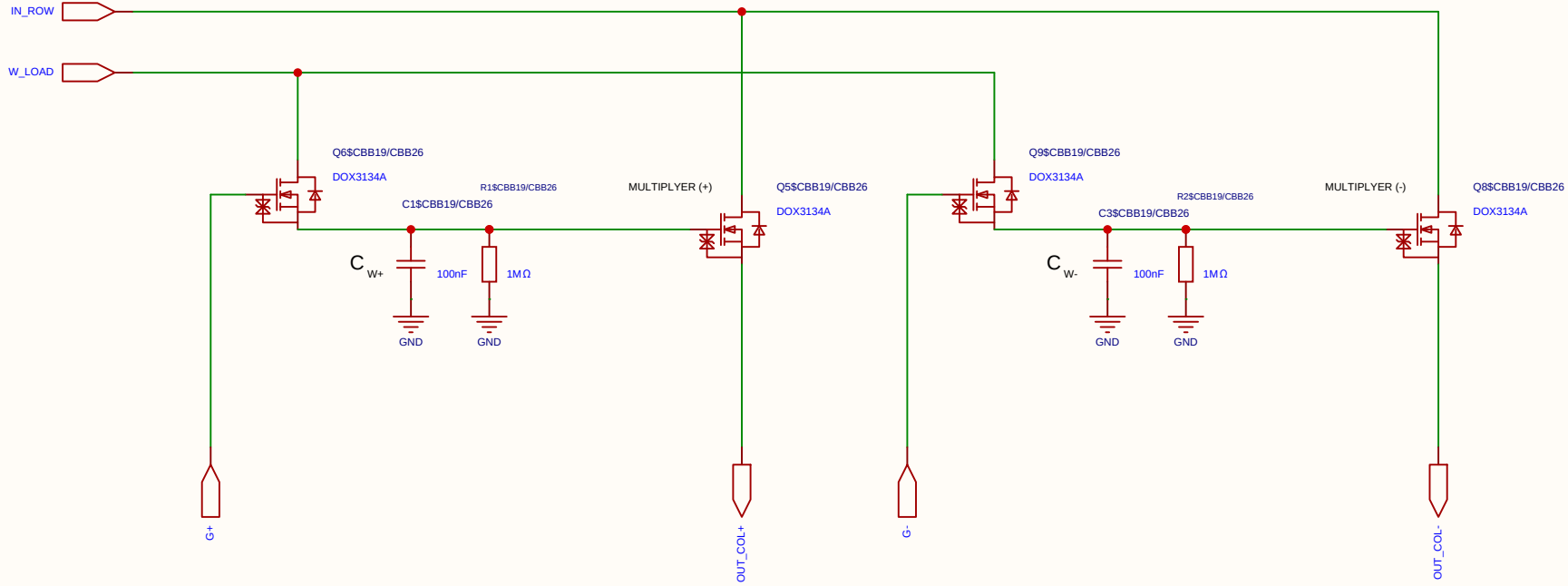
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

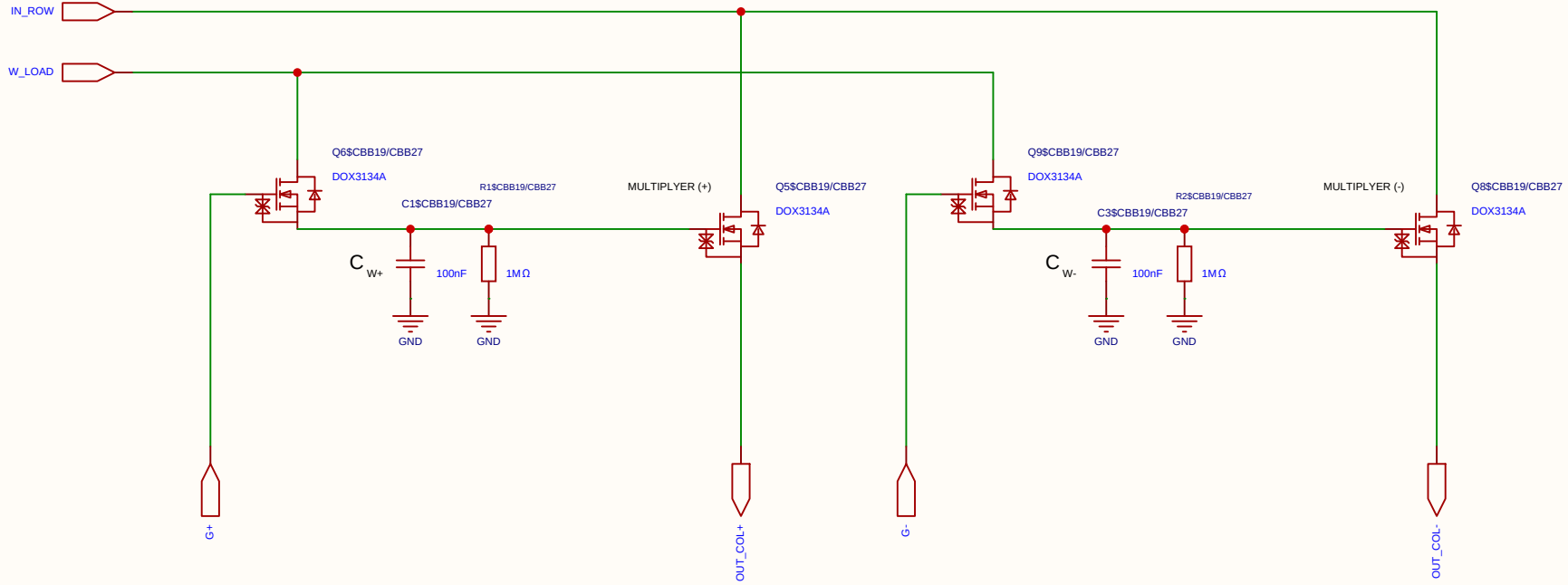
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

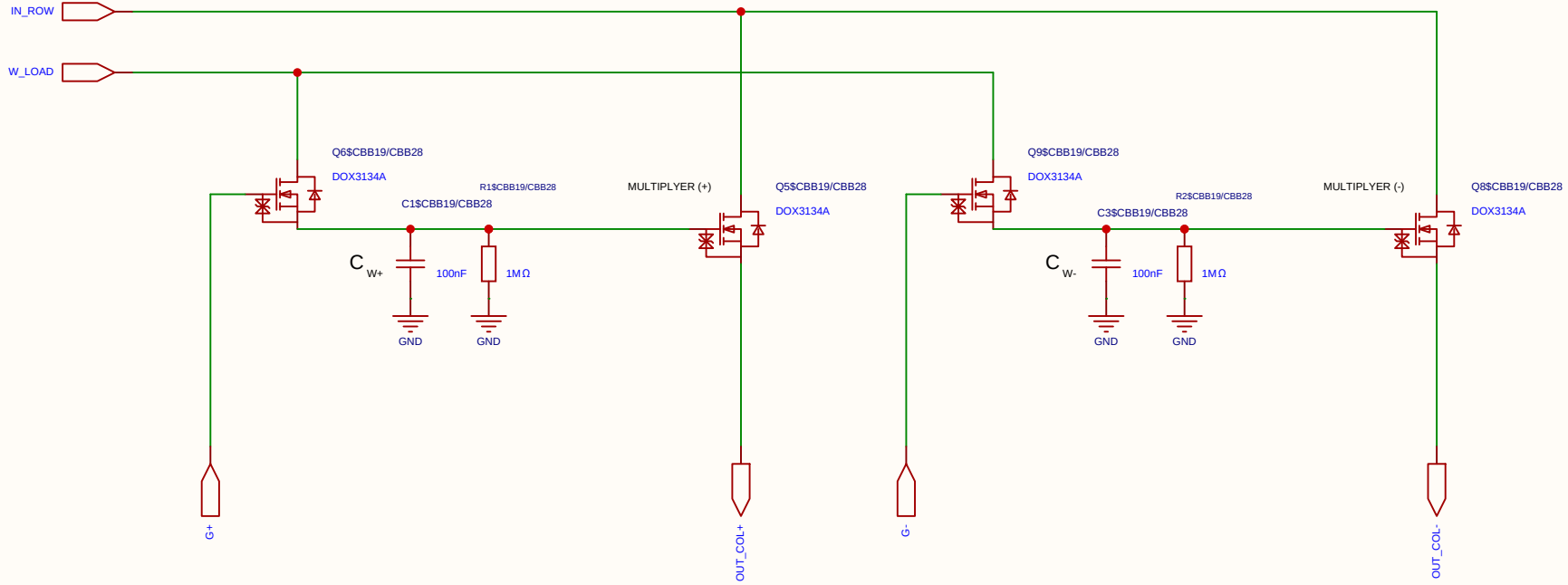
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C		Create at	2026-06-20
			Update at	2026-06-28
Board			Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel

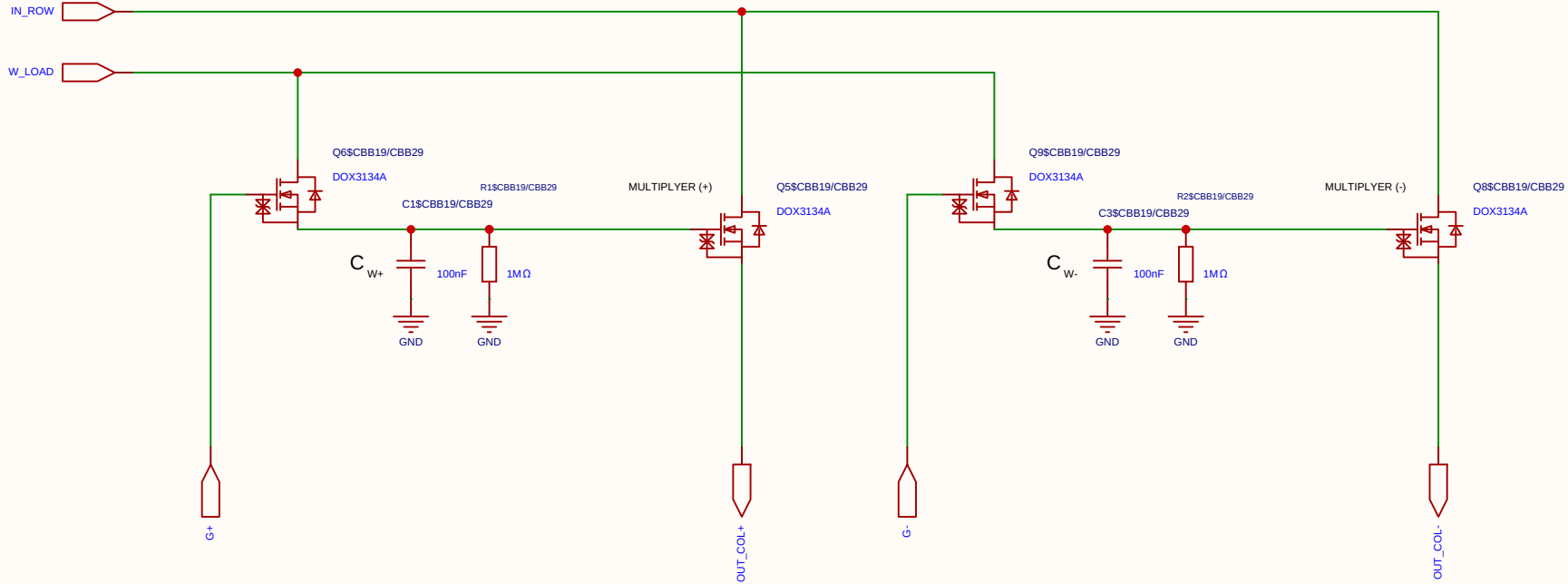
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

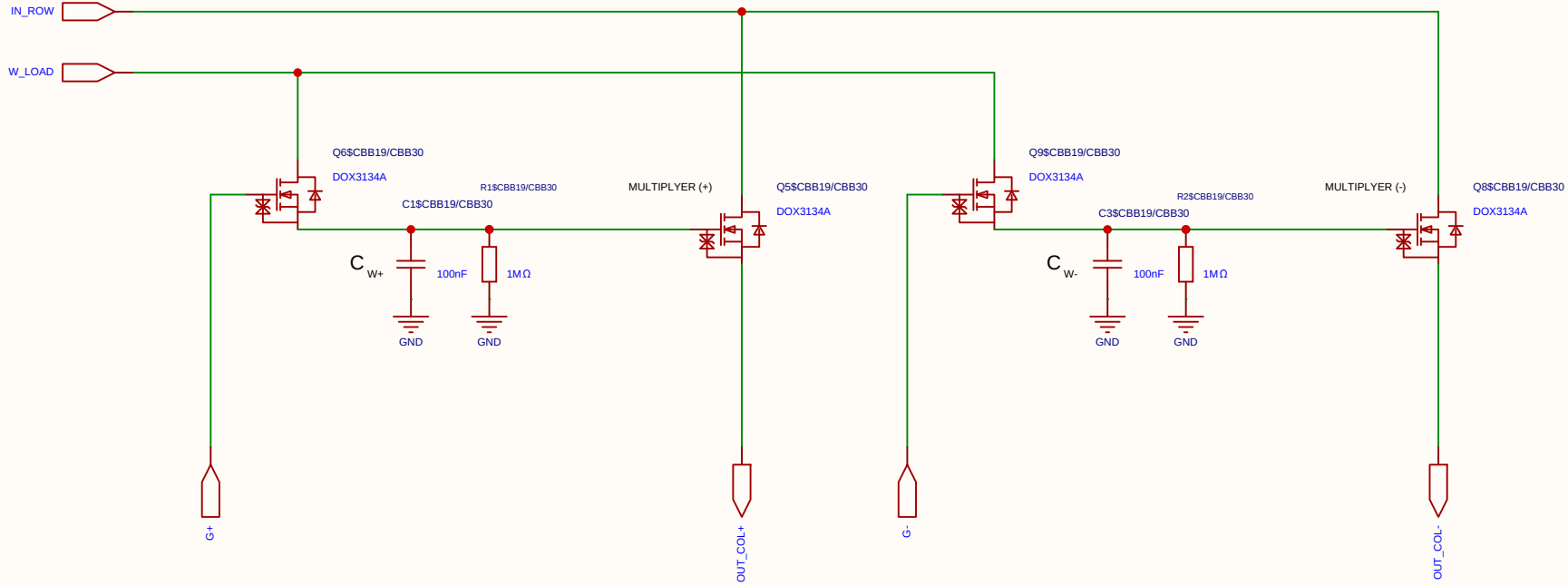
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

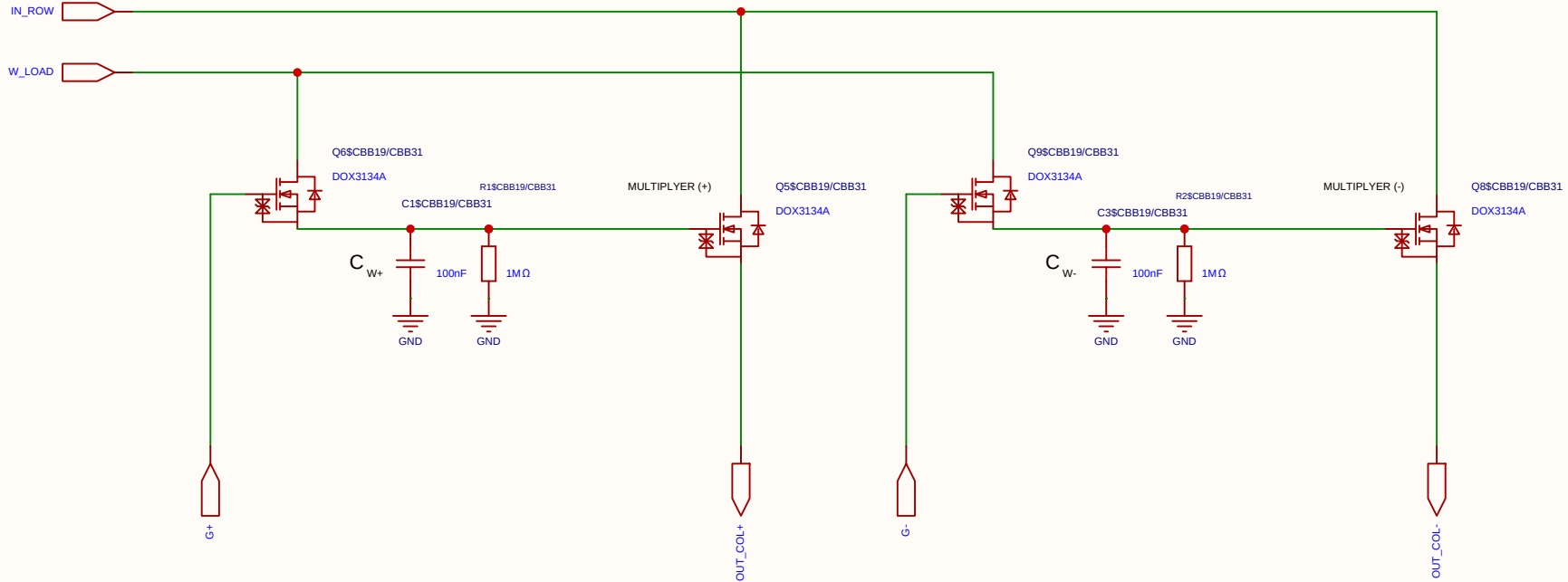
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

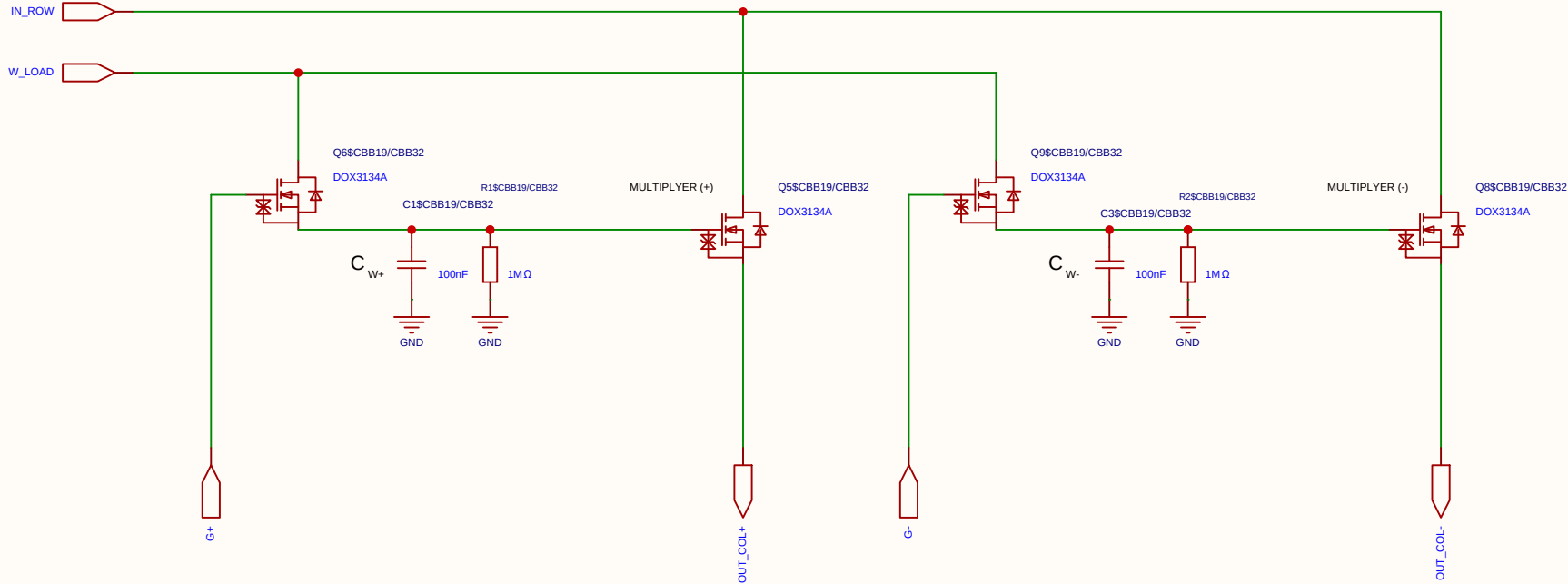
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

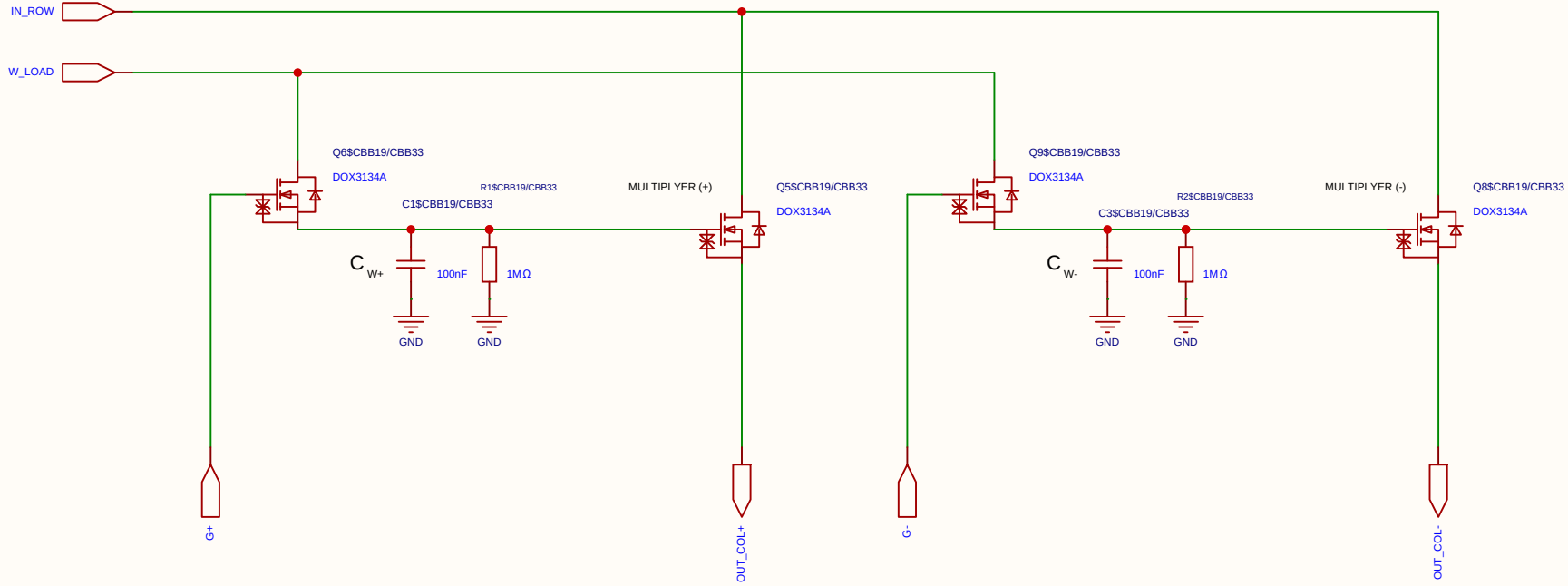
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

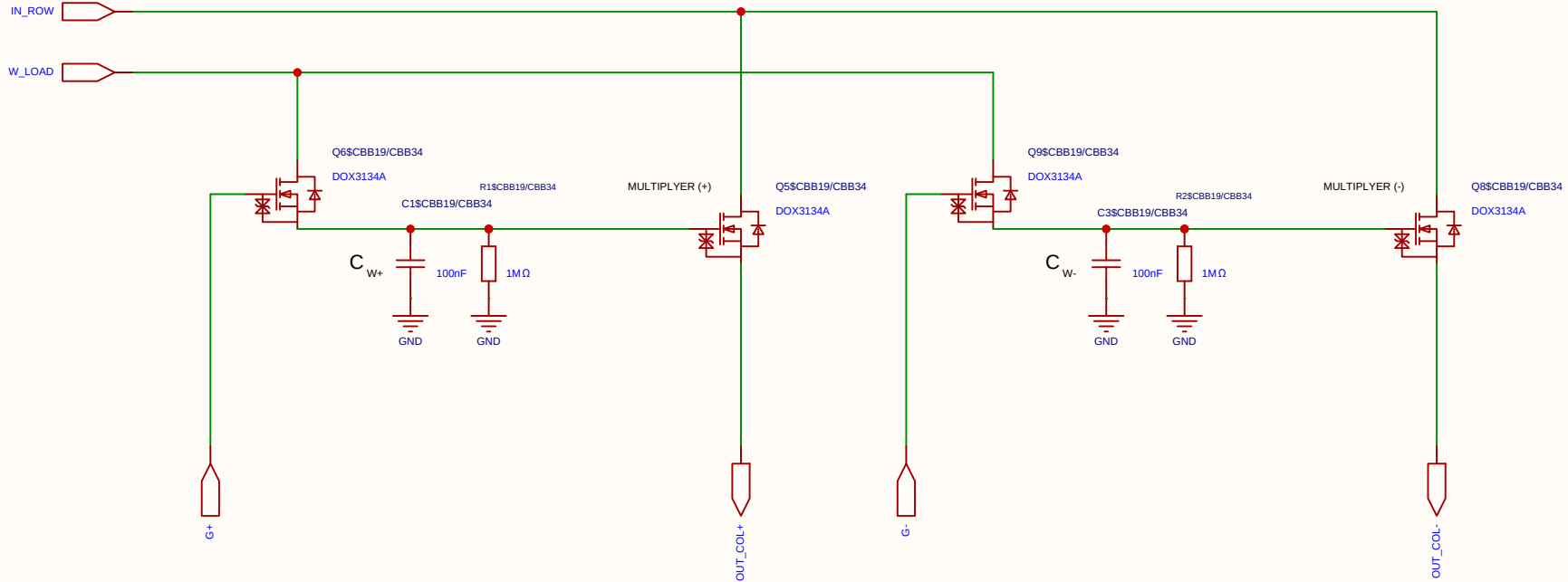
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

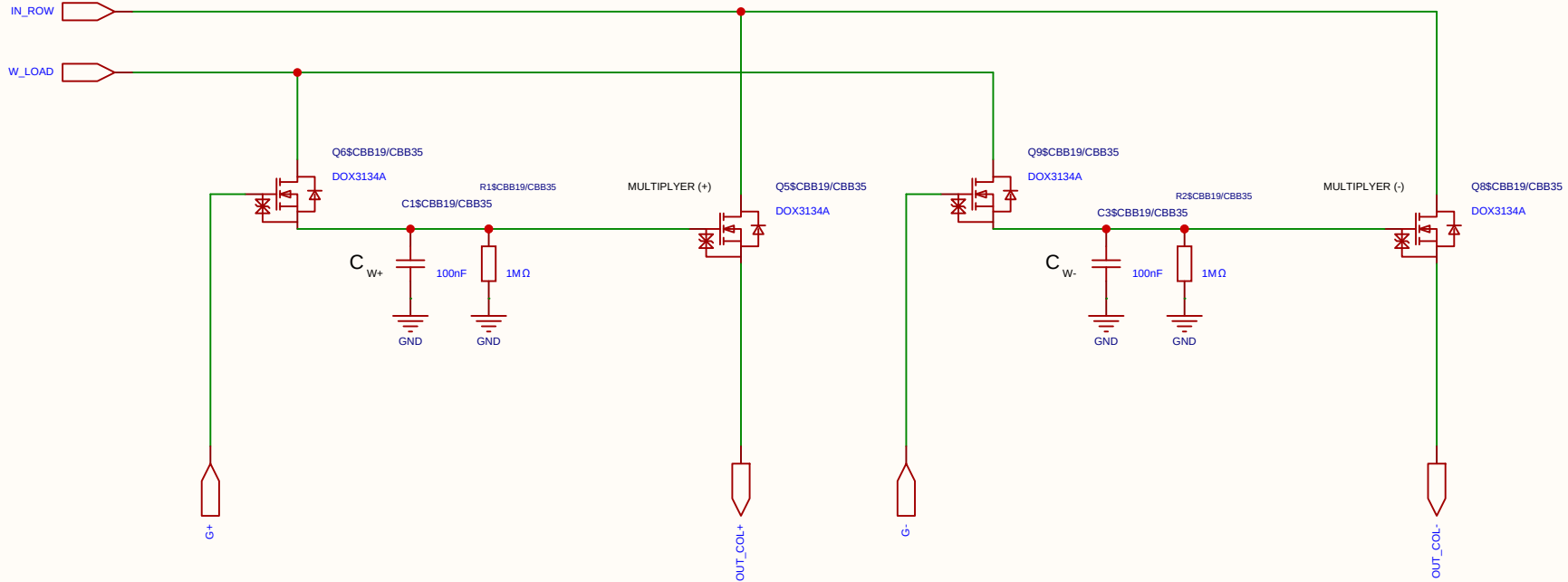
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

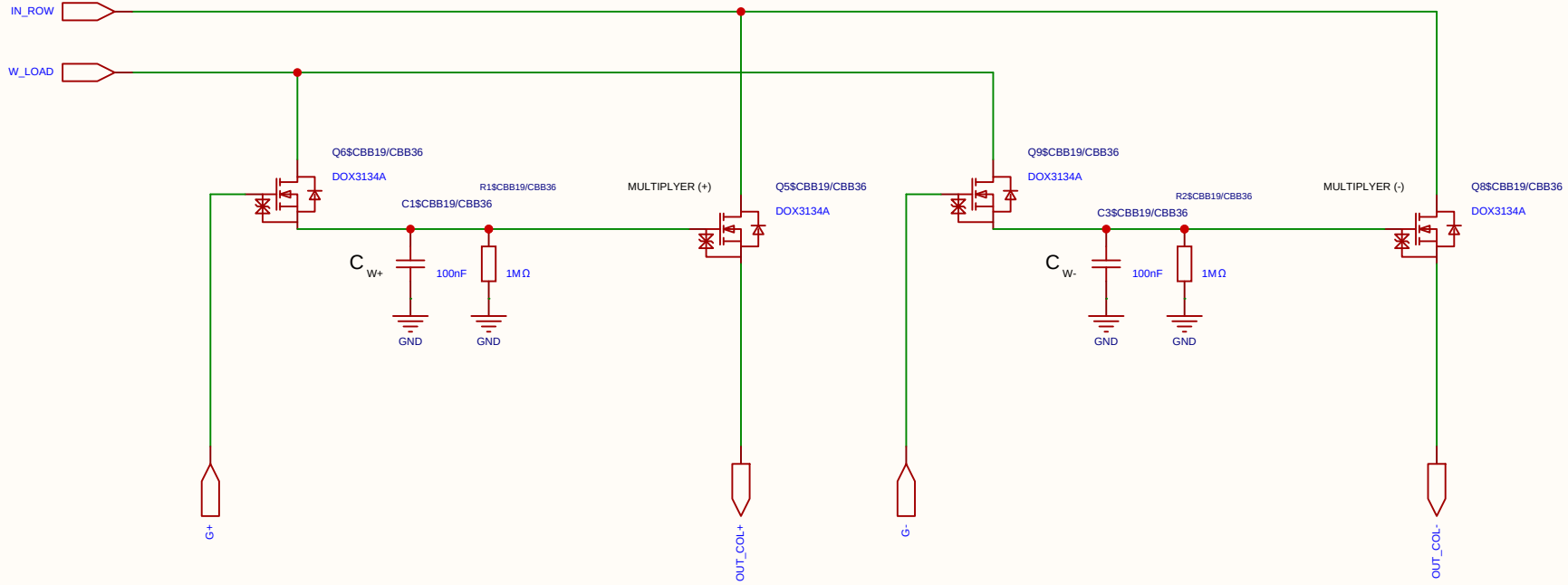
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

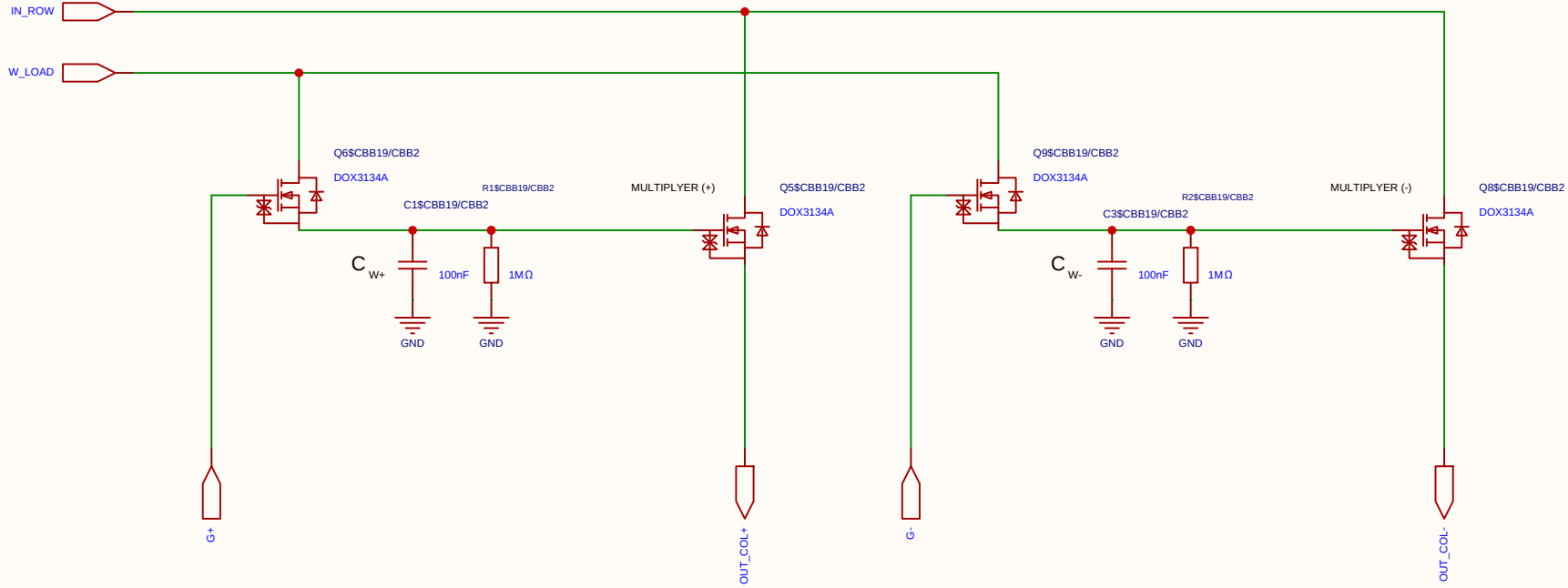
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

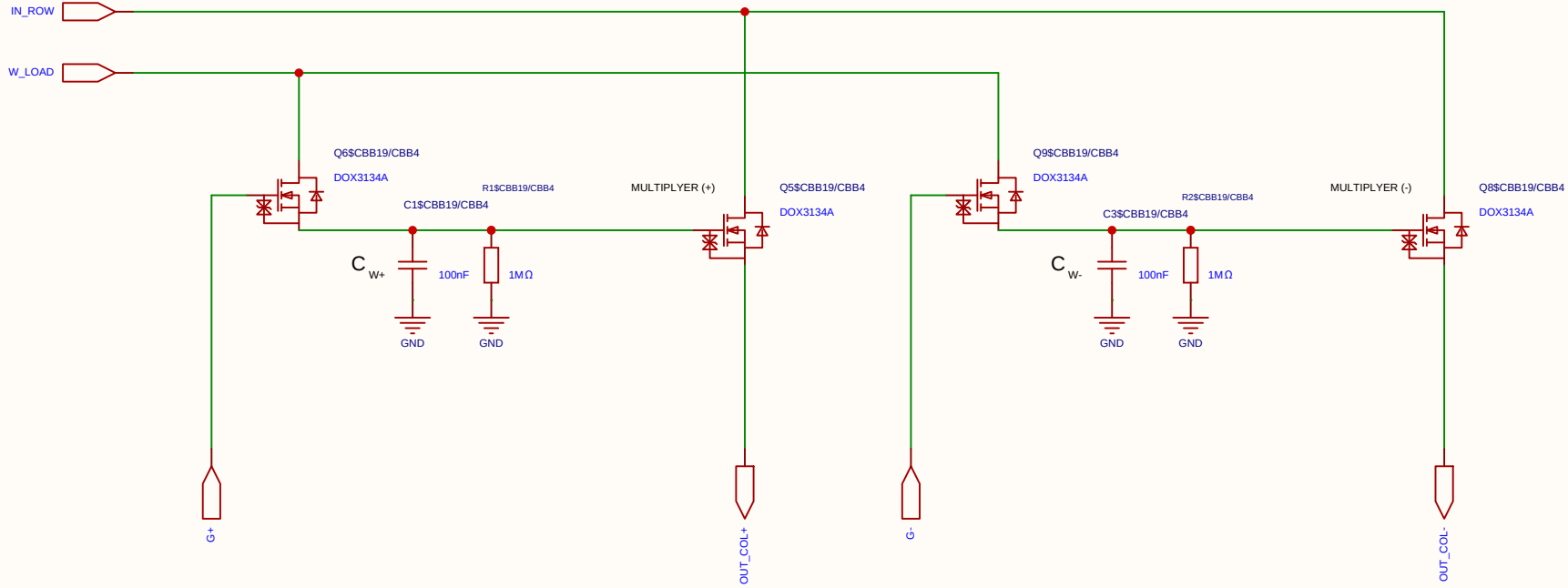
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel

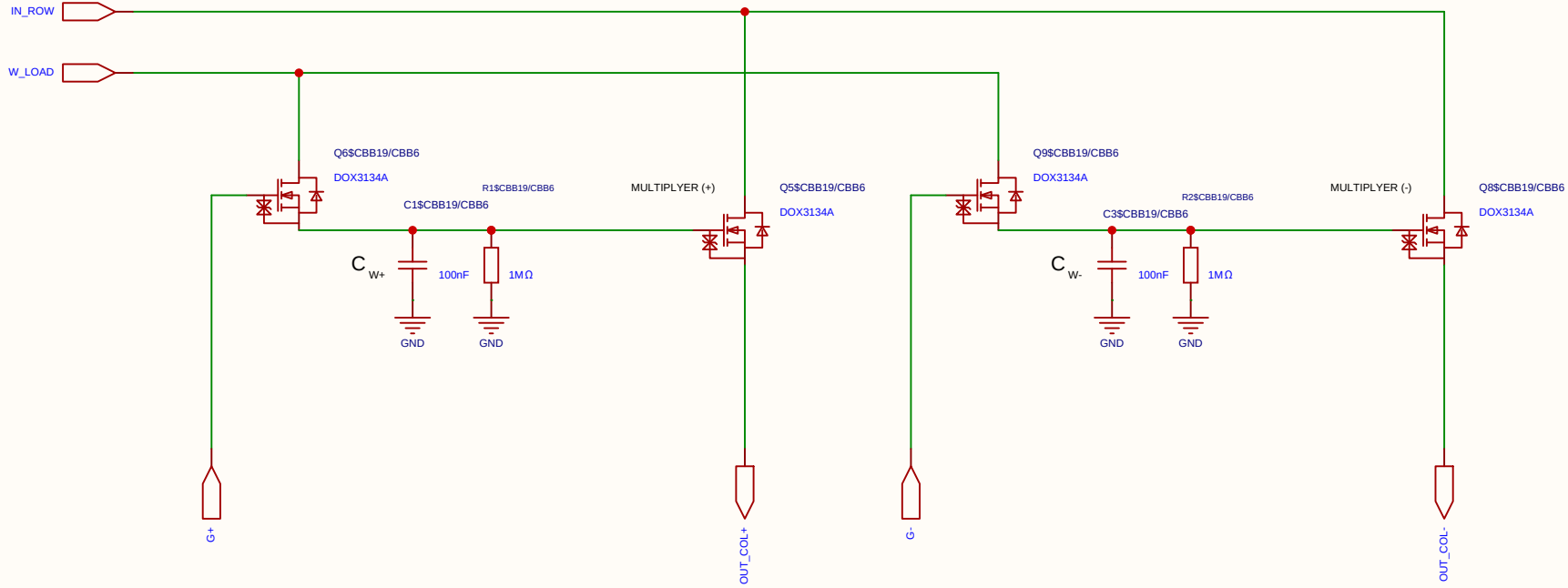
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C		Create at	2026-06-20
			Update at	2026-06-28
Board			Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel

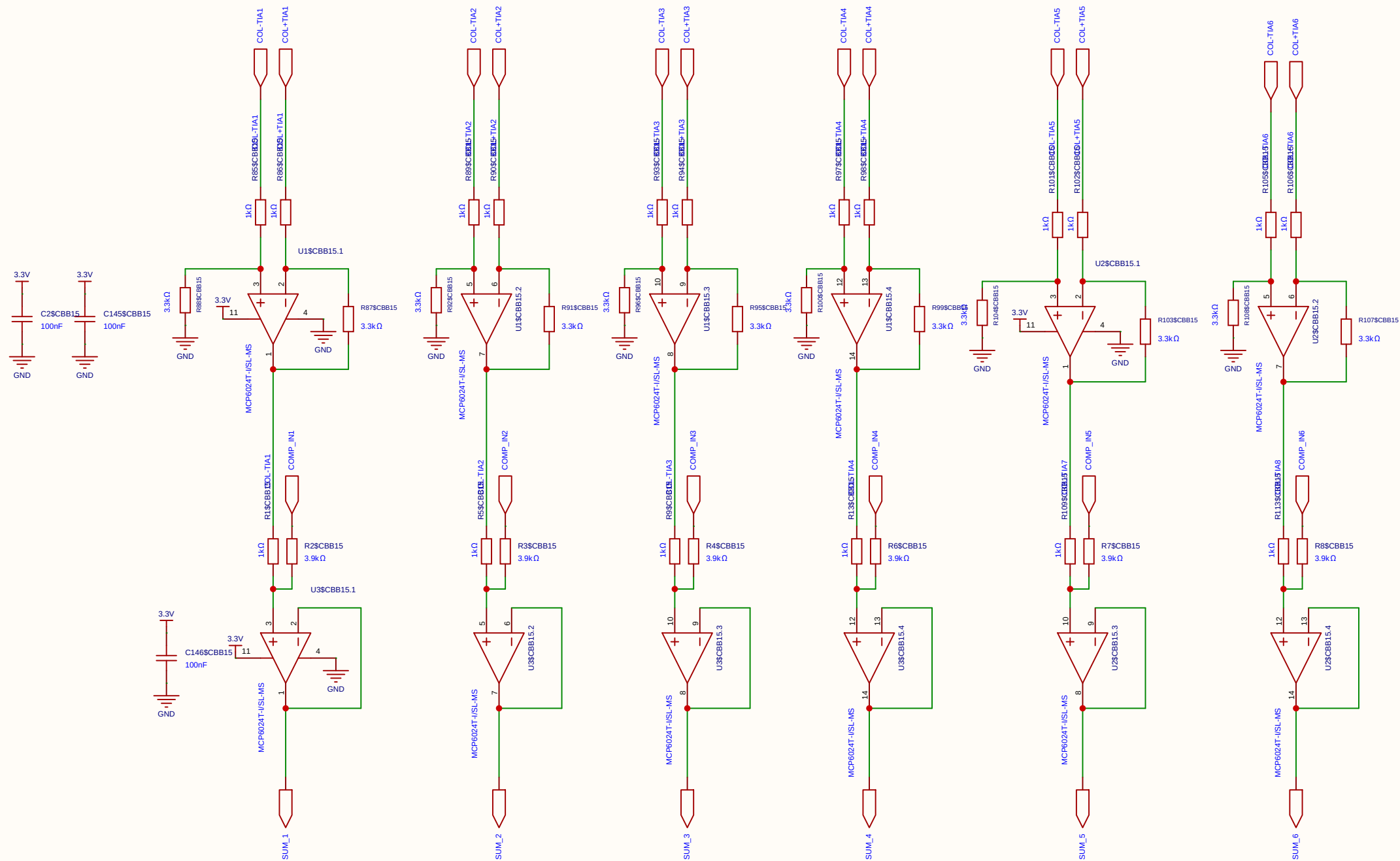
Differential 2x 2T1C



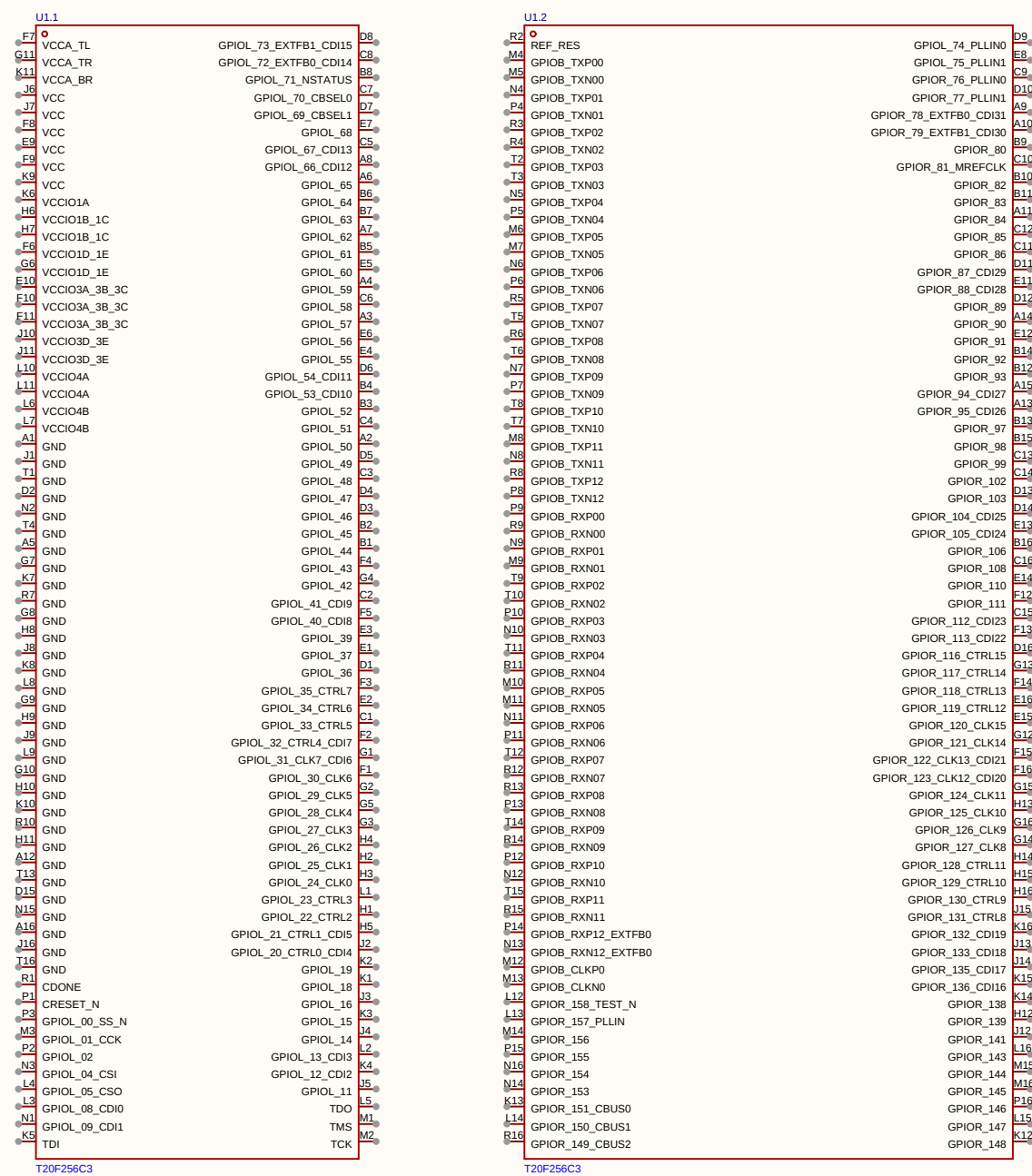
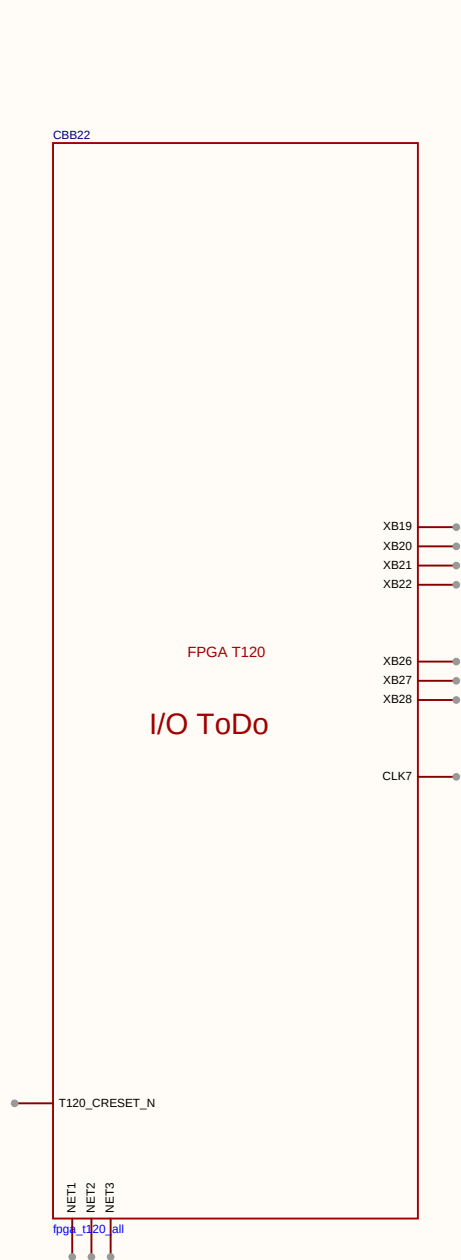
Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

• **Calibration Example:** To match a 3.3 V ADC, the resistors are configured for a Gain of 6.6×.

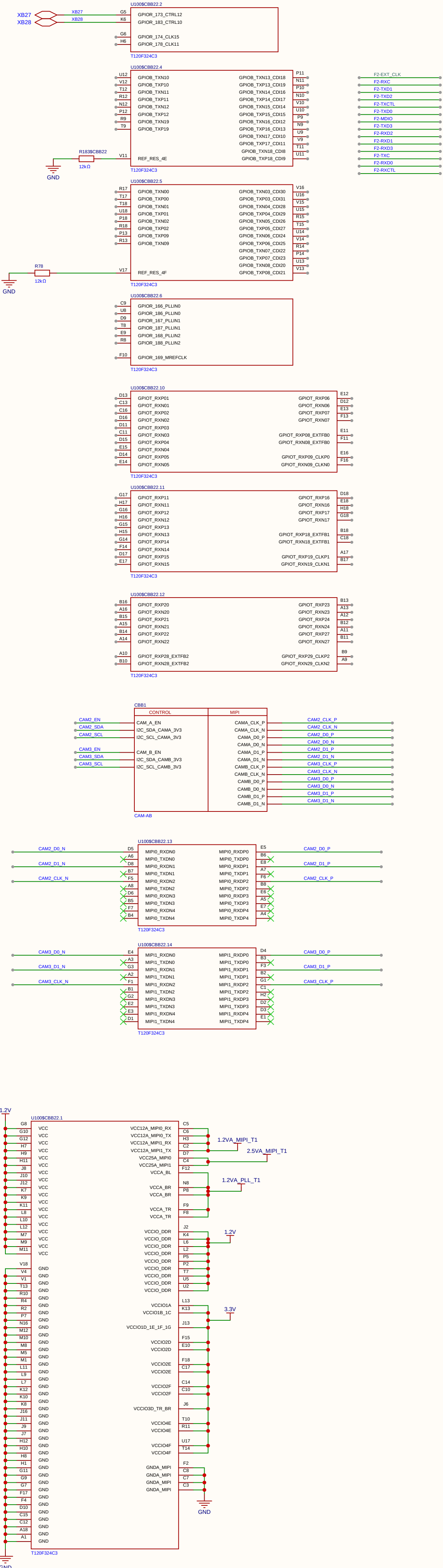
$V_{ADC} = 6.6 \times (1.65 \text{ V} - V_{out_TIA})$



Schematic	SUM35		Create at	2026-06-19
			Update at	2026-06-19
Board			Page	sum35
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

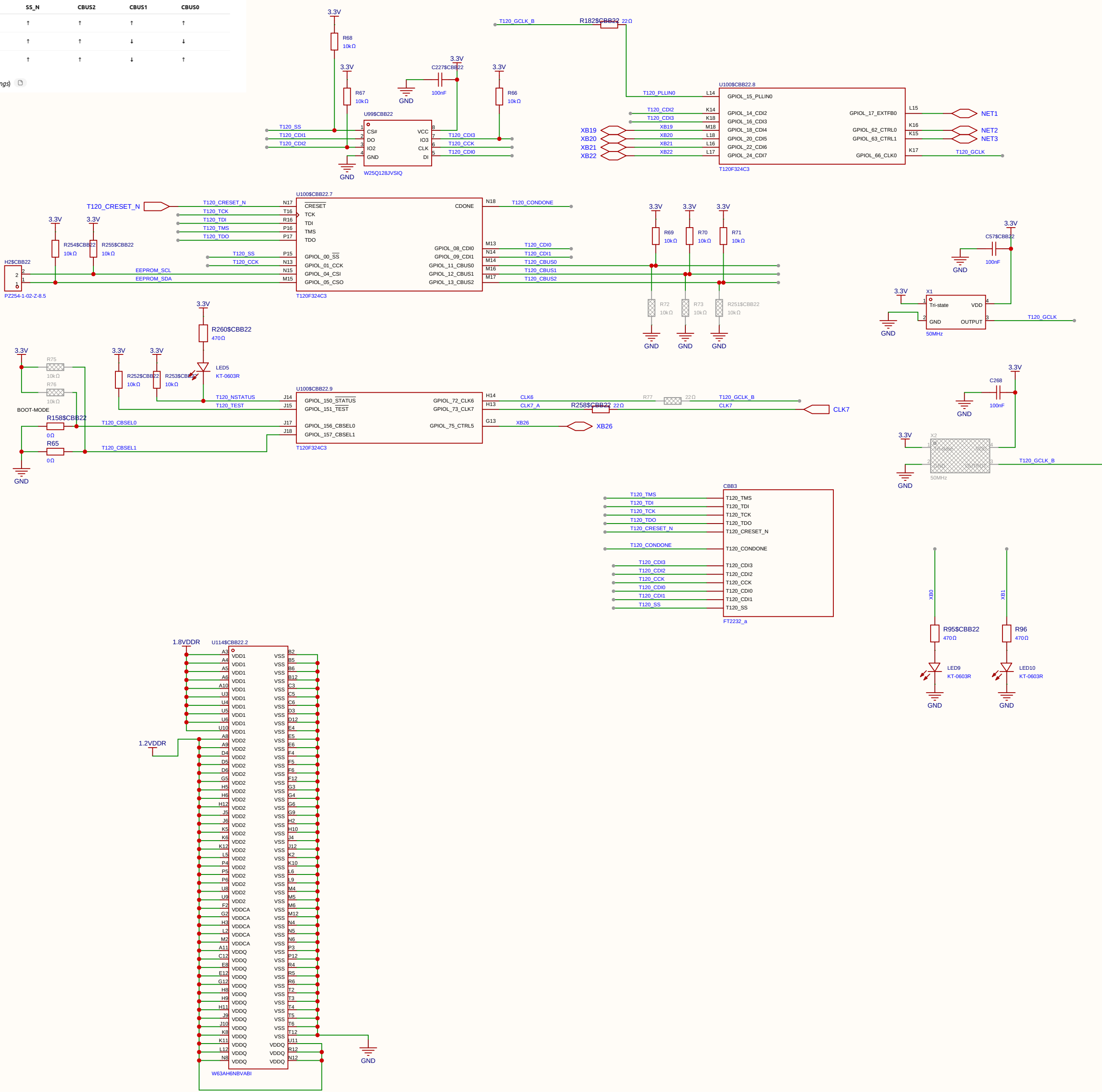


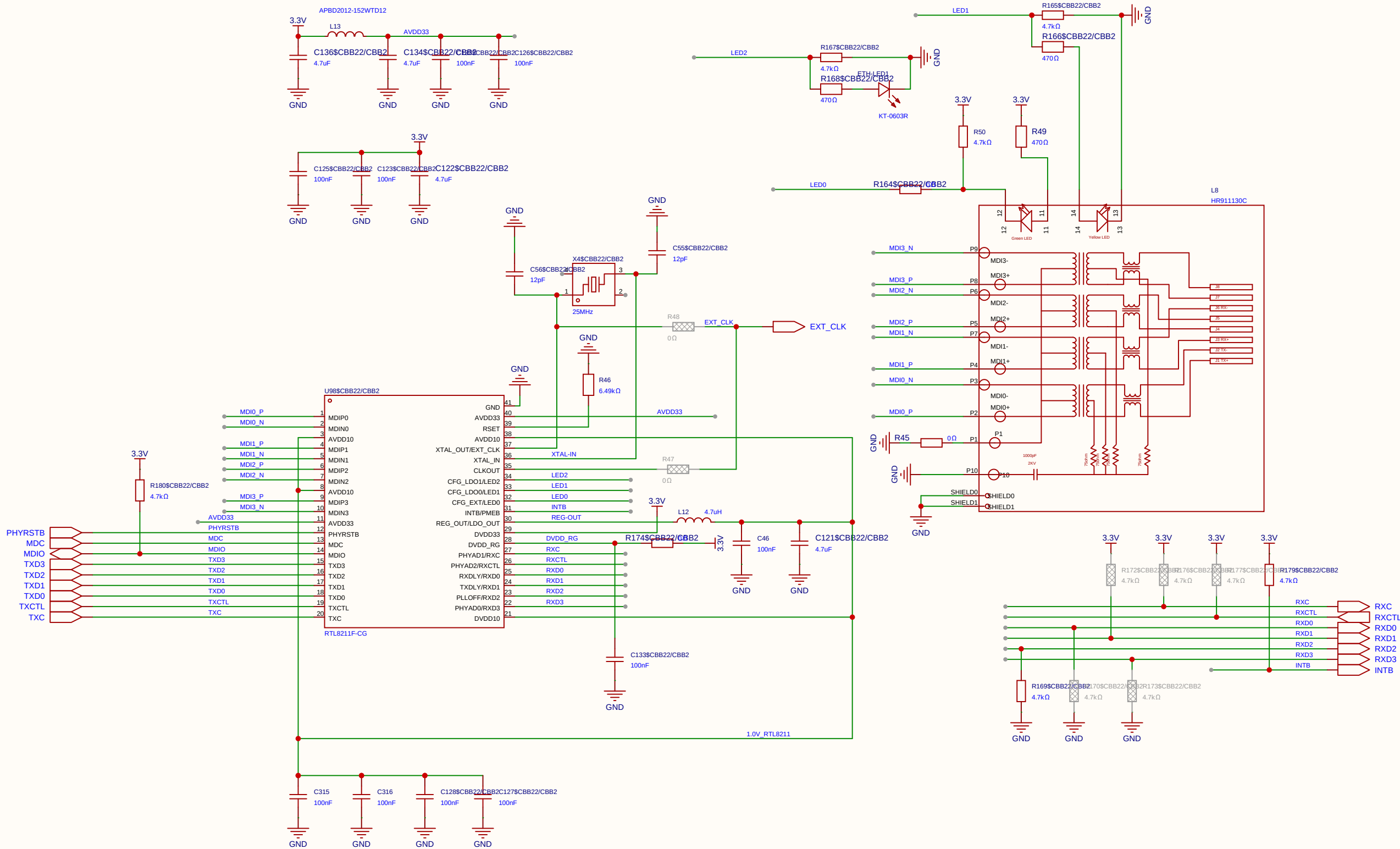
Schematic	Inverted_MAML			Create at	2026-06-25
Board	Analog_Matrix			Update at	2026-06-25
Drawn	Inverted_MAML_6x6				
Reviewed					
	Version	Size	Page 3 Total 3		
	V1.0	A4	EasyEDA.com		



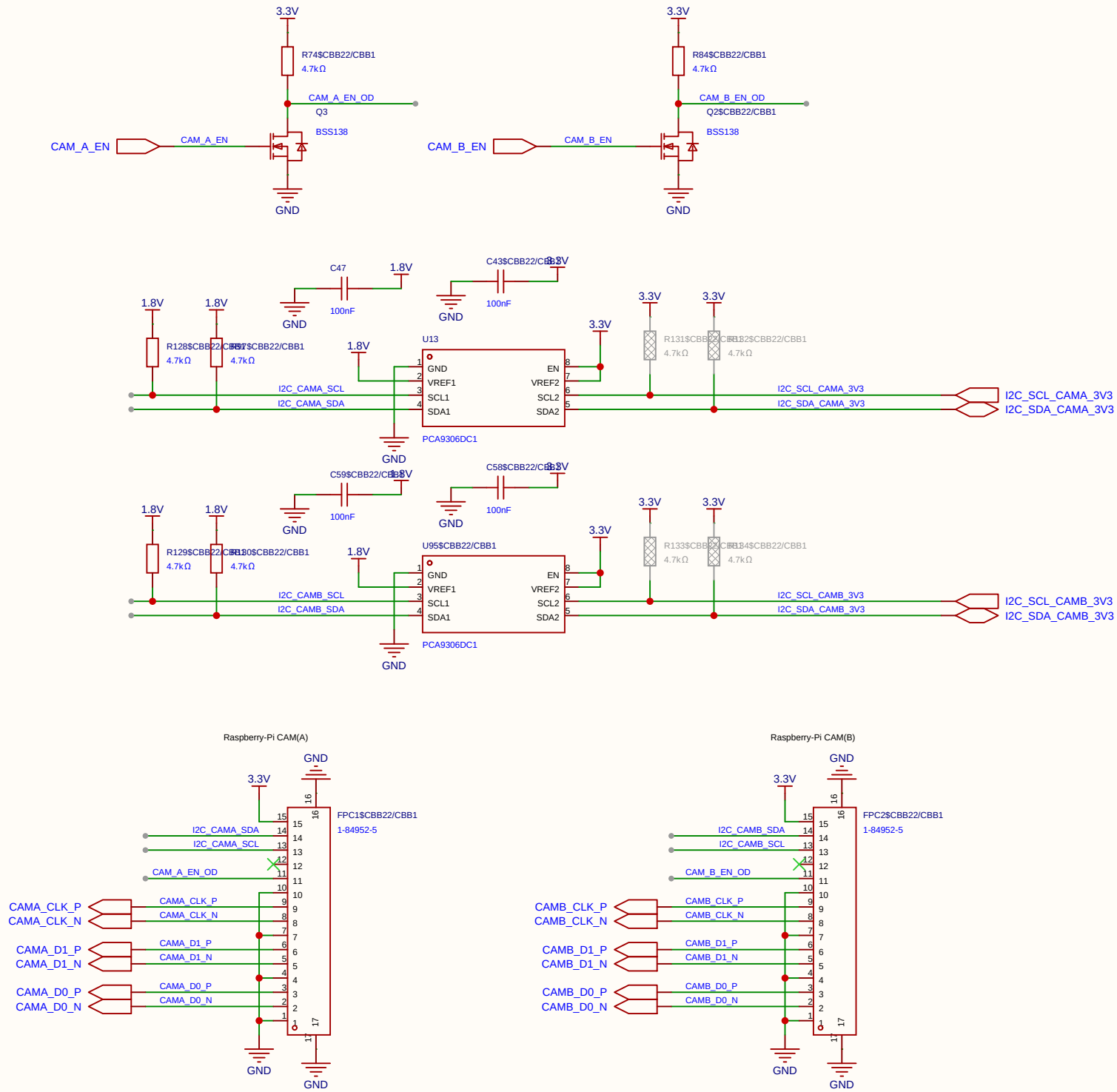
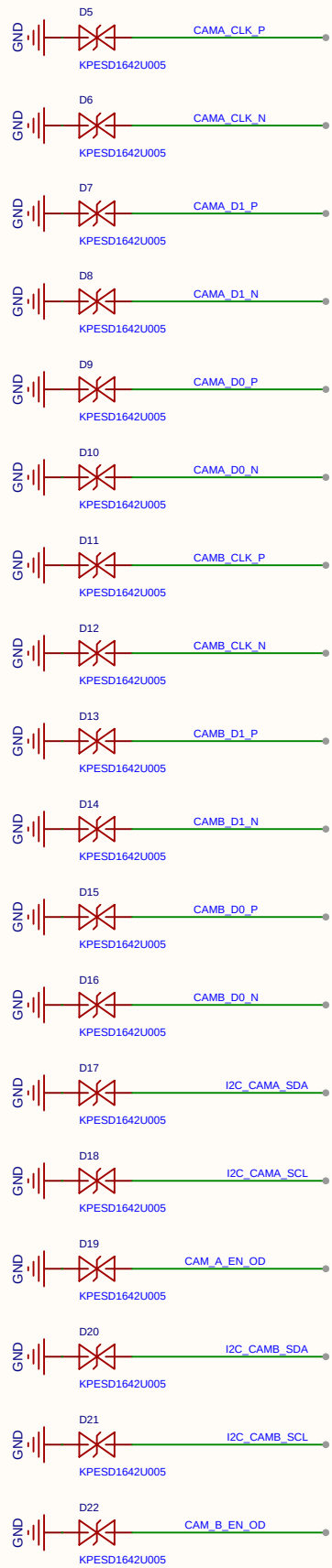
Wdrh	TEST_N	S1_N	CRU52	CRU51	CRU50
x1	1	1	1	1	1
x2	1	1	1	1	1
x4	1	1	1	1	1

(Table 5, SPI Hardware Settings)





Schematic	ETH_PHY			Create at	2026-06-25
				Update at	2026-06-25
Board				Page	P1
Drawn		Inverted_MAML_6x6			
Reviewed					
				Page 1 Total 1	
		V1.0	A4	EasyEDA.com	



Schematic	CAM-AB			Create at	2026-06-25
				Update at	2026-06-25
Board				Page	P1
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